Solver Specification and Verification Report:

`highOrderManufacturedFDAImplicitPETSc`

Manufactured monodomain-like PDE–ODE solver with high-order finite volumes, Picard, diagonal-I_{ion}, and JFNK nonlinear coupling on a PETSc Krylov backend

Pablo

July 1, 2026

Abstract

This report documents the numerical formulation implemented in:

`highOrderManufacturedFDAImplicitPETSc`.

The solver is a method of manufactured solutions (MMS) analogue of the electro-activation solver: it solves a monodomain-like reaction–diffusion PDE for the transmembrane potential V_m, evolves local state ODEs for (u_1, u_2, u_3), evaluates a nonlinear ionic-current surrogate $I_{\text{ion}}(V_m, u_1, u_2, u_3)$, and allows the same nonlinear coupling strategies used in the physiological solver: `Picard`, `diagonalIion`, and `JFNK`. The PDE can be discretised with the standard cell-centred finite-volume operator or with a high-order Local Regression Estimators (LRE) reconstruction. The nonlinear source can be evaluated at cell centres or integrated from cell quadrature points. In the default high-order source path the state ODEs are advanced once per cell and then reconstructed to the ionic quadrature points; a legacy option can instead advance the state ODE independently at each quadrature point. Time integration is a θ method with `backwardEuler` ($\theta = 1$) or `crankNicolson` ($\theta = 1/2$), and the cell-average mass operator can be `lumped` or high-order `consistent`. Linear voltage systems are solved through a thin PETSc wrapper that caches the KSP, matrix and ILU(T) symbolic factorisation across nonlinear iterations and time steps; an Eigen fallback (`SparseLU/BiCGSTAB`) is also available. JFNK uses a persistent matrix-free `MatShell`+KSP pair, defaulting to restarted GMRES, and an optional preconditioner derived from $\mathbf{A}$ with a local I_{ion} linearisation (`diagonalIion`). The sparse-matrix assemblies of $\mathbf{M}$ and $\mathbf{K}$ are chunked over cells/faces to bound the transient triplet-vector peak on large 3D unstructured meshes. The current implementation also exposes an automatic memory-optimisation mode with adaptive stiffness triplet reservation, compact consistent-mass assembly, conditional allocation of ionic quadrature-point fields, and allocator trimming around large setup phases. The results section is data driven: tables are generated from the current tutorial outputs and fitted directly from the aggregated `errors_vs_N` files.

Contents

1 Purpose and notation **2**

2 Governing equations **3**
- 2.1 Electrophysiology motivation . . . 3
- 2.2 Manufactured PDE–ODE system . . . 3
- 2.3 Exact solution and the role of β . . . 4

3 Spatial discretisation **4**
- 3.1 Finite-volume integral form . . . 4
- 3.2 Standard cell-centred operator . . . 5
- 3.3 High-order LRE operator . . . 5
- 3.4 Lumped and consistent mass matrices . . . 6
- 3.5 High-order averaging of I_{ion} and states . . . 6

4 Time integration **7**
4.1 The θ method . . . 7
4.2 ODE states inside one PDE time step . . . 7

5 Nonlinear coupled solve **7**
5.1 Residual and diagnostics . . . 7
5.2 Picard fixed-point method . . . 8
5.3 Diagonal-I_{ion} linearisation . . . 8
5.4 Jacobian-free Newton–Krylov . . . 9
5.5 Armijo line search and rollback . . . 10

6 Linear-algebra backend and implementation notes **10**
6.1 JFNK preconditioner: `diagonalIion` ILU(T) . . . 11
6.2 Chunked assembly of $\mathbf{M}$ and $\mathbf{K}$ . . . 12
6.3 Memory layout in the time loop . . . 12

7 Theoretical accuracy and practical error floors **13**

8 Solver controls **14**

9 Output diagnostics **16**

10 Error Measures and Convergence Rates **16**

11 Results **17**
11.1 Two-dimensional manufactured problem . . . 17
11.1.1 `Picard` . . . 18
11.1.2 `diagonalIion` . . . 23
11.1.3 `JFNK` . . . 28
11.2 Two-dimensional results: comments . . . 34
11.3 Three-dimensional manufactured problem . . . 34
11.3.1 `Picard` . . . 34
11.3.2 `diagonalIion` . . . 40
11.3.3 `JFNK` . . . 45

12 Notes **51**

1 Purpose and notation

The MMS problem is not a physiological ionic model. Its role is to provide an exact solution for a PDE–ODE–source structure that is deliberately close to the electro-activation solver. If the same finite-volume operator, same source quadrature, same state ODE integration, same θ time discretisation, and same nonlinear coupling are used here and in the electro solver, then the observed MMS convergence gives evidence for the numerical order of those shared building blocks.

Symbol	Meaning	Location in solver
$\Omega \subset \mathbb{R}^d$	computational domain, $d = 1, 2, 3$	OpenFOAM mesh
K_i	finite-volume cell	`mesh.cells()`
$\lvert K_i \rvert$	cell volume	`mesh.V()`
$V_m(\mathbf{x}, t)$	manufactured transmembrane potential	`Vm`
$\mathbf{u} = (u_1, u_2, u_3)$	manufactured state variables	`u1,u2,u3`
$I_{\text{ion}}(V_m, \mathbf{u})$	nonlinear source surrogate	`Iion`
χ and C_m	monodomain scaling constants	`cardiacProperties`
$\mathbf{D}$	conductivity tensor	`conductivity`
$\mathbf{M}$	cell-average mass operator	`M` (moved into `BImplicit` after $\mathbf{A}/\mathbf{B}$
$\mathbf{K}$	volume-normalised diffusion/stiffness operator	`K`
$\mathbf{L} = (\chi C_m)^{-1}\mathbf{K}$	scaled diffusion operator	absorbed into $\mathbf{A}/\mathbf{B}$ (no separate st
$\mathbf{A}, \mathbf{B}$	implicit θ-scheme matrices	`AImplicit,BImplicit`
$\mathsf{F}(\mathsf{x})$	nonlinear residual for voltage candidate x	`residualFor`
$\mathbf{P}$	persistent JFNK preconditioner matrix $\approx \mathbf{A} - \theta\,\mathrm{diag}(\mathbf{d})$	`jfnkP`

The cell-centred vector of voltages at time t^n is denoted $\mathsf{V}^n \in \mathbb{R}^{N_c}$. A nonlinear iteration inside one time step uses the superscript k, e.g. V^k is a candidate for V^{n+1}. Quadrature point q in cell K_i is denoted $\mathbf{x}_{iq}$ with positive weight ω_{iq}.

2 Governing equations

2.1 Electrophysiology motivation

The physiological monodomain equation can be written, in one common convention, as

$$\chi C_m \frac{\partial V_m}{\partial t} = \nabla \cdot (\mathbf{D} \nabla V_m) - \chi C_m \, I_{\text{ion}}(V_m, \mathbf{s}) + I_{\text{ext}}, \tag{1}$$

where $\mathbf{s}$ are ionic states and I_{ion} is often stored as a voltage rate with units $\mathrm{V\,s^{-1}}$. Dividing by χC_m gives

$$\frac{\partial V_m}{\partial t} = \frac{1}{\chi C_m} \nabla \cdot (\mathbf{D} \nabla V_m) - I_{\text{ion}}(V_m, \mathbf{s}) + \frac{I_{\text{ext}}}{\chi C_m}. \tag{2}$$

The manufactured solver replaces the physiological states by three smooth MMS states and replaces the cell-model ionic current by a constructed nonlinear function. The purpose is not to reproduce cellular electrophysiology, but to exercise the same numerical responsibilities: diffusion, local ODEs, nonlinear source, quadrature, implicit time stepping, and nonlinear iteration.

2.2 Manufactured PDE–ODE system

The solver advances

$$\frac{\partial V_m}{\partial t} = \frac{1}{\chi C_m} \nabla \cdot (\mathbf{D} \nabla V_m) - I_{\text{ion}}(V_m, u_1, u_2, u_3), \tag{3}$$

$$\dot{u}_1 = (u_1 + u_3 - V_m)^2 u_2^2 + \frac{1}{2}(u_1 + u_3 - V_m) u_2^2 (V_m - u_3), \tag{4}$$

$$\dot{u}_2 = -(u_1 + u_3 - V_m) u_2^3, \tag{5}$$

$$\dot{u}_3 = 0. \tag{6}$$

The nonlinear ionic-current surrogate is

$$I_{\text{ion}}(V_m, u_1, u_2, u_3) = -\frac{1}{2}(u_1 + u_3 - V_m) u_2^2 (V_m - u_3) + \frac{\beta}{\chi C_m}(V_m - u_3). \tag{7}$$

The PDE source used in the matrix right-hand side is

$$S(V_m, \mathbf{u}) = -I_{\text{ion}}(V_m, \mathbf{u}). \tag{8}$$

Thus the term called `Iion` in the output is the nonlinear manufactured current, while the term passed into the voltage equation is its negative.

2.3 Exact solution and the role of β

The exact fields are built from two smooth spatial functions F and G:

$$V_m^\star(\mathbf{x}, t) = \sqrt{1+t}\, F(\mathbf{x}), \tag{9}$$

$$u_1^\star(\mathbf{x}, t) = (1+t)G(\mathbf{x}) + V_m^\star(\mathbf{x}, t), \tag{10}$$

$$u_2^\star(\mathbf{x}, t) = \frac{1}{(1+t)\sqrt{G(\mathbf{x})}}, \tag{11}$$

$$u_3^\star(\mathbf{x}, t) = 0. \tag{12}$$

The solver uses

$$F = \begin{cases} \cos(\pi x), & d = 1, \\ \cos(\pi x)\cos(2\pi y), & d = 2, \\ \cos(\pi x)\cos(2\pi y)\cos(3\pi z), & d = 3, \end{cases} \qquad G = \begin{cases} 1 + x, & d = 1, \\ 1 + xy^2, & d = 2, \\ 1 + xy^2 z^3, & d = 3. \end{cases} \tag{13}$$

The constant β closes the analytical balance between the Laplacian of F and the conductivity tensor. For a diagonal constant conductivity sampled from `conductivity[0]`,

$$\beta = \begin{cases} -\pi^2 D_{xx}, & d = 1, \\ -\pi^2 (D_{xx} + 4D_{yy}), & d = 2, \\ -\pi^2 (D_{xx} + 4D_{yy} + 9D_{zz}), & d = 3. \end{cases} \tag{14}$$

This choice makes the manufactured source compatible with the exact trigonometric mode under the selected conductivity tensor. Boundary values for V_m, u_1, u_2, and u_3 are overwritten with the exact solution where appropriate, so the verification error is a discretisation error rather than a boundary-data error.

3 Spatial discretisation

3.1 Finite-volume integral form

Integrating (3) over a control volume K_i and dividing by the cell volume gives the cell-average balance used by the implementation:

$$\frac{1}{|K_i|}\int_{K_i} \frac{\partial V_m}{\partial t}\, dV = \frac{1}{\chi C_m |K_i|}\int_{\partial K_i} \mathbf{n}\cdot\mathbf{D}\nabla V_m\, dA + \frac{1}{|K_i|}\int_{K_i} S(V_m, \mathbf{u})\, dV. \tag{15}$$

After spatial discretisation the semi-discrete cell-average problem has the matrix form

$$\mathbf{M}\frac{d\mathsf{V}}{dt} = \mathbf{L}\mathsf{V} + \mathsf{S}(\mathsf{V}, \mathbf{u}), \qquad \mathbf{L} = \frac{1}{\chi C_m}\mathbf{K}. \tag{16}$$

Here $\mathbf{K}\mathsf{V}$ approximates the volume-normalised divergence $(1/|K_i|)\int_{\partial K_i} \mathbf{n}\cdot\mathbf{D}\nabla V_m\, dA$, and S_i is a cell-average source rate, not a volume-integrated source. Consequently the lumped finite-volume mass used by the code is the identity matrix.

3.2 Standard cell-centred operator

When `useHighOrder_Vm=false`, the solver assembles the standard cell-centred finite-volume diffusion operator. For an internal face f between cells P and N, the orthogonal part is of the form

$$\frac{1}{|K_P|}\int_f \mathbf{n}\cdot\mathbf{D}\nabla V_m\,dA \approx \frac{a_f}{|K_P|}(V_N - V_P), \tag{17}$$

where a_f contains the face area, face-normal direction, conductivity tensor, and distance between cell centres. Boundary faces use the OpenFOAM patch type. `zeroGradient` and `empty` patches give no-flux contributions; fixed value patches contribute known boundary terms.

For smooth solutions on regular orthogonal meshes, this standard finite-volume operator is second-order accurate in space for diffusion. On skewed or non-orthogonal meshes the effective order depends on the mesh quality and the non-orthogonal treatment.

3.3 High-order LRE operator

When `useHighOrder_Vm=true`, the solver uses Local Regression Estimators (LRE) stencils. Around each cell centre $\mathbf{x}_i$, a polynomial reconstruction of degree N is fitted from neighbouring cell values:

$$V_m(\mathbf{x}_i + \mathbf{r}) \approx V_i + \nabla V_i\cdot\mathbf{r} + \frac{1}{2}\mathbf{r}^T H_i\mathbf{r} + \frac{1}{6}T_i[\mathbf{r},\mathbf{r},\mathbf{r}] + \cdots. \tag{18}$$

The dictionary entries `N`, `Nn`, `k`, and `maxStencilSize` control the polynomial degree, target number of stencil neighbours, radial weight exponent, and maximum stencil size. The solver can evaluate reconstructed values, gradients, Hessians, and third-order tensors, depending on the selected polynomial order.

Boundary patches are classified automatically before the LRE objects are constructed. `fixedValue` and `fixedVoltage` patches are inserted into the LRE stencils as boundary ghost points so that the manufactured boundary value can be used. `zeroGradient` patches are passed to LRE as `mirrorPatches`, which doubles the local stencil by reflecting neighbour locations across the boundary plane and preserves the full reconstruction order at homogeneous Neumann boundaries. `empty`, processor, and unsupported patches are left out of the explicit include/mirror lists. A patch cannot be both included as a Dirichlet ghost patch and mirrored; the LRE constructor checks this and aborts if the dictionary is inconsistent.

The high-order dictionaries are read from `highOrderCoeffs/LRECoeffs_Vm` for the diffusion/mass reconstruction and from `highOrderCoeffs/LRECoeffs_Iion` for source/state quadrature. The entry `LRECoeffs_Iion` is required when high-order ionic quadrature is enabled. The name `LRECoeffs_states` no longer aliases the ionic quadrature coefficients; it is reserved for the optional state-reconstruction interpolator used by `stateIntegrationMode=cellCentredReconstruct`.

The high-order stiffness matrix is assembled by evaluating reconstructed gradients at face quadrature points:

$$(\mathbf{K}\mathsf{V})_i \approx \frac{1}{|K_i|}\sum_{f\subset\partial K_i}\sigma_{if}|f|\sum_{q\in f}\omega_{fq}\,\mathbf{n}_f\cdot\mathbf{D}_{\mathrm{own}(f)}\nabla_h V_m(\mathbf{x}_{fq}), \tag{19}$$

where σ_{if} is the owner/neighbour orientation sign. For the manufactured verification cases the conductivity is constant, so using the owner-cell tensor in this face reconstruction is equivalent to any consistent face interpolation. Fixed-value boundaries contribute known exact-boundary terms to a separate vector $\mathsf{b}(t)$. The resulting operator is sparse but has a wider stencil than the standard finite-volume Laplacian. For smooth fields, good stencils, sufficiently accurate quadrature, and non-degenerate meshes, the expected spatial consistency improves with the LRE polynomial degree. In practice the observed order is the minimum of: reconstruction order, quadrature order, time-integration order, ODE tolerance floor, nonlinear residual floor, and mesh/stencil conditioning. If the error is controlled only by the interpolation/reconstruction of V_m, then the expected behaviour is the usual polynomial one: linear reconstruction (`p1`) gives second-order accuracy, quadratic reconstruction (`p2`) gives third-order accuracy, and cubic reconstruction (`p3`) gives fourth-order accuracy.

3.4 Lumped and consistent mass matrices

The `lumped` mass operator is the identity:

$$\mathbf{M}_{ij}^{\text{lump}} = \delta_{ij}. \tag{20}$$

This follows from the volume-normalised finite-volume form in (15): the unknowns are advanced as cell-average rates, so the volume factor has already been divided out of both the diffusion operator and the source. The identity mass is cheap, robust, positive, and coincides with the usual one-degree-of-freedom cell-centred update. If high-order V_m is disabled, the solver uses this diagonal mass even if the input asks for `consistent`, because there is no high-order reconstruction operator to integrate.

The `consistent` mass matrix is available when `useHighOrder_Vm=true` and is assembled with cell quadrature and the LRE reconstruction coefficients. For quadrature point $\mathbf{x}_{iq}$ in cell K_i, write the local Taylor/LRE reconstruction as

$$V_{mh}(\mathbf{x}_{iq}) = \sum_j c_{ijq} V_j, \tag{21}$$

where the non-zero coefficients c_{ijq} are restricted to the cell stencil of K_i and include the self coefficient. The implemented consistent cell-average mass is then

$$\mathbf{M}_{ij}^{\text{cons}} \approx \sum_{q \in K_i} \bar{\omega}_{iq}\, c_{ijq}, \qquad \bar{\omega}_{iq} = \frac{\omega_{iq}}{\sum_{q'} \omega_{iq'}}. \tag{22}$$

It is therefore a high-order averaging/projection operator induced by the LRE reconstruction, not a Galerkin mass matrix with products of basis functions. It better matches the high-order spatial approximation, but it is sparse rather than diagonal and increases the cost and conditioning sensitivity of the implicit solve.

3.5 High-order averaging of I_{ion} and states

The cell-average source rate can be approximated either by a cell-centred rule or by quadrature. Since $S = -I_{\text{ion}}$, when `useHighOrder_Iion=false` or when the mesh is effectively one-dimensional, the solver uses the cell-centred rate

$$\mathsf{S}_i \approx -I_{\text{ion}}(V_i, u_{1,i}, u_{2,i}, u_{3,i}). \tag{23}$$

When `useHighOrder_Iion=true` and $d > 1$, the voltage and states are evaluated at ionic quadrature points and the source rate is averaged back to the cell:

$$\mathsf{S}_i \approx -\sum_{q \in K_i} \bar{\omega}_{iq}\, I_{\text{ion}}\left(V_{mh}(\mathbf{x}_{iq}), u_{1,iq}, u_{2,iq}, u_{3,iq}\right). \tag{24}$$

These quadrature-point scalar fields are allocated only for multi-dimensional high-order ionic integration, i.e. when `useHighOrder_Iion=true` and $d > 1$. In all other cases the solver still reports the nominal number of ionic integration points for auditability, but the allocated quadrature-point storage is zero and the source/state update stays cell-centred.

There are two state-update modes for the high-order ionic path:

- `cellCentredReconstruct` (default) advances the ODE once per cell from the time-step-start states using the cell-centred old voltage and the current cell-centred voltage candidate. The updated cell-centred states are then reconstructed to ionic quadrature points with the separate `LREC oeffs_states` interpolator before evaluating I_{ion}. This is the default because it costs one ODE integration per cell.

- `gaussPointODE` reconstructs the old and candidate voltages to each ionic quadrature point and advances an independent state ODE at every quadrature point. The quadrature-point states are then averaged back to cell centres. This legacy path costs one ODE integration per quadrature point.

4 Time integration

4.1 The θ method

Let $\Delta t = t^{n+1} - t^n$. The solver maps `backwardEuler` to $\theta = 1$ and `crankNicolson` to $\theta = 1/2$. Applying the θ method to (16) gives

$$\mathbf{M}\frac{\mathsf{V}^{n+1} - \mathsf{V}^n}{\Delta t} = \mathbf{L}\left(\theta\mathsf{V}^{n+1} + (1-\theta)\mathsf{V}^n\right) + \theta\mathsf{S}^{n+1} + (1-\theta)\mathsf{S}^n + \theta\mathsf{b}^{n+1} + (1-\theta)\mathsf{b}^n. \tag{25}$$

Equivalently,

$$\mathbf{A}\mathsf{V}^{n+1} = \mathbf{B}\mathsf{V}^n + \theta\mathsf{S}^{n+1} + (1-\theta)\mathsf{S}^n + \theta\mathsf{b}^{n+1} + (1-\theta)\mathsf{b}^n, \tag{26}$$

$$\mathbf{A} = \frac{1}{\Delta t}\mathbf{M} - \theta\mathbf{L}, \qquad \mathbf{B} = \frac{1}{\Delta t}\mathbf{M} + (1-\theta)\mathbf{L}. \tag{27}$$

In the implementation, $\mathbf{L}$ is never stored as a separate sparse matrix. The solver builds $\mathbf{M}$ and $\mathbf{K}$ once (with the chunked assemblies described in §6.2) and constructs $\mathbf{A}$, $\mathbf{B}$ directly from the relation $\mathbf{L} = (\chi C_m)^{-1}\mathbf{K}$ via expression-template arithmetic on the sparse matrices: $\mathbf{A} \leftarrow \mathbf{M}/\Delta t - (\theta/\chi C_m)\mathbf{K}$ and $\mathbf{B} \leftarrow \mathbf{M}/\Delta t + ((1-\theta)/\chi C_m)\mathbf{K}$. After $\mathbf{A}$ is copied from $\mathbf{M}$, the storage of $\mathbf{M}$ is moved into $\mathbf{B}$ to avoid keeping a third copy of the mass matrix alive throughout the run.

`backwardEuler` is first order in time and strongly damping. It is usually the most robust choice for stiff or poorly converged nonlinear problems. `crankNicolson` is second order in time for smooth solutions and converged nonlinear/source terms, but is less dissipative. Its observed order can drop if the nonlinear iterations, state ODE tolerance, or spatial error dominate.

4.2 ODE states inside one PDE time step

For a voltage candidate $V_m^{n+1,k}$, the state ODEs are advanced from the same time-step-start states $\mathbf{u}^n$. Inside the ODE integrator, voltage is interpolated linearly between the old and candidate values:

$$V_m(\tau) = (1-\alpha)V_m^n + \alpha V_m^{n+1,k}, \qquad \alpha = \frac{\tau}{\Delta t}, \quad 0 \le \tau \le \Delta t. \tag{28}$$

The available state solvers are:

`stateODESolver`	Method	Nominal order
`Euler` or `forwardEuler`	explicit Euler	1
`RK4`	classical Runge–Kutta	4
`RKF45` or `RKT45`	adaptive embedded Fehlberg pair	4/5

For `RKF45`, the fifth-order candidate is accepted and the fourth-order embedded estimate supplies the local error indicator

$$e = \max_j \frac{|u_j^{(5)} - u_j^{(4)}|}{\texttt{stateODEAbsTol} + \texttt{stateODERelTol}\max(|u_j^{(5)}|, |u_j^{(4)}|)}. \tag{29}$$

A sub-step is accepted if $e \le 1$ or if the minimum step has been reached; the next sub-step is scaled by a safety factor proportional to $e^{-1/4}$. In convergence studies, the ODE tolerances must be tight enough that the state integration error does not mask the spatial or PDE time-integration order.

5 Nonlinear coupled solve

5.1 Residual and diagnostics

For any voltage candidate x, the solver evaluates the corresponding state fields and source by calling `evaluateNonlinearFields`. This means: reconstruct V_m to I_{ion} quadrature points if required,

reset/evolve states from t^n to t^{n+1} using x, compute I_{ion}, and assemble the source. The coupled voltage residual is

$$\mathsf{F}(\mathsf{x}) = \mathbf{A}\mathsf{x} - \left[\mathbf{B}\mathsf{V}^n + \theta\left(\mathsf{S}(\mathsf{x}) + \mathsf{b}^{n+1}\right) + (1-\theta)\left(\mathsf{S}^n + \mathsf{b}^n\right)\right], \tag{30}$$

where b^n and b^{n+1} are known boundary/source contributions from the manufactured boundary treatment.

The solver records several relative diagnostics:

$$R_{\text{cpl}} = \frac{\|\mathsf{F}(\mathsf{x})\|_2}{\|\mathsf{r}(\mathsf{x})\|_2 + \epsilon}, \qquad R_{V_m} = \frac{\left\|\mathsf{V}^{k+1} - \mathsf{V}^k\right\|_2}{\|\mathsf{V}^k\|_2 + \epsilon}, \tag{31}$$

$$R_{I_{\text{ion}}} = \frac{\left\|\mathsf{I}_{\text{ion}}^{k+1} - \mathsf{I}_{\text{ion}}^{k}\right\|_2}{\left\|\mathsf{I}_{\text{ion}}^{k}\right\|_2 + \epsilon}, \qquad R_{u_j} = \frac{\left\|\mathsf{u}_j^{k+1} - \mathsf{u}_j^{k}\right\|_2}{\left\|\mathsf{u}_j^{k}\right\|_2 + \epsilon}. \tag{32}$$

Here ϵ is the OpenFOAM `SMALL` safeguard. When high-order I_{ion} quadrature uses `gaussPointODE`, state residuals are measured at the quadrature points. In the default `cellCentredReconstruct` mode, and in the low-order source path, the authoritative states are cell-centred and the state residuals are measured at cell centres. The maximum state residual is

$$R_{\mathbf{u}} = \max(R_{u_1}, R_{u_2}, R_{u_3}). \tag{33}$$

The residual history is written to `postProcessing/highOrderManufacturedFDAImplicitPETSc/nonlinearResiduals.dat` when `writeNonlinearResiduals=true`.

5.2 Picard fixed-point method

Picard treats the nonlinear source as lagged during each linear voltage solve. Given V^k, the solver computes $\mathsf{S}^k = \mathsf{S}(\mathsf{V}^k, \mathbf{u}(\mathsf{V}^k))$ and then solves

$$\mathbf{A}\widehat{\mathsf{V}}^{k+1} = \mathbf{B}\mathsf{V}^n + \theta(\mathsf{S}^k + \mathsf{b}^{n+1}) + (1-\theta)(\mathsf{S}^n + \mathsf{b}^n). \tag{34}$$

The relaxed update is

$$\mathsf{V}^{k+1} = \mathsf{V}^k + \omega\left(\widehat{\mathsf{V}}^{k+1} - \mathsf{V}^k\right), \qquad \omega = \texttt{nonlinearRelaxation}. \tag{35}$$

Convergence requires at least `minNonlinearIterations`; in the Picard branch the active criteria are $R_{V_m} \leq$ `nonlinearVmTolerance` and all state residuals below `nonlinearStatesTolerance`. If Picard exhausts the iteration budget, the current implementation accepts the last Picard iterate and prints a warning, matching the legacy `highOrderManufacturedFDAImplicit` behaviour. The source vector used for the Picard residual diagnostic is the lagged source from the current corrector evaluation; the fully coupled residual is not a Picard stopping criterion in this branch.

5.3 Diagonal-I_{ion} linearisation

`diagonalIon` keeps the same scalar voltage unknowns but adds an approximate row-wise source sensitivity to the matrix. The implementation perturbs all internal voltage unknowns by the same scalar `diagonalIonEpsilon`, re-imposes the exact boundary values, and uses the result as a diagonal surrogate:

$$d_i \approx \frac{S_i(\mathsf{V}^k + \varepsilon\mathbf{1}) - S_i(\mathsf{V}^k)}{\varepsilon}, \qquad \varepsilon = \texttt{diagonalIonEpsilon}. \tag{36}$$

The linearised source is

$$S_i(\mathsf{V}^{k+1}) \approx S_i(\mathsf{V}^k) + d_i\left(V_i^{k+1} - V_i^k\right). \tag{37}$$

Moving the unknown part to the left gives

$$\left(\mathbf{A} - \theta\,\mathrm{diag}(\mathbf{d})\right)\widehat{\mathsf{V}}^{k+1} = \mathsf{r}(\mathsf{S}^k) - \theta\,\mathbf{d} \odot \mathsf{V}^k, \tag{38}$$

Algorithm 1 Picard nonlinear coupling

Input: Old solution $\mathsf{V}^n, \mathbf{u}^n$, source S^n

Build $\mathbf{A}$ and $\mathbf{B}$ from $\mathbf{M}$, $\mathbf{L}$, θ, and Δt
Initialise V^0 from V^n and impose exact boundary values at t^{n+1}
for $k = 0, \ldots, N_{\mathrm{nl}} - 1$ **do**
    Evaluate states and source S^k from V^k
    Solve $\mathbf{A}\widehat{\mathsf{V}}^{k+1} = \mathsf{r}(\mathsf{S}^k)$
    Relax $\mathsf{V}^{k+1} = \mathsf{V}^k + \omega(\widehat{\mathsf{V}}^{k+1} - \mathsf{V}^k)$
    Apply exact boundary values; the source is refreshed at the next Picard corrector
    Compute R_{V_m}, R_{u_1}, R_{u_2}, R_{u_3}, $R_{I_{\mathrm{ion}}}$
    if minimum iterations reached and Picard tolerances hold **then**
        accept and exit loop
    end if
end for

Output: Last Picard iterate is accepted; a warning is printed if not converged

where $\odot$ is componentwise multiplication and

$$\mathsf{r}(\mathsf{S}^k) = \mathbf{B}\mathsf{V}^n + \theta\left(\mathsf{S}^k + \mathsf{b}^{n+1}\right) + (1-\theta)\left(\mathsf{S}^n + \mathsf{b}^n\right). \tag{39}$$

This is the same right-hand side used by the Picard voltage solve; the `diagonalIion` branch only moves the local linearised source contribution to the left-hand side. This is not a full Newton Jacobian. For a strictly cell-local source it approximates the local voltage sensitivity, including the effect of re-integrating the states for the perturbed candidate. With high-order reconstruction/quadrature active, it should be read as a diagonal surrogate for a uniform-direction row sensitivity: off-diagonal Jacobian couplings from reconstruction and state reconstruction are not assembled. It is often more implicit than Picard while remaining cheaper and simpler than JFNK.

Algorithm 2 `diagonalIion` nonlinear coupling

Input: Old solution $\mathsf{V}^n, \mathbf{u}^n$, source S^n

Initialise V^0 from V^n and impose exact boundary values at t^{n+1}
for $k = 0, \ldots, N_{\mathrm{nl}} - 1$ **do**
    Evaluate S^k and I_{ion}^k from V^k
    Perturb all internal V_m values by `diagonalIionEpsilon`, re-apply boundary values, and re-evaluate source
    Form the diagonal surrogate d_i by the row-wise finite difference
    Solve $(\mathbf{A} - \theta\,\mathrm{diag}(\mathbf{d}))\,\widehat{\mathsf{V}}^{k+1} = \mathsf{r}(\mathsf{S}^k) - \theta\mathbf{d} \odot \mathsf{V}^k$
    Relax the voltage update
    Re-evaluate source/states at the relaxed voltage
    Check coupled, voltage, I_{ion}, and optional state criteria
    if minimum iterations reached and diagonal tolerances hold **then**
        accept and exit loop
    end if
end for
if not converged and `nonlinearAcceptUnconverged=false` **then**
    roll back V_m, I_{ion}, and states to the beginning of the time step
end if

5.4 Jacobian-free Newton–Krylov

JFNK seeks a zero of (30):

$$\mathsf{F}(\mathsf{x}) = 0. \tag{40}$$

Newton's method would solve

$$\mathbf{J}(\mathsf{x}_k)\delta_k = -\mathsf{F}(\mathsf{x}_k), \qquad \mathsf{x}_{k+1} = \mathsf{x}_k + \alpha\delta_k, \tag{41}$$

but the full Jacobian is not assembled. Instead, a Krylov method only needs products $\mathbf{Jv}$, approximated by the finite difference

$$\mathbf{J}(\mathsf{x})\mathbf{v} \approx \frac{\mathsf{F}(\mathsf{x}+\epsilon\mathbf{v}) - \mathsf{F}(\mathsf{x})}{\epsilon}, \qquad \epsilon = \texttt{jfnkEpsilon}\,\frac{1+\|\mathsf{x}\|_2}{\|\mathbf{v}\|_2 + \epsilon_{\text{mach}}}. \tag{42}$$

Each matrix-free Krylov product therefore performs a full nonlinear residual evaluation: ODE solve, source evaluation, source quadrature if active, and matrix-vector application. This is the main cost of JFNK.

With the PETSc backend, the matrix-free solve is delegated to the persistent shell `KSP`; the default `jfnkPetscKspType=gmres` gives restarted GMRES with restart length `jfnkPetscRestart`, but other PETSc KSP types can be selected through the dictionary or the PETSc options database. With the Eigen fallback, the in-file implementation uses restarted GMRES(m), where $m =$ `jfnkMaxKrylovIterations` and the number of restarts is `jfnkMaxRestarts`; its Arnoldi basis is orthogonalised in the code and the least-squares projected problem is solved with Eigen's column-pivoted Householder QR. The configured Krylov solve stops when its relative residual is below `jfnkLinearTolerance` or when the iteration/restart budget is exhausted.

5.5 Armijo line search and rollback

After the Krylov solve proposes a Newton correction δ_k, the solver optionally uses Armijo backtracking. Starting from $\alpha =$ `nonlinearRelaxation`, it accepts the trial step if

$$\|\mathsf{F}(\mathsf{x}_k + \alpha\delta_k)\|_2^2 \le (1 - 2c\alpha)\,\|\mathsf{F}(\mathsf{x}_k)\|_2^2, \qquad c = \texttt{jfnkArmijoC}. \tag{43}$$

If the condition fails, α is halved until either the condition is met, `jfnkLineSearchMaxIter` is reached, or `jfnkLineSearchAlphaMin` is reached. This globalises Newton by discouraging steps that increase the nonlinear residual.

For `JFNK` and `diagonalIion`, if the nonlinear solver reaches the maximum number of iterations without convergence, the default behaviour is a rollback: V_m, I_{ion}, and the state fields are restored to the beginning of the time step. Setting `nonlinearAcceptUnconverged=true` disables this rollback and accepts the last iterate. Picard is the exception: to preserve the legacy manufactured implicit behaviour, an unconverged Picard iteration accepts the last Picard iterate with a warning.

6 Linear-algebra backend and implementation notes

The PDE-level operators $\mathbf{A}$, $\mathbf{B}$, $\mathbf{K}$, and the JFNK preconditioner $\mathbf{P}$ are stored as Eigen `SparseMatrix<scalar,RowMaj` while the heavy linear solves run through PETSc by default. The solver wraps the PETSc API in two small classes that exist in the anonymous namespace of the translation unit:

- `PetscKspMatrixSolver` owns a (matrix, KSP, preconditioner) triple. The first call to `reset()` translates an Eigen `SparseMatrix` into a PETSc `MatSeqAIJ`, builds the `KSP`, and triggers the symbolic ILU(T) factorisation. Subsequent updates call `updateValues()`: this zeros the PETSc `Mat` in place (`MatZeroEntries`), refreshes its numerical entries (`MatSetValue`/`MatAssemblyEnd`), and asks the PETSc preconditioner to recompute its numerical factor while reusing the existing symbolic structure (`KSPSetReusePreconditioner(PETSC_FALSE)` followed by `KSPSetUp`). This means $\mathbf{P}$ can be refreshed many times per time step without freeing and re-allocating the ILU(T) factor.

- `PetscShellKspSolver` wraps a matrix-free shell-matrix used by JFNK. A persistent `Mat` of type `MatShell`, a `KSP`, the right-hand side and solution vectors, and the two callback contexts (`matVec`,

Algorithm 3 Jacobian-free Newton–Krylov nonlinear coupling

Input: Old solution $\mathsf{V}^n, \mathbf{u}^n$, old source S^n

Build initial guess x_0 from V^n or extrapolation
for $k = 0, \ldots, N_{\mathrm{nl}} - 1$ **do**
  Evaluate $\mathsf{F}(\mathsf{x}_k)$, states, source, and I_{ion}
  if minimum iterations reached and nonlinear tolerances hold **then**
    accept x_k and exit loop
  end if
  Define matrix-free product $\mathbf{v} \mapsto [\mathsf{F}(\mathsf{x}_k + \epsilon\mathbf{v}) - \mathsf{F}(\mathsf{x}_k)]/\epsilon$
  Use the configured matrix-free Krylov solve to approximate $\mathbf{J}\delta_k = -\mathsf{F}(\mathsf{x}_k)$
  if `jfnkLineSearch=true` **then**
    Set $\alpha \leftarrow$ `nonlinearRelaxation`
    while Armijo condition (43) fails and budget remains **do**
      $\alpha \leftarrow \max(\alpha/2, \texttt{jfnkLineSearchAlphaMin})$
    end while
  else
    $\alpha \leftarrow$ `nonlinearRelaxation`
  end if
  Update $\mathsf{x}_{k+1} = \mathsf{x}_k + \alpha\delta_k$
  Re-evaluate states/source at x_{k+1} and write residual diagnostics
  if minimum iterations reached and nonlinear tolerances hold **then**
    accept x_{k+1} and exit loop
  end if
end for
if not converged and `nonlinearAcceptUnconverged=false` **then**
  roll back V_m, I_{ion}, and states to their time-step-start values
end if

`applyPreconditioner`) are allocated once outside the time loop. Each Newton iteration only refreshes the `std::function` payloads of those contexts and calls `KSPSolve`; the PETSc `KSP` object and its work storage persist across solves. Numerically this is identical to constructing a fresh shell matrix per Newton iteration, but it avoids allocating and freeing roughly $40\, m\, N_c$ scalars per JFNK call (with $m =$ `jfnkPetscRestart` and N_c the number of cells).

For users who prefer to stay in pure Eigen, `linearSolverBackend` can be set to `Eigen`. In that mode the voltage solve uses `Eigen::SparseLU` or `Eigen::BiCGSTAB` and the JFNK inner GMRES uses the in-file implementations `solveGMRES` (no preconditioner) or `solveLeftPreconditionedGMRES` with the Eigen `IncompleteLUT`. The two backends are numerically equivalent for the same matrix and tolerances: they differ only in the linear solver implementation, not in the PDE residual.

PETSc type names are normalised before the KSP/PC objects are created. KSP aliases such as `PETSc`, `GMRES`, `KSPGMRES`, and `JFNK` select PETSc `gmres`; `BiCGSTAB/BCGS` select `bcgs`; and `SparseLU/LU` select `preonly`. PC aliases `off`, `false`, `none`, and `noPC` select PETSc `none`; `ILUT` selects `ilu`; `SparseLU/LU` select `lu`; and the common misspelling `jakobi` is accepted as `jacobi`. If `petscUseOptions=true`, PETSc's options database can still override solver details through the configured options prefix.

6.1 JFNK preconditioner: `diagonalIion` ILU(T)

When `jfnkPreconditioner=diagonalIion`, the inner GMRES applies a left preconditioner $\mathbf{P}^{-1}$ defined implicitly by an ILU(T) factorisation of

$$\mathbf{P} = \mathbf{A} - \theta\,\mathrm{diag}(\mathbf{d}), \qquad d_i = \frac{S_i(\mathsf{x} + \varepsilon\mathbf{1}) - S_i(\mathsf{x})}{\varepsilon}, \tag{44}$$

which is exactly the matrix that the `diagonalIion` nonlinear method would solve. The preconditioner solver type (`jfnkPreconditionerKspType`, `jfnkPreconditionerPcType`) defaults to a direct application

of the ILU(T) factor (`preonly+ilu`). The two ILU(T) controls `jfnkPreconditionerDropTolerance` and `jfnkPreconditionerFillFactor` are passed verbatim to PETSc's `PCFactorSetDropTolerance` and `PCFactorSetFill`. The preconditioner is refreshed every `jfnkPreconditionerUpdateFrequency` Newton iterations inside a time step. Across time steps it stays alive through the cached `PetscKspMatrixSolver` above. Selecting `jfnkPreconditioner=diffusion` uses $\mathbf{A}$ as the preconditioner matrix without the diagonal source correction. Selecting `jfnkPreconditioner=none` disables the left preconditioner and runs the matrix-free Krylov solve without the shell preconditioner.

The persistent matrix $\mathbf{P} =$ `jfnkP` is allocated once. The first preconditioner update copies $\mathbf{A}$ into $\mathbf{P}$ (full sparse copy); all subsequent updates only overwrite the value buffer `P.valuePtr()` from `AImplicit.valuePtr()` via `std::copy` and then add the diagonal correction in place. This is correct because the sparsity pattern of $\mathbf{A}$ is invariant in time, and the `addDiagonalToMatrix` helper only touches diagonal entries that already exist in $\mathbf{A}$ (the identity/lumped mass path and the LRE consistent-mass self coefficients provide the diagonal entries used by the correction).

6.2 Chunked assembly of M and K

The high-order stiffness matrix is the dominant transient memory cost of the setup phase. On a 3D unstructured tetrahedral mesh with ∼290k cells, p_3 LRE reconstruction, ∼60-cell stencils, and stabilisation active, the monolithic assembly accumulates of order 4×10^8 Eigen triplets in a single `std::vector<Triplet>`. Because the initial reservation cannot match this exactly without an extra pre-pass, the vector would grow by capacity-doubling, briefly holding both the old and new buffer in memory.

To avoid that transient peak, the solver assembles both $\mathbf{M}$ and $\mathbf{K}$ in fixed-size chunks. For $\mathbf{K}$ the chunks are blocks of `faceChunk = 50 000` internal faces; for $\mathbf{M}$ they are blocks of `cellChunk = 50 000` cells. Each chunk fills a triplet buffer of bounded size, builds a chunk-local sparse matrix $\mathbf{K}_{\text{local}}$ (resp. $\mathbf{M}_{\text{local}}$) via `setFromTriplets`, and folds it into the accumulated matrix:

$$\mathbf{K} \leftarrow \mathbf{K} + \mathbf{K}_{\text{local}}, \qquad \mathbf{M} \leftarrow \mathbf{M} + \mathbf{M}_{\text{local}}, \tag{45}$$

where the first chunk uses assignment instead of accumulation. After the chunked sweep over internal faces, the boundary-face contributions are appended through the same flush mechanism. The triplet buffer is cleared between chunks and explicitly released (`std::vector<Triplet>().swap(triplets)`) at the end of the routine.

For the stiffness matrix, the chunk triplet reservation is controlled by `stiffnessTripletsPerFaceReserve`. With `memoryOptimization=off` this keeps the conservative historical value of 800 triplets per face. With `memoryOptimization=on`, or with `memoryOptimization=auto` on large 3D high-order cases, the reserve is selected from `memoryOptimizationFluxTripletsPerFace` for flux-only assemblies and from `memoryOptimizationStabilisedTripletsPerFace` when stabilisation is active. This changes only the capacity of the temporary `std::vector`; the set of triplets that pass `addTripletIfNeeded(...)` is unchanged.

For the consistent high-order mass matrix, `compactMassAssembly` can accumulate all quadrature-point contributions to a cell row in a small local buffer before emitting triplets. The compact path therefore emits one triplet per final row/stencil entry instead of one triplet per quadrature contribution. Mathematically it assembles the same mass matrix; only the local floating-point summation order is changed. The first non-empty mass and stiffness chunks are moved into the global Eigen sparse matrix with `swap()`, avoiding an extra full sparse copy during setup.

Numerically the chunked assembly is identical to the monolithic one: sparse matrix addition is associative and commutative, and the per-triplet filter `addTripletIfNeeded(...)` is independent of the chunking because it inspects each triplet value before insertion.

6.3 Memory layout in the time loop

Several sparse matrices and PETSc objects live across the time loop, so the solver is careful about which copies are kept:

- $\mathbf{M}$ is moved (not copied) into $\mathbf{B}$ once the implicit matrices are assembled. After this point $\mathbf{M}$ is empty and only $\mathbf{A}$ and $\mathbf{B}$ hold the time-discrete operators.
- In the Picard/`diagonalIion` branches, the modified system matrix `ACurrent` $= \mathbf{A} - \theta\,\mathrm{diag}(\mathbf{d})$ is allocated once outside the corrector loop; subsequent corrector iterations refresh its value buffer from $\mathbf{A}$ via `std::copy` and re-apply the diagonal correction in place. The branch is also skipped entirely for pure Picard when the PETSc cached solver is used, since that branch never needs a separately stored linearised matrix.
- The Picard solve uses a single `PetscKspMatrixSolver` cached across the run via `reset()` on the first call. Because $\mathbf{A}$ is time-constant in the Picard branch, no further numerical updates are needed.
- The JFNK preconditioner solver uses the `updateValues()` fast path described above, so the ILU(T) factor is built once symbolically and only its numerical entries are recomputed per refresh.
- The seven ionic quadrature-point scalar fields (`VmOld`, `VmGuess`, source, `Iion`, and three states) are sized with `allocatedIionIntegrationPoints`. This is zero unless high-order ionic integration is actually active in $d > 1$, so pure cell-centred source runs do not pay for unused quadrature-point storage.
- The diagnostic high-order flux/laplacian reconstruction is not evaluated on every time step. The solver always forms the numerical laplacian used in the RHS diagnostic from the already assembled matrix expression $\mathbf{K}\mathsf{V} + \mathsf{b}$. The expensive `computeHighOrderLaplacian` path that fills `fluxVm_HO` is only run when an output time needs the flux field, and once more during final post-processing.
- On glibc systems the solver tightens the allocator before any large allocation: `M_ARENA_MAX` is set to 2 to prevent the OpenMP ODE thread pool from forcing dozens of arenas, `M_MMAP_THRESHOLD` and `M_TRIM_THRESHOLD` are lowered to 64 KiB so that mid-sized Eigen temporaries during the LRE QR factorisations are placed in `mmap`'d regions that are returned to the OS on free. When `memoryOptimizationTrimHeap=true` and the memory optimisation mode is effective, `malloc_trim(0)` is called after the mass-matrix assembly, after the stiffness-matrix assembly, and after $\mathbf{A}/\mathbf{B}$ are built; this releases the heap padding left behind by the very large number of small Eigen allocations performed during LRE setup.
- When `profileTimings=true`, setup also prints peak-RSS checkpoints before and after mass assembly, before and after stiffness assembly, and after implicit matrix construction. These checkpoints report the monotone `getrusage(RUSAGE_SELF).ru_maxrss` peak, not the instantaneous RSS.

These choices do not change any numerical output; they only change peak resident memory and the cost of repeated linear solves.

7 Theoretical accuracy and practical error floors

For a smooth manufactured solution and exact nonlinear solves, the intended orders are:

Component	Expected order	Main caveat
standard FV diffusion	second order in space on regular orthogonal meshes	mesh quality
LRE high-order reconstruction	increases with polynomial degree N	stencil quality and quadrature
`backwardEuler`	first order in time	numerical damping
`crankNicolson`	second order in time	requires converged nonlinear/source terms
`Euler` states	first order in time	rarely appropriate for high-order studies
`RK4` states	fourth order for smooth ODEs	fixed step over PDE Δt
`RKF45` states	adaptive fourth/fifth-order pair	tolerance sets error floor
Picard/JFNK/diagonal	nonlinear iteration, not spatial order by itself	residual tolerance must be small enough

The measured MMS order is the order of the entire chain. It can be limited by the largest of the following errors: spatial reconstruction, mass matrix approximation, source quadrature, PDE time stepping, state ODE integration, linear solver tolerance, nonlinear residual tolerance, and floating-point or conditioning effects. In particular, a high-order spatial method will not show its nominal order unless Δt, ODE tolerances, and nonlinear tolerances are sufficiently tight.

8 Solver controls

The numerical controls are read from `constant/spatialIntegrationProperties` and `constant/timeIntegrationProp` the tutorial `run_convergence.py` writes these dictionaries.

Control	Meaning
`useHighOrder_Vm`	Selects standard FV or LRE high-order diffusion/mass reconstruction for V_m.
`useHighOrder_Iion`	Selects cell-centred source evaluation or high-order ionic quadrature in $d > 1$. State placement is then controlled by `stateIntegrationMode`.
`useHighOrder_states`	Legacy alias used only if `useHighOrder_Iion` is absent.
`implicitScheme`	`backwardEuler` or `crankNicolson`; sets θ.
`massMatrix`	`lumped` identity cell-average mass operator or LRE `consistent` high-order averaging mass.
`highOrderCoeffs/stabilisationAlpha`	Optional positive face-stabilisation strength for the high-order diffusion operator; defaults to zero.
`highOrderCoeffs/LRECoeffs_Vm`	LRE settings `N`, `Nn`, `k`, and `maxStencilSize` for V_m diffusion and the consistent mass matrix.
`highOrderCoeffs/LRECoeffs_Iion`	Required LRE settings `N`, `Nn`, `k`, and `maxStencilSize` for ionic quadrature.
`highOrderCoeffs/LRECoeffs_states`	Required only for `stateIntegrationMode=cellCentredReconstruct` with high-order ionic quadrature; controls reconstruction of cell-centred states to ionic quadrature points.
`stateIntegrationMode`	`cellCentredReconstruct` (default, one ODE solve per cell plus state reconstruction) or `gaussPointODE` (legacy, one ODE solve per ionic quadrature point).
`linearSolverBackend`	`PETSc` (default) routes the voltage solve through PETSc; `Eigen` keeps the original in-process Eigen path.
`linearSolver`	Eigen sparse solver name when the Eigen backend is selected: `BiCGSTAB` or `SparseLU`.
`tolerance`, `maxIterations`	Linear PDE solver stopping tolerance and iteration cap.
`petscLinearKspType`, `petscLinearPcType`, `petscLinearRestart`	PETSc KSP type (`gmres`, `bcgs`, `preonly`, ...), preconditioner type (`ilu`, `lu`, `none`, ...) and GMRES restart length for the Picard voltage solve.
`petscLinearOptionsPrefix`, `petscUseOptions`	Options-database prefix and switch that lets the user override KSP/PC settings from the PETSc options database.
`nonlinearMethod`	`Picard`, `diagonalIion`, or `JFNK`.
`nonlinearIterations`	Maximum nonlinear iterations per PDE time step.
`minNonlinearIterations`	Minimum nonlinear iterations before convergence can be accepted.
`nonlinearTolerance`	Coupled residual tolerance for JFNK/diagonal branches.
`nonlinearVmTolerance`	Relative L_2 tolerance for the voltage update.
`nonlinearStatesTolerance`	Relative L_2 tolerance for state updates.
`nonlinearRequireStatesConvergence`	If true, state residuals are part of the JFNK/diagonal stopping test.

Control	Meaning
`nonlinearAcceptUnconverged`	If false, JFNK/diagonal roll back on nonlinear failure.
`nonlinearIionTolerance`	Relative L_2 tolerance for I_{ion} changes.
`nonlinearRelaxation`	Picard relaxation or initial Newton step length.
`jfnkLinearSolverBackend`	`PETSc` (default) for the persistent shell-Krylov solver or `Eigen` for the in-file GMRES implementations.
`jfnkMaxKrylovIterations`	GMRES restart length m (also the Krylov budget for the matrix-free solve).
`jfnkMaxRestarts`	Number of restarted GMRES cycles.
`jfnkLinearTolerance`	GMRES relative residual tolerance.
`jfnkPetscKspType`, `jfnkPetscPcType`, `jfnkPetscRestart`, `jfnkPetscOptionsPrefix`	PETSc KSP/PC settings and options-database prefix used by the persistent shell-Krylov solver.
`jfnkEpsilon`	Scaling of the finite-difference Jacobian-vector perturbation.
`jfnkInitGuessOrder`	JFNK-only initial guess: constant old voltage for order 0, or linear extrapolation from the previous two committed voltages for order ≥ 1.
`jfnkInitGuessVmMin/Max`	Clamp for extrapolated guess and optionally for ODE input.
`jfnkClampODEInput`	If true, clamps voltage passed to state ODE evaluations.
`jfnkLineSearch`, `jfnkLineSearchMaxIter`, `jfnkLineSearchAlphaMin`, `jfnkArmijoC`	Armijo backtracking controls.
`jfnkPreconditioner`	Left preconditioner for the JFNK Krylov solve: `none`, `diffusion` (alias `linear`, `AImplicit`) or `diagonalIion` (alias `diagonal`, `localDiagonal`).
`jfnkPreconditionerKspType`, `jfnkPreconditionerPcType`	KSP and PC types for the preconditioner solve, typically `preonly+ilu` to apply the cached ILU(T) factor directly.
`jfnkPreconditionerDropTolerance`, `jfnkPreconditionerFillFactor`	PETSc ILU(T) drop tolerance and fill factor for the JFNK preconditioner factorisation.
`jfnkPreconditionerTolerance`, `jfnkPreconditionerMaxIterations`	Inner KSP tolerance and iteration cap for the preconditioner solve (used by iterative inner solvers; the default `preonly` ignores them).
`jfnkPreconditionerUpdateFrequency`, `jfnkPreconditionerOptionsPrefix`	How often the preconditioner matrix and factor are refreshed (every n Newton iterations) and the PETSc options-database prefix.
`diagonalIionEpsilon`	Finite-difference perturbation for diagonal source derivative.
`stateODESolver`	`RKF45`, `RKT45`, `RK4`, or `Euler`.
`stateODEInitialStep`, `stateODEAbsTol`, `stateODERelTol`, `stateODEMaxSteps`	Adaptive ODE step and tolerance controls. The same values can fall back to `initialODEStep`, `absTol`, `relTol`, and `maxSteps` in `timeIntegrationProperties`; `stateODETolerance` and `stateODEMaxSubSteps` are retained as legacy fallbacks.
`stateODEUseOpenMP`, `stateODENumThreads`, `stateODEOpenMPThreshold`	Enables OpenMP parallelisation of the per-cell/per-quadrature-point state ODE integration, sets the thread count and the cell threshold above which the parallel region is entered.

Control	Meaning
`memoryOptimization`	`off`, `on`, or `auto`. In `auto`, memory optimisations are effective only for 3D high-order V_m runs with at least `memoryOptimizationCellThreshold` cells.
`memoryOptimizationCellThreshold`	Cell-count threshold used by `memoryOptimization=auto`; default 8000.
`memoryOptimizationAdaptiveTripletReserve`	If effective, selects a smaller or stabilised stiffness triplet reserve instead of the conservative 800 triplets per face.
`memoryOptimizationFluxTripletsPerFace`, `memoryOptimizationStabilisedTripletsPerFace`	Triplets-per-face reserve used for high-order stiffness assembly in flux-only and stabilised cases. Values are clamped to at least 16.
`memoryOptimizationTrimHeap`	Enables gated `malloc_trim(0)` calls after large setup phases when memory optimisation is effective.
`memoryOptimizationCompactMassAssembly`	Enables compact row-wise triplet emission for the consistent high-order mass matrix when memory optimisation is effective.
`profileTimings`	Records per-phase wall-clock timings (assembly, GMRES, preconditioner setup/apply, sparse solves) for the final summary.
`writeNonlinearResiduals`	Writes per-iteration nonlinear residual diagnostics.

9 Output diagnostics

At the final time, the solver writes cell-centred fields, manufactured exact fields, error fields, reconstructed/quadrature diagnostics when available, and a text summary. The summary includes mesh size, dimension, Δt, time scheme, mass matrix, high-order settings for V_m and I_{ion}, selected nonlinear method, relevant tolerances, ODE controls, nonlinear convergence statistics, effective memory-optimisation settings, allocated ionic quadrature-point storage, wall-clock timings, and peak resident memory. These diagnostics are intended to make each convergence point auditable: the numerical method used to generate a data point is recorded alongside the error data.

10 Error Measures and Convergence Rates

The solver writes cell-centred volume-weighted L_1, L_2, and L_∞ errors for V_m, u_1, and u_2. For a generic field $q \in \{V_m, u_1, u_2\}$, let q_i be the numerical value stored at the centre of cell K_i, $q^\star(\mathbf{x}, t)$ the manufactured exact solution, and

$$e_i(q) = q_i - q^\star(\mathbf{C}_i, t), \qquad |\Omega_h| = \sum_i |K_i|. \tag{46}$$

The cell-centred errors written by the solver are

$$E_1^C(q) = \frac{1}{|\Omega_h|} \sum_i |K_i|\, |e_i(q)|, \tag{47}$$

$$E_2^C(q) = \left[\frac{1}{|\Omega_h|} \sum_i |K_i|\, e_i(q)^2 \right]^{1/2}, \tag{48}$$

$$E_\infty^C(q) = \max_i |e_i(q)|. \tag{49}$$

These are volume-weighted mean/RMS errors, not unnormalised integral norms.

The solver also writes reconstructed quadrature errors, denoted by the superscript G. Let $\mathbf{x}_{iq}$ be a quadrature point in cell K_i and let ω_{iq} be its normalised quadrature weight, so that $m_{iq} = |K_i|\omega_{iq}$

is the physical measure associated with that point. For voltage, q_{iq}^G is the LRE/Taylor reconstructed voltage $V_{m,h}(\mathbf{x}_{iq})$; for the states, when high-order ionic integration is active, q_{iq}^G is the state value used at the ionic quadrature point. In the default `cellCentredReconstruct` mode this value is reconstructed from the advanced cell-centred state; in the legacy `gaussPointODE` mode it is the state advanced directly at that quadrature point. The quadrature-point error is

$$e_{iq}^G(q) = q_{iq}^G - q^\star(\mathbf{x}_{iq}, t), \qquad |\Omega_h|_G = \sum_i \sum_q m_{iq}. \tag{50}$$

The corresponding quadrature errors are

$$E_1^G(q) = \frac{1}{|\Omega_h|_G} \sum_i \sum_q m_{iq}\, |e_{iq}^G(q)|, \tag{51}$$

$$E_2^G(q) = \left[\frac{1}{|\Omega_h|_G} \sum_i \sum_q m_{iq}\, \left(e_{iq}^G(q)\right)^2 \right]^{1/2}, \tag{52}$$

$$E_\infty^G(q) = \max_{i,q} |e_{iq}^G(q)|. \tag{53}$$

If the corresponding high-order diagnostic is inactive, the report maps the cell-centred errors into the same output columns.

For the quadrature diagnostics, the exact-solution RMS norm and the relative percentage error are

$$\|q^\star\|_{2,G} = \left[\frac{1}{|\Omega_h|_G} \sum_i \sum_q m_{iq}\, \left(q^\star(\mathbf{x}_{iq}, t)\right)^2 \right]^{1/2}, \qquad E_{\mathrm{rel},2}^G(q) = 100\, \frac{E_2^G(q)}{\max(\|q^\star\|_{2,G}, \mathtt{SMALL})}. \tag{54}$$

Pairwise rates use consecutive mesh sizes,

$$p_{N_1,N_2} = \frac{\log(E_{N_1}/E_{N_2})}{\log(h_{N_1}/h_{N_2})}, \qquad h_N = 1/N. \tag{55}$$

The convergence tables below recompute log–log fits directly from the `errors_vs_N` files. The 2D tables use $N = 10$–100 and $N = 70$–100; the 3D tables use $N = 10$–30 and $N = 20$–30.

11 Results

This section collects the current convergence tables generated from `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/`. All result blocks in this section use `crankNicolson` time integration, the `consistent` mass matrix, and `RKF45` for the state ODEs. The tables are split by dimension, nonlinear method, mesh family, stabilisation parameter, and time step. Their first column records the actual `Vm-states-Iion` triplet used in each run.

11.1 Two-dimensional manufactured problem

The tables in this subsection are generated directly from the current `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0` result tree. The method-specific roots are `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/2D_CN_cons_Picard_RKF45`, `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/2D_CN_cons_diagonalIion_RKF45`, `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/2D_CN_cons_JFNK_RKF45`. All entries use `crankNicolson`, the `consistent` mass matrix, and `RKF45` for the state ODEs. The first column reports `Vm-states-Iion`: the V_m reconstruction degree, the state treatment (`CCp*`, `GP`, or `na`), and the I_{ion} quadrature/reconstruction setting. Fits use $N = 10$–100 for the broad mesh range and $N = 70$–100 for the fine-mesh range.

11.1.1 `Picard`

$\Delta t = 10^{-2}$

Table 2: 2D manufactured results for `Picard`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.992	1.992	2.135	2.135	2.134	2.134	1.985	1.985	2.500	2.500	2.497	2.497	5.143	16406.2
`NO--na--NO`	hexa	0.1	1.849	1.849	1.612	1.612	1.618	1.618	2.278	2.278	2.159	2.159	2.163	2.163	18.689	18355.6
`NO--na--NO`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	66.651	17158.0
`NO--na--NO`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	118.52	20786.3
`NO--CCp1--p1`	hexa	0.0	1.993	1.993	2.136	2.354	2.134	2.333	1.985	1.985	2.505	2.289	2.497	2.242	17.995	16908.0
`NO--CCp1--p1`	hexa	0.1	1.849	1.849	1.611	2.354	1.618	2.310	2.278	2.278	2.159	2.289	2.163	2.225	37.501	18456.9
`NO--CCp1--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	126.46	17892.6
`NO--CCp1--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	158.68	20895.8
`NO--CCp1--p2`	hexa	0.0	1.993	1.993	2.140	2.229	2.134	2.184	1.985	1.985	2.513	2.150	2.495	2.094	21.941	17118.3
`NO--CCp1--p2`	hexa	0.1	1.849	1.849	1.609	2.229	1.619	2.181	2.278	2.278	2.159	2.149	2.163	2.091	32.920	18586.4
`NO--CCp1--p2`	triangular unstr	0.0	0.730	0.730	0.778	2.111	0.784	1.164	0.862	0.862	1.037	2.101	1.107	1.150	133.37	18167.6
`NO--CCp1--p2`	triangular unstr	0.1	0.722	0.722	0.772	2.112	0.777	1.172	0.871	0.871	1.027	2.105	1.097	1.143	153.34	21019.5
`NO--CCp1--p3`	hexa	0.0	1.993	1.993	2.140	2.247	2.134	2.211	1.985	1.985	2.513	2.167	2.495	2.112	20.564	17226.5
`NO--CCp1--p3`	hexa	0.1	1.849	1.849	1.609	2.247	1.619	2.206	2.278	2.278	2.159	2.166	2.163	2.109	25.935	18641.4
`NO--CCp1--p3`	triangular unstr	0.0	0.730	0.730	0.778	2.139	0.784	1.132	0.862	0.862	1.037	2.131	1.107	1.144	115.55	18286.4
`NO--CCp1--p3`	triangular unstr	0.1	0.722	0.722	0.772	2.140	0.777	1.139	0.871	0.871	1.027	2.135	1.097	1.136	153.47	21092.0
`NO--CCp2--p1`	hexa	0.0	1.992	1.992	2.141	3.042	2.132	2.624	1.985	1.985	2.518	2.999	2.491	2.604	25.070	17413.8
`NO--CCp2--p1`	hexa	0.1	1.849	1.849	1.607	3.038	1.618	1.948	2.278	2.278	2.158	2.987	2.162	2.187	42.260	18557.9
`NO--CCp2--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	120.75	19093.8
`NO--CCp2--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	180.69	20978.0
`NO--CCp2--p2`	hexa	0.0	1.991	1.991	2.145	3.063	2.121	2.765	1.986	1.986	2.519	3.002	2.444	2.630	29.600	17632.8
`NO--CCp2--p2`	hexa	0.1	1.851	1.851	1.600	3.061	1.624	2.068	2.274	2.274	2.155	2.996	2.162	2.210	30.224	18673.4
`NO--CCp2--p2`	triangular unstr	0.0	0.730	0.730	0.778	1.941	0.784	0.776	0.862	0.862	1.037	1.095	1.107	1.097	117.04	19335.1
`NO--CCp2--p2`	triangular unstr	0.1	0.722	0.722	0.772	1.957	0.777	0.773	0.871	0.871	1.027	1.090	1.097	1.087	177.40	21128.6
`NO--CCp2--p3`	hexa	0.0	1.991	1.991	2.145	3.058	2.121	2.763	1.986	1.986	2.519	3.001	2.444	2.632	28.256	17729.6
`NO--CCp2--p3`	hexa	0.1	1.851	1.851	1.600	3.056	1.624	2.069	2.274	2.274	2.155	2.995	2.162	2.211	42.963	18800.3
`NO--CCp2--p3`	triangular unstr	0.0	0.730	0.730	0.778	1.952	0.784	0.776	0.862	0.862	1.037	1.098	1.107	1.096	112.55	19472.0
`NO--CCp2--p3`	triangular unstr	0.1	0.722	0.722	0.772	1.968	0.777	0.773	0.871	0.871	1.027	1.093	1.097	1.087	173.29	21226.5
`NO--CCp3--p1`	hexa	0.0	1.992	1.992	2.139	4.063	2.132	2.165	1.985	1.985	2.518	3.183	2.491	2.491	40.748	18769.0
`NO--CCp3--p1`	hexa	0.1	1.849	1.849	1.605	3.118	1.618	1.661	2.278	2.278	2.158	2.340	2.163	2.189	52.484	19843.9
`NO--CCp3--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	131.85	21560.9
`NO--CCp3--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	207.53	23322.9
`NO--CCp3--p2`	hexa	0.0	1.992	1.992	2.141	4.277	2.120	2.217	1.985	1.985	2.519	4.233	2.444	2.444	34.667	18981.8
`NO--CCp3--p2`	hexa	0.1	1.851	1.851	1.594	3.421	1.622	1.722	2.274	2.274	2.155	2.589	2.162	2.221	54.473	20039.5
`NO--CCp3--p2`	triangular unstr	0.0	0.730	0.730	0.778	1.071	0.784	0.796	0.862	0.862	1.037	1.037	1.107	1.109	165.34	21845.6
`NO--CCp3--p2`	triangular unstr	0.1	0.722	0.722	0.772	1.081	0.777	0.798	0.871	0.871	1.027	1.028	1.097	1.100	208.25	23561.6
`NO--CCp3--p3`	hexa	0.0	1.992	1.992	2.141	4.293	2.120	2.214	1.985	1.985	2.519	4.254	2.444	2.444	47.414	19082.2
`NO--CCp3--p3`	hexa	0.1	1.851	1.851	1.594	3.409	1.622	1.721	2.274	2.274	2.155	2.578	2.162	2.221	56.993	20145.0
`NO--CCp3--p3`	triangular unstr	0.0	0.730	0.730	0.778	1.054	0.784	0.796	0.862	0.862	1.037	1.037	1.107	1.109	180.29	21975.3
`NO--CCp3--p3`	triangular unstr	0.1	0.722	0.722	0.772	1.064	0.777	0.797	0.871	0.871	1.027	1.028	1.097	1.100	205.65	23684.7
`NO--GP--p1`	hexa	0.0	1.992	1.992	2.000	2.057	2.005	2.061	1.985	1.985	2.002	2.031	2.023	2.031	6.823	16540.8
`NO--GP--p1`	hexa	0.1	1.849	1.849	2.000	2.049	1.997	2.053	2.278	2.278	2.002	2.039	2.030	2.039	21.520	18420.0
`NO--GP--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	72.359	17303.3
`NO--GP--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	128.31	20871.0
`NO--GP--p2`	hexa	0.0	1.992	1.992	1.999	2.015	2.002	2.017	1.986	1.986	2.001	2.010	2.011	2.010	14.302	16794.5
`NO--GP--p2`	hexa	0.1	1.851	1.851	1.999	2.015	2.001	2.017	2.274	2.274	2.001	2.011	2.013	2.011	26.791	18563.5
`NO--GP--p2`	triangular unstr	0.0	0.730	0.730	1.878	1.418	1.102	1.411	0.862	0.862	1.678	1.219	1.112	1.269	71.948	17559.7
`NO--GP--p2`	triangular unstr	0.1	0.722	0.722	1.880	1.427	1.107	1.420	0.871	0.871	1.682	1.220	1.104	1.270	132.64	21009.7
`NO--GP--p3`	hexa	0.0	1.992	1.992	1.999	2.019	2.002	2.021	1.986	1.986	2.001	2.011	2.011	2.011	34.842	16938.5
`NO--GP--p3`	hexa	0.1	1.851	1.851	1.999	2.019	2.001	2.021	2.274	2.274	2.001	2.013	2.013	2.013	36.836	18613.1
`NO--GP--p3`	triangular unstr	0.0	0.730	0.730	1.878	1.389	1.102	1.384	0.862	0.862	1.678	1.206	1.112	1.258	78.186	17682.8
`NO--GP--p3`	triangular unstr	0.1	0.722	0.722	1.881	1.398	1.107	1.392	0.871	0.871	1.682	1.207	1.104	1.259	129.33	21069.0
`p1--na--NO`	hexa	0.0	1.911	2.146	1.616	1.616	1.626	1.626	2.612	2.168	2.381	2.381	2.387	2.387	116.43	17750.5
`p1--na--NO`	hexa	0.1	1.971	2.163	1.677	1.677	1.687	1.687	2.614	2.141	2.416	2.416	2.422	2.422	93.581	19067.1
`p1--na--NO`	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	566.18	19578.9
`p1--na--NO`	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	636.61	21518.4
`p1--CCp1--p1`	hexa	0.0	1.911	2.146	1.616	2.349	1.626	1.955	2.612	2.168	2.381	2.294	2.387	2.369	114.17	18231.1
`p1--CCp1--p1`	hexa	0.1	1.971	2.163	1.677	2.351	1.687	2.045	2.614	2.141	2.416	2.293	2.422	2.383	93.147	19257.6
`p1--CCp1--p1`	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	591.55	20179.6
`p1--CCp1--p1`	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	609.37	21594.2
`p1--CCp1--p2`	hexa	0.0	1.911	2.146	1.616	2.228	1.626	2.057	2.612	2.168	2.381	2.153	2.387	2.232	116.12	18427.4
`p1--CCp1--p2`	hexa	0.1	1.971	2.163	1.676	2.228	1.687	2.104	2.614	2.141	2.416	2.152	2.422	2.208	104.61	19367.0
`p1--CCp1--p2`	triangular unstr	0.0	2.271	2.126	2.004	2.102	2.002	2.036	2.384	2.115	2.552	2.063	2.549	2.285	668.38	20456.4
`p1--CCp1--p2`	triangular unstr	0.1	2.302	2.106	2.092	2.102	2.090	2.066	2.222	2.054	2.524	2.061	2.524	2.190	625.36	21732.8
`p1--CCp1--p3`	hexa	0.0	1.911	2.146	1.616	2.246	1.626	2.063	2.612	2.168	2.381	2.170	2.387	2.255	83.912	18517.0
`p1--CCp1--p3`	hexa	0.1	1.971	2.163	1.676	2.246	1.687	2.115	2.614	2.141	2.416	2.169	2.422	2.234	97.852	19438.2
`p1--CCp1--p3`	triangular unstr	0.0	2.271	2.126	2.004	2.127	2.002	2.040	2.384	2.115	2.552	2.080	2.549	2.315	725.98	20581.6
`p1--CCp1--p3`	triangular unstr	0.1	2.302	2.106	2.092	2.127	2.090	2.075	2.222	2.054	2.524	2.078	2.524	2.217	606.33	21809.8
`p1--CCp2--p1`	hexa	0.0	1.911	2.146	1.616	2.904	1.626	1.665	2.612	2.168	2.381	2.783	2.387	2.395	91.324	18731.6
`p1--CCp2--p1`	hexa	0.1	1.971	2.163	1.676	2.954	1.687	1.736	2.614	2.141	2.416	2.859	2.422	2.431	109.98	19338.5
`p1--CCp2--p1`	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	797.98	21404.6

			Fit $N = 10$–100						Fit $N = 70$–100							
Vm-states-Ion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p1--CCp2--p1	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	616.74	21869.5
p1--CCp2--p2	hexa	0.0	1.912	2.146	1.616	2.990	1.627	1.688	2.612	2.168	2.381	2.882	2.387	2.400	98.916	18950.6
p1--CCp2--p2	hexa	0.1	1.971	2.163	1.676	3.018	1.687	1.765	2.613	2.141	2.416	2.931	2.422	2.437	105.47	19476.9
p1--CCp2--p2	triangular unstr	0.0	2.271	2.126	2.004	2.677	2.002	1.977	2.384	2.115	2.552	2.669	2.549	2.528	768.73	21638.0
p1--CCp2--p2	triangular unstr	0.1	2.301	2.106	2.092	2.799	2.090	2.062	2.222	2.054	2.524	2.735	2.523	2.511	678.26	22074.3
p1--CCp2--p3	hexa	0.0	1.912	2.146	1.616	2.987	1.627	1.688	2.612	2.168	2.381	2.884	2.387	2.400	102.86	19056.1
p1--CCp2--p3	hexa	0.1	1.971	2.163	1.676	3.015	1.687	1.765	2.613	2.141	2.416	2.932	2.422	2.437	100.90	19565.9
p1--CCp2--p3	triangular unstr	0.0	2.271	2.126	2.004	2.681	2.002	1.977	2.384	2.115	2.552	2.672	2.549	2.527	746.78	21749.9
p1--CCp2--p3	triangular unstr	0.1	2.301	2.106	2.092	2.802	2.090	2.062	2.222	2.054	2.524	2.740	2.523	2.510	743.82	22186.1
p1--CCp3--p1	hexa	0.0	1.911	2.146	1.616	2.034	1.626	1.630	2.612	2.168	2.381	2.378	2.387	2.407	125.49	20075.4
p1--CCp3--p1	hexa	0.1	1.971	2.163	1.677	2.165	1.687	1.692	2.614	2.141	2.416	2.409	2.422	2.443	124.88	20548.1
p1--CCp3--p1	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	680.89	23892.8
p1--CCp3--p1	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	773.90	24267.6
p1--CCp3--p2	hexa	0.0	1.912	2.146	1.617	2.239	1.627	1.634	2.612	2.168	2.381	2.377	2.387	2.426	131.82	20280.0
p1--CCp3--p2	hexa	0.1	1.971	2.163	1.677	2.394	1.688	1.698	2.613	2.141	2.416	2.402	2.422	2.464	138.65	20735.8
p1--CCp3--p2	triangular unstr	0.0	2.271	2.126	2.004	2.031	2.002	1.964	2.384	2.115	2.552	2.527	2.549	2.528	642.44	24134.8
p1--CCp3--p2	triangular unstr	0.1	2.301	2.106	2.092	2.141	2.090	2.052	2.222	2.054	2.524	2.511	2.523	2.516	762.00	24520.7
p1--CCp3--p3	hexa	0.0	1.912	2.146	1.617	2.225	1.627	1.634	2.612	2.168	2.381	2.376	2.387	2.426	131.02	20374.9
p1--CCp3--p3	hexa	0.1	1.971	2.163	1.677	2.379	1.688	1.697	2.613	2.141	2.416	2.402	2.422	2.463	141.29	20835.1
p1--CCp3--p3	triangular unstr	0.0	2.271	2.126	2.004	2.027	2.002	1.964	2.384	2.115	2.552	2.527	2.549	2.528	639.63	24254.8
p1--CCp3--p3	triangular unstr	0.1	2.301	2.106	2.092	2.136	2.090	2.052	2.222	2.054	2.524	2.512	2.523	2.516	799.58	24633.1
p1--GP--p1	hexa	0.0	1.911	2.146	1.998	2.154	1.838	2.160	2.612	2.168	2.007	2.316	2.258	2.316	81.360	17778.0
p1--GP--p1	hexa	0.1	1.971	2.163	1.999	2.221	1.898	2.227	2.614	2.141	2.005	2.319	2.234	2.318	86.674	19080.5
p1--GP--p1	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	713.46	19611.5
p1--GP--p1	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	644.65	21555.8
p1--GP--p2	hexa	0.0	1.912	2.146	1.999	2.182	1.951	2.181	2.612	2.168	2.002	2.161	2.105	2.157	87.844	17999.5
p1--GP--p2	hexa	0.1	1.971	2.163	1.999	2.201	1.975	2.199	2.613	2.141	2.001	2.156	2.080	2.151	94.921	19190.6
p1--GP--p2	triangular unstr	0.0	2.271	2.126	1.992	2.069	2.012	2.068	2.384	2.115	2.004	2.184	2.284	2.176	648.18	19866.6
p1--GP--p2	triangular unstr	0.1	2.301	2.106	1.992	2.089	2.038	2.087	2.222	2.054	2.002	2.117	2.186	2.113	717.87	21700.1
p1--GP--p3	hexa	0.0	1.912	2.146	1.999	2.196	1.951	2.195	2.612	2.168	2.002	2.177	2.105	2.173	90.087	18115.0
p1--GP--p3	hexa	0.1	1.971	2.163	1.999	2.217	1.975	2.215	2.613	2.141	2.001	2.172	2.080	2.167	100.92	19245.1
p1--GP--p3	triangular unstr	0.0	2.271	2.126	1.992	2.082	2.012	2.081	2.384	2.115	2.004	2.219	2.284	2.210	575.95	19986.2
p1--GP--p3	triangular unstr	0.1	2.301	2.106	1.992	2.109	2.038	2.106	2.222	2.054	2.002	2.144	2.186	2.139	677.12	21763.5
p2--na--NO	hexa	0.0	2.127	2.567	2.176	2.176	2.176	2.176	2.016	2.179	2.139	2.139	2.139	2.139	85.517	18507.1
p2--na--NO	hexa	0.1	2.128	2.607	2.180	2.180	2.181	2.181	2.018	2.209	2.153	2.153	2.153	2.153	87.941	20275.2
p2--na--NO	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.162	2.162	1.982	2.068	2.261	2.261	2.263	2.263	593.35	21544.9
p2--na--NO	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	591.36	24379.3
p2--CCp1--p1	hexa	0.0	2.149	2.608	2.201	2.356	2.203	2.337	2.019	2.209	2.154	2.292	2.155	2.244	119.60	19444.1
p2--CCp1--p1	hexa	0.1	2.151	2.651	2.208	2.356	2.211	2.338	2.021	2.247	2.171	2.291	2.172	2.246	129.42	20862.0
p2--CCp1--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.162	2.162	1.982	2.068	2.261	2.261	2.263	2.263	628.11	22471.6
p2--CCp1--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	644.03	24808.0
p2--CCp1--p2	hexa	0.0	2.212	2.708	2.272	2.230	2.283	2.188	2.026	2.303	2.199	2.150	2.206	2.095	100.91	19646.5
p2--CCp1--p2	hexa	0.1	2.223	2.759	2.291	2.229	2.304	2.188	2.030	2.364	2.229	2.150	2.238	2.095	112.47	20964.3
p2--CCp1--p2	triangular unstr	0.0	2.118	2.532	2.273	2.101	2.279	2.040	1.977	2.149	2.394	2.057	2.419	2.018	643.37	22685.6
p2--CCp1--p2	triangular unstr	0.1	2.104	2.578	2.283	2.101	2.292	2.039	1.972	2.185	2.455	2.057	2.480	2.018	583.69	24950.0
p2--CCp1--p3	hexa	0.0	2.212	2.708	2.272	2.248	2.283	2.215	2.026	2.303	2.199	2.167	2.206	2.114	110.96	19740.6
p2--CCp1--p3	hexa	0.1	2.223	2.759	2.291	2.248	2.304	2.214	2.030	2.364	2.229	2.167	2.238	2.114	104.35	21027.4
p2--CCp1--p3	triangular unstr	0.0	2.118	2.532	2.273	2.126	2.279	2.050	1.977	2.149	2.394	2.074	2.419	2.023	641.34	22793.8
p2--CCp1--p3	triangular unstr	0.1	2.104	2.578	2.283	2.126	2.292	2.050	1.972	2.185	2.455	2.074	2.480	2.023	584.37	25015.4
p2--CCp2--p1	hexa	0.0	2.149	2.608	2.203	3.039	2.203	2.289	2.019	2.210	2.155	2.989	2.155	2.169	125.93	19924.8
p2--CCp2--p1	hexa	0.1	2.152	2.651	2.211	3.040	2.211	2.310	2.021	2.247	2.173	2.991	2.172	2.190	99.702	20884.2
p2--CCp2--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.162	2.162	1.982	2.068	2.261	2.261	2.263	2.263	658.27	23562.2
p2--CCp2--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	597.89	24901.6
p2--CCp2--p2	hexa	0.0	2.212	2.708	2.284	3.063	2.283	2.480	2.026	2.303	2.207	3.000	2.206	2.250	104.23	20117.0
p2--CCp2--p2	hexa	0.1	2.224	2.760	2.305	3.063	2.305	2.528	2.030	2.365	2.239	3.001	2.239	2.294	102.48	21009.3
p2--CCp2--p2	triangular unstr	0.0	2.118	2.532	2.286	3.012	2.280	2.428	1.977	2.149	2.413	2.995	2.420	2.443	661.07	23788.5
p2--CCp2--p2	triangular unstr	0.1	2.104	2.579	2.300	3.012	2.293	2.467	1.972	2.186	2.481	2.996	2.483	2.512	607.64	25036.1
p2--CCp2--p3	hexa	0.0	2.212	2.708	2.284	3.058	2.283	2.480	2.026	2.303	2.207	2.999	2.206	2.251	108.61	20220.7
p2--CCp2--p3	hexa	0.1	2.224	2.760	2.305	3.058	2.305	2.527	2.030	2.365	2.239	3.000	2.239	2.296	106.87	21070.0
p2--CCp2--p3	triangular unstr	0.0	2.118	2.532	2.286	3.010	2.280	2.425	1.977	2.149	2.413	2.995	2.420	2.444	657.26	23893.4
p2--CCp2--p3	triangular unstr	0.1	2.104	2.579	2.300	3.010	2.293	2.465	1.972	2.186	2.481	2.996	2.483	2.512	604.31	25115.3
p2--CCp3--p1	hexa	0.0	2.149	2.608	2.203	3.175	2.203	2.220	2.019	2.210	2.155	2.123	2.155	2.154	123.23	21147.6
p2--CCp3--p1	hexa	0.1	2.152	2.651	2.211	3.250	2.211	2.227	2.021	2.247	2.173	2.151	2.172	2.171	123.06	21550.1
p2--CCp3--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.162	2.162	1.982	2.068	2.261	2.261	2.263	2.263	701.21	26008.8
p2--CCp3--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	636.67	26829.5
p2--CCp3--p2	hexa	0.0	2.212	2.708	2.284	3.846	2.283	2.347	2.026	2.303	2.207	2.397	2.206	2.204	130.67	21307.4
p2--CCp3--p2	hexa	0.1	2.224	2.760	2.305	3.936	2.304	2.369	2.030	2.365	2.239	2.584	2.239	2.237	126.33	21721.4
p2--CCp3--p2	triangular unstr	0.0	2.118	2.532	2.286	3.422	2.279	2.345	1.977	2.149	2.413	2.437	2.420	2.419	706.14	26226.4
p2--CCp3--p2	triangular unstr	0.1	2.104	2.579	2.300	3.528	2.293	2.358	1.972	2.186	2.481	2.550	2.483	2.482	647.27	27058.1
p2--CCp3--p3	hexa	0.0	2.212	2.708	2.284	3.839	2.283	2.346	2.026	2.303	2.207	2.351	2.206	2.204	133.61	21423.4
p2--CCp3--p3	hexa	0.1	2.224	2.760	2.305	3.933	2.304	2.368	2.030	2.365	2.239	2.531	2.239	2.237	131.80	21828.1
p2--CCp3--p3	triangular unstr	0.0	2.118	2.532	2.286	3.389	2.279	2.343	1.977	2.149	2.413	2.404	2.420	2.419	687.20	26345.2
p2--CCp3--p3	triangular unstr	0.1	2.104	2.579	2.300	3.497	2.293	2.356	1.972	2.186	2.481	2.509	2.483	2.482	651.90	27158.1
p2--GP--p1	hexa	0.0	2.149	2.608	2.004	2.716	2.109	2.720	2.019	2.210	2.002	2.416	2.034	2.422	113.49	18606.1
p2--GP--p1	hexa	0.1	2.152	2.651	2.004	2.757	2.107	2.760	2.021	2.247	2.002	2.470	2.033	2.476	95.066	20320.7
p2--GP--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.162	2.162	1.982	2.068	2.261	2.261	2.263	2.263	613.49	21589.3
p2--GP--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	601.41	24401.3
p2--GP--p2	hexa	0.0	2.212	2.708	2.004	2.942	2.118	2.941	2.026	2.303	2.001	2.735	2.020	2.738	101.01	18797.0
p2--GP--p2	hexa	0.1	2.224	2.760	2.004	2.969	2.116	2.968	2.030	2.365	2.001	2.794	2.019	2.798	97.950	20436.8
p2--GP--p2	triangular unstr	0.0	2.118	2.532	1.993	2.860	2.044	2.862	1.977	2.149	2.000	2.751	2.026	2.758	607.86	21808.2
p2--GP--p2	triangular unstr	0.1	2.104	2.579	1.993	2.895	2.042	2.896	1.972	2.186	2.000	2.820	2.025	2.823	595.60	24545.1
p2--GP--p3	hexa	0.0	2.212	2.708	2.004	2.940	2.118	2.940	2.026	2.303	2.001	2.739	2.020	2.743	115.81	18890.9

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`p2--GP--p3`	hexa	0.1	2.224	2.760	2.004	2.967	2.117	2.966	2.030	2.365	2.001	2.798	2.019	2.801	105.75	20496.0
`p2--GP--p3`	triangular unstr	0.0	2.118	2.532	1.993	2.863	2.044	2.865	1.977	2.149	2.000	2.758	2.026	2.764	600.35	21930.4
`p2--GP--p3`	triangular unstr	0.1	2.104	2.579	1.993	2.897	2.042	2.898	1.972	2.186	2.000	2.825	2.025	2.829	571.33	24606.0
`p3--na--NO`	hexa	0.0	2.121	2.481	2.256	2.256	2.258	2.258	1.999	2.037	2.472	2.472	2.469	2.469	139.64	21801.9
`p3--na--NO`	hexa	0.1	2.104	2.476	2.238	2.238	2.240	2.240	1.996	2.034	2.468	2.468	2.465	2.465	131.88	24501.4
`p3--na--NO`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.346	2.346	1.972	1.988	3.183	3.183	3.173	3.173	702.42	27521.8
`p3--na--NO`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.183	3.183	3.173	3.173	668.22	31503.9
`p3--CCp1--p1`	hexa	0.0	2.180	2.629	2.347	2.353	2.360	2.339	2.003	2.055	2.608	2.289	2.617	2.247	148.33	22964.6
`p3--CCp1--p1`	hexa	0.1	2.155	2.624	2.323	2.353	2.335	2.339	1.998	2.051	2.600	2.289	2.609	2.247	150.54	25237.0
`p3--CCp1--p1`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.346	2.346	1.972	1.988	3.183	3.183	3.173	3.173	731.36	28009.0
`p3--CCp1--p1`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.183	3.183	3.173	3.173	731.11	31964.1
`p3--CCp1--p2`	hexa	0.0	2.672	3.361	2.672	2.229	2.497	2.184	2.142	2.316	1.404	2.150	1.266	2.093	149.27	23080.2
`p3--CCp1--p2`	hexa	0.1	2.615	3.360	2.623	2.229	2.442	2.184	2.103	2.281	1.305	2.150	1.208	2.093	163.93	25346.8
`p3--CCp1--p2`	triangular unstr	0.0	2.439	2.843	2.196	2.101	2.045	2.038	1.999	2.026	0.132	2.057	0.463	2.016	761.76	28151.6
`p3--CCp1--p2`	triangular unstr	0.1	2.369	2.829	2.121	2.101	1.975	2.038	1.995	2.023	0.125	2.057	0.459	2.016	716.74	32105.2
`p3--CCp1--p3`	hexa	0.0	2.672	3.361	2.672	2.247	2.497	2.210	2.142	2.316	1.404	2.166	1.266	2.111	142.46	23133.2
`p3--CCp1--p3`	hexa	0.1	2.615	3.360	2.623	2.247	2.442	2.210	2.103	2.281	1.305	2.166	1.208	2.111	148.27	25409.8
`p3--CCp1--p3`	triangular unstr	0.0	2.439	2.843	2.196	2.126	2.045	2.048	1.999	2.026	0.132	2.073	0.463	2.021	736.86	28242.9
`p3--CCp1--p3`	triangular unstr	0.1	2.369	2.829	2.120	2.126	1.975	2.048	1.995	2.023	0.125	2.073	0.459	2.021	707.29	32170.5
`p3--CCp2--p1`	hexa	0.0	2.181	2.631	2.362	3.042	2.362	2.762	2.003	2.056	2.629	2.999	2.620	2.746	146.87	23022.2
`p3--CCp2--p1`	hexa	0.1	2.156	2.626	2.337	3.042	2.337	2.758	1.998	2.051	2.621	2.999	2.611	2.740	142.23	25237.2
`p3--CCp2--p1`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.346	2.346	1.972	1.988	3.183	3.183	3.173	3.173	743.90	28372.6
`p3--CCp2--p1`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.183	3.183	3.173	3.173	753.95	31971.9
`p3--CCp2--p2`	hexa	0.0	3.520	3.971	2.698	3.063	2.707	3.071	4.171	4.124	0.669	3.002	0.640	2.724	147.94	23158.8
`p3--CCp2--p2`	hexa	0.1	3.547	4.002	2.648	3.063	2.656	3.069	4.105	4.100	0.514	3.002	0.488	2.716	151.97	25350.9
`p3--CCp2--p2`	triangular unstr	0.0	3.749	3.924	2.081	3.013	2.060	2.855	2.166	2.959	0.011	2.997	0.007	1.543	744.77	28535.6
`p3--CCp2--p2`	triangular unstr	0.1	3.693	3.923	1.982	3.013	1.961	2.852	1.803	2.888	0.004	2.997	0.001	1.542	693.97	32108.5
`p3--CCp2--p3`	hexa	0.0	3.520	3.971	2.698	3.058	2.707	3.064	4.171	4.124	0.669	3.001	0.640	2.728	155.77	23227.6
`p3--CCp2--p3`	hexa	0.1	3.547	4.002	2.648	3.058	2.656	3.062	4.105	4.100	0.514	3.001	0.488	2.721	148.70	25410.3
`p3--CCp2--p3`	triangular unstr	0.0	3.748	3.924	2.081	3.011	2.059	2.848	2.167	2.959	0.011	2.996	0.007	1.566	753.06	28621.7
`p3--CCp2--p3`	triangular unstr	0.1	3.692	3.923	1.982	3.011	1.960	2.845	1.803	2.888	0.004	2.996	0.001	1.564	686.54	32165.2
`p3--CCp3--p1`	hexa	0.0	2.181	2.631	2.362	4.082	2.362	2.385	2.003	2.056	2.629	3.391	2.620	2.620	168.18	23706.2
`p3--CCp3--p1`	hexa	0.1	2.156	2.626	2.337	4.082	2.337	2.361	1.998	2.051	2.621	3.389	2.611	2.611	160.28	25271.4
`p3--CCp3--p1`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.346	2.346	1.972	1.988	3.183	3.183	3.173	3.173	798.28	30639.9
`p3--CCp3--p1`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.183	3.183	3.173	3.173	715.06	32100.9
`p3--CCp3--p2`	hexa	0.0	3.520	3.971	2.705	4.177	2.708	2.869	4.171	4.124	0.670	3.889	0.642	0.729	173.31	23875.3
`p3--CCp3--p2`	hexa	0.1	3.547	4.002	2.656	4.177	2.658	2.839	4.105	4.100	0.515	3.888	0.489	0.577	154.17	25374.4
`p3--CCp3--p2`	triangular unstr	0.0	3.749	3.924	2.085	3.906	2.061	2.111	2.166	2.959	0.011	2.694	0.007	0.004	784.14	30867.0
`p3--CCp3--p2`	triangular unstr	0.1	3.693	3.924	1.986	3.906	1.962	2.027	1.803	2.888	0.004	2.693	0.001	-0.001	721.74	32251.5
`p3--CCp3--p3`	hexa	0.0	3.520	3.971	2.705	4.182	2.708	2.865	4.171	4.124	0.670	3.874	0.642	0.728	184.56	23970.5
`p3--CCp3--p3`	hexa	0.1	3.547	4.002	2.656	4.182	2.658	2.835	4.105	4.100	0.515	3.873	0.489	0.577	160.92	25445.4
`p3--CCp3--p3`	triangular unstr	0.0	3.749	3.924	2.084	3.889	2.061	2.107	2.167	2.959	0.011	2.591	0.007	0.004	784.89	30979.2
`p3--CCp3--p3`	triangular unstr	0.1	3.693	3.924	1.986	3.889	1.962	2.022	1.803	2.888	0.004	2.591	0.001	-0.001	719.74	32318.2
`p3--GP--p1`	hexa	0.0	2.181	2.631	2.002	2.762	1.992	2.758	2.003	2.056	2.002	2.661	2.008	2.651	139.74	21832.7
`p3--GP--p1`	hexa	0.1	2.156	2.626	2.002	2.756	1.991	2.752	1.998	2.051	2.002	2.653	2.008	2.642	137.21	24543.6
`p3--GP--p1`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.346	2.346	1.972	1.988	3.183	3.183	3.173	3.173	723.30	27552.4
`p3--GP--p1`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.183	3.183	3.173	3.173	697.04	31547.6
`p3--GP--p2`	hexa	0.0	3.520	3.971	2.000	3.744	1.999	3.748	4.171	4.124	2.001	1.457	2.006	1.395	127.20	21964.0
`p3--GP--p2`	hexa	0.1	3.547	4.002	1.999	3.757	1.998	3.761	4.105	4.100	2.001	1.347	2.006	1.283	134.32	24618.0
`p3--GP--p2`	triangular unstr	0.0	3.749	3.924	1.992	2.844	2.005	2.843	2.166	2.959	2.000	-0.092	2.012	-0.097	721.62	27688.9
`p3--GP--p2`	triangular unstr	0.1	3.693	3.924	1.992	2.836	2.005	2.836	1.803	2.888	2.000	-0.098	2.012	-0.103	725.45	31682.7
`p3--GP--p3`	hexa	0.0	3.520	3.971	2.000	3.738	1.999	3.742	4.171	4.124	2.001	1.419	2.006	1.354	135.58	22012.0
`p3--GP--p3`	hexa	0.1	3.547	4.002	2.000	3.751	1.999	3.755	4.105	4.100	2.001	1.304	2.006	1.237	142.03	24707.6
`p3--GP--p3`	triangular unstr	0.0	3.749	3.924	1.992	2.812	2.005	2.811	2.166	2.959	2.000	-0.108	2.012	-0.114	732.26	27747.5
`p3--GP--p3`	triangular unstr	0.1	3.693	3.924	1.992	2.803	2.005	2.803	1.803	2.888	2.000	-0.115	2.012	-0.120	691.27	31746.0

$\Delta t = 10^{-3}$

Table 3: 2D manufactured results for `Picard`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.997	1.997	1.998	1.998	1.998	1.998	2.000	2.000	2.004	2.004	2.004	2.004	37.411	16434.0
`NO--na--NO`	hexa	0.1	1.851	1.851	1.607	1.607	1.613	1.613	2.294	2.294	2.140	2.140	2.142	2.142	60.356	18355.3
`NO--na--NO`	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	356.55	17094.1
`NO--na--NO`	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	409.76	20788.7
`NO--CCp1--p1`	hexa	0.0	1.997	1.997	1.999	2.354	1.998	2.333	2.000	2.000	2.004	2.290	2.004	2.242	65.100	16893.3
`NO--CCp1--p1`	hexa	0.1	1.851	1.851	1.608	2.354	1.613	2.310	2.294	2.294	2.140	2.290	2.142	2.226	103.65	18467.3
`NO--CCp1--p1`	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	466.11	17834.6
`NO--CCp1--p1`	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	445.29	20897.4
`NO--CCp1--p2`	hexa	0.0	1.997	1.997	2.000	2.229	1.999	2.184	2.000	2.000	2.004	2.150	2.004	2.094	77.641	17123.0
`NO--CCp1--p2`	hexa	0.1	1.851	1.851	1.608	2.229	1.614	2.181	2.295	2.295	2.140	2.150	2.142	2.092	75.766	18575.5
`NO--CCp1--p2`	triangular unstr	0.0	0.798	0.798	0.777	2.111	0.783	1.163	0.963	0.963	1.038	2.102	1.107	1.150	488.02	18105.3
`NO--CCp1--p2`	triangular unstr	0.1	0.785	0.785	0.771	2.112	0.776	1.171	0.953	0.953	1.027	2.105	1.097	1.143	494.01	21026.6
`NO--CCp1--p3`	hexa	0.0	1.997	1.997	2.000	2.247	1.999	2.211	2.000	2.000	2.004	2.167	2.004	2.113	82.027	17228.8

			Fit $N = 10$–100						Fit $N = 70$–100							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
NO--CCp1--p3	hexa	0.1	1.851	1.851	1.608	2.247	1.614	2.206	2.295	2.295	2.140	2.167	2.142	2.110	106.15	18649.7
NO--CCp1--p3	triangular unstr	0.0	0.798	0.798	0.777	2.139	0.783	1.131	0.963	0.963	1.038	2.131	1.107	1.144	510.96	18236.0
NO--CCp1--p3	triangular unstr	0.1	0.785	0.785	0.771	2.140	0.776	1.138	0.953	0.953	1.027	2.135	1.097	1.137	489.56	21093.5
NO--CCp2--p1	hexa	0.0	1.997	1.997	2.000	3.042	1.997	2.510	1.999	1.999	2.004	2.998	2.004	2.157	98.814	17419.8
NO--CCp2--p1	hexa	0.1	1.851	1.851	1.603	3.038	1.613	1.942	2.294	2.294	2.140	2.987	2.142	2.167	132.60	18553.2
NO--CCp2--p1	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	577.70	19034.2
NO--CCp2--p1	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	569.23	20972.7
NO--CCp2--p2	hexa	0.0	1.995	1.995	2.003	3.063	1.997	2.671	1.999	1.999	2.004	3.002	2.003	2.246	122.24	17643.1
NO--CCp2--p2	hexa	0.1	1.853	1.853	1.597	3.061	1.619	2.063	2.289	2.289	2.140	2.996	2.141	2.192	110.37	18698.7
NO--CCp2--p2	triangular unstr	0.0	0.799	0.799	0.777	1.940	0.783	0.775	0.963	0.963	1.038	1.095	1.107	1.097	591.70	19305.3
NO--CCp2--p2	triangular unstr	0.1	0.786	0.786	0.771	1.955	0.776	0.772	0.953	0.953	1.027	1.090	1.097	1.087	634.90	21133.8
NO--CCp2--p3	hexa	0.0	1.995	1.995	2.003	3.058	1.997	2.670	1.999	1.999	2.004	3.001	2.003	2.251	137.62	17753.7
NO--CCp2--p3	hexa	0.1	1.853	1.853	1.597	3.056	1.619	2.064	2.289	2.289	2.140	2.995	2.141	2.193	154.38	18785.2
NO--CCp2--p3	triangular unstr	0.0	0.799	0.799	0.777	1.950	0.783	0.775	0.963	0.963	1.038	1.098	1.107	1.097	611.98	19426.7
NO--CCp2--p3	triangular unstr	0.1	0.786	0.786	0.771	1.966	0.776	0.772	0.953	0.953	1.027	1.092	1.097	1.087	607.74	21224.9
NO--CCp3--p1	hexa	0.0	1.997	1.997	1.998	3.944	1.997	2.030	1.999	1.999	2.004	2.505	2.004	2.004	226.62	18786.9
NO--CCp3--p1	hexa	0.1	1.851	1.851	1.600	3.114	1.613	1.657	2.294	2.294	2.140	2.324	2.142	2.170	182.05	19809.8
NO--CCp3--p1	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	734.01	21534.7
NO--CCp3--p1	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	741.32	23329.8
NO--CCp3--p2	hexa	0.0	1.996	1.996	1.998	4.244	1.996	2.092	1.999	1.999	2.004	3.862	2.003	2.002	226.96	19000.2
NO--CCp3--p2	hexa	0.1	1.853	1.853	1.590	3.420	1.617	1.718	2.289	2.289	2.140	2.581	2.141	2.203	247.54	20029.0
NO--CCp3--p2	triangular unstr	0.0	0.799	0.799	0.777	1.070	0.783	0.795	0.963	0.963	1.038	1.038	1.107	1.110	748.16	21804.4
NO--CCp3--p2	triangular unstr	0.1	0.786	0.786	0.771	1.080	0.776	0.797	0.953	0.953	1.027	1.028	1.097	1.100	751.27	23567.3
NO--CCp3--p3	hexa	0.0	1.996	1.996	1.998	4.257	1.996	2.090	1.999	1.999	2.004	3.849	2.003	2.002	216.69	19095.7
NO--CCp3--p3	hexa	0.1	1.853	1.853	1.590	3.408	1.617	1.717	2.289	2.289	2.140	2.570	2.141	2.203	194.93	20116.4
NO--CCp3--p3	triangular unstr	0.0	0.799	0.799	0.777	1.053	0.783	0.795	0.963	0.963	1.038	1.038	1.107	1.110	738.56	21921.5
NO--CCp3--p3	triangular unstr	0.1	0.786	0.786	0.771	1.062	0.776	0.796	0.953	0.953	1.027	1.028	1.097	1.100	782.87	23682.0
NO--GP--p1	hexa	0.0	1.997	1.997	1.999	2.050	1.998	2.053	2.000	2.000	2.000	2.007	2.000	2.007	50.315	16582.5
NO--GP--p1	hexa	0.1	1.851	1.851	1.999	2.042	1.990	2.046	2.294	2.294	2.000	2.015	2.008	2.016	73.314	18444.8
NO--GP--p1	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	405.08	17260.0
NO--GP--p1	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	469.96	20883.1
NO--GP--p2	hexa	0.0	1.997	1.997	1.999	2.013	1.998	2.015	1.999	1.999	2.000	2.002	2.000	2.002	110.15	16824.2
NO--GP--p2	hexa	0.1	1.853	1.853	1.999	2.013	1.998	2.014	2.290	2.290	2.000	2.003	2.001	2.003	107.80	18552.9
NO--GP--p2	triangular unstr	0.0	0.799	0.799	1.878	1.417	1.101	1.410	0.963	0.963	1.677	1.218	1.112	1.269	425.12	17491.6
NO--GP--p2	triangular unstr	0.1	0.786	0.786	1.880	1.426	1.105	1.419	0.953	0.953	1.680	1.219	1.104	1.270	468.71	21015.9
NO--GP--p3	hexa	0.0	1.997	1.997	1.999	2.016	1.999	2.018	1.999	1.999	2.000	2.002	2.000	2.002	113.09	16926.6
NO--GP--p3	hexa	0.1	1.853	1.853	1.999	2.016	1.998	2.018	2.290	2.290	2.000	2.004	2.002	2.004	139.45	18620.6
NO--GP--p3	triangular unstr	0.0	0.799	0.799	1.878	1.388	1.101	1.383	0.963	0.963	1.677	1.205	1.112	1.258	469.91	17633.3
NO--GP--p3	triangular unstr	0.1	0.786	0.786	1.880	1.397	1.106	1.391	0.953	0.953	1.680	1.206	1.104	1.258	503.51	21076.0
p1--na--NO	hexa	0.0	1.912	2.147	1.616	1.616	1.626	1.626	2.611	2.168	2.380	2.380	2.385	2.385	119.56	17760.4
p1--na--NO	hexa	0.1	1.971	2.163	1.677	1.677	1.687	1.687	2.615	2.141	2.414	2.414	2.419	2.419	138.64	19053.5
p1--na--NO	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	1039.5	19596.0
p1--na--NO	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	884.21	21530.4
p1--CCp1--p1	hexa	0.0	1.912	2.147	1.616	2.350	1.626	1.955	2.611	2.168	2.380	2.295	2.385	2.369	145.59	18215.8
p1--CCp1--p1	hexa	0.1	1.971	2.163	1.677	2.351	1.687	2.045	2.615	2.141	2.414	2.294	2.419	2.382	175.98	19252.1
p1--CCp1--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	1116.5	20203.4
p1--CCp1--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	961.19	21599.8
p1--CCp1--p2	hexa	0.0	1.912	2.147	1.616	2.228	1.626	2.057	2.611	2.168	2.380	2.154	2.385	2.233	191.20	18422.0
p1--CCp1--p2	hexa	0.1	1.971	2.163	1.676	2.228	1.686	2.104	2.615	2.141	2.414	2.153	2.419	2.209	145.60	19364.9
p1--CCp1--p2	triangular unstr	0.0	2.292	2.131	2.004	2.103	2.003	2.036	2.600	2.164	2.545	2.063	2.542	2.283	1189.9	20458.9
p1--CCp1--p2	triangular unstr	0.1	2.345	2.112	2.092	2.103	2.089	2.066	2.552	2.095	2.516	2.061	2.515	2.188	956.58	21743.2
p1--CCp1--p3	hexa	0.0	1.912	2.147	1.616	2.246	1.626	2.063	2.611	2.168	2.380	2.171	2.385	2.256	167.32	18527.7
p1--CCp1--p3	hexa	0.1	1.971	2.163	1.676	2.246	1.686	2.115	2.615	2.141	2.414	2.170	2.419	2.234	167.15	19424.6
p1--CCp1--p3	triangular unstr	0.0	2.292	2.131	2.004	2.127	2.003	2.040	2.600	2.164	2.545	2.081	2.542	2.312	1277.0	20590.8
p1--CCp1--p3	triangular unstr	0.1	2.345	2.112	2.092	2.128	2.089	2.076	2.552	2.095	2.516	2.078	2.515	2.216	992.42	21805.8
p1--CCp2--p1	hexa	0.0	1.912	2.147	1.616	2.903	1.626	1.665	2.611	2.168	2.380	2.782	2.385	2.394	178.87	18729.5
p1--CCp2--p1	hexa	0.1	1.971	2.163	1.676	2.953	1.687	1.736	2.615	2.141	2.414	2.858	2.419	2.428	180.49	19321.9
p1--CCp2--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	1330.2	21406.7
p1--CCp2--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	1045.8	21863.4
p1--CCp2--p2	hexa	0.0	1.912	2.147	1.616	2.990	1.627	1.689	2.611	2.168	2.380	2.882	2.385	2.400	175.61	18938.8
p1--CCp2--p2	hexa	0.1	1.972	2.163	1.676	3.018	1.687	1.765	2.615	2.141	2.414	2.931	2.419	2.435	216.23	19466.0
p1--CCp2--p2	triangular unstr	0.0	2.292	2.131	2.004	2.676	2.003	1.978	2.600	2.164	2.546	2.665	2.542	2.522	1361.8	21636.3
p1--CCp2--p2	triangular unstr	0.1	2.345	2.112	2.092	2.799	2.089	2.063	2.552	2.095	2.516	2.731	2.515	2.504	1202.3	22085.7
p1--CCp2--p3	hexa	0.0	1.912	2.147	1.616	2.987	1.627	1.688	2.611	2.168	2.380	2.884	2.385	2.400	163.85	19042.7
p1--CCp2--p3	hexa	0.1	1.972	2.163	1.676	3.015	1.687	1.765	2.615	2.141	2.414	2.931	2.419	2.435	227.61	19539.8
p1--CCp2--p3	triangular unstr	0.0	2.292	2.131	2.004	2.681	2.003	1.978	2.600	2.164	2.546	2.668	2.542	2.522	1316.6	21754.9
p1--CCp2--p3	triangular unstr	0.1	2.345	2.112	2.092	2.802	2.089	2.062	2.552	2.095	2.516	2.736	2.515	2.504	1276.9	22203.2
p1--CCp3--p1	hexa	0.0	1.912	2.147	1.617	2.034	1.626	1.630	2.611	2.168	2.380	2.378	2.385	2.406	243.76	20052.7
p1--CCp3--p1	hexa	0.1	1.971	2.163	1.677	2.164	1.687	1.692	2.615	2.141	2.414	2.408	2.419	2.441	315.44	20522.5
p1--CCp3--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	1178.4	23901.7
p1--CCp3--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	1484.4	24289.4
p1--CCp3--p2	hexa	0.0	1.913	2.147	1.617	2.239	1.627	1.635	2.611	2.168	2.380	2.377	2.385	2.426	259.12	20250.1
p1--CCp3--p2	hexa	0.1	1.972	2.163	1.677	2.394	1.688	1.698	2.615	2.141	2.414	2.402	2.419	2.463	300.54	20715.8
p1--CCp3--p2	triangular unstr	0.0	2.292	2.131	2.004	2.031	2.003	1.965	2.600	2.164	2.546	2.521	2.542	2.523	1225.7	24140.6
p1--CCp3--p2	triangular unstr	0.1	2.345	2.112	2.092	2.141	2.089	2.052	2.552	2.095	2.516	2.506	2.515	2.510	1474.8	24516.2
p1--CCp3--p3	hexa	0.0	1.913	2.147	1.617	2.224	1.627	1.634	2.611	2.168	2.380	2.376	2.385	2.426	282.39	20356.2
p1--CCp3--p3	hexa	0.1	1.972	2.163	1.677	2.379	1.688	1.698	2.615	2.141	2.414	2.401	2.419	2.462	288.53	20809.4
p1--CCp3--p3	triangular unstr	0.0	2.292	2.131	2.004	2.028	2.003	1.965	2.600	2.164	2.546	2.522	2.542	2.523	1239.2	24244.2
p1--CCp3--p3	triangular unstr	0.1	2.345	2.112	2.092	2.136	2.089	2.052	2.552	2.095	2.516	2.506	2.515	2.510	1459.8	24637.2
p1--GP--p1	hexa	0.0	1.912	2.147	1.997	2.152	1.836	2.158	2.611	2.168	2.005	2.309	2.249	2.308	123.49	17807.9
p1--GP--p1	hexa	0.1	1.971	2.163	1.999	2.219	1.895	2.225	2.615	2.141	2.003	2.309	2.222	2.308	138.59	19090.8
p1--GP--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	1204.8	19634.4

			Fit $N = 10$–100						Fit $N = 70$–100							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p1--GP--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	1079.4	21558.4
p1--GP--p2	hexa	0.0	1.913	2.147	1.998	2.181	1.949	2.179	2.611	2.168	2.001	2.155	2.096	2.151	180.40	18008.2
p1--GP--p2	hexa	0.1	1.972	2.163	1.999	2.199	1.972	2.197	2.615	2.141	2.001	2.150	2.071	2.145	209.63	19212.9
p1--GP--p2	triangular unstr	0.0	2.292	2.131	1.991	2.067	2.009	2.066	2.600	2.164	2.002	2.176	2.268	2.168	849.42	19867.6
p1--GP--p2	triangular unstr	0.1	2.345	2.112	1.992	2.087	2.034	2.085	2.552	2.095	2.000	2.109	2.167	2.105	1197.1	21695.8
p1--GP--p3	hexa	0.0	1.913	2.147	1.999	2.194	1.949	2.193	2.611	2.168	2.001	2.170	2.096	2.166	249.46	18130.7
p1--GP--p3	hexa	0.1	1.972	2.163	1.999	2.215	1.973	2.213	2.615	2.141	2.001	2.165	2.071	2.160	247.86	19286.1
p1--GP--p3	triangular unstr	0.0	2.292	2.131	1.992	2.080	2.009	2.079	2.600	2.164	2.002	2.209	2.268	2.200	895.96	19987.1
p1--GP--p3	triangular unstr	0.1	2.345	2.112	1.992	2.106	2.034	2.104	2.552	2.095	2.000	2.134	2.167	2.129	1063.9	21763.2
p2--na--NO	hexa	0.0	2.128	2.568	2.142	2.142	2.143	2.143	2.020	2.183	2.025	2.025	2.026	2.026	159.22	18524.9
p2--na--NO	hexa	0.1	2.129	2.608	2.143	2.143	2.144	2.144	2.022	2.213	2.027	2.027	2.028	2.028	156.57	20279.8
p2--na--NO	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	832.28	21545.1
p2--na--NO	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	894.20	24372.9
p2--CCp1--p1	hexa	0.0	2.150	2.609	2.164	2.356	2.166	2.334	2.023	2.214	2.029	2.293	2.030	2.232	185.32	19424.4
p2--CCp1--p1	hexa	0.1	2.153	2.652	2.167	2.356	2.170	2.335	2.026	2.251	2.031	2.293	2.032	2.235	174.20	20861.9
p2--CCp1--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	1017.2	22478.0
p2--CCp1--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1104.4	24819.2
p2--CCp1--p2	hexa	0.0	2.213	2.709	2.224	2.230	2.235	2.188	2.031	2.308	2.039	2.151	2.041	2.094	197.88	19638.1
p2--CCp1--p2	hexa	0.1	2.225	2.760	2.235	2.230	2.248	2.187	2.036	2.369	2.043	2.151	2.046	2.094	201.35	20980.1
p2--CCp1--p2	triangular unstr	0.0	2.121	2.535	2.161	2.102	2.166	2.040	1.990	2.162	2.003	2.058	2.012	2.018	1037.7	22673.4
p2--CCp1--p2	triangular unstr	0.1	2.108	2.582	2.154	2.101	2.160	2.039	1.987	2.199	1.998	2.058	2.003	2.018	1032.1	24958.1
p2--CCp1--p3	hexa	0.0	2.213	2.709	2.224	2.248	2.235	2.214	2.031	2.308	2.039	2.168	2.041	2.113	174.28	19730.4
p2--CCp1--p3	hexa	0.1	2.225	2.760	2.235	2.248	2.248	2.214	2.036	2.369	2.043	2.168	2.046	2.113	208.11	21044.0
p2--CCp1--p3	triangular unstr	0.0	2.122	2.535	2.161	2.127	2.166	2.050	1.990	2.162	2.003	2.074	2.012	2.023	1010.9	22791.0
p2--CCp1--p3	triangular unstr	0.1	2.108	2.582	2.154	2.126	2.160	2.050	1.987	2.199	1.998	2.074	2.003	2.023	1065.4	25021.3
p2--CCp2--p1	hexa	0.0	2.150	2.609	2.166	3.039	2.166	2.253	2.023	2.214	2.029	2.986	2.030	2.044	213.58	19889.8
p2--CCp2--p1	hexa	0.1	2.153	2.653	2.170	3.040	2.170	2.269	2.026	2.251	2.032	2.989	2.032	2.050	217.22	20885.5
p2--CCp2--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	1138.4	23567.7
p2--CCp2--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1144.6	24899.3
p2--CCp2--p2	hexa	0.0	2.214	2.709	2.236	3.063	2.235	2.434	2.032	2.308	2.041	2.999	2.041	2.086	212.93	20102.2
p2--CCp2--p2	hexa	0.1	2.226	2.761	2.250	3.063	2.249	2.475	2.037	2.371	2.046	3.000	2.047	2.105	224.72	21011.9
p2--CCp2--p2	triangular unstr	0.0	2.122	2.536	2.172	3.012	2.166	2.317	1.990	2.162	2.003	2.991	2.012	2.038	1164.8	23788.5
p2--CCp2--p2	triangular unstr	0.1	2.108	2.583	2.166	3.012	2.160	2.339	1.987	2.200	1.999	2.993	2.002	2.039	1153.8	25037.3
p2--CCp2--p3	hexa	0.0	2.214	2.709	2.236	3.058	2.235	2.434	2.032	2.308	2.041	2.998	2.041	2.088	212.31	20200.3
p2--CCp2--p3	hexa	0.1	2.226	2.761	2.250	3.058	2.249	2.474	2.037	2.371	2.046	2.999	2.047	2.106	218.90	21067.4
p2--CCp2--p3	triangular unstr	0.0	2.122	2.536	2.172	3.009	2.166	2.314	1.990	2.162	2.003	2.991	2.012	2.039	1149.3	23888.6
p2--CCp2--p3	triangular unstr	0.1	2.108	2.583	2.166	3.010	2.160	2.336	1.987	2.200	1.999	2.993	2.002	2.040	1185.5	25120.3
p2--CCp3--p1	hexa	0.0	2.150	2.609	2.166	3.135	2.166	2.183	2.023	2.214	2.029	1.992	2.030	2.029	277.42	21135.1
p2--CCp3--p1	hexa	0.1	2.153	2.653	2.170	3.206	2.170	2.186	2.026	2.251	2.032	2.002	2.032	2.031	304.67	21532.0
p2--CCp3--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	1287.9	26014.3
p2--CCp3--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1342.9	26843.0
p2--CCp3--p2	hexa	0.0	2.214	2.709	2.236	3.796	2.235	2.299	2.032	2.308	2.041	2.178	2.041	2.039	293.32	21317.7
p2--CCp3--p2	hexa	0.1	2.226	2.761	2.249	3.882	2.249	2.314	2.037	2.371	2.046	2.315	2.046	2.044	304.93	21712.7
p2--CCp3--p2	triangular unstr	0.0	2.122	2.536	2.172	3.303	2.166	2.231	1.990	2.162	2.003	1.988	2.012	2.010	1272.6	26231.3
p2--CCp3--p2	triangular unstr	0.1	2.108	2.583	2.166	3.390	2.160	2.225	1.987	2.200	1.999	2.010	2.002	2.001	1357.1	27057.7
p2--CCp3--p3	hexa	0.0	2.214	2.709	2.236	3.789	2.235	2.298	2.032	2.308	2.041	2.133	2.041	2.039	296.50	21401.5
p2--CCp3--p3	hexa	0.1	2.226	2.761	2.249	3.877	2.249	2.313	2.037	2.371	2.046	2.262	2.046	2.044	308.41	21792.9
p2--CCp3--p3	triangular unstr	0.0	2.122	2.536	2.172	3.267	2.166	2.229	1.990	2.162	2.003	1.957	2.012	2.010	1416.7	26349.1
p2--CCp3--p3	triangular unstr	0.1	2.108	2.583	2.166	3.356	2.160	2.223	1.987	2.200	1.999	1.972	2.002	2.001	1354.9	27168.7
p2--GP--p1	hexa	0.0	2.150	2.609	2.003	2.689	2.102	2.694	2.023	2.214	2.001	2.304	2.011	2.312	136.92	18575.6
p2--GP--p1	hexa	0.1	2.153	2.653	2.003	2.728	2.100	2.732	2.026	2.251	2.001	2.350	2.011	2.358	160.65	20323.0
p2--GP--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	913.79	21572.9
p2--GP--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	933.55	24422.7
p2--GP--p2	hexa	0.0	2.214	2.709	2.003	2.924	2.114	2.923	2.032	2.308	2.000	2.644	2.007	2.650	182.72	18784.6
p2--GP--p2	hexa	0.1	2.226	2.761	2.003	2.952	2.112	2.951	2.037	2.371	2.000	2.706	2.007	2.711	232.36	20423.3
p2--GP--p2	triangular unstr	0.0	2.122	2.536	1.992	2.805	2.037	2.807	1.990	2.162	1.998	2.489	2.003	2.499	912.84	21817.9
p2--GP--p2	triangular unstr	0.1	2.108	2.583	1.992	2.840	2.035	2.842	1.987	2.200	1.998	2.553	2.002	2.561	1016.7	24550.7
p2--GP--p3	hexa	0.0	2.214	2.709	2.004	2.923	2.114	2.922	2.032	2.308	2.000	2.650	2.007	2.656	213.98	18882.5
p2--GP--p3	hexa	0.1	2.226	2.761	2.004	2.950	2.113	2.950	2.037	2.371	2.000	2.711	2.007	2.716	266.03	20500.2
p2--GP--p3	triangular unstr	0.0	2.122	2.536	1.993	2.809	2.037	2.812	1.990	2.162	1.998	2.500	2.003	2.510	938.35	21942.5
p2--GP--p3	triangular unstr	0.1	2.108	2.583	1.993	2.843	2.035	2.846	1.987	2.200	1.998	2.564	2.002	2.572	1037.7	24620.3
p3--na--NO	hexa	0.0	2.125	2.485	2.132	2.132	2.135	2.135	2.013	2.050	2.025	2.025	2.024	2.024	173.20	21790.9
p3--na--NO	hexa	0.1	2.108	2.481	2.114	2.114	2.117	2.117	2.010	2.048	2.020	2.020	2.020	2.020	164.48	24491.5
p3--na--NO	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	964.35	27515.3
p3--na--NO	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	994.75	31523.2
p3--CCp1--p1	hexa	0.0	2.185	2.634	2.188	2.354	2.201	2.339	2.020	2.072	2.038	2.290	2.040	2.248	237.07	22963.0
p3--CCp1--p1	hexa	0.1	2.160	2.629	2.164	2.354	2.176	2.339	2.016	2.068	2.031	2.290	2.032	2.248	235.69	25220.2
p3--CCp1--p1	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1256.8	28007.2
p3--CCp1--p1	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1171.8	31951.8
p3--CCp1--p2	hexa	0.0	2.620	3.309	2.640	2.229	2.603	2.184	2.117	2.267	2.264	2.150	2.211	2.094	255.61	23079.3
p3--CCp1--p2	hexa	0.1	2.564	3.307	2.592	2.229	2.554	2.184	2.084	2.237	2.206	2.150	2.160	2.094	240.70	25330.0
p3--CCp1--p2	triangular unstr	0.0	2.396	2.791	2.510	2.101	2.460	2.038	2.003	2.026	2.041	2.058	2.014	2.017	1207.2	28162.8
p3--CCp1--p2	triangular unstr	0.1	2.330	2.777	2.443	2.101	2.397	2.038	1.999	2.022	2.031	2.058	2.006	2.017	1169.7	32100.9
p3--CCp1--p3	hexa	0.0	2.620	3.309	2.640	2.247	2.603	2.210	2.117	2.267	2.264	2.167	2.211	2.112	275.00	23143.0
p3--CCp1--p3	hexa	0.1	2.564	3.307	2.592	2.247	2.554	2.210	2.084	2.237	2.206	2.167	2.160	2.112	262.22	25414.1
p3--CCp1--p3	triangular unstr	0.0	2.396	2.791	2.509	2.126	2.460	2.048	2.003	2.026	2.041	2.074	2.014	2.023	1189.6	28235.2
p3--CCp1--p3	triangular unstr	0.1	2.330	2.777	2.443	2.126	2.397	2.048	2.000	2.023	2.031	2.074	2.006	2.023	1252.4	32160.5
p3--CCp2--p1	hexa	0.0	2.187	2.637	2.200	3.042	2.203	2.643	2.020	2.073	2.041	2.999	2.040	2.257	293.35	23015.5
p3--CCp2--p1	hexa	0.1	2.162	2.631	2.175	3.042	2.177	2.638	2.016	2.069	2.033	2.999	2.033	2.252	285.17	25238.3
p3--CCp2--p1	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1278.2	28367.6
p3--CCp2--p1	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1352.5	31980.0
p3--CCp2--p2	hexa	0.0	3.539	3.992	3.284	3.063	3.303	3.110	4.323	4.278	4.078	3.002	4.085	3.055	277.42	23151.1

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`p3--CCp2--p2`	hexa	0.1	3.570	4.025	3.312	3.063	3.332	3.109	4.298	4.271	4.078	3.002	4.086	3.049	272.86	25356.0
`p3--CCp2--p2`	triangular unstr	0.0	4.057	4.096	3.842	3.013	3.861	3.085	4.476	4.165	4.249	2.997	4.256	3.023	1294.1	28527.1
`p3--CCp2--p2`	triangular unstr	0.1	4.077	4.105	3.880	3.013	3.901	3.082	4.460	4.139	4.196	2.997	4.211	3.022	1435.7	32105.5
`p3--CCp2--p3`	hexa	0.0	3.539	3.992	3.284	3.058	3.303	3.102	4.323	4.278	4.078	3.001	4.085	3.050	262.11	23226.5
`p3--CCp2--p3`	hexa	0.1	3.570	4.025	3.312	3.058	3.332	3.101	4.298	4.271	4.078	3.001	4.086	3.044	289.51	25404.4
`p3--CCp2--p3`	triangular unstr	0.0	4.057	4.096	3.842	3.011	3.861	3.072	4.476	4.165	4.250	2.997	4.257	3.019	1323.3	28624.5
`p3--CCp2--p3`	triangular unstr	0.1	4.077	4.105	3.880	3.011	3.901	3.070	4.460	4.139	4.196	2.997	4.211	3.017	1410.4	32177.0
`p3--CCp3--p1`	hexa	0.0	2.186	2.637	2.200	3.993	2.202	2.225	2.020	2.073	2.041	2.812	2.040	2.040	326.31	23721.1
`p3--CCp3--p1`	hexa	0.1	2.161	2.631	2.174	3.993	2.177	2.201	2.016	2.069	2.033	2.809	2.033	2.033	356.50	25270.9
`p3--CCp3--p1`	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1358.5	30641.3
`p3--CCp3--p1`	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1550.3	32104.3
`p3--CCp3--p2`	hexa	0.0	3.538	3.993	3.290	4.231	3.304	3.441	4.323	4.279	4.079	4.284	4.086	4.100	333.92	23866.2
`p3--CCp3--p2`	hexa	0.1	3.569	4.026	3.319	4.231	3.333	3.485	4.299	4.271	4.081	4.283	4.087	4.101	389.68	25377.7
`p3--CCp3--p2`	triangular unstr	0.0	4.058	4.096	3.846	4.109	3.862	3.934	4.476	4.165	4.252	4.049	4.256	4.239	1394.4	30860.8
`p3--CCp3--p2`	triangular unstr	0.1	4.078	4.105	3.884	4.108	3.903	3.988	4.460	4.139	4.200	4.049	4.211	4.188	1577.0	32238.8
`p3--CCp3--p3`	hexa	0.0	3.538	3.993	3.290	4.241	3.304	3.438	4.323	4.279	4.079	4.302	4.086	4.100	367.11	23951.7
`p3--CCp3--p3`	hexa	0.1	3.569	4.026	3.319	4.241	3.333	3.481	4.299	4.271	4.081	4.300	4.087	4.100	393.42	25437.6
`p3--CCp3--p3`	triangular unstr	0.0	4.057	4.096	3.846	4.114	3.862	3.932	4.477	4.165	4.252	4.050	4.257	4.240	1419.5	30971.1
`p3--CCp3--p3`	triangular unstr	0.1	4.077	4.105	3.884	4.114	3.903	3.986	4.460	4.139	4.200	4.050	4.211	4.189	1585.1	32323.1
`p3--GP--p1`	hexa	0.0	2.187	2.637	2.001	2.598	1.989	2.596	2.020	2.073	2.000	2.068	1.999	2.066	266.14	21856.4
`p3--GP--p1`	hexa	0.1	2.161	2.631	2.001	2.592	1.988	2.589	2.016	2.069	2.000	2.060	1.999	2.059	237.47	24532.6
`p3--GP--p1`	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1040.7	27557.2
`p3--GP--p1`	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1082.2	31544.0
`p3--GP--p2`	hexa	0.0	3.538	3.992	1.999	4.003	1.997	4.010	4.323	4.279	2.000	4.194	2.001	4.195	256.36	21952.8
`p3--GP--p2`	hexa	0.1	3.569	4.026	1.999	4.033	1.997	4.040	4.299	4.271	2.000	4.212	2.000	4.213	293.86	24648.2
`p3--GP--p2`	triangular unstr	0.0	4.057	4.096	1.992	4.077	2.002	4.083	4.476	4.165	1.998	4.204	2.003	4.205	1039.0	27685.6
`p3--GP--p2`	triangular unstr	0.1	4.077	4.105	1.992	4.094	2.001	4.098	4.460	4.139	1.998	4.195	2.003	4.198	1148.6	31684.3
`p3--GP--p3`	hexa	0.0	3.538	3.992	1.999	4.000	1.998	4.008	4.323	4.279	2.000	4.198	2.001	4.200	311.89	22022.3
`p3--GP--p3`	hexa	0.1	3.569	4.026	1.999	4.031	1.997	4.039	4.299	4.271	2.000	4.217	2.001	4.219	334.26	24706.1
`p3--GP--p3`	triangular unstr	0.0	4.057	4.096	1.992	4.079	2.002	4.086	4.476	4.165	1.998	4.227	2.003	4.229	1166.7	27749.1
`p3--GP--p3`	triangular unstr	0.1	4.077	4.105	1.992	4.098	2.002	4.103	4.460	4.139	1.998	4.219	2.003	4.223	1198.0	31741.2

11.1.2 `diagonalIion`

$\Delta t = 10^{-2}$

Table 4: 2D manufactured results for `diagonalIion`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.992	1.992	2.135	2.135	2.134	2.134	1.985	1.985	2.500	2.500	2.497	2.497	10.461	16447.3
`NO--na--NO`	hexa	0.1	1.849	1.849	1.612	1.612	1.618	1.618	2.278	2.278	2.159	2.159	2.163	2.163	34.701	18361.8
`NO--na--NO`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	85.422	17192.2
`NO--na--NO`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	158.63	20812.7
`NO--CCp1--p1`	hexa	0.0	1.993	1.993	2.136	2.354	2.134	2.333	1.985	1.985	2.504	2.289	2.497	2.242	26.905	16918.2
`NO--CCp1--p1`	hexa	0.1	1.849	1.849	1.611	2.354	1.618	2.310	2.278	2.278	2.159	2.289	2.163	2.225	53.309	18484.4
`NO--CCp1--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	152.08	17929.7
`NO--CCp1--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	202.40	20927.9
`NO--CCp1--p2`	hexa	0.0	1.993	1.993	2.139	2.229	2.134	2.184	1.985	1.985	2.513	2.150	2.495	2.094	36.302	17151.2
`NO--CCp1--p2`	hexa	0.1	1.849	1.849	1.609	2.229	1.619	2.181	2.278	2.278	2.159	2.149	2.163	2.091	56.018	18610.3
`NO--CCp1--p2`	triangular unstr	0.0	0.730	0.730	0.778	2.111	0.784	1.164	0.862	0.862	1.037	2.101	1.107	1.150	128.42	18180.6
`NO--CCp1--p2`	triangular unstr	0.1	0.722	0.722	0.772	2.112	0.777	1.172	0.871	0.871	1.027	2.105	1.097	1.143	193.06	21050.2
`NO--CCp1--p3`	hexa	0.0	1.993	1.993	2.139	2.247	2.134	2.211	1.985	1.985	2.513	2.167	2.495	2.112	39.682	17266.4
`NO--CCp1--p3`	hexa	0.1	1.849	1.849	1.609	2.247	1.619	2.206	2.278	2.278	2.159	2.166	2.163	2.109	53.827	18659.0
`NO--CCp1--p3`	triangular unstr	0.0	0.730	0.730	0.778	2.139	0.784	1.132	0.862	0.862	1.037	2.131	1.107	1.144	158.11	18311.1
`NO--CCp1--p3`	triangular unstr	0.1	0.722	0.722	0.772	2.140	0.777	1.139	0.871	0.871	1.027	2.135	1.097	1.136	210.31	21126.0
`NO--CCp2--p1`	hexa	0.0	1.992	1.992	2.141	3.042	2.132	2.624	1.985	1.985	2.517	2.999	2.491	2.603	40.777	17394.3
`NO--CCp2--p1`	hexa	0.1	1.849	1.849	1.607	3.038	1.618	1.948	2.278	2.278	2.158	2.987	2.162	2.187	57.797	18603.4
`NO--CCp2--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	173.84	19021.5
`NO--CCp2--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	225.85	21033.4
`NO--CCp2--p2`	hexa	0.0	1.991	1.991	2.145	3.063	2.120	2.764	1.986	1.986	2.519	3.002	2.444	2.629	42.048	17646.0
`NO--CCp2--p2`	hexa	0.1	1.851	1.851	1.600	3.061	1.624	2.068	2.274	2.274	2.155	2.996	2.162	2.210	59.158	18838.7
`NO--CCp2--p2`	triangular unstr	0.0	0.730	0.730	0.778	1.941	0.784	0.776	0.862	0.862	1.037	1.095	1.107	1.097	162.86	19271.0
`NO--CCp2--p2`	triangular unstr	0.1	0.722	0.722	0.772	1.957	0.777	0.773	0.871	0.871	1.027	1.090	1.097	1.087	250.97	21225.4
`NO--CCp2--p3`	hexa	0.0	1.991	1.991	2.145	3.058	2.120	2.763	1.986	1.986	2.519	3.001	2.444	2.632	40.253	17775.1
`NO--CCp2--p3`	hexa	0.1	1.851	1.851	1.600	3.056	1.624	2.069	2.274	2.274	2.155	2.995	2.162	2.211	57.600	18965.1
`NO--CCp2--p3`	triangular unstr	0.0	0.730	0.730	0.778	1.952	0.784	0.776	0.862	0.862	1.037	1.098	1.107	1.096	171.94	19401.9
`NO--CCp2--p3`	triangular unstr	0.1	0.722	0.722	0.772	1.968	0.777	0.773	0.871	0.871	1.027	1.093	1.097	1.087	226.38	21364.4
`NO--CCp3--p1`	hexa	0.0	1.992	1.992	2.139	4.063	2.132	2.165	1.985	1.985	2.517	3.183	2.491	2.491	55.598	18727.0
`NO--CCp3--p1`	hexa	0.1	1.849	1.849	1.605	3.118	1.618	1.661	2.278	2.278	2.158	2.340	2.163	2.189	77.383	19869.3
`NO--CCp3--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	206.33	21447.4
`NO--CCp3--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	261.62	23304.6
`NO--CCp3--p2`	hexa	0.0	1.992	1.992	2.141	4.277	2.120	2.216	1.985	1.985	2.519	4.233	2.444	2.443	71.803	19000.5
`NO--CCp3--p2`	hexa	0.1	1.851	1.851	1.594	3.421	1.622	1.722	2.274	2.274	2.155	2.589	2.162	2.221	75.873	20100.4

			Fit $N = 10$–100						Fit $N = 70$–100							
Vm-states-Ion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
NO--CCp3--p2	triangular unstr	0.0	0.730	0.730	0.778	1.071	0.784	0.796	0.862	0.862	1.037	1.037	1.107	1.109	214.00	21742.7
NO--CCp3--p2	triangular unstr	0.1	0.722	0.722	0.772	1.081	0.777	0.798	0.871	0.871	1.027	1.028	1.097	1.100	282.58	23548.4
NO--CCp3--p3	hexa	0.0	1.992	1.992	2.141	4.293	2.120	2.214	1.985	1.985	2.519	4.254	2.444	2.443	60.643	19121.6
NO--CCp3--p3	hexa	0.1	1.851	1.851	1.594	3.409	1.622	1.721	2.274	2.274	2.155	2.578	2.162	2.221	67.099	20203.7
NO--CCp3--p3	triangular unstr	0.0	0.730	0.730	0.778	1.054	0.784	0.796	0.862	0.862	1.037	1.037	1.107	1.109	243.59	21897.1
NO--CCp3--p3	triangular unstr	0.1	0.722	0.722	0.772	1.064	0.777	0.797	0.871	0.871	1.027	1.028	1.097	1.100	198.26	23661.4
NO--GP--p1	hexa	0.0	1.992	1.992	2.000	2.057	2.005	2.061	1.985	1.985	2.002	2.031	2.023	2.031	13.699	16534.5
NO--GP--p1	hexa	0.1	1.849	1.849	2.000	2.049	1.997	2.053	2.278	2.278	2.002	2.039	2.030	2.039	45.175	18416.0
NO--GP--p1	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	87.018	17338.7
NO--GP--p1	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	165.88	20895.6
NO--GP--p2	hexa	0.0	1.992	1.992	1.999	2.015	2.002	2.017	1.986	1.986	2.001	2.010	2.011	2.010	28.704	16813.7
NO--GP--p2	hexa	0.1	1.851	1.851	1.999	2.015	2.001	2.017	2.274	2.274	2.001	2.011	2.013	2.011	52.057	18561.0
NO--GP--p2	triangular unstr	0.0	0.730	0.730	1.878	1.418	1.102	1.411	0.862	0.862	1.678	1.219	1.112	1.269	133.70	17598.5
NO--GP--p2	triangular unstr	0.1	0.722	0.722	1.880	1.427	1.107	1.420	0.871	0.871	1.682	1.220	1.104	1.270	133.85	21034.4
NO--GP--p3	hexa	0.0	1.992	1.992	1.999	2.019	2.002	2.021	1.986	1.986	2.001	2.011	2.011	2.011	47.744	16969.7
NO--GP--p3	hexa	0.1	1.851	1.851	1.999	2.019	2.001	2.021	2.274	2.274	2.001	2.013	2.013	2.013	50.120	18613.3
NO--GP--p3	triangular unstr	0.0	0.730	0.730	1.878	1.389	1.102	1.384	0.862	0.862	1.678	1.206	1.112	1.258	71.347	17710.6
NO--GP--p3	triangular unstr	0.1	0.722	0.722	1.881	1.398	1.107	1.392	0.871	0.871	1.682	1.207	1.104	1.259	143.32	21099.4
p1--na--NO	hexa	0.0	1.911	2.146	1.616	1.616	1.626	1.626	2.612	2.168	2.381	2.381	2.387	2.387	80.953	17858.1
p1--na--NO	hexa	0.1	1.971	2.163	1.677	1.677	1.687	1.687	2.614	2.141	2.416	2.416	2.422	2.422	94.616	19059.8
p1--na--NO	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	590.16	19872.9
p1--na--NO	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	559.07	21538.7
p1--CCp1--p1	hexa	0.0	1.911	2.146	1.616	2.349	1.626	1.955	2.612	2.168	2.381	2.294	2.387	2.369	91.258	18293.1
p1--CCp1--p1	hexa	0.1	1.971	2.163	1.677	2.351	1.687	2.045	2.614	2.141	2.416	2.293	2.422	2.383	104.24	19267.6
p1--CCp1--p1	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	623.89	20378.7
p1--CCp1--p1	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	591.30	21752.6
p1--CCp1--p2	hexa	0.0	1.911	2.146	1.616	2.228	1.626	2.057	2.612	2.168	2.381	2.153	2.387	2.232	91.419	18505.1
p1--CCp1--p2	hexa	0.1	1.971	2.163	1.676	2.228	1.687	2.104	2.614	2.141	2.416	2.152	2.422	2.208	107.48	19395.6
p1--CCp1--p2	triangular unstr	0.0	2.271	2.126	2.004	2.102	2.002	2.036	2.384	2.115	2.552	2.063	2.549	2.285	611.35	20633.4
p1--CCp1--p2	triangular unstr	0.1	2.302	2.106	2.092	2.102	2.090	2.066	2.222	2.054	2.524	2.061	2.524	2.190	595.05	21917.4
p1--CCp1--p3	hexa	0.0	1.911	2.146	1.616	2.246	1.626	2.063	2.612	2.168	2.381	2.170	2.387	2.255	97.575	18602.7
p1--CCp1--p3	hexa	0.1	1.971	2.163	1.676	2.246	1.687	2.115	2.614	2.141	2.416	2.169	2.422	2.234	111.94	19454.3
p1--CCp1--p3	triangular unstr	0.0	2.271	2.126	2.004	2.127	2.002	2.040	2.384	2.115	2.552	2.080	2.549	2.315	608.61	20768.4
p1--CCp1--p3	triangular unstr	0.1	2.302	2.106	2.092	2.127	2.090	2.075	2.222	2.054	2.524	2.078	2.524	2.217	593.84	21991.6
p1--CCp2--p1	hexa	0.0	1.911	2.146	1.616	2.904	1.626	1.665	2.612	2.168	2.381	2.783	2.387	2.395	109.67	18769.5
p1--CCp2--p1	hexa	0.1	1.971	2.163	1.676	2.954	1.687	1.736	2.614	2.141	2.416	2.859	2.422	2.431	122.16	19370.8
p1--CCp2--p1	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	661.75	21487.3
p1--CCp2--p1	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	625.71	22098.7
p1--CCp2--p2	hexa	0.0	1.912	2.146	1.616	2.990	1.627	1.688	2.612	2.168	2.381	2.882	2.387	2.400	120.42	18967.0
p1--CCp2--p2	hexa	0.1	1.971	2.163	1.676	3.018	1.687	1.765	2.613	2.141	2.416	2.931	2.422	2.437	122.15	19561.4
p1--CCp2--p2	triangular unstr	0.0	2.271	2.126	2.004	2.677	2.002	1.977	2.384	2.115	2.552	2.669	2.549	2.528	649.54	21723.7
p1--CCp2--p2	triangular unstr	0.1	2.301	2.106	2.092	2.799	2.090	2.062	2.222	2.054	2.524	2.735	2.523	2.511	631.25	22285.7
p1--CCp2--p3	hexa	0.0	1.912	2.146	1.616	2.987	1.627	1.688	2.612	2.168	2.381	2.884	2.387	2.400	121.12	19064.7
p1--CCp2--p3	hexa	0.1	1.971	2.163	1.676	3.015	1.687	1.765	2.613	2.141	2.416	2.932	2.422	2.437	117.46	19617.3
p1--CCp2--p3	triangular unstr	0.0	2.271	2.126	2.004	2.681	2.002	1.977	2.384	2.115	2.552	2.672	2.549	2.527	650.20	21848.2
p1--CCp2--p3	triangular unstr	0.1	2.301	2.106	2.092	2.802	2.090	2.062	2.222	2.054	2.524	2.740	2.523	2.510	629.57	22409.5
p1--CCp3--p1	hexa	0.0	1.911	2.146	1.616	2.034	1.626	1.630	2.612	2.168	2.381	2.378	2.387	2.407	148.41	20023.7
p1--CCp3--p1	hexa	0.1	1.971	2.163	1.677	2.165	1.687	1.692	2.614	2.141	2.416	2.409	2.422	2.443	137.46	20539.4
p1--CCp3--p1	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	694.57	23794.5
p1--CCp3--p1	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	677.84	24272.0
p1--CCp3--p2	hexa	0.0	1.912	2.146	1.617	2.239	1.627	1.634	2.612	2.168	2.381	2.377	2.387	2.426	135.39	20235.5
p1--CCp3--p2	hexa	0.1	1.971	2.163	1.677	2.394	1.688	1.698	2.613	2.141	2.416	2.402	2.422	2.464	139.88	20752.4
p1--CCp3--p2	triangular unstr	0.0	2.271	2.126	2.004	2.031	2.002	1.964	2.384	2.115	2.552	2.527	2.549	2.528	738.52	24032.2
p1--CCp3--p2	triangular unstr	0.1	2.301	2.106	2.092	2.141	2.090	2.052	2.222	2.054	2.524	2.511	2.523	2.516	676.05	24507.1
p1--CCp3--p3	hexa	0.0	1.912	2.146	1.617	2.225	1.627	1.634	2.612	2.168	2.381	2.376	2.387	2.426	146.52	20334.8
p1--CCp3--p3	hexa	0.1	1.971	2.163	1.677	2.379	1.688	1.697	2.613	2.141	2.416	2.402	2.422	2.463	136.92	20848.2
p1--CCp3--p3	triangular unstr	0.0	2.271	2.126	2.004	2.027	2.002	1.964	2.384	2.115	2.552	2.527	2.549	2.528	771.47	24158.5
p1--CCp3--p3	triangular unstr	0.1	2.301	2.106	2.092	2.136	2.090	2.052	2.222	2.054	2.524	2.512	2.523	2.516	663.06	24640.8
p1--GP--p1	hexa	0.0	1.911	2.146	1.998	2.154	1.838	2.160	2.612	2.168	2.007	2.316	2.258	2.316	85.742	17904.5
p1--GP--p1	hexa	0.1	1.971	2.163	1.999	2.221	1.898	2.227	2.614	2.141	2.005	2.319	2.234	2.318	106.80	19111.6
p1--GP--p1	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	598.75	19898.8
p1--GP--p1	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	560.43	21570.5
p1--GP--p2	hexa	0.0	1.912	2.146	1.999	2.182	1.951	2.181	2.612	2.168	2.002	2.161	2.105	2.157	101.45	18110.1
p1--GP--p2	hexa	0.1	1.971	2.163	1.999	2.201	1.975	2.199	2.613	2.141	2.001	2.156	2.080	2.151	122.27	19231.4
p1--GP--p2	triangular unstr	0.0	2.271	2.126	1.992	2.069	2.012	2.068	2.384	2.115	2.004	2.184	2.284	2.176	625.13	20140.2
p1--GP--p2	triangular unstr	0.1	2.301	2.106	1.992	2.089	2.038	2.087	2.222	2.054	2.002	2.117	2.186	2.113	570.64	21716.5
p1--GP--p3	hexa	0.0	1.912	2.146	1.999	2.196	1.951	2.195	2.612	2.168	2.002	2.177	2.105	2.173	108.74	18227.1
p1--GP--p3	hexa	0.1	1.971	2.163	1.999	2.217	1.975	2.215	2.613	2.141	2.001	2.172	2.080	2.167	117.87	19282.9
p1--GP--p3	triangular unstr	0.0	2.271	2.126	1.992	2.082	2.012	2.081	2.384	2.115	2.004	2.219	2.284	2.210	663.03	20261.8
p1--GP--p3	triangular unstr	0.1	2.301	2.106	1.992	2.109	2.038	2.106	2.222	2.054	2.002	2.144	2.186	2.139	563.90	21789.6
p2--na--NO	hexa	0.0	2.127	2.567	2.176	2.176	2.176	2.176	2.016	2.179	2.139	2.139	2.139	2.139	98.709	18691.7
p2--na--NO	hexa	0.1	2.128	2.607	2.180	2.180	2.181	2.181	2.018	2.209	2.153	2.153	2.153	2.153	115.37	20296.1
p2--na--NO	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.161	2.161	1.982	2.068	2.260	2.260	2.263	2.263	620.05	21912.9
p2--na--NO	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	587.89	24376.3
p2--CCp1--p1	hexa	0.0	2.149	2.608	2.201	2.356	2.203	2.337	2.019	2.209	2.154	2.292	2.154	2.244	117.40	19455.1
p2--CCp1--p1	hexa	0.1	2.151	2.651	2.208	2.356	2.211	2.338	2.021	2.247	2.171	2.291	2.172	2.246	119.24	20864.2
p2--CCp1--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.161	2.161	1.982	2.068	2.260	2.260	2.263	2.263	727.82	22654.0
p2--CCp1--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	615.73	24950.3
p2--CCp1--p2	hexa	0.0	2.212	2.708	2.272	2.230	2.283	2.188	2.026	2.303	2.199	2.150	2.206	2.095	123.45	19688.0
p2--CCp1--p2	hexa	0.1	2.223	2.759	2.291	2.229	2.304	2.188	2.030	2.364	2.229	2.150	2.238	2.095	121.64	20990.5
p2--CCp1--p2	triangular unstr	0.0	2.118	2.532	2.273	2.101	2.279	2.040	1.977	2.149	2.394	2.057	2.419	2.018	726.78	22876.9

			Fit $N = 10$–100						Fit $N = 70$–100							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p2--CCp1--p2	triangular unstr	0.1	2.104	2.578	2.283	2.101	2.292	2.039	1.972	2.185	2.455	2.057	2.479	2.018	596.61	25104.2
p2--CCp1--p3	hexa	0.0	2.212	2.708	2.272	2.248	2.283	2.215	2.026	2.303	2.199	2.167	2.206	2.114	132.77	19802.8
p2--CCp1--p3	hexa	0.1	2.223	2.759	2.291	2.248	2.304	2.214	2.030	2.364	2.229	2.167	2.238	2.114	120.06	21058.9
p2--CCp1--p3	triangular unstr	0.0	2.118	2.532	2.273	2.126	2.279	2.050	1.977	2.149	2.394	2.074	2.419	2.023	777.40	22967.1
p2--CCp1--p3	triangular unstr	0.1	2.104	2.578	2.283	2.126	2.292	2.050	1.972	2.185	2.455	2.074	2.479	2.023	627.52	25163.3
p2--CCp2--p1	hexa	0.0	2.149	2.608	2.203	3.039	2.203	2.289	2.019	2.210	2.155	2.989	2.155	2.169	140.24	19917.8
p2--CCp2--p1	hexa	0.1	2.152	2.651	2.211	3.040	2.211	2.310	2.021	2.247	2.173	2.991	2.172	2.189	125.89	20909.1
p2--CCp2--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.161	2.161	1.982	2.068	2.260	2.260	2.263	2.263	754.03	23653.5
p2--CCp2--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	617.64	25105.4
p2--CCp2--p2	hexa	0.0	2.212	2.708	2.284	3.063	2.283	2.480	2.026	2.303	2.206	3.000	2.206	2.250	141.14	20114.9
p2--CCp2--p2	hexa	0.1	2.224	2.760	2.305	3.063	2.305	2.527	2.030	2.365	2.239	3.001	2.239	2.294	134.56	21038.7
p2--CCp2--p2	triangular unstr	0.0	2.118	2.532	2.286	3.012	2.279	2.428	1.977	2.149	2.413	2.995	2.420	2.443	727.80	23875.1
p2--CCp2--p2	triangular unstr	0.1	2.104	2.579	2.300	3.012	2.293	2.467	1.972	2.186	2.481	2.996	2.482	2.511	613.39	25257.0
p2--CCp2--p3	hexa	0.0	2.212	2.708	2.284	3.058	2.283	2.480	2.026	2.303	2.206	2.999	2.206	2.251	140.66	20243.0
p2--CCp2--p3	hexa	0.1	2.224	2.760	2.305	3.058	2.305	2.527	2.030	2.365	2.239	3.000	2.239	2.296	143.18	21098.7
p2--CCp2--p3	triangular unstr	0.0	2.118	2.532	2.286	3.010	2.279	2.425	1.977	2.149	2.413	2.995	2.420	2.443	725.90	23969.0
p2--CCp2--p3	triangular unstr	0.1	2.104	2.579	2.300	3.010	2.293	2.465	1.972	2.186	2.481	2.996	2.482	2.512	622.84	25335.9
p2--CCp3--p1	hexa	0.0	2.149	2.608	2.203	3.175	2.203	2.220	2.019	2.210	2.155	2.123	2.155	2.154	167.65	21118.3
p2--CCp3--p1	hexa	0.1	2.152	2.651	2.211	3.250	2.211	2.227	2.021	2.247	2.173	2.151	2.172	2.171	198.67	21530.3
p2--CCp3--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.161	2.161	1.982	2.068	2.260	2.260	2.263	2.263	773.41	25977.5
p2--CCp3--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	668.09	26830.7
p2--CCp3--p2	hexa	0.0	2.212	2.708	2.284	3.846	2.283	2.347	2.026	2.303	2.206	2.397	2.206	2.204	162.87	21297.1
p2--CCp3--p2	hexa	0.1	2.224	2.760	2.305	3.936	2.304	2.369	2.030	2.365	2.239	2.584	2.238	2.237	184.34	21724.7
p2--CCp3--p2	triangular unstr	0.0	2.118	2.532	2.286	3.422	2.279	2.345	1.977	2.149	2.413	2.437	2.420	2.419	793.59	26150.4
p2--CCp3--p2	triangular unstr	0.1	2.104	2.579	2.300	3.528	2.293	2.358	1.972	2.186	2.481	2.550	2.482	2.482	653.69	27025.5
p2--CCp3--p3	hexa	0.0	2.212	2.708	2.284	3.839	2.283	2.346	2.026	2.303	2.206	2.351	2.206	2.204	159.34	21389.3
p2--CCp3--p3	hexa	0.1	2.224	2.760	2.305	3.933	2.304	2.368	2.030	2.365	2.239	2.531	2.238	2.237	169.65	21784.7
p2--CCp3--p3	triangular unstr	0.0	2.118	2.532	2.286	3.389	2.279	2.343	1.977	2.149	2.413	2.404	2.420	2.419	803.40	26267.3
p2--CCp3--p3	triangular unstr	0.1	2.104	2.579	2.300	3.497	2.293	2.356	1.972	2.186	2.481	2.509	2.482	2.482	596.37	27166.6
p2--GP--p1	hexa	0.0	2.149	2.608	2.004	2.716	2.109	2.720	2.019	2.210	2.002	2.416	2.034	2.422	114.99	18745.8
p2--GP--p1	hexa	0.1	2.152	2.651	2.004	2.757	2.107	2.760	2.021	2.247	2.002	2.470	2.033	2.476	131.63	20347.1
p2--GP--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.161	2.161	1.982	2.068	2.260	2.260	2.263	2.263	643.71	21946.4
p2--GP--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	626.12	24414.7
p2--GP--p2	hexa	0.0	2.212	2.708	2.004	2.942	2.118	2.941	2.026	2.303	2.001	2.734	2.020	2.738	133.29	18953.3
p2--GP--p2	hexa	0.1	2.224	2.760	2.004	2.969	2.116	2.968	2.030	2.365	2.001	2.794	2.019	2.797	154.46	20458.3
p2--GP--p2	triangular unstr	0.0	2.118	2.532	1.993	2.860	2.044	2.862	1.977	2.149	2.000	2.751	2.026	2.757	655.40	22216.4
p2--GP--p2	triangular unstr	0.1	2.104	2.579	1.993	2.895	2.042	2.896	1.972	2.186	2.000	2.820	2.025	2.823	649.16	24557.3
p2--GP--p3	hexa	0.0	2.212	2.708	2.004	2.940	2.118	2.940	2.026	2.303	2.001	2.739	2.020	2.743	129.78	19057.3
p2--GP--p3	hexa	0.1	2.224	2.760	2.004	2.967	2.117	2.966	2.030	2.365	2.001	2.798	2.019	2.801	196.31	20537.1
p2--GP--p3	triangular unstr	0.0	2.118	2.532	1.993	2.863	2.044	2.865	1.977	2.149	2.000	2.758	2.026	2.764	683.61	22289.2
p2--GP--p3	triangular unstr	0.1	2.104	2.579	1.993	2.897	2.041	2.898	1.972	2.186	2.000	2.825	2.025	2.829	626.99	24625.8
p3--na--NO	hexa	0.0	2.121	2.481	2.256	2.256	2.258	2.258	1.999	2.037	2.472	2.472	2.469	2.469	127.96	21803.9
p3--na--NO	hexa	0.1	2.104	2.476	2.238	2.238	2.240	2.240	1.996	2.034	2.467	2.467	2.464	2.464	175.14	24520.5
p3--na--NO	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.345	2.345	1.972	1.988	3.183	3.183	3.172	3.172	725.87	27537.3
p3--na--NO	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.182	3.182	3.171	3.171	700.89	31526.0
p3--CCp1--p1	hexa	0.0	2.180	2.629	2.347	2.353	2.360	2.339	2.003	2.055	2.608	2.289	2.617	2.247	157.29	22965.0
p3--CCp1--p1	hexa	0.1	2.155	2.624	2.323	2.353	2.335	2.339	1.998	2.051	2.600	2.289	2.608	2.247	167.80	25249.5
p3--CCp1--p1	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.345	2.345	1.972	1.988	3.183	3.183	3.172	3.172	853.69	28339.2
p3--CCp1--p1	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.182	3.182	3.171	3.171	638.38	31988.0
p3--CCp1--p2	hexa	0.0	2.672	3.361	2.672	2.229	2.497	2.184	2.142	2.316	1.405	2.150	1.266	2.093	166.44	23100.5
p3--CCp1--p2	hexa	0.1	2.615	3.360	2.623	2.229	2.442	2.184	2.103	2.281	1.306	2.150	1.209	2.093	170.38	25367.0
p3--CCp1--p2	triangular unstr	0.0	2.439	2.843	2.197	2.101	2.046	2.038	1.999	2.026	0.132	2.057	0.463	2.016	850.23	28492.7
p3--CCp1--p2	triangular unstr	0.1	2.369	2.829	2.121	2.101	1.976	2.038	1.995	2.023	0.125	2.057	0.459	2.016	645.23	32105.2
p3--CCp1--p3	hexa	0.0	2.672	3.361	2.672	2.247	2.497	2.210	2.142	2.316	1.405	2.166	1.266	2.111	161.31	23166.0
p3--CCp1--p3	hexa	0.1	2.615	3.360	2.623	2.247	2.442	2.210	2.103	2.281	1.306	2.166	1.209	2.111	173.48	25411.9
p3--CCp1--p3	triangular unstr	0.0	2.439	2.843	2.196	2.126	2.045	2.048	1.999	2.026	0.132	2.073	0.463	2.021	861.85	28573.6
p3--CCp1--p3	triangular unstr	0.1	2.369	2.829	2.121	2.126	1.976	2.048	1.995	2.023	0.125	2.073	0.459	2.021	920.66	32166.9
p3--CCp2--p1	hexa	0.0	2.181	2.631	2.362	3.042	2.362	2.762	2.003	2.055	2.629	2.999	2.619	2.745	171.36	23027.8
p3--CCp2--p1	hexa	0.1	2.156	2.626	2.337	3.042	2.337	2.758	1.998	2.051	2.620	2.999	2.611	2.740	201.57	25251.4
p3--CCp2--p1	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.345	2.345	1.972	1.988	3.183	3.183	3.172	3.172	902.08	28536.7
p3--CCp2--p1	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.182	3.182	3.171	3.171	846.35	31989.9
p3--CCp2--p2	hexa	0.0	3.520	3.971	2.698	3.063	2.707	3.071	4.171	4.123	0.669	3.002	0.641	2.724	156.32	23188.1
p3--CCp2--p2	hexa	0.1	3.547	4.002	2.648	3.063	2.657	3.069	4.104	4.100	0.514	3.002	0.488	2.717	238.50	25382.0
p3--CCp2--p2	triangular unstr	0.0	3.748	3.924	2.081	3.013	2.060	2.855	2.165	2.958	0.011	2.997	0.007	1.544	936.56	28757.3
p3--CCp2--p2	triangular unstr	0.1	3.692	3.923	1.983	3.013	1.961	2.852	1.802	2.887	0.004	2.997	0.001	1.543	822.29	32135.0
p3--CCp2--p3	hexa	0.0	3.520	3.971	2.698	3.058	2.707	3.064	4.171	4.123	0.669	3.001	0.641	2.729	152.33	23274.7
p3--CCp2--p3	hexa	0.1	3.547	4.002	2.648	3.058	2.657	3.062	4.104	4.100	0.514	3.001	0.488	2.721	241.40	25437.1
p3--CCp2--p3	triangular unstr	0.0	3.748	3.923	2.081	3.011	2.060	2.849	2.165	2.958	0.011	2.996	0.007	1.567	944.25	28825.7
p3--CCp2--p3	triangular unstr	0.1	3.692	3.923	1.982	3.011	1.961	2.846	1.802	2.887	0.004	2.996	0.001	1.565	806.35	32197.5
p3--CCp3--p1	hexa	0.0	2.181	2.631	2.362	4.082	2.362	2.385	2.003	2.056	2.629	3.391	2.619	2.619	193.31	23618.3
p3--CCp3--p1	hexa	0.1	2.156	2.626	2.337	4.082	2.337	2.361	1.998	2.051	2.620	3.388	2.611	2.611	189.17	25269.6
p3--CCp3--p1	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.345	2.345	1.972	1.988	3.183	3.183	3.172	3.172	873.64	30657.9
p3--CCp3--p1	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.182	3.182	3.171	3.171	823.45	32119.3
p3--CCp3--p2	hexa	0.0	3.520	3.971	2.705	4.177	2.709	2.869	4.171	4.124	0.671	3.889	0.642	0.730	196.25	23779.8
p3--CCp3--p2	hexa	0.1	3.547	4.002	2.656	4.177	2.658	2.840	4.105	4.100	0.516	3.888	0.490	0.578	239.00	25414.8
p3--CCp3--p2	triangular unstr	0.0	3.749	3.924	2.085	3.907	2.062	2.112	2.165	2.958	0.011	2.695	0.007	0.004	844.77	30890.6
p3--CCp3--p2	triangular unstr	0.1	3.693	3.924	1.987	3.906	1.963	2.027	1.802	2.887	0.004	2.694	0.001	-0.001	758.28	32277.2
p3--CCp3--p3	hexa	0.0	3.520	3.971	2.705	4.182	2.709	2.866	4.171	4.124	0.671	3.875	0.642	0.729	178.20	23856.2
p3--CCp3--p3	hexa	0.1	3.547	4.002	2.656	4.182	2.658	2.835	4.105	4.100	0.516	3.873	0.490	0.577	215.29	25464.6
p3--CCp3--p3	triangular unstr	0.0	3.749	3.924	2.085	3.889	2.061	2.108	2.165	2.958	0.011	2.592	0.007	0.004	905.41	30993.9
p3--CCp3--p3	triangular unstr	0.1	3.693	3.924	1.986	3.889	1.963	2.022	1.802	2.887	0.004	2.591	0.001	-0.001	718.57	32342.4
p3--GP--p1	hexa	0.0	2.181	2.631	2.002	2.762	1.992	2.757	2.003	2.056	2.002	2.661	2.008	2.650	151.23	21847.0

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`p3--GP--p1`	hexa	0.1	2.156	2.626	2.002	2.756	1.991	2.752	1.998	2.051	2.002	2.653	2.008	2.642	196.90	24560.6
`p3--GP--p1`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.345	2.345	1.972	1.988	3.183	3.183	3.172	3.172	792.64	27566.0
`p3--GP--p1`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.182	3.182	3.171	3.171	711.74	31533.8
`p3--GP--p2`	hexa	0.0	3.520	3.971	2.000	3.744	1.999	3.748	4.171	4.124	2.001	1.458	2.006	1.396	178.12	21950.5
`p3--GP--p2`	hexa	0.1	3.547	4.002	1.999	3.757	1.998	3.761	4.105	4.100	2.001	1.348	2.006	1.284	181.72	24663.2
`p3--GP--p2`	triangular unstr	0.0	3.749	3.924	1.992	2.844	2.005	2.843	2.165	2.958	2.000	-0.092	2.012	-0.097	974.21	27701.8
`p3--GP--p2`	triangular unstr	0.1	3.693	3.924	1.992	2.836	2.005	2.836	1.802	2.887	2.000	-0.098	2.012	-0.103	907.50	31675.3
`p3--GP--p3`	hexa	0.0	3.520	3.971	2.000	3.738	1.999	3.742	4.171	4.124	2.001	1.420	2.006	1.355	206.65	22042.1
`p3--GP--p3`	hexa	0.1	3.547	4.002	2.000	3.751	1.999	3.755	4.105	4.100	2.001	1.305	2.006	1.239	227.33	24736.5
`p3--GP--p3`	triangular unstr	0.0	3.748	3.924	1.992	2.812	2.005	2.811	2.165	2.958	2.000	-0.108	2.012	-0.114	802.37	27772.6
`p3--GP--p3`	triangular unstr	0.1	3.693	3.923	1.992	2.803	2.005	2.803	1.802	2.887	2.000	-0.115	2.012	-0.120	733.20	31735.5

$\Delta t = 10^{-3}$

Table 5: 2D manufactured results for `diagonalIion`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.997	1.997	1.998	1.998	1.998	1.998	2.000	2.000	2.004	2.004	2.004	2.004	66.387	16461.7
`NO--na--NO`	hexa	0.1	1.851	1.851	1.607	1.607	1.613	1.613	2.294	2.294	2.140	2.140	2.142	2.142	121.55	18342.1
`NO--na--NO`	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	375.50	17145.7
`NO--na--NO`	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	537.35	20800.8
`NO--CCp1--p1`	hexa	0.0	1.997	1.997	1.999	2.354	1.998	2.333	2.000	2.000	2.004	2.290	2.004	2.242	104.98	16905.3
`NO--CCp1--p1`	hexa	0.1	1.851	1.851	1.608	2.354	1.613	2.310	2.294	2.294	2.140	2.290	2.142	2.226	158.70	18479.0
`NO--CCp1--p1`	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	486.71	17867.2
`NO--CCp1--p1`	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	595.77	20904.5
`NO--CCp1--p2`	hexa	0.0	1.997	1.997	2.000	2.229	1.999	2.184	2.000	2.000	2.004	2.150	2.004	2.094	115.96	17132.7
`NO--CCp1--p2`	hexa	0.1	1.851	1.851	1.608	2.229	1.614	2.181	2.295	2.295	2.140	2.150	2.142	2.092	177.55	18601.7
`NO--CCp1--p2`	triangular unstr	0.0	0.798	0.798	0.777	2.111	0.783	1.163	0.963	0.963	1.038	2.102	1.107	1.150	554.31	18116.1
`NO--CCp1--p2`	triangular unstr	0.1	0.785	0.785	0.771	2.112	0.776	1.171	0.953	0.953	1.027	2.105	1.097	1.143	726.29	21044.1
`NO--CCp1--p3`	hexa	0.0	1.997	1.997	2.000	2.247	1.999	2.211	2.000	2.000	2.004	2.167	2.004	2.113	115.51	17272.0
`NO--CCp1--p3`	hexa	0.1	1.851	1.851	1.608	2.247	1.614	2.206	2.295	2.295	2.140	2.167	2.142	2.110	210.02	18665.7
`NO--CCp1--p3`	triangular unstr	0.0	0.798	0.798	0.777	2.139	0.783	1.131	0.963	0.963	1.038	2.131	1.107	1.144	522.54	18249.6
`NO--CCp1--p3`	triangular unstr	0.1	0.785	0.785	0.771	2.140	0.776	1.138	0.953	0.953	1.027	2.135	1.097	1.137	605.11	21107.3
`NO--CCp2--p1`	hexa	0.0	1.997	1.997	2.000	3.042	1.997	2.510	1.999	1.999	2.004	2.998	2.003	2.157	163.49	17393.5
`NO--CCp2--p1`	hexa	0.1	1.851	1.851	1.603	3.038	1.613	1.942	2.294	2.294	2.140	2.987	2.142	2.167	258.60	18604.2
`NO--CCp2--p1`	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	633.81	18972.1
`NO--CCp2--p1`	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	848.91	21040.1
`NO--CCp2--p2`	hexa	0.0	1.995	1.995	2.003	3.063	1.997	2.671	1.999	1.999	2.004	3.002	2.003	2.246	198.35	17648.5
`NO--CCp2--p2`	hexa	0.1	1.853	1.853	1.597	3.061	1.619	2.063	2.289	2.289	2.140	2.996	2.141	2.192	269.27	18821.5
`NO--CCp2--p2`	triangular unstr	0.0	0.799	0.799	0.777	1.940	0.783	0.775	0.963	0.963	1.038	1.095	1.107	1.097	779.96	19260.5
`NO--CCp2--p2`	triangular unstr	0.1	0.786	0.786	0.771	1.955	0.776	0.772	0.953	0.953	1.027	1.090	1.097	1.087	754.35	21228.6
`NO--CCp2--p3`	hexa	0.0	1.995	1.995	2.003	3.058	1.997	2.670	1.999	1.999	2.004	3.001	2.003	2.251	219.51	17774.7
`NO--CCp2--p3`	hexa	0.1	1.853	1.853	1.597	3.056	1.619	2.064	2.289	2.289	2.140	2.995	2.141	2.193	258.90	18968.1
`NO--CCp2--p3`	triangular unstr	0.0	0.799	0.799	0.777	1.950	0.783	0.775	0.963	0.963	1.038	1.098	1.107	1.097	634.20	19406.0
`NO--CCp2--p3`	triangular unstr	0.1	0.786	0.786	0.771	1.966	0.776	0.772	0.953	0.953	1.027	1.092	1.097	1.087	913.75	21372.5
`NO--CCp3--p1`	hexa	0.0	1.997	1.997	1.998	3.944	1.997	2.030	1.999	1.999	2.004	2.505	2.004	2.004	334.52	18754.3
`NO--CCp3--p1`	hexa	0.1	1.851	1.851	1.600	3.114	1.613	1.657	2.294	2.294	2.140	2.324	2.142	2.170	378.04	19859.4
`NO--CCp3--p1`	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	1067.1	21482.2
`NO--CCp3--p1`	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	1049.8	23311.5
`NO--CCp3--p2`	hexa	0.0	1.996	1.996	1.998	4.244	1.996	2.092	1.999	1.999	2.004	3.862	2.003	2.002	342.67	19008.4
`NO--CCp3--p2`	hexa	0.1	1.853	1.853	1.590	3.420	1.617	1.718	2.289	2.289	2.140	2.581	2.141	2.203	351.87	20077.6
`NO--CCp3--p2`	triangular unstr	0.0	0.799	0.799	0.777	1.070	0.783	0.795	0.963	0.963	1.038	1.038	1.107	1.110	1034.3	21759.2
`NO--CCp3--p2`	triangular unstr	0.1	0.786	0.786	0.771	1.080	0.776	0.797	0.953	0.953	1.027	1.028	1.097	1.100	1164.8	23565.8
`NO--CCp3--p3`	hexa	0.0	1.996	1.996	1.998	4.257	1.996	2.090	1.999	1.999	2.004	3.849	2.003	2.002	383.46	19125.2
`NO--CCp3--p3`	hexa	0.1	1.853	1.853	1.590	3.408	1.617	1.717	2.289	2.289	2.140	2.570	2.141	2.203	369.57	20194.4
`NO--CCp3--p3`	triangular unstr	0.0	0.799	0.799	0.777	1.053	0.783	0.795	0.963	0.963	1.038	1.038	1.107	1.110	824.24	21903.0
`NO--CCp3--p3`	triangular unstr	0.1	0.786	0.786	0.771	1.062	0.776	0.796	0.953	0.953	1.027	1.028	1.097	1.100	1012.6	23686.4
`NO--GP--p1`	hexa	0.0	1.997	1.997	1.999	2.050	1.998	2.053	2.000	2.000	2.000	2.007	2.000	2.007	83.484	16618.9
`NO--GP--p1`	hexa	0.1	1.851	1.851	1.999	2.042	1.990	2.046	2.294	2.294	2.000	2.015	2.008	2.016	169.69	18446.5
`NO--GP--p1`	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	499.64	17283.5
`NO--GP--p1`	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	657.51	20901.9
`NO--GP--p2`	hexa	0.0	1.997	1.997	1.999	2.013	1.998	2.015	1.999	1.999	2.000	2.002	2.000	2.002	202.57	16847.9
`NO--GP--p2`	hexa	0.1	1.853	1.853	1.999	2.013	1.998	2.014	2.290	2.290	2.000	2.003	2.001	2.003	267.50	18578.8
`NO--GP--p2`	triangular unstr	0.0	0.799	0.799	1.878	1.417	1.101	1.410	0.963	0.963	1.677	1.218	1.112	1.269	347.95	17526.6
`NO--GP--p2`	triangular unstr	0.1	0.786	0.786	1.880	1.426	1.105	1.419	0.953	0.953	1.680	1.219	1.104	1.270	548.75	21024.5
`NO--GP--p3`	hexa	0.0	1.997	1.997	1.999	2.016	1.999	2.018	1.999	1.999	2.000	2.002	2.000	2.002	233.61	16957.7
`NO--GP--p3`	hexa	0.1	1.853	1.853	1.999	2.016	1.998	2.018	2.290	2.290	2.000	2.004	2.002	2.004	260.87	18624.0
`NO--GP--p3`	triangular unstr	0.0	0.799	0.799	1.878	1.388	1.101	1.383	0.963	0.963	1.677	1.205	1.112	1.258	405.61	17643.7
`NO--GP--p3`	triangular unstr	0.1	0.786	0.786	1.880	1.397	1.106	1.391	0.953	0.953	1.680	1.206	1.104	1.258	700.56	21097.0
`p1--na--NO`	hexa	0.0	1.912	2.147	1.616	1.616	1.626	1.626	2.611	2.168	2.380	2.380	2.385	2.385	161.28	17839.1
`p1--na--NO`	hexa	0.1	1.971	2.163	1.677	1.677	1.687	1.687	2.615	2.141	2.414	2.414	2.419	2.419	189.26	19067.4
`p1--na--NO`	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	773.26	19883.4
`p1--na--NO`	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	976.30	21554.6
`p1--CCp1--p1`	hexa	0.0	1.912	2.147	1.616	2.350	1.626	1.955	2.611	2.168	2.380	2.295	2.385	2.369	231.15	18280.8

			Fit $N = 10$–100						Fit $N = 70$–100							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p1--CCp1--p1	hexa	0.1	1.971	2.163	1.677	2.351	1.687	2.045	2.615	2.141	2.414	2.294	2.419	2.382	248.72	19264.1
p1--CCp1--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	973.13	20403.7
p1--CCp1--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	1046.9	21754.1
p1--CCp1--p2	hexa	0.0	1.912	2.147	1.616	2.228	1.626	2.057	2.611	2.168	2.380	2.154	2.385	2.233	236.12	18508.2
p1--CCp1--p2	hexa	0.1	1.971	2.163	1.676	2.228	1.686	2.104	2.615	2.141	2.414	2.153	2.419	2.209	259.07	19395.4
p1--CCp1--p2	triangular unstr	0.0	2.292	2.131	2.004	2.103	2.003	2.036	2.600	2.164	2.545	2.063	2.542	2.283	1044.4	20638.8
p1--CCp1--p2	triangular unstr	0.1	2.345	2.112	2.092	2.103	2.089	2.066	2.552	2.095	2.516	2.061	2.515	2.188	1052.1	21916.4
p1--CCp1--p3	hexa	0.0	1.912	2.147	1.616	2.246	1.626	2.063	2.611	2.168	2.380	2.171	2.385	2.256	236.80	18612.1
p1--CCp1--p3	hexa	0.1	1.971	2.163	1.676	2.246	1.686	2.115	2.615	2.141	2.414	2.170	2.419	2.234	258.99	19447.7
p1--CCp1--p3	triangular unstr	0.0	2.292	2.131	2.004	2.127	2.003	2.040	2.600	2.164	2.545	2.081	2.542	2.312	1049.2	20766.1
p1--CCp1--p3	triangular unstr	0.1	2.345	2.112	2.092	2.128	2.089	2.076	2.552	2.095	2.516	2.078	2.515	2.216	1107.6	21983.2
p1--CCp2--p1	hexa	0.0	1.912	2.147	1.616	2.903	1.626	1.665	2.611	2.168	2.380	2.782	2.385	2.394	263.19	18761.8
p1--CCp2--p1	hexa	0.1	1.971	2.163	1.676	2.953	1.687	1.736	2.615	2.141	2.414	2.858	2.419	2.428	293.62	19371.8
p1--CCp2--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	1109.4	21489.3
p1--CCp2--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	1214.9	22111.9
p1--CCp2--p2	hexa	0.0	1.912	2.147	1.616	2.990	1.627	1.689	2.611	2.168	2.380	2.882	2.385	2.400	271.94	18974.1
p1--CCp2--p2	hexa	0.1	1.972	2.163	1.676	3.018	1.687	1.765	2.615	2.141	2.414	2.931	2.419	2.435	314.85	19557.6
p1--CCp2--p2	triangular unstr	0.0	2.292	2.131	2.004	2.676	2.003	1.978	2.600	2.164	2.546	2.665	2.542	2.522	1119.2	21722.9
p1--CCp2--p2	triangular unstr	0.1	2.345	2.112	2.092	2.799	2.089	2.063	2.552	2.095	2.516	2.731	2.515	2.504	1251.5	22294.3
p1--CCp2--p3	hexa	0.0	1.912	2.147	1.616	2.987	1.627	1.688	2.611	2.168	2.380	2.884	2.385	2.400	289.83	19073.9
p1--CCp2--p3	hexa	0.1	1.972	2.163	1.676	3.015	1.687	1.765	2.615	2.141	2.414	2.931	2.419	2.435	328.75	19618.5
p1--CCp2--p3	triangular unstr	0.0	2.292	2.131	2.004	2.681	2.003	1.978	2.600	2.164	2.546	2.668	2.542	2.522	1128.9	21852.8
p1--CCp2--p3	triangular unstr	0.1	2.345	2.112	2.092	2.802	2.089	2.062	2.552	2.095	2.516	2.736	2.515	2.504	1283.7	22413.7
p1--CCp3--p1	hexa	0.0	1.912	2.147	1.617	2.034	1.626	1.630	2.611	2.168	2.380	2.378	2.385	2.406	395.23	20040.1
p1--CCp3--p1	hexa	0.1	1.971	2.163	1.677	2.164	1.687	1.692	2.615	2.141	2.414	2.408	2.419	2.441	436.47	20506.7
p1--CCp3--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	1416.4	23811.3
p1--CCp3--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	1506.6	24280.8
p1--CCp3--p2	hexa	0.0	1.913	2.147	1.617	2.239	1.627	1.635	2.611	2.168	2.380	2.377	2.385	2.426	409.23	20223.1
p1--CCp3--p2	hexa	0.1	1.972	2.163	1.677	2.394	1.688	1.698	2.615	2.141	2.414	2.402	2.419	2.463	432.50	20725.2
p1--CCp3--p2	triangular unstr	0.0	2.292	2.131	2.004	2.031	2.003	1.965	2.600	2.164	2.546	2.521	2.542	2.523	1390.3	24059.1
p1--CCp3--p2	triangular unstr	0.1	2.345	2.112	2.092	2.141	2.089	2.052	2.552	2.095	2.516	2.506	2.515	2.510	1558.2	24526.5
p1--CCp3--p3	hexa	0.0	1.913	2.147	1.617	2.224	1.627	1.634	2.611	2.168	2.380	2.376	2.385	2.426	418.15	20356.2
p1--CCp3--p3	hexa	0.1	1.972	2.163	1.677	2.379	1.688	1.698	2.615	2.141	2.414	2.401	2.419	2.462	447.84	20842.8
p1--CCp3--p3	triangular unstr	0.0	2.292	2.131	2.004	2.028	2.003	1.965	2.600	2.164	2.546	2.522	2.542	2.523	1368.8	24169.1
p1--CCp3--p3	triangular unstr	0.1	2.345	2.112	2.092	2.136	2.089	2.052	2.552	2.095	2.516	2.506	2.515	2.510	1450.6	24662.7
p1--GP--p1	hexa	0.0	1.912	2.147	1.997	2.152	1.836	2.158	2.611	2.168	2.005	2.309	2.249	2.308	204.40	17929.4
p1--GP--p1	hexa	0.1	1.971	2.163	1.999	2.219	1.895	2.225	2.615	2.141	2.003	2.309	2.222	2.308	241.70	19121.1
p1--GP--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	842.03	19912.6
p1--GP--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	1027.9	21576.0
p1--GP--p2	hexa	0.0	1.913	2.147	1.998	2.181	1.949	2.179	2.611	2.168	2.001	2.155	2.096	2.151	320.49	18117.7
p1--GP--p2	hexa	0.1	1.972	2.163	1.999	2.199	1.972	2.197	2.615	2.141	2.001	2.150	2.071	2.145	327.66	19246.6
p1--GP--p2	triangular unstr	0.0	2.292	2.131	1.991	2.067	2.009	2.066	2.600	2.164	2.002	2.176	2.268	2.168	931.04	20140.0
p1--GP--p2	triangular unstr	0.1	2.345	2.112	1.992	2.087	2.034	2.085	2.552	2.095	2.000	2.109	2.167	2.105	1145.7	21718.7
p1--GP--p3	hexa	0.0	1.913	2.147	1.999	2.194	1.949	2.193	2.611	2.168	2.001	2.170	2.096	2.166	334.71	18237.7
p1--GP--p3	hexa	0.1	1.972	2.163	1.999	2.215	1.973	2.213	2.615	2.141	2.001	2.165	2.071	2.160	331.70	19282.7
p1--GP--p3	triangular unstr	0.0	2.292	2.131	1.992	2.080	2.009	2.079	2.600	2.164	2.002	2.209	2.268	2.200	969.28	20260.7
p1--GP--p3	triangular unstr	0.1	2.345	2.112	1.992	2.106	2.034	2.104	2.552	2.095	2.000	2.134	2.167	2.129	1166.4	21785.8
p2--na--NO	hexa	0.0	2.128	2.568	2.142	2.142	2.143	2.143	2.020	2.183	2.025	2.025	2.026	2.026	177.84	18691.0
p2--na--NO	hexa	0.1	2.129	2.608	2.143	2.143	2.144	2.144	2.022	2.213	2.027	2.027	2.028	2.028	199.53	20243.9
p2--na--NO	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	880.50	21921.3
p2--na--NO	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1011.0	24393.0
p2--CCp1--p1	hexa	0.0	2.150	2.609	2.164	2.356	2.166	2.334	2.023	2.214	2.029	2.293	2.030	2.232	253.21	19462.0
p2--CCp1--p1	hexa	0.1	2.153	2.652	2.167	2.356	2.170	2.335	2.026	2.251	2.031	2.293	2.032	2.235	289.62	20876.6
p2--CCp1--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	1108.5	22641.8
p2--CCp1--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1210.2	24958.0
p2--CCp1--p2	hexa	0.0	2.213	2.709	2.224	2.230	2.235	2.188	2.031	2.308	2.039	2.151	2.041	2.094	256.27	19682.3
p2--CCp1--p2	hexa	0.1	2.225	2.760	2.235	2.230	2.248	2.187	2.036	2.369	2.043	2.151	2.046	2.094	306.21	20992.6
p2--CCp1--p2	triangular unstr	0.0	2.121	2.535	2.161	2.102	2.166	2.040	1.990	2.162	2.003	2.058	2.012	2.018	1086.9	22876.5
p2--CCp1--p2	triangular unstr	0.1	2.108	2.582	2.154	2.101	2.160	2.039	1.987	2.199	1.998	2.058	2.003	2.018	1327.0	25104.0
p2--CCp1--p3	hexa	0.0	2.213	2.709	2.224	2.248	2.235	2.214	2.031	2.308	2.039	2.168	2.041	2.113	260.37	19802.0
p2--CCp1--p3	hexa	0.1	2.225	2.760	2.235	2.248	2.248	2.214	2.036	2.369	2.043	2.168	2.046	2.113	317.77	21046.5
p2--CCp1--p3	triangular unstr	0.0	2.121	2.535	2.161	2.127	2.166	2.050	1.990	2.162	2.003	2.074	2.012	2.023	1172.1	22980.5
p2--CCp1--p3	triangular unstr	0.1	2.108	2.582	2.154	2.126	2.160	2.050	1.987	2.199	1.998	2.074	2.003	2.023	1310.8	25177.5
p2--CCp2--p1	hexa	0.0	2.150	2.609	2.166	3.039	2.166	2.253	2.023	2.214	2.029	2.986	2.030	2.044	293.23	19904.3
p2--CCp2--p1	hexa	0.1	2.153	2.653	2.170	3.040	2.170	2.269	2.026	2.251	2.032	2.989	2.032	2.050	347.11	20900.2
p2--CCp2--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	1269.2	23668.3
p2--CCp2--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1460.7	25105.2
p2--CCp2--p2	hexa	0.0	2.214	2.709	2.236	3.063	2.235	2.434	2.032	2.308	2.041	2.999	2.041	2.086	302.60	20118.8
p2--CCp2--p2	hexa	0.1	2.226	2.761	2.250	3.063	2.249	2.475	2.037	2.371	2.046	3.000	2.047	2.105	348.76	21015.7
p2--CCp2--p2	triangular unstr	0.0	2.122	2.536	2.172	3.012	2.166	2.317	1.990	2.162	2.003	2.991	2.012	2.038	1209.5	23871.8
p2--CCp2--p2	triangular unstr	0.1	2.108	2.583	2.166	3.012	2.160	2.339	1.987	2.200	1.999	2.993	2.002	2.039	1466.0	25271.6
p2--CCp2--p3	hexa	0.0	2.214	2.709	2.236	3.058	2.235	2.434	2.032	2.308	2.041	2.998	2.041	2.087	297.91	20232.9
p2--CCp2--p3	hexa	0.1	2.226	2.761	2.250	3.058	2.249	2.474	2.037	2.371	2.046	2.999	2.047	2.106	414.45	21086.6
p2--CCp2--p3	triangular unstr	0.0	2.122	2.536	2.172	3.009	2.166	2.314	1.990	2.162	2.003	2.991	2.012	2.039	1212.5	23983.9
p2--CCp2--p3	triangular unstr	0.1	2.108	2.583	2.166	3.010	2.160	2.336	1.987	2.200	1.999	2.993	2.002	2.040	1487.1	25347.9
p2--CCp3--p1	hexa	0.0	2.150	2.609	2.166	3.135	2.166	2.183	2.023	2.214	2.029	1.992	2.030	2.029	420.59	21104.2
p2--CCp3--p1	hexa	0.1	2.153	2.653	2.170	3.206	2.170	2.186	2.026	2.251	2.032	2.002	2.032	2.031	577.79	21515.3
p2--CCp3--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	1433.7	25987.5
p2--CCp3--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1728.1	26852.1
p2--CCp3--p2	hexa	0.0	2.214	2.709	2.236	3.796	2.235	2.299	2.032	2.308	2.041	2.178	2.041	2.039	452.14	21274.2
p2--CCp3--p2	hexa	0.1	2.226	2.761	2.249	3.882	2.249	2.314	2.037	2.371	2.046	2.315	2.046	2.044	616.38	21717.9
p2--CCp3--p2	triangular unstr	0.0	2.122	2.536	2.172	3.302	2.166	2.231	1.990	2.162	2.003	1.988	2.012	2.010	1455.9	26171.3

			Fit $N = 10$–100						Fit $N = 70$–100							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p2--CCp3--p2	triangular unstr	0.1	2.108	2.583	2.166	3.390	2.160	2.225	1.987	2.200	1.999	2.010	2.002	2.001	1726.2	27073.5
p2--CCp3--p3	hexa	0.0	2.214	2.709	2.236	3.789	2.235	2.298	2.032	2.308	2.041	2.133	2.041	2.039	449.72	21383.6
p2--CCp3--p3	hexa	0.1	2.226	2.761	2.249	3.877	2.249	2.313	2.037	2.371	2.046	2.262	2.046	2.044	645.79	21799.0
p2--CCp3--p3	triangular unstr	0.0	2.122	2.536	2.172	3.267	2.166	2.229	1.990	2.162	2.003	1.957	2.012	2.010	1428.8	26265.6
p2--CCp3--p3	triangular unstr	0.1	2.108	2.583	2.166	3.356	2.160	2.223	1.987	2.200	1.999	1.972	2.002	2.001	1337.8	27180.5
p2--GP--p1	hexa	0.0	2.150	2.609	2.003	2.689	2.102	2.694	2.023	2.214	2.001	2.304	2.011	2.312	229.19	18746.2
p2--GP--p1	hexa	0.1	2.153	2.653	2.003	2.728	2.100	2.732	2.026	2.251	2.001	2.350	2.011	2.358	250.07	20328.7
p2--GP--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	944.25	21969.6
p2--GP--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1097.3	24415.8
p2--GP--p2	hexa	0.0	2.214	2.709	2.003	2.924	2.114	2.923	2.032	2.308	2.000	2.644	2.007	2.650	315.69	18954.4
p2--GP--p2	hexa	0.1	2.226	2.761	2.003	2.952	2.112	2.951	2.037	2.371	2.000	2.706	2.007	2.711	388.09	20460.4
p2--GP--p2	triangular unstr	0.0	2.122	2.536	1.992	2.805	2.037	2.807	1.990	2.162	1.998	2.489	2.003	2.499	1056.9	22218.4
p2--GP--p2	triangular unstr	0.1	2.108	2.583	1.992	2.840	2.035	2.842	1.987	2.200	1.998	2.553	2.002	2.561	1200.0	24559.1
p2--GP--p3	hexa	0.0	2.214	2.709	2.004	2.923	2.114	2.922	2.032	2.308	2.000	2.650	2.007	2.656	377.79	19059.3
p2--GP--p3	hexa	0.1	2.226	2.761	2.004	2.950	2.113	2.950	2.037	2.371	2.000	2.711	2.007	2.716	499.25	20511.8
p2--GP--p3	triangular unstr	0.0	2.122	2.536	1.993	2.809	2.037	2.812	1.990	2.162	1.998	2.500	2.003	2.510	1067.6	22304.9
p2--GP--p3	triangular unstr	0.1	2.108	2.583	1.993	2.843	2.035	2.846	1.987	2.200	1.998	2.564	2.002	2.572	1257.2	24623.2
p3--na--NO	hexa	0.0	2.125	2.485	2.132	2.132	2.135	2.135	2.013	2.050	2.025	2.025	2.024	2.024	245.34	21804.9
p3--na--NO	hexa	0.1	2.108	2.481	2.114	2.114	2.117	2.117	2.010	2.048	2.020	2.020	2.020	2.020	377.83	24497.7
p3--na--NO	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1086.6	27536.0
p3--na--NO	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	999.45	31524.1
p3--CCp1--p1	hexa	0.0	2.185	2.634	2.188	2.354	2.201	2.339	2.020	2.072	2.038	2.290	2.040	2.248	337.38	22971.3
p3--CCp1--p1	hexa	0.1	2.160	2.629	2.164	2.354	2.176	2.339	2.016	2.068	2.031	2.290	2.032	2.248	538.47	25276.2
p3--CCp1--p1	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1249.8	28336.6
p3--CCp1--p1	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1225.3	31983.9
p3--CCp1--p2	hexa	0.0	2.620	3.309	2.640	2.229	2.603	2.184	2.117	2.267	2.264	2.150	2.211	2.094	357.27	23092.7
p3--CCp1--p2	hexa	0.1	2.564	3.307	2.592	2.229	2.554	2.184	2.084	2.237	2.206	2.150	2.160	2.094	569.95	25382.7
p3--CCp1--p2	triangular unstr	0.0	2.396	2.791	2.510	2.101	2.460	2.038	2.003	2.026	2.041	2.058	2.014	2.017	1285.8	28499.6
p3--CCp1--p2	triangular unstr	0.1	2.330	2.777	2.443	2.101	2.397	2.038	1.999	2.022	2.031	2.058	2.006	2.017	1349.2	32108.1
p3--CCp1--p3	hexa	0.0	2.620	3.309	2.640	2.247	2.603	2.210	2.117	2.267	2.264	2.167	2.211	2.112	395.31	23152.0
p3--CCp1--p3	hexa	0.1	2.564	3.307	2.592	2.247	2.554	2.210	2.084	2.237	2.206	2.167	2.160	2.112	560.65	25436.5
p3--CCp1--p3	triangular unstr	0.0	2.396	2.791	2.509	2.126	2.460	2.048	2.003	2.026	2.041	2.074	2.014	2.023	1443.9	28577.5
p3--CCp1--p3	triangular unstr	0.1	2.330	2.777	2.443	2.126	2.397	2.048	2.000	2.023	2.031	2.074	2.006	2.023	1751.9	32158.6
p3--CCp2--p1	hexa	0.0	2.187	2.637	2.200	3.042	2.203	2.642	2.020	2.073	2.041	2.999	2.040	2.257	414.93	23046.5
p3--CCp2--p1	hexa	0.1	2.162	2.631	2.175	3.042	2.177	2.638	2.016	2.069	2.033	2.999	2.033	2.251	538.68	25254.3
p3--CCp2--p1	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1635.1	28534.6
p3--CCp2--p1	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1687.7	31978.8
p3--CCp2--p2	hexa	0.0	3.539	3.992	3.284	3.063	3.303	3.110	4.323	4.278	4.078	3.002	4.085	3.055	444.41	23186.7
p3--CCp2--p2	hexa	0.1	3.570	4.025	3.312	3.063	3.332	3.109	4.298	4.271	4.078	3.002	4.086	3.049	516.67	25373.1
p3--CCp2--p2	triangular unstr	0.0	4.057	4.096	3.842	3.013	3.861	3.085	4.475	4.165	4.251	2.997	4.258	3.023	1695.1	28749.0
p3--CCp2--p2	triangular unstr	0.1	4.077	4.105	3.880	3.013	3.901	3.082	4.459	4.138	4.198	2.997	4.214	3.022	1613.4	32141.5
p3--CCp2--p3	hexa	0.0	3.539	3.992	3.284	3.058	3.303	3.102	4.323	4.278	4.078	3.001	4.085	3.050	450.49	23294.3
p3--CCp2--p3	hexa	0.1	3.570	4.025	3.312	3.058	3.332	3.101	4.298	4.271	4.078	3.001	4.086	3.044	521.37	25432.6
p3--CCp2--p3	triangular unstr	0.0	4.057	4.096	3.842	3.011	3.861	3.072	4.476	4.165	4.251	2.997	4.259	3.019	1574.2	28818.3
p3--CCp2--p3	triangular unstr	0.1	4.077	4.105	3.880	3.011	3.901	3.070	4.459	4.138	4.198	2.997	4.214	3.017	1663.2	32205.2
p3--CCp3--p1	hexa	0.0	2.186	2.637	2.200	3.993	2.202	2.225	2.020	2.073	2.041	2.812	2.040	2.040	549.23	23615.9
p3--CCp3--p1	hexa	0.1	2.161	2.631	2.174	3.993	2.177	2.201	2.016	2.069	2.033	2.809	2.033	2.032	659.38	25283.6
p3--CCp3--p1	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1648.9	30662.0
p3--CCp3--p1	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1792.5	32119.3
p3--CCp3--p2	hexa	0.0	3.538	3.993	3.290	4.231	3.304	3.441	4.323	4.279	4.079	4.284	4.086	4.100	543.80	23800.5
p3--CCp3--p2	hexa	0.1	3.569	4.026	3.319	4.231	3.333	3.485	4.299	4.271	4.081	4.283	4.087	4.101	701.99	25398.9
p3--CCp3--p2	triangular unstr	0.0	4.058	4.096	3.846	4.109	3.863	3.934	4.476	4.165	4.253	4.049	4.258	4.241	1698.7	30891.7
p3--CCp3--p2	triangular unstr	0.1	4.078	4.105	3.885	4.109	3.903	3.989	4.460	4.139	4.202	4.049	4.214	4.191	1679.5	32279.4
p3--CCp3--p3	hexa	0.0	3.538	3.993	3.290	4.241	3.304	3.438	4.323	4.279	4.079	4.302	4.086	4.100	549.65	23864.2
p3--CCp3--p3	hexa	0.1	3.569	4.026	3.319	4.241	3.333	3.481	4.299	4.271	4.081	4.300	4.087	4.100	753.98	25461.3
p3--CCp3--p3	triangular unstr	0.0	4.057	4.096	3.846	4.114	3.862	3.932	4.476	4.165	4.254	4.050	4.259	4.242	1879.3	31010.9
p3--CCp3--p3	triangular unstr	0.1	4.077	4.105	3.884	4.114	3.903	3.986	4.460	4.139	4.202	4.050	4.214	4.192	1453.5	32339.0
p3--GP--p1	hexa	0.0	2.187	2.637	2.001	2.598	1.989	2.596	2.020	2.073	2.000	2.068	1.999	2.066	358.46	21855.3
p3--GP--p1	hexa	0.1	2.161	2.631	2.001	2.592	1.988	2.589	2.016	2.069	2.000	2.060	1.999	2.059	534.78	24560.3
p3--GP--p1	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1511.9	27558.5
p3--GP--p1	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1438.6	31546.0
p3--GP--p2	hexa	0.0	3.538	3.992	1.999	4.003	1.997	4.010	4.323	4.279	2.000	4.194	2.001	4.195	430.39	21968.9
p3--GP--p2	hexa	0.1	3.569	4.026	1.999	4.033	1.997	4.040	4.299	4.271	2.000	4.212	2.000	4.213	512.98	24660.5
p3--GP--p2	triangular unstr	0.0	4.057	4.096	1.992	4.077	2.002	4.083	4.476	4.165	1.998	4.204	2.003	4.204	1565.6	27688.4
p3--GP--p2	triangular unstr	0.1	4.077	4.105	1.992	4.094	2.001	4.098	4.460	4.138	1.998	4.195	2.003	4.198	1452.0	31683.3
p3--GP--p3	hexa	0.0	3.538	3.992	1.999	4.000	1.998	4.008	4.323	4.279	2.000	4.198	2.001	4.200	451.44	22029.4
p3--GP--p3	hexa	0.1	3.569	4.026	1.999	4.031	1.997	4.039	4.299	4.271	2.000	4.217	2.001	4.219	669.66	24733.5
p3--GP--p3	triangular unstr	0.0	4.057	4.096	1.992	4.079	2.002	4.086	4.476	4.165	1.998	4.227	2.003	4.229	1653.7	27773.2
p3--GP--p3	triangular unstr	0.1	4.077	4.105	1.992	4.098	2.002	4.103	4.460	4.138	1.998	4.218	2.003	4.222	1605.1	31748.6

11.1.3 JFNK

$\Delta t = 10^{-2}$

Table 6: 2D manufactured results for `JFNK`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.992	1.992	2.135	2.135	2.134	2.134	1.985	1.985	2.500	2.500	2.497	2.497	9.266	16550.6
`NO--na--NO`	hexa	0.1	1.849	1.849	1.612	1.612	1.618	1.618	2.278	2.278	2.159	2.159	2.163	2.163	30.513	18377.9
`NO--na--NO`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	78.833	17294.1
`NO--na--NO`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	105.57	20811.1
`NO--CCp1--p1`	hexa	0.0	1.993	1.993	2.136	2.354	2.134	2.333	1.985	1.985	2.504	2.289	2.497	2.242	19.227	17065.9
`NO--CCp1--p1`	hexa	0.1	1.849	1.849	1.611	2.354	1.618	2.310	2.278	2.278	2.159	2.289	2.163	2.225	34.874	18494.9
`NO--CCp1--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	170.83	18064.6
`NO--CCp1--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	182.33	20922.1
`NO--CCp1--p2`	hexa	0.0	1.993	1.993	2.139	2.229	2.134	2.184	1.985	1.985	2.513	2.150	2.495	2.094	31.171	17365.6
`NO--CCp1--p2`	hexa	0.1	1.849	1.849	1.609	2.229	1.619	2.181	2.278	2.278	2.159	2.149	2.163	2.091	39.662	18652.2
`NO--CCp1--p2`	triangular unstr	0.0	0.730	0.730	0.778	2.111	0.784	1.164	0.862	0.862	1.037	2.101	1.107	1.150	175.21	18431.4
`NO--CCp1--p2`	triangular unstr	0.1	0.722	0.722	0.772	2.112	0.777	1.172	0.871	0.871	1.027	2.105	1.097	1.143	210.74	21075.5
`NO--CCp1--p3`	hexa	0.0	1.993	1.993	2.139	2.247	2.134	2.211	1.985	1.985	2.513	2.167	2.495	2.112	30.669	17519.9
`NO--CCp1--p3`	hexa	0.1	1.849	1.849	1.609	2.247	1.619	2.206	2.278	2.278	2.159	2.166	2.163	2.109	39.208	18733.1
`NO--CCp1--p3`	triangular unstr	0.0	0.730	0.730	0.778	2.139	0.784	1.132	0.862	0.862	1.037	2.131	1.107	1.144	153.75	18596.0
`NO--CCp1--p3`	triangular unstr	0.1	0.722	0.722	0.772	2.140	0.777	1.139	0.871	0.871	1.027	2.135	1.097	1.136	206.41	21132.4
`NO--CCp2--p1`	hexa	0.0	1.992	1.992	2.141	3.042	2.132	2.624	1.985	1.985	2.517	2.999	2.491	2.603	37.647	17579.3
`NO--CCp2--p1`	hexa	0.1	1.849	1.849	1.607	3.038	1.618	1.948	2.278	2.278	2.158	2.987	2.162	2.187	43.859	18732.4
`NO--CCp2--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	293.34	19273.3
`NO--CCp2--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	279.89	21288.3
`NO--CCp2--p2`	hexa	0.0	1.991	1.991	2.145	3.063	2.120	2.764	1.986	1.986	2.519	3.002	2.444	2.629	49.227	17885.7
`NO--CCp2--p2`	hexa	0.1	1.851	1.851	1.600	3.061	1.624	2.068	2.274	2.274	2.155	2.996	2.162	2.210	46.755	19025.5
`NO--CCp2--p2`	triangular unstr	0.0	0.730	0.730	0.778	1.941	0.784	0.776	0.862	0.862	1.037	1.095	1.107	1.097	252.35	19638.5
`NO--CCp2--p2`	triangular unstr	0.1	0.722	0.722	0.772	1.957	0.777	0.773	0.871	0.871	1.027	1.090	1.097	1.087	265.88	21625.1
`NO--CCp2--p3`	hexa	0.0	1.991	1.991	2.145	3.058	2.120	2.763	1.986	1.986	2.519	3.001	2.444	2.632	48.850	18015.0
`NO--CCp2--p3`	hexa	0.1	1.851	1.851	1.600	3.056	1.624	2.069	2.274	2.274	2.155	2.995	2.162	2.211	51.083	19169.8
`NO--CCp2--p3`	triangular unstr	0.0	0.730	0.730	0.778	1.952	0.784	0.776	0.862	0.862	1.037	1.098	1.107	1.096	337.96	19809.5
`NO--CCp2--p3`	triangular unstr	0.1	0.722	0.722	0.772	1.968	0.777	0.773	0.871	0.871	1.027	1.093	1.097	1.087	316.88	21810.7
`NO--CCp3--p1`	hexa	0.0	1.992	1.992	2.139	4.063	2.132	2.165	1.985	1.985	2.517	3.183	2.491	2.491	67.019	18928.3
`NO--CCp3--p1`	hexa	0.1	1.849	1.849	1.605	3.118	1.618	1.661	2.278	2.278	2.158	2.340	2.163	2.189	85.420	20094.7
`NO--CCp3--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	435.49	21779.7
`NO--CCp3--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	353.60	23790.8
`NO--CCp3--p2`	hexa	0.0	1.992	1.992	2.141	4.277	2.120	2.216	1.985	1.985	2.519	4.233	2.444	2.443	71.573	19210.9
`NO--CCp3--p2`	hexa	0.1	1.851	1.851	1.594	3.421	1.622	1.722	2.274	2.274	2.155	2.589	2.162	2.221	74.356	20366.7
`NO--CCp3--p2`	triangular unstr	0.0	0.730	0.730	0.778	1.071	0.784	0.796	0.862	0.862	1.037	1.037	1.107	1.109	463.11	22135.9
`NO--CCp3--p2`	triangular unstr	0.1	0.722	0.722	0.772	1.081	0.777	0.798	0.871	0.871	1.027	1.028	1.097	1.100	362.07	24132.6
`NO--CCp3--p3`	hexa	0.0	1.992	1.992	2.141	4.293	2.120	2.214	1.985	1.985	2.519	4.254	2.444	2.443	61.808	19363.5
`NO--CCp3--p3`	hexa	0.1	1.851	1.851	1.594	3.409	1.622	1.721	2.274	2.274	2.155	2.578	2.162	2.221	74.433	20513.0
`NO--CCp3--p3`	triangular unstr	0.0	0.730	0.730	0.778	1.054	0.784	0.796	0.862	0.862	1.037	1.037	1.107	1.109	399.21	22286.3
`NO--CCp3--p3`	triangular unstr	0.1	0.722	0.722	0.772	1.064	0.777	0.797	0.871	0.871	1.027	1.028	1.097	1.100	347.65	24284.2
`NO--GP--p1`	hexa	0.0	1.992	1.992	2.000	2.057	2.005	2.061	1.985	1.985	2.002	2.031	2.023	2.031	13.090	16727.8
`NO--GP--p1`	hexa	0.1	1.849	1.849	2.000	2.049	1.997	2.053	2.278	2.278	2.002	2.039	2.030	2.039	24.074	18448.5
`NO--GP--p1`	triangular unstr	0.0	0.730	0.730	0.778	0.778	0.784	0.784	0.862	0.862	1.037	1.037	1.107	1.107	95.302	17470.9
`NO--GP--p1`	triangular unstr	0.1	0.722	0.722	0.772	0.772	0.777	0.777	0.871	0.871	1.027	1.027	1.097	1.097	132.10	20899.6
`NO--GP--p2`	hexa	0.0	1.992	1.992	1.999	2.015	2.002	2.017	1.986	1.986	2.001	2.010	2.011	2.010	29.398	17030.0
`NO--GP--p2`	hexa	0.1	1.851	1.851	1.999	2.015	2.001	2.017	2.274	2.274	2.001	2.011	2.013	2.011	38.154	18577.5
`NO--GP--p2`	triangular unstr	0.0	0.730	0.730	1.878	1.418	1.102	1.411	0.862	0.862	1.678	1.219	1.112	1.269	195.38	17820.7
`NO--GP--p2`	triangular unstr	0.1	0.722	0.722	1.880	1.427	1.107	1.420	0.871	0.871	1.682	1.220	1.104	1.270	175.66	21048.0
`NO--GP--p3`	hexa	0.0	1.992	1.992	1.999	2.019	2.002	2.021	1.986	1.986	2.001	2.011	2.011	2.011	48.439	17196.3
`NO--GP--p3`	hexa	0.1	1.851	1.851	1.999	2.019	2.001	2.021	2.274	2.274	2.001	2.013	2.013	2.013	49.173	18656.2
`NO--GP--p3`	triangular unstr	0.0	0.730	0.730	1.878	1.389	1.102	1.384	0.862	0.862	1.678	1.206	1.112	1.258	201.79	17995.4
`NO--GP--p3`	triangular unstr	0.1	0.722	0.722	1.881	1.398	1.107	1.392	0.871	0.871	1.682	1.207	1.104	1.259	233.70	21114.5
`p1--na--NO`	hexa	0.0	1.911	2.146	1.616	1.616	1.626	1.626	2.612	2.168	2.381	2.381	2.387	2.387	77.310	17878.7
`p1--na--NO`	hexa	0.1	1.971	2.163	1.677	1.677	1.687	1.687	2.614	2.141	2.416	2.416	2.422	2.422	99.193	19092.2
`p1--na--NO`	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	627.40	19950.2
`p1--na--NO`	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	630.44	21567.4
`p1--CCp1--p1`	hexa	0.0	1.911	2.146	1.616	2.349	1.626	1.955	2.612	2.168	2.381	2.294	2.387	2.369	87.842	18383.4
`p1--CCp1--p1`	hexa	0.1	1.971	2.163	1.677	2.351	1.687	2.045	2.614	2.141	2.416	2.293	2.422	2.383	100.72	19298.5
`p1--CCp1--p1`	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	726.75	20592.0
`p1--CCp1--p1`	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	626.27	21799.9
`p1--CCp1--p2`	hexa	0.0	1.911	2.146	1.616	2.228	1.626	2.057	2.612	2.168	2.381	2.153	2.387	2.232	87.717	18666.6
`p1--CCp1--p2`	hexa	0.1	1.971	2.163	1.676	2.228	1.687	2.104	2.614	2.141	2.416	2.152	2.422	2.208	93.410	19426.6
`p1--CCp1--p2`	triangular unstr	0.0	2.271	2.126	2.004	2.102	2.002	2.036	2.384	2.115	2.552	2.063	2.549	2.285	669.65	20937.5
`p1--CCp1--p2`	triangular unstr	0.1	2.302	2.106	2.092	2.102	2.090	2.066	2.222	2.054	2.524	2.061	2.524	2.190	586.52	22037.9
`p1--CCp1--p3`	hexa	0.0	1.911	2.146	1.616	2.246	1.626	2.063	2.612	2.168	2.381	2.170	2.387	2.255	91.858	18812.1
`p1--CCp1--p3`	hexa	0.1	1.971	2.163	1.676	2.246	1.687	2.115	2.614	2.141	2.416	2.169	2.422	2.234	97.556	19515.0
`p1--CCp1--p3`	triangular unstr	0.0	2.271	2.126	2.004	2.127	2.002	2.040	2.384	2.115	2.552	2.080	2.549	2.315	696.72	21106.2
`p1--CCp1--p3`	triangular unstr	0.1	2.302	2.106	2.092	2.127	2.090	2.075	2.222	2.054	2.524	2.078	2.524	2.217	580.06	22150.4
`p1--CCp2--p1`	hexa	0.0	1.911	2.146	1.616	2.904	1.626	1.665	2.612	2.168	2.381	2.783	2.387	2.395	101.76	18900.9
`p1--CCp2--p1`	hexa	0.1	1.971	2.163	1.676	2.954	1.687	1.736	2.614	2.141	2.416	2.859	2.422	2.431	100.97	19452.8
`p1--CCp2--p1`	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	734.25	21779.0
`p1--CCp2--p1`	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	603.52	22409.0
`p1--CCp2--p2`	hexa	0.0	1.912	2.146	1.616	2.990	1.627	1.688	2.612	2.168	2.381	2.882	2.387	2.400	116.50	19176.4
`p1--CCp2--p2`	hexa	0.1	1.971	2.163	1.676	3.018	1.687	1.765	2.613	2.141	2.416	2.931	2.422	2.437	100.84	19694.5
`p1--CCp2--p2`	triangular unstr	0.0	2.271	2.126	2.004	2.677	2.002	1.977	2.384	2.115	2.552	2.669	2.549	2.528	729.70	22122.6
`p1--CCp2--p2`	triangular unstr	0.1	2.301	2.106	2.092	2.799	2.090	2.062	2.222	2.054	2.524	2.735	2.523	2.511	628.08	22707.1

			Fit $N = 10$–100						Fit $N = 70$–100							
Vm-states-Ion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p1--CCp2--p3	hexa	0.0	1.912	2.146	1.616	2.987	1.627	1.688	2.612	2.168	2.381	2.884	2.387	2.400	116.33	19324.3
p1--CCp2--p3	hexa	0.1	1.971	2.163	1.676	3.015	1.687	1.765	2.613	2.141	2.416	2.932	2.422	2.437	104.74	19816.2
p1--CCp2--p3	triangular unstr	0.0	2.271	2.126	2.004	2.681	2.002	1.977	2.384	2.115	2.552	2.672	2.549	2.527	750.46	22285.7
p1--CCp2--p3	triangular unstr	0.1	2.301	2.106	2.092	2.802	2.090	2.062	2.222	2.054	2.524	2.740	2.523	2.510	622.87	22859.8
p1--CCp3--p1	hexa	0.0	1.911	2.146	1.616	2.034	1.626	1.630	2.612	2.168	2.381	2.378	2.387	2.407	147.06	20233.9
p1--CCp3--p1	hexa	0.1	1.971	2.163	1.677	2.165	1.687	1.692	2.614	2.141	2.416	2.409	2.422	2.443	125.22	20735.8
p1--CCp3--p1	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	800.43	24287.5
p1--CCp3--p1	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	683.45	24737.2
p1--CCp3--p2	hexa	0.0	1.912	2.146	1.617	2.239	1.627	1.634	2.612	2.168	2.381	2.377	2.387	2.426	138.21	20509.6
p1--CCp3--p2	hexa	0.1	1.971	2.163	1.677	2.394	1.688	1.698	2.613	2.141	2.416	2.402	2.422	2.464	125.55	21018.0
p1--CCp3--p2	triangular unstr	0.0	2.271	2.126	2.004	2.031	2.002	1.964	2.384	2.115	2.552	2.527	2.549	2.528	782.76	24626.8
p1--CCp3--p2	triangular unstr	0.1	2.301	2.106	2.092	2.141	2.090	2.052	2.222	2.054	2.524	2.511	2.523	2.516	735.35	25077.5
p1--CCp3--p3	hexa	0.0	1.912	2.146	1.617	2.225	1.627	1.634	2.612	2.168	2.381	2.376	2.387	2.426	133.40	20650.3
p1--CCp3--p3	hexa	0.1	1.971	2.163	1.677	2.379	1.688	1.697	2.613	2.141	2.416	2.402	2.422	2.463	133.26	21151.4
p1--CCp3--p3	triangular unstr	0.0	2.271	2.126	2.004	2.027	2.002	1.964	2.384	2.115	2.552	2.527	2.549	2.528	745.42	24787.5
p1--CCp3--p3	triangular unstr	0.1	2.301	2.106	2.092	2.136	2.090	2.052	2.222	2.054	2.524	2.512	2.523	2.516	764.59	25234.9
p1--GP--p1	hexa	0.0	1.911	2.146	1.998	2.154	1.838	2.160	2.612	2.168	2.007	2.316	2.258	2.316	81.233	17974.1
p1--GP--p1	hexa	0.1	1.971	2.163	1.999	2.221	1.898	2.227	2.614	2.141	2.005	2.319	2.234	2.318	102.12	19138.4
p1--GP--p1	triangular unstr	0.0	2.271	2.126	2.004	2.004	2.002	2.002	2.384	2.115	2.552	2.552	2.549	2.549	564.30	20001.6
p1--GP--p1	triangular unstr	0.1	2.301	2.106	2.092	2.092	2.090	2.090	2.222	2.054	2.524	2.524	2.524	2.524	637.92	21601.4
p1--GP--p2	hexa	0.0	1.912	2.146	1.999	2.182	1.951	2.181	2.612	2.168	2.002	2.161	2.105	2.157	98.467	18270.7
p1--GP--p2	hexa	0.1	1.971	2.163	1.999	2.201	1.975	2.199	2.613	2.141	2.001	2.156	2.080	2.151	101.53	19255.2
p1--GP--p2	triangular unstr	0.0	2.271	2.126	1.992	2.069	2.012	2.068	2.384	2.115	2.004	2.184	2.284	2.176	612.59	20336.8
p1--GP--p2	triangular unstr	0.1	2.301	2.106	1.992	2.089	2.038	2.087	2.222	2.054	2.002	2.117	2.186	2.113	623.45	21725.5
p1--GP--p3	hexa	0.0	1.912	2.146	1.999	2.196	1.951	2.195	2.612	2.168	2.002	2.177	2.105	2.173	102.50	18421.3
p1--GP--p3	hexa	0.1	1.971	2.163	1.999	2.217	1.975	2.215	2.613	2.141	2.001	2.172	2.080	2.167	109.22	19318.6
p1--GP--p3	triangular unstr	0.0	2.271	2.126	1.992	2.082	2.012	2.081	2.384	2.115	2.004	2.219	2.284	2.210	633.00	20513.5
p1--GP--p3	triangular unstr	0.1	2.301	2.106	1.992	2.109	2.038	2.106	2.222	2.054	2.002	2.144	2.186	2.139	616.82	21804.6
p2--na--NO	hexa	0.0	2.127	2.567	2.176	2.176	2.176	2.176	2.016	2.179	2.139	2.139	2.139	2.139	93.266	18711.6
p2--na--NO	hexa	0.1	2.128	2.607	2.180	2.180	2.181	2.181	2.018	2.209	2.153	2.153	2.153	2.153	97.913	20308.0
p2--na--NO	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.161	2.161	1.982	2.068	2.260	2.260	2.263	2.263	599.62	21949.5
p2--na--NO	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	576.14	24413.9
p2--CCp1--p1	hexa	0.0	2.149	2.608	2.201	2.356	2.203	2.337	2.019	2.209	2.154	2.292	2.154	2.244	121.17	19564.2
p2--CCp1--p1	hexa	0.1	2.151	2.651	2.208	2.356	2.211	2.338	2.021	2.247	2.171	2.291	2.172	2.246	108.40	20884.4
p2--CCp1--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.161	2.161	1.982	2.068	2.260	2.260	2.263	2.263	632.48	22835.8
p2--CCp1--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	681.24	25000.8
p2--CCp1--p2	hexa	0.0	2.212	2.708	2.272	2.230	2.283	2.188	2.026	2.303	2.199	2.150	2.206	2.095	117.83	19824.1
p2--CCp1--p2	hexa	0.1	2.223	2.759	2.291	2.229	2.304	2.188	2.030	2.364	2.229	2.150	2.238	2.095	120.32	21014.9
p2--CCp1--p2	triangular unstr	0.0	2.118	2.532	2.273	2.101	2.279	2.040	1.977	2.149	2.394	2.057	2.419	2.018	655.10	23158.1
p2--CCp1--p2	triangular unstr	0.1	2.104	2.578	2.283	2.101	2.292	2.039	1.972	2.186	2.455	2.057	2.479	2.018	606.77	25195.1
p2--CCp1--p3	hexa	0.0	2.212	2.708	2.272	2.248	2.283	2.215	2.026	2.303	2.199	2.167	2.206	2.114	118.26	19938.4
p2--CCp1--p3	hexa	0.1	2.223	2.759	2.291	2.248	2.304	2.214	2.030	2.364	2.229	2.167	2.238	2.114	132.47	21079.0
p2--CCp1--p3	triangular unstr	0.0	2.118	2.532	2.273	2.126	2.279	2.050	1.977	2.149	2.394	2.074	2.419	2.023	655.96	23321.8
p2--CCp1--p3	triangular unstr	0.1	2.104	2.578	2.283	2.126	2.292	2.050	1.972	2.186	2.455	2.074	2.479	2.023	592.54	25287.0
p2--CCp2--p1	hexa	0.0	2.149	2.608	2.203	3.039	2.203	2.289	2.019	2.210	2.155	2.989	2.155	2.169	126.07	20011.9
p2--CCp2--p1	hexa	0.1	2.152	2.651	2.211	3.040	2.211	2.310	2.021	2.247	2.173	2.991	2.172	2.189	126.42	20929.3
p2--CCp2--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.161	2.161	1.982	2.068	2.260	2.260	2.263	2.263	714.55	23917.2
p2--CCp2--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	625.61	25225.6
p2--CCp2--p2	hexa	0.0	2.212	2.708	2.284	3.063	2.283	2.480	2.026	2.303	2.206	3.000	2.206	2.250	132.79	20290.6
p2--CCp2--p2	hexa	0.1	2.224	2.760	2.305	3.063	2.305	2.527	2.030	2.365	2.239	3.001	2.239	2.294	132.29	21085.1
p2--CCp2--p2	triangular unstr	0.0	2.118	2.532	2.286	3.012	2.279	2.428	1.977	2.149	2.413	2.995	2.420	2.443	735.63	24244.0
p2--CCp2--p2	triangular unstr	0.1	2.104	2.579	2.300	3.012	2.293	2.467	1.972	2.186	2.481	2.996	2.482	2.511	603.12	25454.5
p2--CCp2--p3	hexa	0.0	2.212	2.708	2.284	3.058	2.283	2.480	2.026	2.303	2.206	2.999	2.206	2.251	135.22	20417.3
p2--CCp2--p3	hexa	0.1	2.224	2.760	2.305	3.058	2.305	2.527	2.030	2.365	2.239	3.000	2.239	2.296	130.50	21160.4
p2--CCp2--p3	triangular unstr	0.0	2.118	2.532	2.286	3.010	2.279	2.425	1.977	2.149	2.413	2.995	2.420	2.443	697.92	24412.9
p2--CCp2--p3	triangular unstr	0.1	2.104	2.579	2.300	3.010	2.293	2.465	1.972	2.186	2.481	2.996	2.482	2.512	598.37	25581.7
p2--CCp3--p1	hexa	0.0	2.149	2.608	2.203	3.175	2.203	2.220	2.019	2.210	2.155	2.123	2.155	2.154	148.20	21250.7
p2--CCp3--p1	hexa	0.1	2.152	2.651	2.211	3.250	2.211	2.227	2.021	2.247	2.173	2.151	2.172	2.171	137.68	21647.8
p2--CCp3--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.161	2.161	1.982	2.068	2.260	2.260	2.263	2.263	753.15	26381.9
p2--CCp3--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	659.73	27285.2
p2--CCp3--p2	hexa	0.0	2.212	2.708	2.284	3.846	2.283	2.347	2.026	2.303	2.206	2.397	2.206	2.204	148.27	21478.1
p2--CCp3--p2	hexa	0.1	2.224	2.760	2.305	3.936	2.304	2.369	2.030	2.365	2.239	2.584	2.238	2.237	157.69	21927.2
p2--CCp3--p2	triangular unstr	0.0	2.118	2.532	2.286	3.422	2.279	2.345	1.977	2.149	2.413	2.437	2.420	2.419	729.00	26704.4
p2--CCp3--p2	triangular unstr	0.1	2.104	2.579	2.300	3.528	2.293	2.358	1.972	2.186	2.481	2.550	2.482	2.482	657.26	27605.0
p2--CCp3--p3	hexa	0.0	2.212	2.708	2.284	3.839	2.283	2.346	2.026	2.303	2.206	2.351	2.206	2.204	159.02	21614.3
p2--CCp3--p3	hexa	0.1	2.224	2.760	2.305	3.933	2.304	2.368	2.030	2.365	2.239	2.531	2.238	2.237	188.88	22066.7
p2--CCp3--p3	triangular unstr	0.0	2.118	2.532	2.286	3.389	2.279	2.343	1.977	2.149	2.413	2.404	2.420	2.419	714.03	26849.9
p2--CCp3--p3	triangular unstr	0.1	2.104	2.579	2.300	3.497	2.293	2.356	1.972	2.186	2.481	2.509	2.482	2.482	644.62	27732.2
p2--GP--p1	hexa	0.0	2.149	2.608	2.004	2.716	2.109	2.720	2.019	2.210	2.002	2.416	2.034	2.422	99.329	18816.7
p2--GP--p1	hexa	0.1	2.152	2.651	2.004	2.757	2.107	2.760	2.021	2.247	2.002	2.470	2.033	2.476	110.50	20350.3
p2--GP--p1	triangular unstr	0.0	2.056	2.374	2.166	2.166	2.161	2.161	1.982	2.068	2.260	2.260	2.263	2.263	581.18	22050.9
p2--GP--p1	triangular unstr	0.1	2.043	2.402	2.162	2.162	2.158	2.158	1.980	2.081	2.286	2.286	2.285	2.285	626.10	24445.3
p2--GP--p2	hexa	0.0	2.212	2.708	2.004	2.942	2.118	2.941	2.026	2.303	2.001	2.734	2.020	2.738	116.93	19113.0
p2--GP--p2	hexa	0.1	2.224	2.760	2.004	2.969	2.116	2.968	2.030	2.365	2.001	2.794	2.019	2.797	111.94	20456.4
p2--GP--p2	triangular unstr	0.0	2.118	2.532	1.993	2.860	2.044	2.862	1.977	2.149	2.000	2.751	2.026	2.757	618.26	22383.5
p2--GP--p2	triangular unstr	0.1	2.104	2.579	1.993	2.895	2.042	2.896	1.972	2.186	2.000	2.820	2.025	2.823	648.71	24583.9
p2--GP--p3	hexa	0.0	2.212	2.708	2.004	2.940	2.118	2.940	2.026	2.303	2.001	2.739	2.020	2.743	129.66	19251.7
p2--GP--p3	hexa	0.1	2.224	2.760	2.004	2.967	2.117	2.966	2.030	2.365	2.001	2.798	2.019	2.801	124.41	20530.1
p2--GP--p3	triangular unstr	0.0	2.118	2.532	1.993	2.863	2.044	2.865	1.977	2.149	2.000	2.758	2.026	2.764	637.22	22551.2
p2--GP--p3	triangular unstr	0.1	2.104	2.579	1.993	2.897	2.041	2.898	1.972	2.186	2.000	2.825	2.025	2.829	691.48	24650.1
p3--na--NO	hexa	0.0	2.121	2.481	2.256	2.256	2.258	2.258	1.999	2.037	2.472	2.472	2.469	2.469	147.82	21837.0
p3--na--NO	hexa	0.1	2.104	2.476	2.238	2.238	2.240	2.240	1.996	2.034	2.467	2.467	2.464	2.464	152.52	24499.4

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`p3--na--NO`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.345	2.345	1.972	1.988	3.183	3.183	3.172	3.172	689.38	27545.2
`p3--na--NO`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.182	3.182	3.171	3.171	760.32	31535.9
`p3--CCp1--p1`	hexa	0.0	2.180	2.629	2.347	2.353	2.360	2.339	2.003	2.055	2.608	2.289	2.617	2.247	167.38	23001.8
`p3--CCp1--p1`	hexa	0.1	2.155	2.624	2.323	2.353	2.335	2.339	1.998	2.051	2.600	2.289	2.608	2.247	204.47	25294.3
`p3--CCp1--p1`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.345	2.345	1.972	1.988	3.183	3.183	3.172	3.172	737.71	28449.0
`p3--CCp1--p1`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.182	3.182	3.171	3.171	623.17	31985.4
`p3--CCp1--p2`	hexa	0.0	2.672	3.361	2.672	2.229	2.497	2.184	2.142	2.316	1.405	2.150	1.266	2.093	152.92	23136.5
`p3--CCp1--p2`	hexa	0.1	2.615	3.360	2.623	2.229	2.442	2.184	2.103	2.281	1.306	2.150	1.209	2.093	202.04	25404.4
`p3--CCp1--p2`	triangular unstr	0.0	2.439	2.843	2.197	2.101	2.046	2.038	1.999	2.026	0.132	2.057	0.463	2.016	745.72	28688.4
`p3--CCp1--p2`	triangular unstr	0.1	2.369	2.829	2.121	2.101	1.976	2.038	1.995	2.022	0.125	2.057	0.459	2.016	632.27	32128.5
`p3--CCp1--p3`	hexa	0.0	2.672	3.361	2.672	2.247	2.497	2.210	2.142	2.316	1.405	2.166	1.266	2.111	153.33	23218.0
`p3--CCp1--p3`	hexa	0.1	2.615	3.360	2.623	2.247	2.442	2.210	2.103	2.281	1.306	2.166	1.209	2.111	191.23	25439.1
`p3--CCp1--p3`	triangular unstr	0.0	2.439	2.843	2.196	2.126	2.045	2.048	1.999	2.026	0.132	2.073	0.463	2.021	752.75	28791.6
`p3--CCp1--p3`	triangular unstr	0.1	2.369	2.829	2.121	2.126	1.976	2.048	1.995	2.023	0.125	2.073	0.459	2.021	640.42	32187.4
`p3--CCp2--p1`	hexa	0.0	2.181	2.631	2.362	3.042	2.362	2.762	2.003	2.055	2.629	2.999	2.619	2.745	158.82	23118.4
`p3--CCp2--p1`	hexa	0.1	2.156	2.626	2.337	3.042	2.337	2.758	1.998	2.051	2.620	2.999	2.611	2.740	161.93	25259.8
`p3--CCp2--p1`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.345	2.345	1.972	1.988	3.183	3.183	3.172	3.172	860.40	28760.1
`p3--CCp2--p1`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.182	3.182	3.171	3.171	830.09	32013.2
`p3--CCp2--p2`	hexa	0.0	3.520	3.971	2.698	3.063	2.707	3.071	4.171	4.123	0.669	3.002	0.641	2.724	159.00	23274.3
`p3--CCp2--p2`	hexa	0.1	3.547	4.002	2.648	3.063	2.657	3.069	4.104	4.100	0.514	3.002	0.488	2.717	162.70	25376.0
`p3--CCp2--p2`	triangular unstr	0.0	3.748	3.924	2.081	3.013	2.060	2.855	2.167	2.959	0.011	2.997	0.007	1.544	906.22	29070.9
`p3--CCp2--p2`	triangular unstr	0.1	3.692	3.923	1.983	3.013	1.961	2.852	1.804	2.888	0.004	2.997	0.001	1.543	824.68	32175.8
`p3--CCp2--p3`	hexa	0.0	3.520	3.971	2.698	3.058	2.707	3.064	4.171	4.123	0.669	3.001	0.641	2.729	145.69	23372.0
`p3--CCp2--p3`	hexa	0.1	3.547	4.002	2.648	3.058	2.657	3.062	4.104	4.100	0.514	3.001	0.488	2.721	166.94	25416.4
`p3--CCp2--p3`	triangular unstr	0.0	3.748	3.924	2.081	3.011	2.060	2.849	2.167	2.959	0.011	2.996	0.007	1.567	891.74	29233.2
`p3--CCp2--p3`	triangular unstr	0.1	3.692	3.923	1.982	3.011	1.961	2.846	1.805	2.888	0.004	2.996	0.001	1.565	810.88	32261.1
`p3--CCp3--p1`	hexa	0.0	2.181	2.631	2.362	4.082	2.362	2.385	2.003	2.056	2.629	3.391	2.619	2.619	164.32	23744.0
`p3--CCp3--p1`	hexa	0.1	2.156	2.626	2.337	4.082	2.337	2.361	1.998	2.051	2.620	3.388	2.611	2.611	209.86	25311.2
`p3--CCp3--p1`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.345	2.345	1.972	1.988	3.183	3.183	3.172	3.172	952.56	31101.1
`p3--CCp3--p1`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.182	3.182	3.171	3.171	832.89	32230.4
`p3--CCp3--p2`	hexa	0.0	3.520	3.971	2.705	4.177	2.709	2.869	4.171	4.124	0.671	3.889	0.642	0.730	181.97	24009.8
`p3--CCp3--p2`	hexa	0.1	3.547	4.002	2.656	4.177	2.658	2.840	4.105	4.100	0.516	3.888	0.490	0.578	236.84	25434.9
`p3--CCp3--p2`	triangular unstr	0.0	3.749	3.924	2.085	3.907	2.062	2.112	2.167	2.959	0.011	2.695	0.007	0.004	829.89	31443.8
`p3--CCp3--p2`	triangular unstr	0.1	3.693	3.924	1.987	3.906	1.963	2.027	1.804	2.888	0.004	2.694	0.001	-0.001	826.54	32405.2
`p3--CCp3--p3`	hexa	0.0	3.520	3.971	2.705	4.182	2.709	2.866	4.171	4.124	0.671	3.875	0.642	0.729	173.66	24132.2
`p3--CCp3--p3`	hexa	0.1	3.547	4.002	2.656	4.182	2.658	2.835	4.105	4.100	0.516	3.873	0.490	0.577	255.78	25497.1
`p3--CCp3--p3`	triangular unstr	0.0	3.749	3.924	2.085	3.889	2.061	2.108	2.167	2.959	0.011	2.592	0.007	0.004	862.24	31610.3
`p3--CCp3--p3`	triangular unstr	0.1	3.693	3.924	1.986	3.889	1.963	2.022	1.804	2.888	0.004	2.591	0.001	-0.001	738.79	32495.2
`p3--GP--p1`	hexa	0.0	2.181	2.631	2.002	2.762	1.992	2.757	2.003	2.056	2.002	2.661	2.008	2.650	148.12	21874.5
`p3--GP--p1`	hexa	0.1	2.156	2.626	2.002	2.756	1.991	2.752	1.998	2.051	2.002	2.653	2.008	2.642	178.17	24556.5
`p3--GP--p1`	triangular unstr	0.0	2.033	2.201	2.348	2.348	2.345	2.345	1.972	1.988	3.183	3.183	3.172	3.172	693.51	27581.7
`p3--GP--p1`	triangular unstr	0.1	2.021	2.196	2.333	2.333	2.331	2.331	1.972	1.987	3.182	3.182	3.171	3.171	800.77	31560.1
`p3--GP--p2`	hexa	0.0	3.520	3.971	2.000	3.744	1.999	3.748	4.171	4.124	2.001	1.458	2.006	1.396	170.50	21996.1
`p3--GP--p2`	hexa	0.1	3.547	4.002	1.999	3.757	1.998	3.761	4.105	4.100	2.001	1.348	2.006	1.284	202.72	24666.6
`p3--GP--p2`	triangular unstr	0.0	3.749	3.924	1.992	2.844	2.005	2.843	2.167	2.959	2.000	-0.092	2.012	-0.097	805.00	27710.0
`p3--GP--p2`	triangular unstr	0.1	3.693	3.924	1.992	2.836	2.005	2.836	1.804	2.888	2.000	-0.098	2.012	-0.103	699.63	31685.8
`p3--GP--p3`	hexa	0.0	3.520	3.971	2.000	3.738	1.999	3.742	4.171	4.124	2.001	1.420	2.006	1.355	174.93	22054.6
`p3--GP--p3`	hexa	0.1	3.547	4.002	2.000	3.751	1.999	3.756	4.105	4.100	2.001	1.305	2.006	1.239	246.90	24749.6
`p3--GP--p3`	triangular unstr	0.0	3.749	3.924	1.992	2.812	2.005	2.811	2.167	2.959	2.000	-0.108	2.012	-0.114	882.90	27779.4
`p3--GP--p3`	triangular unstr	0.1	3.693	3.923	1.992	2.803	2.005	2.803	1.804	2.888	2.000	-0.115	2.012	-0.120	886.85	31746.3

$\Delta t = 10^{-3}$

Table 7: 2D manufactured results for `JFNK`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.997	1.997	1.998	1.998	1.998	1.998	2.000	2.000	2.004	2.004	2.004	2.004	62.774	16499.9
`NO--na--NO`	hexa	0.1	1.851	1.851	1.607	1.607	1.613	1.613	2.294	2.294	2.140	2.140	2.142	2.142	104.37	18378.4
`NO--na--NO`	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	229.82	17289.7
`NO--na--NO`	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	335.59	20799.8
`NO--CCp1--p1`	hexa	0.0	1.997	1.997	1.999	2.354	1.998	2.333	2.000	2.000	2.004	2.290	2.004	2.242	99.950	17020.5
`NO--CCp1--p1`	hexa	0.1	1.851	1.851	1.608	2.354	1.613	2.310	2.294	2.294	2.140	2.290	2.142	2.226	139.82	18502.9
`NO--CCp1--p1`	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	463.94	18085.7
`NO--CCp1--p1`	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	503.50	20923.6
`NO--CCp1--p2`	hexa	0.0	1.997	1.997	2.000	2.229	1.999	2.184	2.000	2.000	2.004	2.150	2.004	2.094	130.30	17331.8
`NO--CCp1--p2`	hexa	0.1	1.851	1.851	1.608	2.229	1.614	2.181	2.295	2.295	2.140	2.150	2.142	2.092	154.64	18665.6
`NO--CCp1--p2`	triangular unstr	0.0	0.798	0.798	0.777	2.111	0.783	1.163	0.963	0.963	1.038	2.102	1.107	1.150	438.67	18426.9
`NO--CCp1--p2`	triangular unstr	0.1	0.785	0.785	0.771	2.112	0.776	1.171	0.953	0.953	1.027	2.105	1.097	1.143	521.01	21072.2
`NO--CCp1--p3`	hexa	0.0	1.997	1.997	2.000	2.247	1.999	2.211	2.000	2.000	2.004	2.167	2.004	2.113	132.84	17490.5
`NO--CCp1--p3`	hexa	0.1	1.851	1.851	1.608	2.247	1.614	2.206	2.295	2.295	2.140	2.167	2.142	2.110	151.61	18741.9
`NO--CCp1--p3`	triangular unstr	0.0	0.798	0.798	0.777	2.139	0.783	1.131	0.963	0.963	1.038	2.131	1.107	1.144	439.70	18599.6
`NO--CCp1--p3`	triangular unstr	0.1	0.785	0.785	0.771	2.140	0.776	1.138	0.953	0.953	1.027	2.135	1.097	1.137	524.48	21149.1
`NO--CCp2--p1`	hexa	0.0	1.997	1.997	2.000	3.042	1.997	2.510	1.999	1.999	2.004	2.998	2.003	2.157	167.05	17530.6
`NO--CCp2--p1`	hexa	0.1	1.851	1.851	1.603	3.038	1.613	1.942	2.294	2.294	2.140	2.987	2.142	2.167	200.77	18748.5

			Fit $N = 10$–100						Fit $N = 70$–100							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
NO--CCp2--p1	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	729.76	19279.0
NO--CCp2--p1	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	767.38	21262.3
NO--CCp2--p2	hexa	0.0	1.995	1.995	2.003	3.063	1.997	2.671	1.999	1.999	2.004	3.002	2.003	2.246	196.75	17862.5
NO--CCp2--p2	hexa	0.1	1.853	1.853	1.597	3.061	1.619	2.063	2.289	2.289	2.140	2.996	2.141	2.192	217.72	19033.8
NO--CCp2--p2	triangular unstr	0.0	0.799	0.799	0.777	1.940	0.783	0.775	0.963	0.963	1.038	1.095	1.107	1.097	840.38	19631.4
NO--CCp2--p2	triangular unstr	0.1	0.786	0.786	0.771	1.955	0.776	0.772	0.953	0.953	1.027	1.090	1.097	1.087	707.36	21596.1
NO--CCp2--p3	hexa	0.0	1.995	1.995	2.003	3.058	1.997	2.670	1.999	1.999	2.004	3.001	2.003	2.251	224.42	17998.3
NO--CCp2--p3	hexa	0.1	1.853	1.853	1.597	3.056	1.619	2.064	2.289	2.289	2.140	2.995	2.141	2.193	226.37	19155.7
NO--CCp2--p3	triangular unstr	0.0	0.799	0.799	0.777	1.950	0.783	0.775	0.963	0.963	1.038	1.098	1.107	1.097	743.74	19802.3
NO--CCp2--p3	triangular unstr	0.1	0.786	0.786	0.771	1.966	0.776	0.772	0.953	0.953	1.027	1.092	1.097	1.087	808.05	21765.1
NO--CCp3--p1	hexa	0.0	1.997	1.997	1.998	3.944	1.997	2.030	1.999	1.999	2.004	2.505	2.004	2.004	362.53	18892.7
NO--CCp3--p1	hexa	0.1	1.851	1.851	1.600	3.114	1.613	1.657	2.294	2.294	2.140	2.324	2.142	2.170	356.97	20106.8
NO--CCp3--p1	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	1284.8	21788.5
NO--CCp3--p1	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	1109.8	23745.7
NO--CCp3--p2	hexa	0.0	1.996	1.996	1.998	4.244	1.996	2.092	1.999	1.999	2.004	3.862	2.003	2.002	388.72	19203.1
NO--CCp3--p2	hexa	0.1	1.853	1.853	1.590	3.420	1.617	1.718	2.289	2.289	2.140	2.581	2.141	2.203	371.88	20400.8
NO--CCp3--p2	triangular unstr	0.0	0.799	0.799	0.777	1.070	0.783	0.795	0.963	0.963	1.038	1.038	1.107	1.110	1354.0	22118.9
NO--CCp3--p2	triangular unstr	0.1	0.786	0.786	0.771	1.080	0.776	0.797	0.953	0.953	1.027	1.028	1.097	1.100	1096.4	24075.7
NO--CCp3--p3	hexa	0.0	1.996	1.996	1.998	4.257	1.996	2.090	1.999	1.999	2.004	3.849	2.003	2.002	412.25	19356.7
NO--CCp3--p3	hexa	0.1	1.853	1.853	1.590	3.408	1.617	1.717	2.289	2.289	2.140	2.570	2.141	2.203	375.60	20540.9
NO--CCp3--p3	triangular unstr	0.0	0.799	0.799	0.777	1.053	0.783	0.795	0.963	0.963	1.038	1.038	1.107	1.110	1357.6	22304.1
NO--CCp3--p3	triangular unstr	0.1	0.786	0.786	0.771	1.062	0.776	0.796	0.953	0.953	1.027	1.028	1.097	1.100	1205.0	24248.2
NO--GP--p1	hexa	0.0	1.997	1.997	1.999	2.050	1.998	2.053	2.000	2.000	2.000	2.007	2.000	2.007	87.918	16718.4
NO--GP--p1	hexa	0.1	1.851	1.851	1.999	2.042	1.990	2.046	2.294	2.294	2.000	2.015	2.008	2.016	112.16	18447.0
NO--GP--p1	triangular unstr	0.0	0.798	0.798	0.777	0.777	0.783	0.783	0.963	0.963	1.038	1.038	1.107	1.107	299.64	17478.6
NO--GP--p1	triangular unstr	0.1	0.785	0.785	0.771	0.771	0.776	0.776	0.953	0.953	1.027	1.027	1.097	1.097	455.19	20904.7
NO--GP--p2	hexa	0.0	1.997	1.997	1.999	2.013	1.998	2.015	1.999	1.999	2.000	2.002	2.000	2.002	219.26	17038.3
NO--GP--p2	hexa	0.1	1.853	1.853	1.999	2.013	1.998	2.014	2.290	2.290	2.000	2.003	2.001	2.003	209.11	18552.2
NO--GP--p2	triangular unstr	0.0	0.799	0.799	1.878	1.417	1.101	1.410	0.963	0.963	1.677	1.218	1.112	1.269	541.47	17817.7
NO--GP--p2	triangular unstr	0.1	0.786	0.786	1.880	1.426	1.105	1.419	0.953	0.953	1.680	1.219	1.104	1.270	539.30	21041.9
NO--GP--p3	hexa	0.0	1.997	1.997	1.999	2.016	1.999	2.018	1.999	1.999	2.000	2.002	2.000	2.002	253.15	17191.9
NO--GP--p3	hexa	0.1	1.853	1.853	1.999	2.016	1.998	2.018	2.290	2.290	2.000	2.004	2.002	2.004	265.83	18636.6
NO--GP--p3	triangular unstr	0.0	0.799	0.799	1.878	1.388	1.101	1.383	0.963	0.963	1.677	1.205	1.112	1.258	777.03	18010.7
NO--GP--p3	triangular unstr	0.1	0.786	0.786	1.880	1.397	1.106	1.391	0.953	0.953	1.680	1.206	1.104	1.258	770.03	21110.3
p1--na--NO	hexa	0.0	1.912	2.147	1.616	1.616	1.626	1.626	2.611	2.168	2.380	2.380	2.385	2.385	150.02	17887.6
p1--na--NO	hexa	0.1	1.971	2.163	1.677	1.677	1.687	1.687	2.615	2.141	2.414	2.414	2.419	2.419	175.68	19071.9
p1--na--NO	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	839.15	19894.4
p1--na--NO	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	966.93	21562.5
p1--CCp1--p1	hexa	0.0	1.912	2.147	1.616	2.350	1.626	1.955	2.611	2.168	2.380	2.295	2.385	2.369	210.62	18392.6
p1--CCp1--p1	hexa	0.1	1.971	2.163	1.677	2.351	1.687	2.045	2.615	2.141	2.414	2.294	2.419	2.382	241.87	19302.2
p1--CCp1--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	886.52	20537.9
p1--CCp1--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	1042.2	21806.4
p1--CCp1--p2	hexa	0.0	1.912	2.147	1.616	2.228	1.626	2.057	2.611	2.168	2.380	2.154	2.385	2.233	214.61	18677.9
p1--CCp1--p2	hexa	0.1	1.971	2.163	1.676	2.228	1.686	2.104	2.615	2.141	2.414	2.153	2.419	2.209	253.57	19440.2
p1--CCp1--p2	triangular unstr	0.0	2.292	2.131	2.004	2.103	2.003	2.036	2.600	2.164	2.545	2.063	2.542	2.283	860.24	20876.2
p1--CCp1--p2	triangular unstr	0.1	2.345	2.112	2.092	2.103	2.089	2.066	2.552	2.095	2.516	2.061	2.515	2.188	998.30	22029.6
p1--CCp1--p3	hexa	0.0	1.912	2.147	1.616	2.246	1.626	2.063	2.611	2.168	2.380	2.171	2.385	2.256	209.58	18812.9
p1--CCp1--p3	hexa	0.1	1.971	2.163	1.676	2.246	1.686	2.115	2.615	2.141	2.414	2.170	2.419	2.234	237.29	19520.2
p1--CCp1--p3	triangular unstr	0.0	2.292	2.131	2.004	2.127	2.003	2.040	2.600	2.164	2.545	2.081	2.542	2.312	847.53	21045.5
p1--CCp1--p3	triangular unstr	0.1	2.345	2.112	2.092	2.128	2.089	2.076	2.552	2.095	2.516	2.078	2.515	2.216	1030.2	22141.1
p1--CCp2--p1	hexa	0.0	1.912	2.147	1.616	2.903	1.626	1.665	2.611	2.168	2.380	2.782	2.385	2.394	238.64	18898.4
p1--CCp2--p1	hexa	0.1	1.971	2.163	1.676	2.953	1.687	1.736	2.615	2.141	2.414	2.858	2.419	2.428	260.96	19460.9
p1--CCp2--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	1038.1	21734.5
p1--CCp2--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	1177.4	22368.2
p1--CCp2--p2	hexa	0.0	1.912	2.147	1.616	2.990	1.627	1.689	2.611	2.168	2.380	2.882	2.385	2.400	247.89	19179.4
p1--CCp2--p2	hexa	0.1	1.972	2.163	1.676	3.018	1.687	1.765	2.615	2.141	2.414	2.931	2.419	2.435	276.80	19673.1
p1--CCp2--p2	triangular unstr	0.0	2.292	2.131	2.004	2.676	2.003	1.978	2.600	2.164	2.546	2.665	2.542	2.522	1028.7	22067.9
p1--CCp2--p2	triangular unstr	0.1	2.345	2.112	2.092	2.799	2.089	2.063	2.552	2.095	2.516	2.731	2.515	2.504	1202.9	22669.4
p1--CCp2--p3	hexa	0.0	1.912	2.147	1.616	2.987	1.627	1.688	2.611	2.168	2.380	2.884	2.385	2.400	262.69	19310.6
p1--CCp2--p3	hexa	0.1	1.972	2.163	1.676	3.015	1.687	1.765	2.615	2.141	2.414	2.931	2.419	2.435	293.94	19820.1
p1--CCp2--p3	triangular unstr	0.0	2.292	2.131	2.004	2.681	2.003	1.978	2.600	2.164	2.546	2.668	2.542	2.522	1032.6	22243.1
p1--CCp2--p3	triangular unstr	0.1	2.345	2.112	2.092	2.802	2.089	2.062	2.552	2.095	2.516	2.736	2.515	2.504	1219.5	22819.4
p1--CCp3--p1	hexa	0.0	1.912	2.147	1.617	2.034	1.626	1.630	2.611	2.168	2.380	2.378	2.385	2.406	388.93	20236.1
p1--CCp3--p1	hexa	0.1	1.971	2.163	1.677	2.164	1.687	1.692	2.615	2.141	2.414	2.408	2.419	2.441	405.12	20745.1
p1--CCp3--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	1377.6	24229.1
p1--CCp3--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	1489.4	24696.5
p1--CCp3--p2	hexa	0.0	1.913	2.147	1.617	2.239	1.627	1.635	2.611	2.168	2.380	2.377	2.385	2.426	398.56	20515.3
p1--CCp3--p2	hexa	0.1	1.972	2.163	1.677	2.394	1.688	1.698	2.615	2.141	2.414	2.402	2.419	2.463	428.87	21005.9
p1--CCp3--p2	triangular unstr	0.0	2.292	2.131	2.004	2.031	2.003	1.965	2.600	2.164	2.546	2.521	2.542	2.523	1494.7	24554.1
p1--CCp3--p2	triangular unstr	0.1	2.345	2.112	2.092	2.141	2.089	2.052	2.552	2.095	2.516	2.506	2.515	2.510	1457.9	25036.9
p1--CCp3--p3	hexa	0.0	1.913	2.147	1.617	2.224	1.627	1.634	2.611	2.168	2.380	2.376	2.385	2.426	407.19	20617.6
p1--CCp3--p3	hexa	0.1	1.972	2.163	1.677	2.379	1.688	1.698	2.615	2.141	2.414	2.401	2.419	2.462	430.20	21137.6
p1--CCp3--p3	triangular unstr	0.0	2.292	2.131	2.004	2.028	2.003	1.965	2.600	2.164	2.546	2.522	2.542	2.523	1516.1	24723.6
p1--CCp3--p3	triangular unstr	0.1	2.345	2.112	2.092	2.136	2.089	2.052	2.552	2.095	2.516	2.506	2.515	2.510	1452.3	25196.1
p1--GP--p1	hexa	0.0	1.912	2.147	1.997	2.152	1.836	2.158	2.611	2.168	2.005	2.309	2.249	2.308	183.83	17987.7
p1--GP--p1	hexa	0.1	1.971	2.163	1.999	2.219	1.895	2.225	2.615	2.141	2.003	2.309	2.222	2.308	184.59	19116.1
p1--GP--p1	triangular unstr	0.0	2.292	2.131	2.004	2.004	2.003	2.003	2.600	2.164	2.545	2.545	2.542	2.542	799.04	19938.2
p1--GP--p1	triangular unstr	0.1	2.345	2.112	2.092	2.092	2.089	2.089	2.552	2.095	2.516	2.516	2.515	2.515	852.76	21573.2
p1--GP--p2	hexa	0.0	1.913	2.147	1.998	2.181	1.949	2.179	2.611	2.168	2.001	2.155	2.096	2.151	300.35	18274.8
p1--GP--p2	hexa	0.1	1.972	2.163	1.999	2.199	1.972	2.197	2.615	2.141	2.001	2.150	2.071	2.145	287.40	19229.6
p1--GP--p2	triangular unstr	0.0	2.292	2.131	1.991	2.067	2.009	2.066	2.600	2.164	2.002	2.176	2.268	2.168	920.70	20295.7

			Fit $N = 10$–100						Fit $N = 70$–100							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p1--GP--p2	triangular unstr	0.1	2.345	2.112	1.992	2.087	2.034	2.085	2.552	2.095	2.000	2.109	2.167	2.105	1012.2	21716.7
p1--GP--p3	hexa	0.0	1.913	2.147	1.999	2.194	1.949	2.193	2.611	2.168	2.001	2.170	2.096	2.166	315.72	18429.8
p1--GP--p3	hexa	0.1	1.972	2.163	1.999	2.215	1.973	2.213	2.615	2.141	2.001	2.165	2.071	2.160	344.67	19322.5
p1--GP--p3	triangular unstr	0.0	2.292	2.131	1.992	2.080	2.009	2.079	2.600	2.164	2.002	2.209	2.268	2.200	1028.1	20451.8
p1--GP--p3	triangular unstr	0.1	2.345	2.112	1.992	2.106	2.034	2.104	2.552	2.095	2.000	2.134	2.167	2.129	1107.5	21798.8
p2--na--NO	hexa	0.0	2.128	2.568	2.142	2.142	2.143	2.143	2.020	2.183	2.025	2.025	2.026	2.026	154.77	18705.6
p2--na--NO	hexa	0.1	2.129	2.608	2.143	2.143	2.144	2.144	2.022	2.213	2.027	2.027	2.028	2.028	177.02	20298.4
p2--na--NO	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	838.03	21912.5
p2--na--NO	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	871.55	24391.2
p2--CCp1--p1	hexa	0.0	2.150	2.609	2.164	2.356	2.166	2.334	2.023	2.214	2.029	2.293	2.030	2.232	219.58	19535.3
p2--CCp1--p1	hexa	0.1	2.153	2.652	2.167	2.356	2.170	2.335	2.026	2.251	2.031	2.293	2.032	2.235	256.56	20889.2
p2--CCp1--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	1146.2	22805.4
p2--CCp1--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1044.9	24997.5
p2--CCp1--p2	hexa	0.0	2.213	2.709	2.224	2.230	2.235	2.188	2.031	2.308	2.039	2.151	2.041	2.094	228.91	19806.9
p2--CCp1--p2	hexa	0.1	2.225	2.760	2.235	2.230	2.248	2.187	2.036	2.369	2.043	2.151	2.046	2.094	242.31	21005.3
p2--CCp1--p2	triangular unstr	0.0	2.121	2.535	2.161	2.102	2.166	2.040	1.990	2.162	2.003	2.058	2.012	2.018	1105.5	23121.4
p2--CCp1--p2	triangular unstr	0.1	2.108	2.582	2.154	2.101	2.160	2.039	1.987	2.199	1.998	2.058	2.003	2.018	1105.9	25192.9
p2--CCp1--p3	hexa	0.0	2.213	2.709	2.224	2.248	2.235	2.214	2.031	2.308	2.039	2.168	2.041	2.113	238.91	19939.7
p2--CCp1--p3	hexa	0.1	2.225	2.760	2.235	2.248	2.248	2.214	2.036	2.369	2.043	2.168	2.046	2.113	238.71	21069.7
p2--CCp1--p3	triangular unstr	0.0	2.122	2.535	2.161	2.127	2.166	2.050	1.990	2.162	2.003	2.074	2.012	2.023	1207.3	23264.0
p2--CCp1--p3	triangular unstr	0.1	2.108	2.582	2.154	2.126	2.160	2.050	1.987	2.199	1.998	2.074	2.003	2.023	1239.9	25275.6
p2--CCp2--p1	hexa	0.0	2.150	2.609	2.166	3.039	2.166	2.253	2.023	2.214	2.029	2.986	2.030	2.044	264.32	20008.4
p2--CCp2--p1	hexa	0.1	2.153	2.653	2.170	3.040	2.170	2.269	2.026	2.251	2.031	2.989	2.032	2.050	266.71	20936.1
p2--CCp2--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	1235.2	23870.7
p2--CCp2--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1321.5	25229.5
p2--CCp2--p2	hexa	0.0	2.214	2.709	2.236	3.063	2.235	2.434	2.032	2.308	2.041	2.999	2.041	2.086	282.98	20284.4
p2--CCp2--p2	hexa	0.1	2.226	2.761	2.250	3.063	2.249	2.475	2.037	2.371	2.046	3.000	2.046	2.105	286.14	21071.6
p2--CCp2--p2	triangular unstr	0.0	2.122	2.536	2.172	3.012	2.166	2.317	1.990	2.162	2.003	2.991	2.012	2.038	1196.2	24206.0
p2--CCp2--p2	triangular unstr	0.1	2.108	2.583	2.166	3.012	2.160	2.339	1.987	2.200	1.999	2.993	2.002	2.039	1391.4	25433.3
p2--CCp2--p3	hexa	0.0	2.214	2.709	2.236	3.058	2.235	2.434	2.032	2.308	2.041	2.998	2.041	2.088	286.51	20421.3
p2--CCp2--p3	hexa	0.1	2.226	2.761	2.250	3.058	2.249	2.474	2.037	2.371	2.046	2.999	2.046	2.106	303.51	21165.0
p2--CCp2--p3	triangular unstr	0.0	2.122	2.536	2.172	3.009	2.166	2.314	1.990	2.162	2.003	2.991	2.012	2.039	1195.3	24362.8
p2--CCp2--p3	triangular unstr	0.1	2.108	2.583	2.166	3.010	2.160	2.336	1.987	2.200	1.999	2.993	2.002	2.040	1316.5	25555.4
p2--CCp3--p1	hexa	0.0	2.150	2.609	2.166	3.135	2.166	2.183	2.023	2.214	2.029	1.992	2.030	2.029	401.90	21240.8
p2--CCp3--p1	hexa	0.1	2.153	2.653	2.170	3.206	2.170	2.186	2.026	2.251	2.031	2.002	2.032	2.031	423.47	21652.0
p2--CCp3--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	1500.9	26335.4
p2--CCp3--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	1643.3	27242.5
p2--CCp3--p2	hexa	0.0	2.214	2.709	2.236	3.796	2.235	2.299	2.032	2.308	2.041	2.178	2.041	2.039	427.34	21488.3
p2--CCp3--p2	hexa	0.1	2.226	2.761	2.249	3.882	2.249	2.314	2.037	2.371	2.046	2.315	2.046	2.044	465.26	21933.4
p2--CCp3--p2	triangular unstr	0.0	2.122	2.536	2.172	3.302	2.166	2.231	1.990	2.162	2.003	1.988	2.012	2.010	1582.1	26653.0
p2--CCp3--p2	triangular unstr	0.1	2.108	2.583	2.166	3.390	2.160	2.225	1.987	2.200	1.999	2.010	2.002	2.001	1728.1	27559.0
p2--CCp3--p3	hexa	0.0	2.214	2.709	2.236	3.789	2.235	2.298	2.032	2.308	2.041	2.133	2.041	2.039	433.79	21626.6
p2--CCp3--p3	hexa	0.1	2.226	2.761	2.249	3.877	2.249	2.313	2.037	2.371	2.046	2.262	2.046	2.044	537.75	22086.9
p2--CCp3--p3	triangular unstr	0.0	2.122	2.536	2.172	3.267	2.166	2.229	1.990	2.162	2.003	1.957	2.012	2.010	1627.4	26819.6
p2--CCp3--p3	triangular unstr	0.1	2.108	2.583	2.166	3.356	2.160	2.223	1.987	2.200	1.999	1.972	2.002	2.001	1692.3	27730.4
p2--GP--p1	hexa	0.0	2.150	2.609	2.003	2.689	2.102	2.694	2.023	2.214	2.001	2.304	2.011	2.312	206.03	18811.8
p2--GP--p1	hexa	0.1	2.153	2.653	2.003	2.728	2.100	2.732	2.026	2.251	2.001	2.350	2.011	2.358	229.20	20346.2
p2--GP--p1	triangular unstr	0.0	2.058	2.376	2.090	2.090	2.086	2.086	1.991	2.077	1.998	1.998	2.002	2.002	935.71	22002.3
p2--GP--p1	triangular unstr	0.1	2.045	2.405	2.079	2.079	2.075	2.075	1.989	2.091	1.996	1.996	1.997	1.997	863.88	24427.9
p2--GP--p2	hexa	0.0	2.214	2.709	2.003	2.924	2.114	2.923	2.032	2.308	2.000	2.644	2.007	2.650	290.14	19087.4
p2--GP--p2	hexa	0.1	2.226	2.761	2.003	2.952	2.112	2.951	2.037	2.371	2.000	2.706	2.007	2.711	316.92	20450.9
p2--GP--p2	triangular unstr	0.0	2.122	2.536	1.992	2.805	2.037	2.807	1.990	2.162	1.998	2.489	2.003	2.499	1036.4	22340.1
p2--GP--p2	triangular unstr	0.1	2.108	2.583	1.992	2.840	2.035	2.842	1.987	2.200	1.998	2.553	2.002	2.561	987.41	24553.8
p2--GP--p3	hexa	0.0	2.214	2.709	2.004	2.923	2.114	2.922	2.032	2.308	2.000	2.650	2.007	2.656	346.39	19235.2
p2--GP--p3	hexa	0.1	2.226	2.761	2.004	2.950	2.113	2.950	2.037	2.371	2.000	2.711	2.007	2.716	366.70	20522.7
p2--GP--p3	triangular unstr	0.0	2.122	2.536	1.993	2.809	2.037	2.812	1.990	2.162	1.998	2.500	2.003	2.510	1128.1	22514.3
p2--GP--p3	triangular unstr	0.1	2.108	2.583	1.993	2.843	2.035	2.846	1.987	2.200	1.998	2.564	2.002	2.572	1084.0	24631.3
p3--na--NO	hexa	0.0	2.125	2.485	2.132	2.132	2.135	2.135	2.013	2.050	2.025	2.025	2.024	2.024	192.01	21809.6
p3--na--NO	hexa	0.1	2.108	2.481	2.114	2.114	2.117	2.117	2.010	2.048	2.020	2.020	2.020	2.020	320.68	24530.8
p3--na--NO	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	968.40	27545.1
p3--na--NO	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	884.29	31519.3
p3--CCp1--p1	hexa	0.0	2.185	2.634	2.188	2.354	2.201	2.339	2.020	2.072	2.038	2.290	2.040	2.248	305.03	23002.7
p3--CCp1--p1	hexa	0.1	2.160	2.629	2.164	2.354	2.176	2.339	2.016	2.068	2.031	2.290	2.032	2.248	392.23	25270.6
p3--CCp1--p1	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1291.4	28435.5
p3--CCp1--p1	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1154.7	31984.8
p3--CCp1--p2	hexa	0.0	2.620	3.309	2.640	2.229	2.603	2.184	2.117	2.267	2.264	2.150	2.211	2.094	324.98	23127.6
p3--CCp1--p2	hexa	0.1	2.564	3.307	2.592	2.229	2.554	2.184	2.084	2.237	2.206	2.150	2.160	2.094	428.72	25401.0
p3--CCp1--p2	triangular unstr	0.0	2.396	2.791	2.510	2.101	2.460	2.038	2.003	2.026	2.041	2.058	2.014	2.017	1333.6	28660.7
p3--CCp1--p2	triangular unstr	0.1	2.330	2.777	2.443	2.101	2.397	2.038	1.999	2.022	2.031	2.058	2.006	2.017	1166.1	32124.5
p3--CCp1--p3	hexa	0.0	2.620	3.309	2.640	2.247	2.603	2.210	2.117	2.267	2.264	2.167	2.211	2.112	333.14	23206.1
p3--CCp1--p3	hexa	0.1	2.564	3.307	2.592	2.247	2.554	2.210	2.084	2.237	2.206	2.167	2.160	2.112	469.38	25461.2
p3--CCp1--p3	triangular unstr	0.0	2.396	2.791	2.509	2.126	2.460	2.048	2.003	2.026	2.041	2.074	2.014	2.023	1348.9	28769.0
p3--CCp1--p3	triangular unstr	0.1	2.330	2.777	2.443	2.126	2.397	2.048	2.000	2.022	2.031	2.074	2.006	2.023	1202.7	32179.6
p3--CCp2--p1	hexa	0.0	2.187	2.637	2.200	3.042	2.203	2.642	2.020	2.073	2.041	2.999	2.040	2.257	391.36	23106.3
p3--CCp2--p1	hexa	0.1	2.162	2.631	2.175	3.042	2.177	2.638	2.016	2.069	2.033	2.999	2.033	2.252	521.75	25281.9
p3--CCp2--p1	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1547.0	28727.0
p3--CCp2--p1	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1854.1	32015.7
p3--CCp2--p2	hexa	0.0	3.539	3.992	3.284	3.063	3.303	3.110	4.324	4.279	4.079	3.002	4.086	3.055	396.06	23282.3
p3--CCp2--p2	hexa	0.1	3.570	4.025	3.311	3.063	3.331	3.109	4.299	4.272	4.076	3.002	4.084	3.049	561.58	25389.0
p3--CCp2--p2	triangular unstr	0.0	4.057	4.096	3.842	3.013	3.861	3.085	4.476	4.164	4.251	2.997	4.258	3.023	1681.5	29025.2
p3--CCp2--p2	triangular unstr	0.1	4.077	4.105	3.880	3.013	3.901	3.082	4.460	4.139	4.199	2.997	4.214	3.022	1659.8	32176.2
p3--CCp2--p3	hexa	0.0	3.539	3.992	3.284	3.058	3.303	3.102	4.324	4.279	4.079	3.001	4.086	3.050	424.70	23377.0

			Fit $N = 10$–100						Fit $N = 70$–100							
`Vm-states-Ion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`p3--CCp2--p3`	hexa	0.1	3.570	4.025	3.311	3.058	3.331	3.101	4.299	4.272	4.076	3.001	4.084	3.044	548.85	25443.2
`p3--CCp2--p3`	triangular unstr	0.0	4.057	4.096	3.842	3.011	3.861	3.072	4.477	4.165	4.252	2.997	4.259	3.019	1625.8	29189.0
`p3--CCp2--p3`	triangular unstr	0.1	4.076	4.105	3.880	3.011	3.901	3.070	4.461	4.139	4.199	2.997	4.214	3.017	1579.9	32250.7
`p3--CCp3--p1`	hexa	0.0	2.186	2.637	2.200	3.993	2.202	2.225	2.020	2.073	2.041	2.812	2.040	2.040	526.58	23777.7
`p3--CCp3--p1`	hexa	0.1	2.161	2.631	2.174	3.993	2.177	2.201	2.016	2.069	2.033	2.809	2.033	2.032	653.85	25298.9
`p3--CCp3--p1`	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1660.4	31065.9
`p3--CCp3--p1`	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1871.3	32210.7
`p3--CCp3--p2`	hexa	0.0	3.538	3.993	3.290	4.231	3.304	3.441	4.324	4.279	4.081	4.284	4.087	4.101	561.34	24015.9
`p3--CCp3--p2`	hexa	0.1	3.569	4.026	3.319	4.231	3.333	3.485	4.299	4.272	4.078	4.283	4.084	4.097	661.30	25419.6
`p3--CCp3--p2`	triangular unstr	0.0	4.058	4.096	3.846	4.109	3.863	3.934	4.476	4.164	4.253	4.049	4.258	4.241	1691.8	31398.8
`p3--CCp3--p2`	triangular unstr	0.1	4.078	4.105	3.884	4.109	3.903	3.988	4.461	4.139	4.202	4.049	4.214	4.191	1831.7	32387.7
`p3--CCp3--p3`	hexa	0.0	3.538	3.993	3.290	4.241	3.304	3.438	4.324	4.279	4.081	4.302	4.087	4.101	533.17	24150.7
`p3--CCp3--p3`	hexa	0.1	3.569	4.026	3.319	4.241	3.333	3.481	4.299	4.272	4.078	4.300	4.084	4.097	510.03	25470.3
`p3--CCp3--p3`	triangular unstr	0.0	4.057	4.096	3.846	4.114	3.862	3.932	4.477	4.165	4.254	4.050	4.259	4.242	1732.8	31562.0
`p3--CCp3--p3`	triangular unstr	0.1	4.077	4.105	3.884	4.114	3.903	3.986	4.461	4.139	4.202	4.050	4.214	4.192	1720.3	32487.7
`p3--GP--p1`	hexa	0.0	2.187	2.637	2.001	2.598	1.989	2.596	2.020	2.073	2.000	2.068	1.999	2.066	290.60	21859.5
`p3--GP--p1`	hexa	0.1	2.161	2.631	2.001	2.592	1.988	2.589	2.016	2.069	2.000	2.060	1.999	2.059	432.06	24569.5
`p3--GP--p1`	triangular unstr	0.0	2.041	2.209	2.061	2.061	2.060	2.060	1.999	2.015	2.008	2.008	2.008	2.008	1230.9	27569.1
`p3--GP--p1`	triangular unstr	0.1	2.029	2.204	2.046	2.046	2.046	2.046	1.999	2.014	2.008	2.008	2.007	2.007	1092.4	31558.3
`p3--GP--p2`	hexa	0.0	3.538	3.992	1.999	4.003	1.997	4.010	4.324	4.279	2.000	4.194	2.001	4.196	381.25	21971.2
`p3--GP--p2`	hexa	0.1	3.569	4.026	1.999	4.033	1.997	4.040	4.299	4.272	2.000	4.212	2.000	4.213	511.96	24681.8
`p3--GP--p2`	triangular unstr	0.0	4.057	4.096	1.992	4.077	2.002	4.083	4.476	4.164	1.998	4.204	2.003	4.204	1724.5	27714.9
`p3--GP--p2`	triangular unstr	0.1	4.077	4.105	1.992	4.094	2.001	4.098	4.461	4.139	1.998	4.195	2.003	4.198	1145.9	31669.6
`p3--GP--p3`	hexa	0.0	3.538	3.992	1.999	4.000	1.998	4.008	4.324	4.279	2.000	4.198	2.001	4.200	440.53	22050.9
`p3--GP--p3`	hexa	0.1	3.569	4.026	1.999	4.031	1.997	4.039	4.299	4.272	2.000	4.217	2.001	4.219	472.62	24729.1
`p3--GP--p3`	triangular unstr	0.0	4.057	4.096	1.992	4.079	2.002	4.086	4.477	4.165	1.998	4.227	2.003	4.229	1466.0	27775.6
`p3--GP--p3`	triangular unstr	0.1	4.077	4.105	1.992	4.098	2.002	4.103	4.461	4.139	1.998	4.219	2.003	4.222	1625.2	31740.2

11.2 Two-dimensional results: comments

The current 2D tables fix the time scheme and mass matrix to `crankNicolson` and `consistent`, and use `RKF45` for the state ODEs. The first column now records the full `Vm-states-Ion` triplet, so configurations that share the same voltage and ionic reconstruction but differ in state treatment are no longer collapsed into the same label.

Under this fixed discretisation, `Picard`, `diagonalIion`, and `JFNK` produce comparable observed convergence orders to the precision reported in the tables. The practical distinction in this MMS sweep is therefore computational cost and nonlinear robustness rather than asymptotic accuracy.

On the structured hexa mesh, the clearest high-order verification remains the higher-order voltage reconstructions paired with compatible state and ionic quadrature settings. The unstructured triangular mesh is more sensitive to the selected `Vm-states-Ion` triplet, the stabilisation parameter α, and the time step.

11.3 Three-dimensional manufactured problem

The tables in this subsection are generated directly from the current `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0` result tree. The method-specific roots are `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/3D_CN_cons_Picard_RKF45`, `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/3D_CN_cons_diagonalIion_RKF45`, `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/3D_CN_cons_JFNK_RKF45`. All entries use `crankNicolson`, the `consistent` mass matrix, and `RKF45` for the state ODEs. The first column reports `Vm-states-Ion`: the V_m reconstruction degree, the state treatment (`CCp*`, `GP`, or `na`), and the I_{ion} quadrature/reconstruction setting. Fits use $N = 10$–30 for the broad mesh range and $N = 20$–30 for the fine-mesh range.

11.3.1 `Picard`

$\Delta t = 10^{-2}$

Table 8: 3D manufactured results for `Picard`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–30						Fit $N = 20$–30							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.985	1.985	1.994	1.994	1.994	1.994	1.993	1.993	2.009	2.009	2.009	2.009	13.701	4034.4
`NO--na--NO`	hexa	0.1	1.436	1.436	1.297	1.297	1.306	1.306	1.411	1.411	1.214	1.214	1.223	1.223	22.097	6312.2
`NO--na--NO`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	21.144	4988.3
`NO--na--NO`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	105.72	12055.9
`NO--CCp1--p1`	hexa	0.0	1.985	1.985	1.998	2.166	1.995	2.246	1.992	1.992	2.010	2.208	2.009	2.189	28.255	4710.5
`NO--CCp1--p1`	hexa	0.1	1.436	1.436	1.287	2.167	1.306	2.230	1.411	1.411	1.205	2.208	1.224	2.159	46.542	6565.3
`NO--CCp1--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	116.61	6199.0
`NO--CCp1--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	186.46	12250.5
`NO--CCp1--p2`	hexa	0.0	1.985	1.985	1.999	2.098	1.994	2.090	1.992	1.992	2.011	2.130	2.009	2.062	31.539	5888.2
`NO--CCp1--p2`	hexa	0.1	1.436	1.436	1.283	2.098	1.307	2.087	1.411	1.411	1.201	2.130	1.224	2.056	44.590	7233.9
`NO--CCp1--p2`	tetra unstr	0.0	0.664	0.664	0.375	1.924	0.373	0.457	0.826	0.826	0.617	1.896	0.616	0.669	128.54	6649.6
`NO--CCp1--p2`	tetra unstr	0.1	0.677	0.677	0.398	1.928	0.396	0.483	0.823	0.823	0.629	1.903	0.628	0.683	199.14	12498.6
`NO--CCp1--p3`	hexa	0.0	1.985	1.985	1.999	2.094	1.994	2.088	1.992	1.992	2.011	2.127	2.009	2.061	35.704	8248.0
`NO--CCp1--p3`	hexa	0.1	1.436	1.436	1.283	2.095	1.307	2.085	1.411	1.411	1.201	2.127	1.224	2.054	57.374	9568.3
`NO--CCp1--p3`	tetra unstr	0.0	0.664	0.664	0.375	1.923	0.373	0.460	0.826	0.826	0.617	1.906	0.616	0.670	123.28	7555.6
`NO--CCp1--p3`	tetra unstr	0.1	0.677	0.677	0.398	1.926	0.396	0.485	0.823	0.823	0.629	1.912	0.628	0.684	217.11	12994.2
`NO--CCp2--p1`	hexa	0.0	1.977	1.977	2.009	3.171	1.987	2.618	1.989	1.989	2.014	3.091	2.005	2.390	37.567	5489.6
`NO--CCp2--p1`	hexa	0.1	1.447	1.447	1.257	3.172	1.318	2.180	1.426	1.426	1.176	3.092	1.239	1.626	56.054	6775.8
`NO--CCp2--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	190.82	9543.3
`NO--CCp2--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	244.84	13294.2
`NO--CCp2--p2`	hexa	0.0	1.969	1.969	2.010	3.247	1.979	2.753	1.985	1.985	2.013	3.146	2.001	2.494	55.354	6662.9
`NO--CCp2--p2`	hexa	0.1	1.457	1.457	1.248	3.248	1.333	2.320	1.441	1.441	1.170	3.147	1.257	1.718	58.989	7953.8
`NO--CCp2--p2`	tetra unstr	0.0	0.664	0.664	0.375	2.080	0.373	0.465	0.826	0.826	0.617	1.273	0.616	0.673	183.95	9999.5
`NO--CCp2--p2`	tetra unstr	0.1	0.677	0.677	0.398	2.128	0.396	0.486	0.823	0.823	0.629	1.333	0.628	0.683	257.01	13767.7
`NO--CCp2--p3`	hexa	0.0	1.969	1.969	2.010	3.252	1.979	2.758	1.985	1.985	2.013	3.149	2.001	2.501	68.040	9017.1
`NO--CCp2--p3`	hexa	0.1	1.457	1.457	1.248	3.253	1.333	2.328	1.441	1.441	1.170	3.150	1.257	1.726	66.357	10307.2
`NO--CCp2--p3`	tetra unstr	0.0	0.664	0.664	0.375	1.992	0.373	0.466	0.826	0.826	0.617	1.241	0.616	0.675	180.62	10904.7
`NO--CCp2--p3`	tetra unstr	0.1	0.677	0.677	0.398	2.039	0.396	0.487	0.823	0.823	0.629	1.295	0.628	0.686	257.54	14649.4
`NO--CCp3--p1`	hexa	0.0	1.980	1.980	2.001	4.033	1.989	2.062	1.991	1.991	2.011	4.072	2.007	2.017	90.356	7390.4
`NO--CCp3--p1`	hexa	0.1	1.452	1.452	1.244	4.015	1.323	1.276	1.429	1.429	1.170	4.028	1.241	1.186	89.988	8677.1
`NO--CCp3--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	358.12	18313.3
`NO--CCp3--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	385.12	22079.8
`NO--CCp3--p2`	hexa	0.0	1.976	1.976	2.000	3.980	1.985	2.107	1.990	1.990	2.011	4.053	2.005	2.022	94.372	8563.2
`NO--CCp3--p2`	hexa	0.1	1.469	1.469	1.233	3.966	1.343	1.305	1.447	1.447	1.163	4.019	1.261	1.187	88.556	9854.5
`NO--CCp3--p2`	tetra unstr	0.0	0.664	0.664	0.375	0.890	0.373	0.418	0.826	0.826	0.617	0.674	0.616	0.657	346.61	18765.0
`NO--CCp3--p2`	tetra unstr	0.1	0.677	0.677	0.398	0.930	0.396	0.441	0.823	0.823	0.629	0.689	0.628	0.670	399.32	22514.1
`NO--CCp3--p3`	hexa	0.0	1.976	1.976	2.000	3.973	1.985	2.108	1.990	1.990	2.011	4.054	2.005	2.022	94.607	10920.3
`NO--CCp3--p3`	hexa	0.1	1.469	1.469	1.233	3.959	1.343	1.305	1.447	1.447	1.163	4.021	1.261	1.186	106.47	12209.7
`NO--CCp3--p3`	tetra unstr	0.0	0.664	0.664	0.375	0.888	0.373	0.432	0.826	0.826	0.617	0.684	0.616	0.665	348.72	19672.6
`NO--CCp3--p3`	tetra unstr	0.1	0.677	0.677	0.398	0.926	0.396	0.454	0.823	0.823	0.629	0.697	0.628	0.677	418.61	23417.4
`NO--GP--p1`	hexa	0.0	1.984	1.984	1.992	2.072	1.992	2.077	1.992	1.992	1.996	2.038	1.997	2.041	28.095	4465.1
`NO--GP--p1`	hexa	0.1	1.459	1.459	1.992	2.073	1.993	2.079	1.429	1.429	1.996	2.038	1.996	2.041	37.380	6564.9
`NO--GP--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	29.082	5234.8
`NO--GP--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	97.186	12251.2
`NO--GP--p2`	hexa	0.0	1.983	1.983	1.984	2.014	1.985	2.018	1.992	1.992	1.992	2.007	1.993	2.009	72.453	5644.5
`NO--GP--p2`	hexa	0.1	1.482	1.482	1.984	2.015	1.985	2.018	1.447	1.447	1.992	2.007	1.993	2.010	84.100	7214.1
`NO--GP--p2`	tetra unstr	0.0	0.664	0.664	1.832	0.861	0.516	0.859	0.826	0.826	1.792	0.867	0.652	0.866	43.013	5699.3
`NO--GP--p2`	tetra unstr	0.1	0.677	0.677	1.844	0.883	0.546	0.881	0.823	0.823	1.812	0.884	0.668	0.883	115.92	12501.1
`NO--GP--p3`	hexa	0.0	1.983	1.983	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.993	2.008	159.70	7997.1
`NO--GP--p3`	hexa	0.1	1.482	1.482	1.984	2.011	1.985	2.015	1.447	1.447	1.992	2.006	1.992	2.008	156.73	9198.0
`NO--GP--p3`	tetra unstr	0.0	0.664	0.664	1.828	0.863	0.513	0.862	0.826	0.826	1.790	0.868	0.651	0.868	71.823	6602.5
`NO--GP--p3`	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.542	0.884	0.823	0.823	1.810	0.885	0.667	0.885	146.28	12996.8
`p1--na--NO`	hexa	0.0	1.967	2.095	1.927	1.927	1.930	1.930	1.919	2.125	1.838	1.838	1.845	1.845	77.496	7145.3
`p1--na--NO`	hexa	0.1	2.067	2.093	2.047	2.047	2.048	2.048	2.066	2.124	2.033	2.033	2.035	2.035	67.389	8698.8
`p1--na--NO`	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	207.91	9888.9
`p1--na--NO`	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	209.36	11566.5
`p1--CCp1--p1`	hexa	0.0	1.967	2.095	1.926	2.166	1.930	2.193	1.919	2.125	1.836	2.207	1.845	2.115	73.944	7263.0
`p1--CCp1--p1`	hexa	0.1	2.067	2.093	2.046	2.167	2.048	2.225	2.066	2.124	2.033	2.208	2.035	2.168	93.736	8854.4
`p1--CCp1--p1`	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	315.13	10862.6
`p1--CCp1--p1`	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	289.28	12081.8
`p1--CCp1--p2`	hexa	0.0	1.967	2.095	1.925	2.098	1.930	2.083	1.919	2.125	1.835	2.130	1.845	2.049	83.398	8101.2
`p1--CCp1--p2`	hexa	0.1	2.067	2.093	2.046	2.098	2.047	2.090	2.066	2.124	2.033	2.130	2.035	2.061	87.516	9506.3
`p1--CCp1--p2`	tetra unstr	0.0	1.474	1.866	1.139	1.973	1.142	1.493	1.771	1.928	1.419	1.984	1.428	1.590	330.99	11305.1
`p1--CCp1--p2`	tetra unstr	0.1	1.613	1.909	1.262	1.973	1.264	1.621	1.889	1.963	1.556	1.983	1.558	1.721	309.50	12482.1
`p1--CCp1--p3`	hexa	0.0	1.967	2.095	1.925	2.094	1.930	2.081	1.919	2.125	1.835	2.127	1.845	2.048	105.81	10456.5
`p1--CCp1--p3`	hexa	0.1	2.067	2.093	2.046	2.094	2.047	2.088	2.066	2.124	2.033	2.127	2.035	2.060	106.65	11155.3
`p1--CCp1--p3`	tetra unstr	0.0	1.474	1.866	1.139	1.965	1.142	1.505	1.771	1.928	1.419	1.982	1.428	1.597	307.94	12243.6
`p1--CCp1--p3`	tetra unstr	0.1	1.613	1.909	1.262	1.965	1.264	1.632	1.889	1.963	1.556	1.982	1.558	1.727	309.45	13293.2
`p1--CCp2--p1`	hexa	0.0	1.966	2.095	1.926	3.169	1.929	2.043	1.919	2.125	1.831	3.089	1.845	1.844	115.71	7558.1
`p1--CCp2--p1`	hexa	0.1	2.065	2.093	2.050	3.170	2.046	2.200	2.066	2.124	2.033	3.090	2.034	2.090	96.811	8857.6
`p1--CCp2--p1`	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	356.84	14195.6
`p1--CCp2--p1`	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	342.43	14883.3
`p1--CCp2--p2`	hexa	0.0	1.965	2.095	1.925	3.244	1.929	2.095	1.919	2.125	1.828	3.143	1.845	1.846	115.21	8770.9
`p1--CCp2--p2`	hexa	0.1	2.064	2.093	2.051	3.245	2.045	2.265	2.065	2.124	2.033	3.145	2.034	2.115	109.97	9580.9
`p1--CCp2--p2`	tetra unstr	0.0	1.475	1.866	1.139	2.917	1.142	1.199	1.771	1.928	1.418	2.913	1.428	1.465	379.66	14646.2
`p1--CCp2--p2`	tetra unstr	0.1	1.613	1.909	1.262	2.928	1.264	1.336	1.889	1.963	1.555	2.938	1.558	1.614	374.84	15337.7

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p1--CCp2--p3	hexa	0.0	1.965	2.095	1.925	3.250	1.929	2.098	1.919	2.125	1.828	3.146	1.845	1.848	108.78	11158.3
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.051	3.251	2.045	2.269	2.065	2.124	2.033	3.148	2.034	2.117	114.59	11791.9
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.139	2.922	1.142	1.211	1.771	1.928	1.418	2.906	1.428	1.476	352.44	15569.3
p1--CCp2--p3	tetra unstr	0.1	1.613	1.909	1.262	2.938	1.264	1.348	1.889	1.963	1.555	2.938	1.558	1.624	367.63	16260.6
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.922	4.067	1.929	1.877	1.919	2.125	1.830	4.138	1.845	1.729	114.34	9452.8
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.046	4.057	2.047	2.042	2.066	2.124	2.032	4.119	2.034	2.007	131.72	9844.0
p1--CCp3--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	499.68	22974.6
p1--CCp3--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	473.73	23656.3
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.920	4.009	1.930	1.853	1.920	2.125	1.827	4.112	1.846	1.682	127.48	10692.9
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.045	4.000	2.047	2.039	2.066	2.124	2.031	4.092	2.034	1.993	142.00	10999.5
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.139	2.936	1.142	1.202	1.771	1.928	1.418	2.110	1.428	1.472	498.40	23422.0
p1--CCp3--p2	tetra unstr	0.1	1.613	1.909	1.262	3.163	1.264	1.332	1.889	1.963	1.555	2.437	1.558	1.614	470.69	24109.9
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.920	4.003	1.930	1.852	1.920	2.125	1.827	4.114	1.846	1.681	162.36	13075.5
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.045	3.992	2.047	2.039	2.066	2.124	2.031	4.094	2.034	1.993	143.37	13660.3
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.139	2.837	1.142	1.240	1.771	1.928	1.418	2.066	1.428	1.496	507.99	24331.7
p1--CCp3--p3	tetra unstr	0.1	1.613	1.909	1.262	3.084	1.264	1.375	1.889	1.963	1.555	2.395	1.558	1.644	490.47	25022.1
p1--GP--p1	hexa	0.0	1.967	2.095	1.992	2.157	1.999	2.164	1.919	2.125	1.997	2.201	2.001	2.201	87.931	7353.8
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.002	2.162	2.066	2.124	1.997	2.199	2.005	2.199	78.820	8903.7
p1--GP--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	222.68	9942.5
p1--GP--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	219.88	11623.1
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.920	2.125	1.993	2.129	1.997	2.128	127.68	8029.7
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.093	1.991	2.096	2.066	2.124	1.993	2.128	1.998	2.126	120.62	9552.9
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.771	1.928	1.953	1.877	1.777	1.880	215.29	10400.8
p1--GP--p2	tetra unstr	0.1	1.613	1.909	1.920	1.897	1.790	1.900	1.889	1.963	1.955	1.921	1.848	1.922	230.12	11946.2
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.920	2.125	1.993	2.126	1.996	2.125	195.10	10159.3
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.066	2.124	1.993	2.124	1.998	2.123	195.10	10948.2
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.874	1.771	1.928	1.951	1.889	1.775	1.892	264.06	11323.2
p1--GP--p3	tetra unstr	0.1	1.613	1.909	1.916	1.900	1.786	1.903	1.889	1.963	1.953	1.927	1.846	1.929	256.05	12695.5
p2--na--NO	hexa	0.0	1.947	3.025	1.944	1.944	1.948	1.948	2.001	2.766	2.003	2.003	2.005	2.005	126.94	25863.3
p2--na--NO	hexa	0.1	1.946	3.053	1.943	1.943	1.947	1.947	2.000	2.807	2.002	2.002	2.005	2.005	127.33	23781.8
p2--na--NO	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	329.21	17527.9
p2--na--NO	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	299.85	21384.0
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.995	2.168	2.003	2.183	2.035	2.884	2.035	2.209	2.040	2.147	162.66	13057.9
p2--CCp1--p1	hexa	0.1	2.008	3.133	2.000	2.167	2.008	2.193	2.037	2.925	2.037	2.209	2.042	2.153	154.56	15809.6
p2--CCp1--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	401.07	18278.8
p2--CCp1--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	409.42	21360.5
p2--CCp1--p2	hexa	0.0	2.040	3.142	2.028	2.098	2.039	2.087	2.059	2.936	2.057	2.131	2.064	2.063	153.06	13705.3
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.037	2.098	2.049	2.088	2.063	2.975	2.062	2.131	2.069	2.063	144.90	16447.5
p2--CCp1--p2	tetra unstr	0.0	2.103	2.703	1.863	1.971	1.864	1.960	2.383	2.718	2.113	1.982	2.121	1.974	369.74	18791.5
p2--CCp1--p2	tetra unstr	0.1	2.347	2.823	2.013	1.971	2.016	1.970	2.727	2.903	2.362	1.982	2.372	1.985	404.73	21713.1
p2--CCp1--p3	hexa	0.0	2.040	3.142	2.028	2.095	2.039	2.085	2.059	2.936	2.057	2.127	2.064	2.061	146.72	15161.4
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.037	2.095	2.049	2.086	2.063	2.975	2.062	2.127	2.069	2.061	168.64	17752.1
p2--CCp1--p3	tetra unstr	0.0	2.103	2.703	1.863	1.963	1.864	1.958	2.383	2.718	2.112	1.981	2.121	1.972	408.34	19711.2
p2--CCp1--p3	tetra unstr	0.1	2.347	2.823	2.013	1.963	2.016	1.968	2.727	2.903	2.362	1.981	2.372	1.983	451.98	22462.7
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.999	3.170	2.004	2.077	2.035	2.884	2.038	3.091	2.040	2.074	151.33	13056.4
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.005	3.171	2.009	2.096	2.037	2.925	2.040	3.091	2.043	2.083	144.64	15812.2
p2--CCp2--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	414.32	21601.2
p2--CCp2--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	459.74	23596.6
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.036	3.246	2.040	2.173	2.059	2.936	2.063	3.146	2.065	2.129	183.26	13759.2
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.046	3.246	2.050	2.209	2.064	2.975	2.068	3.146	2.070	2.147	160.11	16458.6
p2--CCp2--p2	tetra unstr	0.0	2.103	2.704	1.863	2.944	1.864	1.932	2.384	2.719	2.114	2.972	2.121	2.120	456.55	22081.1
p2--CCp2--p2	tetra unstr	0.1	2.349	2.824	2.014	2.945	2.016	2.103	2.732	2.904	2.367	2.972	2.373	2.405	446.99	24005.8
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.036	3.251	2.040	2.176	2.059	2.936	2.063	3.149	2.065	2.130	193.86	15535.7
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.046	3.251	2.050	2.212	2.064	2.975	2.068	3.149	2.070	2.149	164.90	17811.0
p2--CCp2--p3	tetra unstr	0.0	2.103	2.704	1.863	2.958	1.864	1.943	2.384	2.719	2.114	2.980	2.121	2.133	428.34	22991.9
p2--CCp2--p3	tetra unstr	0.1	2.349	2.824	2.014	2.959	2.016	2.110	2.732	2.904	2.367	2.981	2.373	2.411	463.22	24901.5
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.999	4.087	2.004	2.022	2.035	2.884	2.038	4.142	2.040	2.059	188.83	13728.6
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.004	4.082	2.009	2.028	2.037	2.925	2.040	4.137	2.043	2.062	176.63	15813.4
p2--CCp3--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	572.02	30371.1
p2--CCp3--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	607.78	31489.2
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.035	4.022	2.040	2.074	2.059	2.936	2.063	4.109	2.065	2.104	182.15	14822.2
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.045	4.017	2.050	2.087	2.064	2.975	2.068	4.103	2.070	2.111	177.15	16475.9
p2--CCp3--p2	tetra unstr	0.0	2.103	2.704	1.863	3.893	1.864	1.896	2.384	2.719	2.114	3.841	2.121	2.127	586.59	30834.3
p2--CCp3--p2	tetra unstr	0.1	2.349	2.824	2.014	3.911	2.016	2.046	2.732	2.904	2.367	3.890	2.373	2.389	634.18	31950.0
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.035	4.014	2.040	2.075	2.059	2.936	2.063	4.110	2.065	2.105	244.85	17100.4
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.045	4.010	2.050	2.087	2.064	2.975	2.068	4.104	2.070	2.112	199.29	18144.8
p2--CCp3--p3	tetra unstr	0.0	2.103	2.704	1.863	3.888	1.864	1.929	2.384	2.719	2.114	3.831	2.121	2.146	611.73	31741.8
p2--CCp3--p3	tetra unstr	0.1	2.349	2.824	2.014	3.907	2.016	2.072	2.732	2.904	2.367	3.883	2.373	2.403	632.63	32867.4
p2--GP--p1	hexa	0.0	2.003	3.107	2.011	3.133	2.845	3.122	2.035	2.884	2.004	3.005	2.449	2.999	164.63	26074.9
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.888	3.131	2.037	2.925	2.004	3.021	2.476	3.015	138.88	23991.3
p2--GP--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	342.41	17573.5
p2--GP--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	330.06	21400.4
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.898	3.215	2.059	2.936	2.001	3.092	2.432	3.083	236.16	26715.1
p2--GP--p2	hexa	0.1	2.050	3.166	2.004	3.236	2.920	3.222	2.064	2.975	2.001	3.105	2.440	3.095	179.04	24639.4
p2--GP--p2	tetra unstr	0.0	2.103	2.704	1.928	2.794	2.100	2.796	2.384	2.719	1.960	2.779	2.102	2.784	342.91	17961.8
p2--GP--p2	tetra unstr	0.1	2.349	2.824	1.928	2.831	2.146	2.833	2.732	2.904	1.960	2.861	2.166	2.866	336.90	21691.1
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.897	3.220	2.059	2.936	2.000	3.095	2.432	3.085	355.85	28015.7
p2--GP--p3	hexa	0.1	2.050	3.166	2.003	3.242	2.920	3.227	2.064	2.975	2.000	3.108	2.439	3.097	288.86	25931.7
p2--GP--p3	tetra unstr	0.0	2.103	2.704	1.924	2.802	2.082	2.802	2.384	2.719	1.958	2.769	2.088	2.773	414.66	18813.0
p2--GP--p3	tetra unstr	0.1	2.349	2.824	1.924	2.847	2.128	2.847	2.732	2.904	1.958	2.864	2.152	2.868	497.14	22280.7
p3--na--NO	hexa	0.0	2.313	3.204	2.308	2.308	2.312	2.312	2.193	2.830	2.197	2.197	2.200	2.200	897.53	77071.6
p3--na--NO	hexa	0.1	2.278	3.207	2.274	2.274	2.278	2.278	2.169	2.826	2.174	2.174	2.176	2.176	755.12	60518.2

			Fit $N = 10$–30						Fit $N = 20$–30							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`p3--na--NO`	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1241.6	45021.5
`p3--na--NO`	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1292.1	53563.8
`p3--CCp1--p1`	hexa	0.0	2.761	3.638	2.698	2.166	2.745	2.264	2.550	3.526	2.508	2.207	2.553	2.199	878.49	44825.6
`p3--CCp1--p1`	hexa	0.1	2.717	3.647	2.659	2.166	2.706	2.261	2.501	3.534	2.465	2.207	2.509	2.198	790.19	45947.9
`p3--CCp1--p1`	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1339.5	45670.1
`p3--CCp1--p1`	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1387.9	53118.8
`p3--CCp1--p2`	hexa	0.0	3.129	3.794	3.046	2.097	3.078	2.094	2.914	3.838	2.857	2.130	2.869	2.063	1000.0	45474.5
`p3--CCp1--p2`	hexa	0.1	3.103	3.804	3.033	2.097	3.064	2.093	2.862	3.851	2.828	2.130	2.835	2.063	818.24	46594.6
`p3--CCp1--p2`	tetra unstr	0.0	3.009	3.711	2.897	1.970	2.893	1.968	3.092	3.661	3.022	1.982	3.021	1.961	1488.2	45933.5
`p3--CCp1--p2`	tetra unstr	0.1	3.120	3.786	3.019	1.970	3.015	1.968	3.099	3.741	3.149	1.982	3.147	1.960	1324.3	53368.7
`p3--CCp1--p3`	hexa	0.0	3.129	3.794	3.046	2.094	3.078	2.092	2.914	3.838	2.857	2.127	2.869	2.062	821.92	46769.5
`p3--CCp1--p3`	hexa	0.1	3.103	3.804	3.033	2.094	3.064	2.091	2.862	3.851	2.828	2.127	2.835	2.062	829.13	47889.4
`p3--CCp1--p3`	tetra unstr	0.0	3.009	3.711	2.897	1.963	2.892	1.966	3.091	3.661	3.021	1.981	3.021	1.960	1619.5	46505.3
`p3--CCp1--p3`	tetra unstr	0.1	3.120	3.786	3.018	1.963	3.015	1.966	3.099	3.741	3.149	1.981	3.147	1.960	1344.9	53870.2
`p3--CCp2--p1`	hexa	0.0	2.781	3.643	2.756	3.172	2.763	2.955	2.572	3.538	2.566	3.092	2.573	2.804	663.90	44829.9
`p3--CCp2--p1`	hexa	0.1	2.738	3.653	2.719	3.172	2.726	2.951	2.523	3.546	2.524	3.092	2.530	2.794	837.36	45948.4
`p3--CCp2--p1`	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1489.8	46456.2
`p3--CCp2--p1`	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1414.7	53257.0
`p3--CCp2--p2`	hexa	0.0	3.332	3.812	3.211	3.247	3.230	3.235	3.260	3.882	3.083	3.147	3.113	3.176	668.73	45478.7
`p3--CCp2--p2`	hexa	0.1	3.350	3.823	3.234	3.247	3.253	3.247	3.275	3.896	3.103	3.147	3.135	3.190	833.85	46594.5
`p3--CCp2--p2`	tetra unstr	0.0	3.196	3.775	2.986	2.942	2.983	2.997	3.461	3.809	3.184	2.970	3.189	3.072	1497.0	46813.3
`p3--CCp2--p2`	tetra unstr	0.1	3.450	3.862	3.163	2.942	3.162	3.045	3.777	3.920	3.434	2.970	3.440	3.120	1290.8	53564.8
`p3--CCp2--p3`	hexa	0.0	3.332	3.812	3.211	3.253	3.230	3.234	3.260	3.882	3.083	3.150	3.113	3.175	678.52	46771.1
`p3--CCp2--p3`	hexa	0.1	3.350	3.823	3.234	3.253	3.253	3.246	3.275	3.896	3.103	3.150	3.135	3.189	812.26	47892.2
`p3--CCp2--p3`	tetra unstr	0.0	3.195	3.776	2.985	2.957	2.983	3.007	3.461	3.809	3.184	2.980	3.188	3.076	1498.5	47514.6
`p3--CCp2--p3`	tetra unstr	0.1	3.450	3.862	3.163	2.957	3.161	3.054	3.777	3.921	3.434	2.980	3.440	3.122	1273.4	54155.3
`p3--CCp3--p1`	hexa	0.0	2.780	3.644	2.753	4.033	2.762	2.775	2.572	3.538	2.566	4.067	2.573	2.597	1165.1	44825.3
`p3--CCp3--p1`	hexa	0.1	2.737	3.653	2.716	4.033	2.725	2.746	2.522	3.545	2.523	4.068	2.530	2.555	804.75	45947.1
`p3--CCp3--p1`	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1615.4	52498.7
`p3--CCp3--p1`	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1395.3	57649.7
`p3--CCp3--p2`	hexa	0.0	3.330	3.812	3.205	3.978	3.228	3.112	3.256	3.881	3.086	4.042	3.110	3.026	962.21	45471.4
`p3--CCp3--p2`	hexa	0.1	3.347	3.823	3.226	3.978	3.250	3.143	3.271	3.895	3.106	4.042	3.131	3.046	818.88	46596.1
`p3--CCp3--p2`	tetra unstr	0.0	3.196	3.775	2.987	3.943	2.983	3.011	3.461	3.808	3.184	3.959	3.188	3.209	1542.9	52918.9
`p3--CCp3--p2`	tetra unstr	0.1	3.450	3.862	3.164	3.942	3.162	3.181	3.777	3.920	3.434	3.959	3.440	3.451	1398.8	58052.9
`p3--CCp3--p3`	hexa	0.0	3.330	3.812	3.205	3.971	3.228	3.112	3.256	3.881	3.086	4.044	3.110	3.027	1042.9	46765.3
`p3--CCp3--p3`	hexa	0.1	3.347	3.823	3.226	3.971	3.250	3.143	3.271	3.895	3.106	4.044	3.131	3.046	864.30	47889.9
`p3--CCp3--p3`	tetra unstr	0.0	3.195	3.776	2.986	3.944	2.982	3.056	3.460	3.809	3.184	3.961	3.188	3.240	1623.8	53767.0
`p3--CCp3--p3`	tetra unstr	0.1	3.449	3.862	3.163	3.944	3.161	3.224	3.777	3.920	3.434	3.961	3.439	3.481	1407.7	58877.6
`p3--GP--p1`	hexa	0.0	2.780	3.644	2.003	3.792	2.717	3.810	2.572	3.538	2.000	3.687	2.355	3.691	930.65	77273.8
`p3--GP--p1`	hexa	0.1	2.737	3.653	2.003	3.799	2.707	3.817	2.522	3.546	2.000	3.695	2.347	3.700	905.37	77345.2
`p3--GP--p1`	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1124.9	45255.4
`p3--GP--p1`	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1323.8	53639.5
`p3--GP--p2`	hexa	0.0	3.331	3.812	1.994	3.893	2.454	3.912	3.257	3.881	1.996	3.958	2.171	3.969	979.58	77917.7
`p3--GP--p2`	hexa	0.1	3.348	3.823	1.994	3.897	2.447	3.916	3.272	3.896	1.996	3.964	2.167	3.975	853.75	77986.4
`p3--GP--p2`	tetra unstr	0.0	3.196	3.775	1.927	3.844	2.032	3.845	3.461	3.808	1.961	3.840	1.977	3.846	1028.3	45508.8
`p3--GP--p2`	tetra unstr	0.1	3.450	3.862	1.927	3.883	2.028	3.884	3.777	3.920	1.961	3.904	1.974	3.909	1038.4	53893.4
`p3--GP--p3`	hexa	0.0	3.331	3.812	1.994	3.886	2.452	3.905	3.257	3.881	1.995	3.960	2.170	3.970	1098.9	79214.1
`p3--GP--p3`	hexa	0.1	3.348	3.823	1.993	3.890	2.446	3.909	3.272	3.896	1.995	3.966	2.167	3.976	769.90	79286.6
`p3--GP--p3`	tetra unstr	0.0	3.195	3.776	1.923	3.842	2.021	3.844	3.460	3.809	1.958	3.837	1.974	3.844	1068.5	46006.7
`p3--GP--p3`	tetra unstr	0.1	3.450	3.862	1.923	3.884	2.017	3.885	3.777	3.921	1.958	3.904	1.971	3.911	1068.8	54392.0

$\Delta t = 10^{-3}$

Table 9: 3D manufactured results for `Picard`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–30						Fit $N = 20$–30							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.985	1.985	1.985	1.985	1.985	1.985	1.993	1.993	1.993	1.993	1.993	1.993	86.614	4035.7
`NO--na--NO`	hexa	0.1	1.438	1.438	1.293	1.293	1.302	1.302	1.413	1.413	1.211	1.211	1.220	1.220	67.106	6315.1
`NO--na--NO`	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	135.14	4989.6
`NO--na--NO`	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	252.56	12057.8
`NO--CCp1--p1`	hexa	0.0	1.986	1.986	1.989	2.167	1.986	2.246	1.993	1.993	1.995	2.208	1.993	2.188	121.46	4714.8
`NO--CCp1--p1`	hexa	0.1	1.438	1.438	1.298	2.167	1.302	2.230	1.413	1.413	1.212	2.208	1.220	2.158	144.23	6566.5
`NO--CCp1--p1`	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	358.87	6202.6
`NO--CCp1--p1`	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	401.30	12253.2
`NO--CCp1--p2`	hexa	0.0	1.986	1.986	1.991	2.098	1.986	2.090	1.993	1.993	1.995	2.130	1.993	2.062	126.95	5894.5
`NO--CCp1--p2`	hexa	0.1	1.438	1.438	1.300	2.098	1.303	2.086	1.413	1.413	1.212	2.130	1.220	2.055	174.32	7235.8
`NO--CCp1--p2`	tetra unstr	0.0	0.663	0.663	0.377	1.924	0.375	0.459	0.829	0.829	0.617	1.896	0.617	0.670	375.15	6653.6
`NO--CCp1--p2`	tetra unstr	0.1	0.677	0.677	0.400	1.928	0.398	0.484	0.828	0.828	0.629	1.903	0.629	0.684	461.73	12500.1
`NO--CCp1--p3`	hexa	0.0	1.986	1.986	1.991	2.094	1.986	2.088	1.993	1.993	1.995	2.127	1.993	2.060	189.07	8249.2
`NO--CCp1--p3`	hexa	0.1	1.438	1.438	1.300	2.095	1.303	2.085	1.413	1.413	1.212	2.127	1.220	2.054	232.83	9568.0
`NO--CCp1--p3`	tetra unstr	0.0	0.663	0.663	0.377	1.922	0.375	0.462	0.829	0.829	0.617	1.906	0.617	0.671	435.28	7561.8
`NO--CCp1--p3`	tetra unstr	0.1	0.677	0.677	0.400	1.926	0.398	0.487	0.828	0.828	0.629	1.912	0.629	0.685	491.97	12998.0
`NO--CCp2--p1`	hexa	0.0	1.977	1.977	2.002	3.171	1.978	2.610	1.989	1.989	1.999	3.091	1.989	2.376	198.39	5489.1
`NO--CCp2--p1`	hexa	0.1	1.447	1.447	1.258	3.172	1.313	2.174	1.428	1.428	1.175	3.092	1.236	1.620	211.28	6772.3

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
NO--CCp2--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	646.90	9551.6
NO--CCp2--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	642.48	13261.1
NO--CCp2--p2	hexa	0.0	1.969	1.969	2.004	3.247	1.970	2.745	1.985	1.985	1.998	3.146	1.985	2.481	282.85	6666.9
NO--CCp2--p2	hexa	0.1	1.457	1.457	1.251	3.248	1.326	2.314	1.443	1.443	1.170	3.147	1.253	1.712	229.83	7952.0
NO--CCp2--p2	tetra unstr	0.0	0.663	0.663	0.377	2.079	0.375	0.467	0.829	0.829	0.617	1.272	0.617	0.673	654.87	10004.3
NO--CCp2--p2	tetra unstr	0.1	0.677	0.677	0.400	2.127	0.398	0.488	0.828	0.828	0.629	1.333	0.629	0.684	716.55	13729.4
NO--CCp2--p3	hexa	0.0	1.969	1.969	2.004	3.252	1.970	2.751	1.985	1.985	1.998	3.149	1.985	2.488	339.01	9025.0
NO--CCp2--p3	hexa	0.1	1.457	1.457	1.251	3.253	1.326	2.323	1.443	1.443	1.170	3.150	1.253	1.720	294.83	10309.6
NO--CCp2--p3	tetra unstr	0.0	0.663	0.663	0.378	1.991	0.375	0.469	0.829	0.829	0.617	1.240	0.617	0.676	675.80	10910.1
NO--CCp2--p3	tetra unstr	0.1	0.677	0.677	0.400	2.038	0.398	0.489	0.828	0.828	0.629	1.295	0.629	0.686	703.70	14607.9
NO--CCp3--p1	hexa	0.0	1.980	1.980	1.993	4.033	1.980	2.052	1.992	1.992	1.995	4.072	1.992	2.002	346.12	7391.3
NO--CCp3--p1	hexa	0.1	1.453	1.453	1.241	4.015	1.318	1.271	1.431	1.431	1.167	4.028	1.238	1.184	328.43	8669.8
NO--CCp3--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	1243.0	18315.6
NO--CCp3--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	1293.8	22044.6
NO--CCp3--p2	hexa	0.0	1.975	1.975	1.992	3.981	1.975	2.098	1.990	1.990	1.995	4.053	1.990	2.007	374.09	8567.8
NO--CCp3--p2	hexa	0.1	1.469	1.469	1.230	3.966	1.336	1.300	1.449	1.449	1.160	4.019	1.258	1.185	407.91	9852.1
NO--CCp3--p2	tetra unstr	0.0	0.663	0.663	0.378	0.891	0.375	0.420	0.829	0.829	0.617	0.675	0.617	0.658	1250.2	18767.3
NO--CCp3--p2	tetra unstr	0.1	0.677	0.677	0.400	0.930	0.398	0.443	0.828	0.828	0.629	0.689	0.629	0.671	1248.8	22474.0
NO--CCp3--p3	hexa	0.0	1.975	1.975	1.992	3.973	1.975	2.099	1.991	1.991	1.995	4.055	1.990	2.007	428.80	10913.1
NO--CCp3--p3	hexa	0.1	1.469	1.469	1.230	3.959	1.336	1.300	1.449	1.449	1.160	4.021	1.258	1.184	503.35	12204.4
NO--CCp3--p3	tetra unstr	0.0	0.663	0.663	0.378	0.888	0.375	0.434	0.829	0.829	0.617	0.684	0.617	0.666	1289.0	19673.6
NO--CCp3--p3	tetra unstr	0.1	0.677	0.677	0.400	0.927	0.398	0.456	0.828	0.828	0.629	0.698	0.629	0.678	1303.2	23376.9
NO--GP--p1	hexa	0.0	1.985	1.985	1.992	2.071	1.992	2.077	1.993	1.993	1.996	2.038	1.996	2.041	168.28	4465.5
NO--GP--p1	hexa	0.1	1.461	1.461	1.992	2.073	1.992	2.079	1.431	1.431	1.996	2.038	1.995	2.041	162.93	6564.0
NO--GP--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	140.66	5234.5
NO--GP--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	230.04	12251.4
NO--GP--p2	hexa	0.0	1.984	1.984	1.984	2.014	1.984	2.018	1.992	1.992	1.992	2.007	1.993	2.009	438.08	5647.7
NO--GP--p2	hexa	0.1	1.483	1.483	1.984	2.014	1.985	2.018	1.450	1.450	1.992	2.007	1.992	2.009	441.07	7215.9
NO--GP--p2	tetra unstr	0.0	0.663	0.663	1.832	0.862	0.518	0.860	0.829	0.829	1.792	0.867	0.653	0.866	253.95	5704.6
NO--GP--p2	tetra unstr	0.1	0.677	0.677	1.843	0.883	0.547	0.882	0.828	0.828	1.812	0.884	0.668	0.883	327.91	12501.2
NO--GP--p3	hexa	0.0	1.984	1.984	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.992	2.008	1011.8	7998.2
NO--GP--p3	hexa	0.1	1.483	1.483	1.984	2.011	1.984	2.015	1.450	1.450	1.992	2.005	1.992	2.008	957.54	9195.7
NO--GP--p3	tetra unstr	0.0	0.663	0.663	1.828	0.864	0.514	0.862	0.829	0.829	1.790	0.869	0.652	0.868	457.00	6601.5
NO--GP--p3	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.543	0.884	0.828	0.828	1.810	0.886	0.667	0.885	564.41	13000.6
p1--na--NO	hexa	0.0	1.967	2.095	1.924	1.924	1.927	1.927	1.920	2.125	1.833	1.833	1.840	1.840	155.87	7144.0
p1--na--NO	hexa	0.1	2.067	2.093	2.044	2.044	2.044	2.044	2.067	2.124	2.028	2.028	2.029	2.029	136.81	8698.4
p1--na--NO	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	398.80	9891.6
p1--na--NO	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	397.09	11569.4
p1--CCp1--p1	hexa	0.0	1.967	2.095	1.924	2.166	1.926	2.192	1.920	2.125	1.833	2.207	1.840	2.114	166.93	7261.7
p1--CCp1--p1	hexa	0.1	2.067	2.093	2.043	2.167	2.044	2.224	2.067	2.124	2.028	2.208	2.029	2.167	166.18	8857.0
p1--CCp1--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	610.94	10859.4
p1--CCp1--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	539.25	12082.9
p1--CCp1--p2	hexa	0.0	1.967	2.095	1.924	2.098	1.926	2.083	1.920	2.125	1.833	2.130	1.840	2.049	201.25	8095.6
p1--CCp1--p2	hexa	0.1	2.067	2.093	2.043	2.098	2.044	2.090	2.067	2.124	2.028	2.130	2.029	2.061	249.86	9499.3
p1--CCp1--p2	tetra unstr	0.0	1.475	1.866	1.142	1.973	1.144	1.493	1.768	1.927	1.422	1.984	1.432	1.592	596.95	11306.5
p1--CCp1--p2	tetra unstr	0.1	1.612	1.909	1.265	1.973	1.267	1.622	1.884	1.962	1.558	1.983	1.561	1.723	658.33	12480.0
p1--CCp1--p3	hexa	0.0	1.967	2.095	1.924	2.094	1.926	2.081	1.920	2.125	1.833	2.127	1.840	2.048	301.30	10445.0
p1--CCp1--p3	hexa	0.1	2.067	2.093	2.043	2.094	2.044	2.088	2.067	2.124	2.028	2.127	2.029	2.059	274.68	11154.7
p1--CCp1--p3	tetra unstr	0.0	1.475	1.866	1.142	1.965	1.144	1.506	1.768	1.927	1.422	1.982	1.432	1.599	640.80	12241.1
p1--CCp1--p3	tetra unstr	0.1	1.612	1.909	1.265	1.965	1.267	1.633	1.884	1.962	1.558	1.982	1.561	1.729	597.66	13293.2
p1--CCp2--p1	hexa	0.0	1.966	2.095	1.924	3.169	1.926	2.040	1.920	2.125	1.827	3.089	1.840	1.840	269.04	7542.1
p1--CCp2--p1	hexa	0.1	2.065	2.093	2.048	3.170	2.043	2.196	2.066	2.124	2.028	3.090	2.028	2.085	273.06	8855.9
p1--CCp2--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	812.43	14195.8
p1--CCp2--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	770.23	14886.6
p1--CCp2--p2	hexa	0.0	1.966	2.095	1.924	3.244	1.925	2.092	1.920	2.125	1.824	3.143	1.840	1.841	289.16	8760.2
p1--CCp2--p2	hexa	0.1	2.064	2.093	2.050	3.245	2.042	2.261	2.065	2.124	2.028	3.145	2.028	2.109	272.25	9575.8
p1--CCp2--p2	tetra unstr	0.0	1.475	1.866	1.141	2.917	1.145	1.201	1.768	1.927	1.422	2.912	1.432	1.469	834.07	14648.6
p1--CCp2--p2	tetra unstr	0.1	1.612	1.909	1.264	2.928	1.267	1.339	1.884	1.962	1.558	2.938	1.561	1.617	805.97	15334.0
p1--CCp2--p3	hexa	0.0	1.966	2.095	1.924	3.250	1.925	2.095	1.920	2.125	1.824	3.146	1.840	1.843	334.46	11158.4
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.050	3.251	2.042	2.265	2.065	2.124	2.028	3.148	2.028	2.111	380.06	11779.2
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.141	2.922	1.145	1.213	1.768	1.927	1.422	2.906	1.432	1.479	862.07	15565.5
p1--CCp2--p3	tetra unstr	0.1	1.612	1.909	1.264	2.938	1.267	1.350	1.884	1.962	1.558	2.938	1.561	1.627	820.54	16257.5
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.919	4.067	1.926	1.874	1.920	2.125	1.826	4.139	1.840	1.725	319.04	9440.7
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.043	4.057	2.043	2.038	2.067	2.124	2.026	4.119	2.029	2.002	353.23	9829.8
p1--CCp3--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	1325.7	22972.5
p1--CCp3--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	1347.4	23653.0
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.917	4.010	1.926	1.850	1.921	2.125	1.823	4.112	1.841	1.677	423.99	10685.6
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.042	4.000	2.043	2.035	2.067	2.124	2.025	4.092	2.029	1.988	428.66	10985.7
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.141	2.935	1.145	1.205	1.768	1.927	1.422	2.111	1.432	1.475	1319.9	23418.0
p1--CCp3--p2	tetra unstr	0.1	1.612	1.909	1.264	3.163	1.267	1.335	1.884	1.962	1.558	2.438	1.561	1.616	1339.2	24104.2
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.917	4.003	1.926	1.849	1.921	2.125	1.823	4.114	1.841	1.676	577.60	13061.6
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.042	3.992	2.043	2.035	2.067	2.124	2.025	4.094	2.029	1.988	535.66	13643.6
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.141	2.836	1.145	1.243	1.768	1.927	1.422	2.067	1.432	1.500	1304.3	24333.6
p1--CCp3--p3	tetra unstr	0.1	1.612	1.909	1.264	3.084	1.267	1.378	1.884	1.962	1.558	2.395	1.561	1.647	1434.8	25014.7
p1--GP--p1	hexa	0.0	1.968	2.095	1.992	2.157	1.999	2.164	1.920	2.125	1.997	2.201	2.001	2.200	254.63	7353.9
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.001	2.162	2.067	2.124	1.997	2.199	2.004	2.199	222.49	8905.5
p1--GP--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	435.69	9944.3
p1--GP--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	427.55	11621.8
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.921	2.125	1.993	2.129	1.996	2.128	481.37	8024.5
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.092	1.991	2.096	2.067	2.124	1.993	2.128	1.998	2.126	472.63	9555.5
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.768	1.927	1.953	1.877	1.777	1.880	493.81	10403.2

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p1--GP--p2	tetra unstr	0.1	1.612	1.909	1.920	1.897	1.790	1.900	1.884	1.962	1.955	1.921	1.849	1.923	531.81	11946.4
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.921	2.125	1.993	2.126	1.996	2.124	900.71	10144.0
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.067	2.124	1.993	2.124	1.998	2.123	963.73	10940.8
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.873	1.768	1.927	1.951	1.889	1.775	1.892	780.16	11337.6
p1--GP--p3	tetra unstr	0.1	1.612	1.909	1.916	1.900	1.786	1.903	1.884	1.962	1.953	1.927	1.847	1.929	679.56	12707.2
p2--na--NO	hexa	0.0	1.947	3.025	1.942	1.942	1.946	1.946	2.001	2.766	2.000	2.000	2.003	2.003	210.78	25865.1
p2--na--NO	hexa	0.1	1.946	3.053	1.941	1.941	1.946	1.946	2.000	2.807	1.999	1.999	2.002	2.002	207.61	23781.5
p2--na--NO	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	654.46	17527.1
p2--na--NO	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	551.02	21382.3
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.993	2.168	2.001	2.183	2.035	2.884	2.031	2.209	2.036	2.145	276.91	13059.6
p2--CCp1--p1	hexa	0.1	2.008	3.133	1.997	2.167	2.006	2.192	2.037	2.925	2.033	2.209	2.038	2.152	269.12	15810.2
p2--CCp1--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	791.52	18277.7
p2--CCp1--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	805.96	21365.2
p2--CCp1--p2	hexa	0.0	2.040	3.141	2.024	2.098	2.036	2.087	2.059	2.936	2.052	2.131	2.060	2.062	289.55	13706.2
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.033	2.098	2.046	2.088	2.064	2.974	2.056	2.131	2.065	2.062	333.13	16453.6
p2--CCp1--p2	tetra unstr	0.0	2.105	2.705	1.863	1.971	1.864	1.960	2.382	2.719	2.113	1.982	2.122	1.974	740.31	18792.7
p2--CCp1--p2	tetra unstr	0.1	2.348	2.824	2.014	1.971	2.018	1.970	2.721	2.901	2.364	1.982	2.374	1.986	866.90	21712.1
p2--CCp1--p3	hexa	0.0	2.040	3.141	2.024	2.095	2.036	2.085	2.059	2.936	2.052	2.127	2.060	2.061	344.81	15168.3
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.033	2.095	2.046	2.086	2.064	2.974	2.056	2.127	2.065	2.061	368.42	17748.2
p2--CCp1--p3	tetra unstr	0.0	2.105	2.705	1.863	1.963	1.864	1.958	2.382	2.719	2.113	1.981	2.122	1.973	819.47	19712.8
p2--CCp1--p3	tetra unstr	0.1	2.347	2.824	2.014	1.963	2.018	1.968	2.721	2.901	2.364	1.981	2.374	1.984	901.98	22464.9
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.998	3.170	2.002	2.075	2.035	2.884	2.035	3.091	2.037	2.071	370.35	13057.8
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.003	3.171	2.007	2.094	2.037	2.925	2.037	3.091	2.039	2.079	306.27	15810.1
p2--CCp2--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	961.03	21600.9
p2--CCp2--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1023.0	23594.7
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.034	3.246	2.037	2.170	2.059	2.936	2.059	3.146	2.061	2.125	420.00	13761.2
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.044	3.246	2.047	2.206	2.064	2.975	2.064	3.146	2.066	2.143	349.30	16459.6
p2--CCp2--p2	tetra unstr	0.0	2.105	2.705	1.863	2.944	1.864	1.932	2.384	2.720	2.116	2.972	2.122	2.121	954.64	22082.1
p2--CCp2--p2	tetra unstr	0.1	2.350	2.825	2.016	2.945	2.018	2.104	2.727	2.903	2.369	2.972	2.376	2.408	1040.0	23997.5
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.034	3.251	2.037	2.173	2.059	2.936	2.059	3.149	2.061	2.126	457.56	15535.8
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.044	3.251	2.047	2.209	2.064	2.975	2.064	3.149	2.066	2.144	401.16	17813.2
p2--CCp2--p3	tetra unstr	0.0	2.105	2.705	1.863	2.958	1.864	1.943	2.384	2.720	2.116	2.980	2.122	2.134	1051.8	22990.9
p2--CCp2--p3	tetra unstr	0.1	2.350	2.825	2.016	2.959	2.018	2.111	2.727	2.903	2.369	2.981	2.376	2.414	1088.7	24901.2
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.997	4.087	2.002	2.020	2.035	2.884	2.035	4.142	2.037	2.056	411.75	13728.9
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.002	4.082	2.007	2.026	2.037	2.925	2.037	4.137	2.039	2.058	401.83	15810.7
p2--CCp3--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1500.5	30375.9
p2--CCp3--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1552.3	31492.0
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.033	4.022	2.037	2.072	2.060	2.936	2.059	4.109	2.061	2.100	520.74	14822.2
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.043	4.017	2.047	2.084	2.064	2.975	2.064	4.103	2.066	2.107	460.62	16473.4
p2--CCp3--p2	tetra unstr	0.0	2.105	2.705	1.863	3.893	1.864	1.897	2.384	2.720	2.116	3.841	2.123	2.129	1501.0	30835.0
p2--CCp3--p2	tetra unstr	0.1	2.350	2.825	2.016	3.911	2.018	2.048	2.727	2.903	2.369	3.890	2.376	2.392	1579.6	31952.0
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.033	4.014	2.037	2.073	2.060	2.936	2.059	4.110	2.061	2.101	681.86	17100.2
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.043	4.010	2.047	2.085	2.064	2.975	2.064	4.104	2.066	2.107	571.45	18143.3
p2--CCp3--p3	tetra unstr	0.0	2.105	2.705	1.863	3.888	1.864	1.929	2.384	2.720	2.116	3.831	2.123	2.148	1532.3	31743.1
p2--CCp3--p3	tetra unstr	0.1	2.350	2.825	2.016	3.907	2.018	2.074	2.727	2.903	2.369	3.884	2.376	2.406	1612.3	32870.0
p2--GP--p1	hexa	0.0	2.004	3.107	2.011	3.133	2.844	3.122	2.035	2.884	2.004	3.004	2.447	2.998	435.91	26067.6
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.886	3.131	2.037	2.925	2.004	3.021	2.473	3.014	313.10	23991.7
p2--GP--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	657.39	17584.1
p2--GP--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	692.09	21405.3
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.897	3.215	2.060	2.936	2.001	3.092	2.431	3.082	759.22	26717.6
p2--GP--p2	hexa	0.1	2.051	3.166	2.004	3.236	2.920	3.221	2.064	2.975	2.001	3.105	2.439	3.094	623.79	24635.8
p2--GP--p2	tetra unstr	0.0	2.105	2.705	1.928	2.794	2.100	2.795	2.384	2.720	1.960	2.779	2.102	2.783	887.93	17964.1
p2--GP--p2	tetra unstr	0.1	2.350	2.825	1.928	2.831	2.146	2.833	2.727	2.903	1.960	2.862	2.167	2.866	776.48	21693.1
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.896	3.220	2.060	2.936	2.000	3.095	2.430	3.085	1431.6	28015.9
p2--GP--p3	hexa	0.1	2.051	3.166	2.003	3.241	2.919	3.226	2.064	2.975	2.000	3.108	2.438	3.097	1111.8	25928.3
p2--GP--p3	tetra unstr	0.0	2.105	2.705	1.924	2.801	2.082	2.802	2.384	2.720	1.958	2.768	2.089	2.773	1108.6	18824.2
p2--GP--p3	tetra unstr	0.1	2.350	2.825	1.924	2.846	2.128	2.847	2.727	2.903	1.958	2.864	2.153	2.868	1250.4	22275.1
p3--na--NO	hexa	0.0	2.313	3.204	2.303	2.303	2.307	2.307	2.193	2.830	2.189	2.189	2.192	2.192	1163.8	77061.2
p3--na--NO	hexa	0.1	2.278	3.207	2.269	2.269	2.273	2.273	2.169	2.827	2.166	2.166	2.168	2.168	1137.9	77131.2
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	1409.3	45038.0
p3--na--NO	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	1785.5	53562.0
p3--CCp1--p1	hexa	0.0	2.758	3.637	2.675	2.166	2.727	2.263	2.547	3.524	2.476	2.207	2.525	2.199	1051.8	44827.2
p3--CCp1--p1	hexa	0.1	2.713	3.646	2.635	2.166	2.687	2.261	2.497	3.532	2.433	2.207	2.481	2.197	965.91	45949.0
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2254.3	45663.0
p3--CCp1--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	1997.6	53125.5
p3--CCp1--p2	hexa	0.0	3.096	3.790	2.987	2.097	3.017	2.094	2.864	3.829	2.773	2.130	2.782	2.063	952.29	45473.8
p3--CCp1--p2	hexa	0.1	3.064	3.800	2.965	2.097	2.994	2.093	2.807	3.842	2.734	2.130	2.739	2.063	981.62	46587.4
p3--CCp1--p2	tetra unstr	0.0	2.979	3.699	2.857	1.970	2.857	1.968	3.035	3.633	2.949	1.982	2.960	1.961	2125.4	45933.6
p3--CCp1--p2	tetra unstr	0.1	3.071	3.771	2.960	1.971	2.963	1.968	3.012	3.707	3.038	1.982	3.053	1.960	2058.2	53371.6
p3--CCp1--p3	hexa	0.0	3.096	3.790	2.987	2.094	3.017	2.092	2.864	3.829	2.773	2.127	2.782	2.062	1048.1	46766.9
p3--CCp1--p3	hexa	0.1	3.064	3.800	2.965	2.094	2.994	2.091	2.807	3.842	2.734	2.127	2.739	2.062	1049.4	47890.8
p3--CCp1--p3	tetra unstr	0.0	2.978	3.699	2.856	1.963	2.857	1.966	3.034	3.633	2.949	1.981	2.960	1.960	2280.9	46502.7
p3--CCp1--p3	tetra unstr	0.1	3.071	3.772	2.959	1.963	2.963	1.966	3.011	3.707	3.037	1.981	3.053	1.960	2033.4	53871.1
p3--CCp2--p1	hexa	0.0	2.782	3.644	2.745	3.172	2.752	2.948	2.574	3.538	2.546	3.092	2.553	2.790	823.62	44830.9
p3--CCp2--p1	hexa	0.1	2.739	3.653	2.708	3.172	2.715	2.943	2.524	3.546	2.503	3.092	2.509	2.778	982.87	45950.0
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2270.3	46460.9
p3--CCp2--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2137.6	53103.6
p3--CCp2--p2	hexa	0.0	3.335	3.812	3.202	3.247	3.220	3.232	3.264	3.882	3.065	3.147	3.095	3.171	867.38	45475.8
p3--CCp2--p2	hexa	0.1	3.352	3.823	3.223	3.247	3.241	3.244	3.280	3.896	3.083	3.147	3.114	3.186	1040.2	46594.9
p3--CCp2--p2	tetra unstr	0.0	3.198	3.777	2.984	2.942	2.981	2.996	3.463	3.810	3.182	2.970	3.187	3.073	2331.7	46813.0
p3--CCp2--p2	tetra unstr	0.1	3.452	3.863	3.162	2.942	3.161	3.045	3.775	3.921	3.433	2.970	3.439	3.121	2028.2	53563.8
p3--CCp2--p3	hexa	0.0	3.335	3.812	3.202	3.253	3.220	3.231	3.264	3.882	3.065	3.150	3.095	3.171	1096.9	46770.9

			Fit $N = 10$–30						Fit $N = 20$–30							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`p3--CCp2--p3`	hexa	0.1	3.352	3.823	3.223	3.253	3.241	3.243	3.279	3.896	3.083	3.150	3.114	3.185	1155.1	47889.2
`p3--CCp2--p3`	tetra unstr	0.0	3.198	3.777	2.984	2.957	2.981	3.007	3.463	3.810	3.182	2.980	3.187	3.076	2405.5	47513.2
`p3--CCp2--p3`	tetra unstr	0.1	3.451	3.863	3.162	2.957	3.160	3.055	3.776	3.921	3.433	2.980	3.439	3.123	2108.3	54156.0
`p3--CCp3--p1`	hexa	0.0	2.781	3.644	2.742	4.034	2.751	2.764	2.573	3.538	2.546	4.068	2.552	2.578	1355.6	44827.4
`p3--CCp3--p1`	hexa	0.1	2.737	3.653	2.705	4.034	2.713	2.735	2.523	3.546	2.502	4.068	2.508	2.535	1142.3	45947.2
`p3--CCp3--p1`	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2810.5	52501.8
`p3--CCp3--p1`	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2501.8	57652.3
`p3--CCp3--p2`	hexa	0.0	3.332	3.813	3.194	3.978	3.216	3.106	3.260	3.882	3.069	4.042	3.091	3.018	1421.8	45470.9
`p3--CCp3--p2`	hexa	0.1	3.349	3.823	3.214	3.978	3.236	3.135	3.275	3.896	3.087	4.042	3.109	3.036	1175.8	46593.2
`p3--CCp3--p2`	tetra unstr	0.0	3.198	3.777	2.985	3.943	2.981	3.010	3.463	3.810	3.182	3.960	3.187	3.210	2790.9	52920.7
`p3--CCp3--p2`	tetra unstr	0.1	3.452	3.863	3.163	3.943	3.161	3.182	3.775	3.921	3.433	3.960	3.438	3.453	2518.8	58052.7
`p3--CCp3--p3`	hexa	0.0	3.332	3.813	3.194	3.971	3.216	3.106	3.260	3.882	3.069	4.044	3.091	3.018	1510.6	46764.8
`p3--CCp3--p3`	hexa	0.1	3.349	3.823	3.214	3.971	3.236	3.136	3.275	3.896	3.087	4.044	3.109	3.037	1445.5	47890.3
`p3--CCp3--p3`	tetra unstr	0.0	3.197	3.777	2.984	3.944	2.980	3.056	3.462	3.810	3.182	3.962	3.186	3.240	2937.0	53766.4
`p3--CCp3--p3`	tetra unstr	0.1	3.451	3.863	3.162	3.944	3.160	3.224	3.775	3.921	3.433	3.962	3.438	3.483	2602.0	58874.8
`p3--GP--p1`	hexa	0.0	2.781	3.644	2.003	3.789	2.716	3.807	2.573	3.538	2.000	3.679	2.354	3.683	1183.8	77275.7
`p3--GP--p1`	hexa	0.1	2.738	3.653	2.003	3.796	2.706	3.814	2.523	3.546	2.000	3.687	2.345	3.692	1122.7	77341.2
`p3--GP--p1`	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	1767.9	45258.5
`p3--GP--p1`	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	1954.2	53638.4
`p3--GP--p2`	hexa	0.0	3.332	3.813	1.994	3.892	2.454	3.910	3.260	3.882	1.996	3.955	2.171	3.965	1548.8	77916.9
`p3--GP--p2`	hexa	0.1	3.350	3.823	1.994	3.896	2.447	3.914	3.275	3.896	1.996	3.961	2.167	3.971	1229.3	77991.0
`p3--GP--p2`	tetra unstr	0.0	3.198	3.777	1.927	3.839	2.032	3.841	3.463	3.810	1.961	3.829	1.977	3.835	1666.6	45510.1
`p3--GP--p2`	tetra unstr	0.1	3.452	3.863	1.927	3.878	2.028	3.879	3.775	3.921	1.961	3.893	1.974	3.898	1704.4	53890.6
`p3--GP--p3`	hexa	0.0	3.332	3.813	1.994	3.884	2.452	3.904	3.260	3.882	1.995	3.956	2.170	3.967	2130.9	79212.1
`p3--GP--p3`	hexa	0.1	3.349	3.823	1.993	3.889	2.446	3.908	3.275	3.896	1.995	3.962	2.167	3.973	1565.6	79288.0
`p3--GP--p3`	tetra unstr	0.0	3.197	3.777	1.923	3.838	2.021	3.840	3.462	3.810	1.958	3.826	1.974	3.833	1853.5	46006.8
`p3--GP--p3`	tetra unstr	0.1	3.451	3.863	1.923	3.879	2.017	3.880	3.775	3.921	1.958	3.893	1.971	3.900	1872.5	54389.9

11.3.2 `diagonalIion`

$\Delta t = 10^{-2}$

Table 10: 3D manufactured results for `diagonalIion`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–30						Fit $N = 20$–30							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.985	1.985	1.994	1.994	1.994	1.994	1.993	1.993	2.009	2.009	2.009	2.009	17.839	4049.9
`NO--na--NO`	hexa	0.1	1.436	1.436	1.297	1.297	1.306	1.306	1.411	1.411	1.214	1.214	1.223	1.223	31.231	6313.3
`NO--na--NO`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	38.256	5022.6
`NO--na--NO`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	152.32	12058.9
`NO--CCp1--p1`	hexa	0.0	1.985	1.985	1.998	2.166	1.995	2.246	1.992	1.992	2.010	2.208	2.009	2.189	33.377	4757.2
`NO--CCp1--p1`	hexa	0.1	1.436	1.436	1.287	2.167	1.306	2.230	1.411	1.411	1.205	2.208	1.224	2.159	60.609	6569.4
`NO--CCp1--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	131.10	6192.3
`NO--CCp1--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	195.78	12251.9
`NO--CCp1--p2`	hexa	0.0	1.985	1.985	1.999	2.098	1.994	2.090	1.992	1.992	2.011	2.130	2.009	2.062	40.973	6194.9
`NO--CCp1--p2`	hexa	0.1	1.436	1.436	1.283	2.098	1.307	2.087	1.411	1.411	1.201	2.130	1.224	2.056	76.370	7373.5
`NO--CCp1--p2`	tetra unstr	0.0	0.664	0.664	0.375	1.924	0.373	0.457	0.826	0.826	0.617	1.896	0.616	0.669	117.69	6648.8
`NO--CCp1--p2`	tetra unstr	0.1	0.677	0.677	0.398	1.928	0.396	0.483	0.823	0.823	0.629	1.903	0.628	0.683	197.93	12502.6
`NO--CCp1--p3`	hexa	0.0	1.985	1.985	1.999	2.094	1.994	2.088	1.992	1.992	2.011	2.127	2.009	2.061	54.810	9025.9
`NO--CCp1--p3`	hexa	0.1	1.436	1.436	1.283	2.095	1.307	2.085	1.411	1.411	1.201	2.127	1.224	2.054	103.38	10166.2
`NO--CCp1--p3`	tetra unstr	0.0	0.664	0.664	0.375	1.923	0.373	0.460	0.826	0.826	0.617	1.906	0.616	0.670	119.98	7667.8
`NO--CCp1--p3`	tetra unstr	0.1	0.677	0.677	0.398	1.926	0.396	0.485	0.823	0.823	0.629	1.912	0.628	0.684	203.25	12998.5
`NO--CCp2--p1`	hexa	0.0	1.977	1.977	2.009	3.171	1.987	2.618	1.989	1.989	2.014	3.091	2.005	2.390	52.204	5530.9
`NO--CCp2--p1`	hexa	0.1	1.447	1.447	1.257	3.172	1.318	2.180	1.426	1.426	1.176	3.092	1.239	1.626	76.123	6923.8
`NO--CCp2--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	202.22	9409.5
`NO--CCp2--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	256.48	13682.7
`NO--CCp2--p2`	hexa	0.0	1.969	1.969	2.010	3.247	1.979	2.753	1.985	1.985	2.013	3.146	2.001	2.494	59.701	6947.9
`NO--CCp2--p2`	hexa	0.1	1.457	1.457	1.248	3.248	1.333	2.320	1.441	1.441	1.170	3.147	1.257	1.718	100.29	8180.6
`NO--CCp2--p2`	tetra unstr	0.0	0.664	0.664	0.375	2.080	0.373	0.465	0.826	0.826	0.617	1.273	0.616	0.673	213.68	9939.8
`NO--CCp2--p2`	tetra unstr	0.1	0.677	0.677	0.398	2.128	0.396	0.486	0.823	0.823	0.629	1.333	0.628	0.683	257.37	14142.1
`NO--CCp2--p3`	hexa	0.0	1.969	1.969	2.010	3.252	1.979	2.758	1.985	1.985	2.013	3.149	2.001	2.501	76.927	9777.2
`NO--CCp2--p3`	hexa	0.1	1.457	1.457	1.248	3.253	1.333	2.328	1.441	1.441	1.170	3.150	1.257	1.726	109.29	10937.0
`NO--CCp2--p3`	tetra unstr	0.0	0.664	0.664	0.375	1.992	0.373	0.466	0.826	0.826	0.617	1.241	0.616	0.675	191.52	11027.8
`NO--CCp2--p3`	tetra unstr	0.1	0.677	0.677	0.398	2.039	0.396	0.487	0.823	0.823	0.629	1.295	0.628	0.686	264.47	15030.7
`NO--CCp3--p1`	hexa	0.0	1.980	1.980	2.001	4.033	1.989	2.062	1.991	1.991	2.011	4.072	2.007	2.017	96.054	7428.4
`NO--CCp3--p1`	hexa	0.1	1.452	1.452	1.244	4.015	1.323	1.276	1.429	1.429	1.170	4.028	1.241	1.186	122.78	8775.2
`NO--CCp3--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	407.78	18177.9
`NO--CCp3--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	422.09	22222.9
`NO--CCp3--p2`	hexa	0.0	1.976	1.976	2.000	3.980	1.985	2.107	1.990	1.990	2.011	4.053	2.005	2.022	104.58	8844.7
`NO--CCp3--p2`	hexa	0.1	1.469	1.469	1.233	3.966	1.343	1.305	1.447	1.447	1.163	4.019	1.261	1.187	157.18	9986.5
`NO--CCp3--p2`	tetra unstr	0.0	0.664	0.664	0.375	0.890	0.373	0.418	0.826	0.826	0.617	0.674	0.616	0.657	397.25	18709.8
`NO--CCp3--p2`	tetra unstr	0.1	0.677	0.677	0.398	0.930	0.396	0.441	0.823	0.823	0.629	0.689	0.628	0.670	427.69	22681.6
`NO--CCp3--p3`	hexa	0.0	1.976	1.976	2.000	3.973	1.985	2.108	1.990	1.990	2.011	4.054	2.005	2.022	125.21	11668.9
`NO--CCp3--p3`	hexa	0.1	1.469	1.469	1.233	3.959	1.343	1.305	1.447	1.447	1.163	4.021	1.261	1.186	159.54	12808.5

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
NO--CCp3--p3	tetra unstr	0.0	0.664	0.664	0.375	0.888	0.373	0.432	0.826	0.826	0.617	0.684	0.616	0.665	428.12	19793.8
NO--CCp3--p3	tetra unstr	0.1	0.677	0.677	0.398	0.926	0.396	0.454	0.823	0.823	0.629	0.697	0.628	0.677	432.58	23574.8
NO--GP--p1	hexa	0.0	1.984	1.984	1.992	2.072	1.992	2.077	1.992	1.992	1.996	2.038	1.997	2.041	59.598	4506.2
NO--GP--p1	hexa	0.1	1.459	1.459	1.992	2.073	1.993	2.079	1.429	1.429	1.996	2.038	1.996	2.041	71.966	6563.7
NO--GP--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	51.433	5256.8
NO--GP--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	147.95	12251.1
NO--GP--p2	hexa	0.0	1.983	1.983	1.984	2.014	1.985	2.018	1.992	1.992	1.992	2.007	1.993	2.009	158.85	5951.4
NO--GP--p2	hexa	0.1	1.482	1.482	1.984	2.015	1.985	2.018	1.447	1.447	1.992	2.007	1.993	2.010	155.23	7217.8
NO--GP--p2	tetra unstr	0.0	0.664	0.664	1.832	0.861	0.516	0.859	0.826	0.826	1.792	0.867	0.652	0.866	77.643	5716.7
NO--GP--p2	tetra unstr	0.1	0.677	0.677	1.844	0.883	0.546	0.881	0.823	0.823	1.812	0.884	0.668	0.883	175.86	12501.7
NO--GP--p3	hexa	0.0	1.983	1.983	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.993	2.008	399.48	8785.1
NO--GP--p3	hexa	0.1	1.482	1.482	1.984	2.011	1.985	2.015	1.447	1.447	1.992	2.006	1.992	2.008	325.33	9791.0
NO--GP--p3	tetra unstr	0.0	0.664	0.664	1.828	0.863	0.513	0.862	0.826	0.826	1.790	0.868	0.651	0.868	138.65	6716.1
NO--GP--p3	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.542	0.884	0.823	0.823	1.810	0.885	0.667	0.885	256.75	12995.9
p1--na--NO	hexa	0.0	1.967	2.095	1.927	1.927	1.930	1.930	1.919	2.125	1.838	1.838	1.845	1.845	76.120	7133.9
p1--na--NO	hexa	0.1	2.067	2.093	2.047	2.047	2.048	2.048	2.066	2.124	2.033	2.033	2.035	2.035	77.548	8693.9
p1--na--NO	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	261.81	10336.8
p1--na--NO	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	251.13	11875.1
p1--CCp1--p1	hexa	0.0	1.967	2.095	1.926	2.166	1.930	2.193	1.919	2.125	1.836	2.207	1.845	2.115	102.19	7284.7
p1--CCp1--p1	hexa	0.1	2.067	2.093	2.046	2.167	2.048	2.225	2.066	2.124	2.033	2.208	2.035	2.168	108.52	8861.8
p1--CCp1--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	326.80	11245.2
p1--CCp1--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	280.71	12509.6
p1--CCp1--p2	hexa	0.0	1.967	2.095	1.925	2.098	1.930	2.083	1.919	2.125	1.835	2.130	1.845	2.049	112.64	8199.6
p1--CCp1--p2	hexa	0.1	2.067	2.093	2.046	2.098	2.047	2.090	2.066	2.124	2.033	2.130	2.035	2.061	102.35	9508.0
p1--CCp1--p2	tetra unstr	0.0	1.474	1.866	1.139	1.973	1.142	1.493	1.771	1.928	1.419	1.984	1.428	1.590	360.40	11702.0
p1--CCp1--p2	tetra unstr	0.1	1.613	1.909	1.262	1.973	1.264	1.621	1.889	1.963	1.556	1.983	1.558	1.721	283.85	12931.0
p1--CCp1--p3	hexa	0.0	1.967	2.095	1.925	2.094	1.930	2.081	1.919	2.125	1.835	2.127	1.845	2.048	126.80	11006.7
p1--CCp1--p3	hexa	0.1	2.067	2.093	2.046	2.094	2.047	2.088	2.066	2.124	2.033	2.127	2.035	2.060	121.17	11450.9
p1--CCp1--p3	tetra unstr	0.0	1.474	1.866	1.139	1.965	1.142	1.505	1.771	1.928	1.419	1.982	1.428	1.597	366.81	12637.4
p1--CCp1--p3	tetra unstr	0.1	1.613	1.909	1.262	1.965	1.264	1.632	1.889	1.963	1.556	1.982	1.558	1.727	288.00	13736.6
p1--CCp2--p1	hexa	0.0	1.966	2.095	1.926	3.169	1.929	2.043	1.919	2.125	1.831	3.089	1.845	1.844	108.14	7668.9
p1--CCp2--p1	hexa	0.1	2.065	2.093	2.050	3.170	2.046	2.200	2.066	2.124	2.033	3.090	2.034	2.090	133.55	8866.9
p1--CCp2--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	383.95	14523.2
p1--CCp2--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	340.65	15430.1
p1--CCp2--p2	hexa	0.0	1.965	2.095	1.925	3.244	1.929	2.095	1.919	2.125	1.828	3.143	1.845	1.846	119.40	8935.3
p1--CCp2--p2	hexa	0.1	2.064	2.093	2.051	3.245	2.045	2.265	2.065	2.124	2.033	3.145	2.034	2.115	124.52	9632.8
p1--CCp2--p2	tetra unstr	0.0	1.475	1.866	1.139	2.917	1.142	1.199	1.771	1.928	1.418	2.913	1.428	1.465	406.79	14966.3
p1--CCp2--p2	tetra unstr	0.1	1.613	1.909	1.262	2.928	1.264	1.336	1.889	1.963	1.555	2.938	1.558	1.614	343.96	15843.4
p1--CCp2--p3	hexa	0.0	1.965	2.095	1.925	3.250	1.929	2.098	1.919	2.125	1.828	3.146	1.845	1.848	138.47	11754.3
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.051	3.251	2.045	2.269	2.065	2.124	2.033	3.148	2.034	2.117	146.88	12096.8
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.139	2.922	1.142	1.211	1.771	1.928	1.418	2.906	1.428	1.476	419.30	15835.9
p1--CCp2--p3	tetra unstr	0.1	1.613	1.909	1.262	2.938	1.264	1.348	1.889	1.963	1.555	2.938	1.558	1.624	349.25	16789.7
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.922	4.067	1.929	1.877	1.919	2.125	1.830	4.138	1.845	1.729	154.68	9571.5
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.046	4.057	2.047	2.042	2.066	2.124	2.032	4.119	2.034	2.007	181.97	9945.3
p1--CCp3--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	587.62	23052.2
p1--CCp3--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	503.46	24024.5
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.920	4.009	1.930	1.853	1.920	2.125	1.827	4.112	1.846	1.682	204.40	10809.0
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.045	4.000	2.047	2.039	2.066	2.124	2.031	4.092	2.034	1.993	200.25	11136.9
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.139	2.936	1.142	1.202	1.771	1.928	1.418	2.110	1.428	1.472	609.85	23479.2
p1--CCp3--p2	tetra unstr	0.1	1.613	1.909	1.262	3.163	1.264	1.332	1.889	1.963	1.555	2.437	1.558	1.614	508.43	24409.8
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.920	4.003	1.930	1.852	1.920	2.125	1.827	4.114	1.846	1.680	228.53	13640.9
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.045	3.992	2.047	2.039	2.066	2.124	2.031	4.094	2.034	1.993	218.65	13990.9
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.139	2.837	1.142	1.240	1.771	1.928	1.418	2.066	1.428	1.496	657.70	24392.8
p1--CCp3--p3	tetra unstr	0.1	1.613	1.909	1.262	3.084	1.264	1.375	1.889	1.963	1.555	2.395	1.558	1.644	515.30	25375.1
p1--GP--p1	hexa	0.0	1.967	2.095	1.992	2.157	1.999	2.164	1.919	2.125	1.997	2.201	2.001	2.201	104.41	7341.0
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.002	2.162	2.066	2.124	1.997	2.199	2.005	2.199	109.30	8899.4
p1--GP--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	247.77	10396.4
p1--GP--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	222.16	11931.7
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.920	2.125	1.993	2.129	1.997	2.128	190.37	8084.3
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.093	1.991	2.096	2.066	2.124	1.993	2.128	1.998	2.126	187.99	9545.7
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.771	1.928	1.953	1.877	1.777	1.880	288.75	10853.6
p1--GP--p2	tetra unstr	0.1	1.613	1.909	1.920	1.897	1.790	1.900	1.889	1.963	1.955	1.921	1.848	1.922	250.54	12308.8
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.920	2.125	1.993	2.126	1.996	2.125	374.96	10772.3
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.066	2.124	1.993	2.124	1.998	2.123	363.94	11158.1
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.874	1.771	1.928	1.951	1.889	1.775	1.892	341.86	11836.5
p1--GP--p3	tetra unstr	0.1	1.613	1.909	1.916	1.900	1.786	1.903	1.889	1.963	1.953	1.927	1.846	1.929	311.35	13265.5
p2--na--NO	hexa	0.0	1.947	3.025	1.944	1.944	1.948	1.948	2.001	2.766	2.003	2.003	2.005	2.005	159.49	25851.3
p2--na--NO	hexa	0.1	1.946	3.053	1.943	1.943	1.947	1.947	2.000	2.807	2.002	2.002	2.005	2.005	153.99	23778.1
p2--na--NO	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	410.45	18237.1
p2--na--NO	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	380.78	21603.7
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.995	2.168	2.003	2.183	2.035	2.884	2.035	2.209	2.040	2.147	169.04	13059.4
p2--CCp1--p1	hexa	0.1	2.008	3.133	2.000	2.167	2.008	2.193	2.037	2.925	2.037	2.209	2.042	2.153	209.15	15810.5
p2--CCp1--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	623.79	19036.2
p2--CCp1--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	448.44	21893.6
p2--CCp1--p2	hexa	0.0	2.040	3.142	2.028	2.098	2.039	2.087	2.059	2.936	2.057	2.131	2.064	2.063	172.60	13704.2
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.037	2.098	2.049	2.088	2.063	2.975	2.062	2.131	2.069	2.063	185.63	16455.6
p2--CCp1--p2	tetra unstr	0.0	2.103	2.703	1.863	1.971	1.864	1.960	2.383	2.718	2.113	1.982	2.121	1.974	614.57	19514.9
p2--CCp1--p2	tetra unstr	0.1	2.347	2.823	2.013	1.971	2.016	1.970	2.727	2.903	2.362	1.982	2.372	1.985	575.02	22293.9
p2--CCp1--p3	hexa	0.0	2.040	3.142	2.028	2.095	2.039	2.085	2.059	2.936	2.057	2.127	2.064	2.061	204.59	15345.8
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.037	2.095	2.049	2.086	2.063	2.975	2.062	2.127	2.069	2.061	214.83	17754.1
p2--CCp1--p3	tetra unstr	0.0	2.103	2.703	1.863	1.963	1.864	1.958	2.383	2.718	2.112	1.981	2.121	1.972	616.65	20468.9

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p2--CCp1--p3	tetra unstr	0.1	2.347	2.823	2.013	1.963	2.016	1.968	2.727	2.903	2.362	1.981	2.372	1.983	557.20	23166.7
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.999	3.170	2.004	2.077	2.035	2.884	2.038	3.091	2.040	2.074	196.47	13056.7
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.005	3.171	2.009	2.096	2.037	2.925	2.040	3.091	2.043	2.083	175.37	15810.1
p2--CCp2--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	632.62	22303.5
p2--CCp2--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	636.46	24373.4
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.036	3.246	2.040	2.173	2.059	2.936	2.063	3.146	2.065	2.129	209.25	13797.8
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.046	3.246	2.050	2.209	2.064	2.975	2.068	3.146	2.070	2.147	206.11	16457.2
p2--CCp2--p2	tetra unstr	0.0	2.103	2.704	1.863	2.944	1.864	1.932	2.384	2.719	2.114	2.972	2.121	2.120	659.62	22775.0
p2--CCp2--p2	tetra unstr	0.1	2.349	2.824	2.014	2.945	2.016	2.103	2.732	2.904	2.367	2.972	2.373	2.405	574.74	24845.4
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.036	3.251	2.040	2.176	2.059	2.936	2.063	3.149	2.065	2.130	232.99	15899.4
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.046	3.251	2.050	2.212	2.064	2.975	2.068	3.149	2.070	2.149	215.34	17754.7
p2--CCp2--p3	tetra unstr	0.0	2.103	2.704	1.863	2.958	1.864	1.943	2.384	2.719	2.114	2.980	2.121	2.133	670.04	23686.6
p2--CCp2--p3	tetra unstr	0.1	2.349	2.824	2.014	2.959	2.016	2.110	2.732	2.904	2.367	2.981	2.373	2.411	579.70	25797.4
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.999	4.087	2.004	2.022	2.035	2.884	2.038	4.142	2.040	2.059	217.71	13837.2
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.004	4.082	2.009	2.028	2.037	2.925	2.040	4.137	2.043	2.062	220.37	15812.2
p2--CCp3--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	810.83	30956.5
p2--CCp3--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	778.61	32393.4
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.035	4.022	2.040	2.074	2.059	2.936	2.063	4.109	2.065	2.104	243.82	14932.0
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.045	4.017	2.050	2.087	2.064	2.975	2.068	4.103	2.070	2.111	245.01	16536.1
p2--CCp3--p2	tetra unstr	0.0	2.103	2.704	1.863	3.893	1.864	1.896	2.384	2.719	2.114	3.841	2.121	2.127	812.04	31354.9
p2--CCp3--p2	tetra unstr	0.1	2.349	2.824	2.014	3.911	2.016	2.046	2.732	2.904	2.367	3.890	2.373	2.389	790.86	32882.0
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.035	4.014	2.040	2.075	2.059	2.936	2.063	4.110	2.065	2.105	256.46	17495.5
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.045	4.010	2.050	2.087	2.064	2.975	2.068	4.104	2.070	2.112	259.96	18178.2
p2--CCp3--p3	tetra unstr	0.0	2.103	2.704	1.863	3.888	1.864	1.929	2.384	2.719	2.114	3.831	2.121	2.146	944.89	32180.7
p2--CCp3--p3	tetra unstr	0.1	2.349	2.824	2.014	3.907	2.016	2.072	2.732	2.904	2.367	3.883	2.373	2.403	819.01	33736.5
p2--GP--p1	hexa	0.0	2.003	3.107	2.011	3.133	2.845	3.122	2.035	2.884	2.004	3.005	2.449	2.999	209.03	26058.5
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.888	3.131	2.037	2.925	2.004	3.021	2.476	3.015	171.87	23988.3
p2--GP--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	419.78	18303.6
p2--GP--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	402.79	21700.4
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.898	3.215	2.059	2.936	2.001	3.092	2.432	3.083	291.55	26710.8
p2--GP--p2	hexa	0.1	2.050	3.166	2.004	3.236	2.920	3.222	2.064	2.975	2.001	3.105	2.440	3.095	265.41	24629.7
p2--GP--p2	tetra unstr	0.0	2.103	2.704	1.928	2.794	2.100	2.796	2.384	2.719	1.960	2.779	2.102	2.784	421.04	18693.3
p2--GP--p2	tetra unstr	0.1	2.349	2.824	1.928	2.831	2.146	2.833	2.732	2.904	1.960	2.861	2.166	2.866	432.63	21997.2
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.897	3.220	2.059	2.936	2.000	3.095	2.432	3.085	524.46	28002.3
p2--GP--p3	hexa	0.1	2.050	3.166	2.003	3.242	2.920	3.227	2.064	2.975	2.000	3.108	2.439	3.097	385.37	25931.2
p2--GP--p3	tetra unstr	0.0	2.103	2.704	1.924	2.802	2.082	2.802	2.384	2.719	1.958	2.769	2.088	2.773	463.12	19698.3
p2--GP--p3	tetra unstr	0.1	2.349	2.824	1.924	2.847	2.128	2.847	2.732	2.904	1.958	2.864	2.152	2.868	486.74	22652.1
p3--na--NO	hexa	0.0	2.313	3.204	2.308	2.308	2.312	2.312	2.193	2.830	2.197	2.197	2.200	2.200	784.33	77066.1
p3--na--NO	hexa	0.1	2.278	3.207	2.274	2.274	2.278	2.278	2.169	2.826	2.174	2.174	2.176	2.176	737.90	76981.0
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1208.8	45193.0
p3--na--NO	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1221.0	53420.2
p3--CCp1--p1	hexa	0.0	2.761	3.638	2.698	2.166	2.745	2.264	2.550	3.526	2.508	2.207	2.552	2.199	959.09	44827.9
p3--CCp1--p1	hexa	0.1	2.717	3.647	2.659	2.166	2.706	2.261	2.501	3.534	2.465	2.207	2.509	2.198	861.81	45950.9
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1763.5	45814.3
p3--CCp1--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1488.6	53115.7
p3--CCp1--p2	hexa	0.0	3.129	3.794	3.046	2.097	3.078	2.094	2.914	3.838	2.857	2.130	2.869	2.063	870.58	45473.1
p3--CCp1--p2	hexa	0.1	3.103	3.804	3.033	2.097	3.064	2.093	2.862	3.851	2.828	2.130	2.835	2.063	929.39	46595.2
p3--CCp1--p2	tetra unstr	0.0	3.009	3.711	2.897	1.970	2.893	1.968	3.092	3.661	3.022	1.982	3.021	1.961	1680.0	46130.8
p3--CCp1--p2	tetra unstr	0.1	3.120	3.786	3.019	1.970	3.015	1.968	3.099	3.741	3.149	1.982	3.147	1.960	1559.0	53372.8
p3--CCp1--p3	hexa	0.0	3.129	3.794	3.046	2.094	3.078	2.092	2.914	3.838	2.857	2.127	2.869	2.062	856.84	46769.8
p3--CCp1--p3	hexa	0.1	3.103	3.804	3.033	2.094	3.064	2.091	2.862	3.851	2.828	2.127	2.835	2.062	942.46	47890.2
p3--CCp1--p3	tetra unstr	0.0	3.009	3.711	2.897	1.963	2.892	1.966	3.091	3.661	3.021	1.981	3.021	1.960	1636.3	46753.4
p3--CCp1--p3	tetra unstr	0.1	3.120	3.786	3.018	1.963	3.015	1.966	3.099	3.741	3.149	1.981	3.147	1.960	1566.4	53951.5
p3--CCp2--p1	hexa	0.0	2.781	3.643	2.755	3.172	2.763	2.955	2.572	3.538	2.566	3.092	2.573	2.804	852.69	44826.7
p3--CCp2--p1	hexa	0.1	2.738	3.653	2.719	3.172	2.726	2.951	2.523	3.546	2.524	3.092	2.530	2.794	884.41	45948.7
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1617.2	47060.5
p3--CCp2--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1545.2	53594.1
p3--CCp2--p2	hexa	0.0	3.332	3.812	3.211	3.247	3.230	3.235	3.260	3.882	3.083	3.147	3.113	3.176	877.41	45474.7
p3--CCp2--p2	hexa	0.1	3.350	3.823	3.234	3.247	3.253	3.247	3.275	3.896	3.103	3.147	3.135	3.190	926.79	46596.9
p3--CCp2--p2	tetra unstr	0.0	3.196	3.775	2.986	2.942	2.983	2.997	3.461	3.809	3.184	2.970	3.189	3.072	1693.7	47445.3
p3--CCp2--p2	tetra unstr	0.1	3.450	3.862	3.163	2.942	3.162	3.045	3.777	3.920	3.434	2.970	3.440	3.120	1589.7	53954.3
p3--CCp2--p3	hexa	0.0	3.332	3.812	3.211	3.253	3.230	3.234	3.260	3.882	3.083	3.150	3.113	3.175	923.11	46766.8
p3--CCp2--p3	hexa	0.1	3.350	3.823	3.234	3.253	3.253	3.246	3.275	3.896	3.103	3.150	3.135	3.189	974.77	47890.8
p3--CCp2--p3	tetra unstr	0.0	3.195	3.776	2.985	2.957	2.983	3.007	3.461	3.809	3.184	2.980	3.188	3.076	1477.0	48215.3
p3--CCp2--p3	tetra unstr	0.1	3.450	3.862	3.163	2.957	3.161	3.054	3.777	3.921	3.434	2.980	3.440	3.122	1530.7	54668.5
p3--CCp3--p1	hexa	0.0	2.780	3.644	2.753	4.033	2.762	2.775	2.572	3.538	2.566	4.067	2.573	2.597	886.39	44826.9
p3--CCp3--p1	hexa	0.1	2.737	3.653	2.716	4.033	2.725	2.746	2.522	3.545	2.523	4.068	2.530	2.555	931.63	45948.9
p3--CCp3--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1735.3	53363.6
p3--CCp3--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1703.5	58670.7
p3--CCp3--p2	hexa	0.0	3.330	3.812	3.205	3.978	3.228	3.112	3.256	3.881	3.086	4.042	3.110	3.026	875.81	45471.7
p3--CCp3--p2	hexa	0.1	3.347	3.823	3.226	3.978	3.250	3.143	3.271	3.895	3.106	4.042	3.131	3.046	1144.0	46591.2
p3--CCp3--p2	tetra unstr	0.0	3.196	3.775	2.987	3.943	2.983	3.011	3.461	3.808	3.184	3.959	3.188	3.209	1657.7	53806.1
p3--CCp3--p2	tetra unstr	0.1	3.450	3.862	3.164	3.942	3.162	3.181	3.777	3.920	3.434	3.959	3.440	3.451	1699.5	59049.0
p3--CCp3--p3	hexa	0.0	3.330	3.812	3.205	3.971	3.228	3.112	3.256	3.881	3.086	4.044	3.110	3.027	945.74	46766.9
p3--CCp3--p3	hexa	0.1	3.347	3.823	3.226	3.971	3.250	3.143	3.271	3.895	3.106	4.044	3.131	3.046	1080.8	47885.5
p3--CCp3--p3	tetra unstr	0.0	3.195	3.776	2.986	3.944	2.982	3.056	3.460	3.809	3.184	3.961	3.188	3.240	1471.9	54513.0
p3--CCp3--p3	tetra unstr	0.1	3.449	3.862	3.163	3.944	3.161	3.224	3.777	3.920	3.434	3.961	3.439	3.481	1691.2	59970.7
p3--GP--p1	hexa	0.0	2.780	3.644	2.003	3.792	2.717	3.810	2.572	3.538	2.000	3.687	2.355	3.691	1136.4	77129.5
p3--GP--p1	hexa	0.1	2.737	3.653	2.003	3.799	2.707	3.817	2.522	3.546	2.000	3.695	2.347	3.700	826.77	77349.6
p3--GP--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1530.2	45303.3
p3--GP--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1458.6	53640.9
p3--GP--p2	hexa	0.0	3.331	3.812	1.994	3.893	2.454	3.912	3.257	3.881	1.996	3.958	2.171	3.969	1024.7	77926.4

			Fit $N = 10$–30						Fit $N = 20$–30							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`p3--GP--p2`	hexa	0.1	3.348	3.823	1.994	3.897	2.447	3.916	3.272	3.896	1.996	3.964	2.167	3.975	846.01	77996.9
`p3--GP--p2`	tetra unstr	0.0	3.196	3.775	1.927	3.844	2.032	3.845	3.461	3.808	1.961	3.840	1.977	3.846	1658.5	45560.0
`p3--GP--p2`	tetra unstr	0.1	3.450	3.862	1.927	3.883	2.028	3.884	3.777	3.920	1.961	3.904	1.974	3.909	1726.4	53889.0
`p3--GP--p3`	hexa	0.0	3.331	3.812	1.994	3.886	2.452	3.905	3.257	3.881	1.995	3.960	2.170	3.970	1232.6	79222.2
`p3--GP--p3`	hexa	0.1	3.348	3.823	1.993	3.890	2.446	3.909	3.272	3.896	1.995	3.966	2.167	3.976	1248.1	79294.2
`p3--GP--p3`	tetra unstr	0.0	3.195	3.776	1.923	3.842	2.021	3.844	3.460	3.809	1.958	3.837	1.974	3.844	1679.4	46150.6
`p3--GP--p3`	tetra unstr	0.1	3.450	3.862	1.923	3.884	2.017	3.885	3.777	3.921	1.958	3.904	1.971	3.911	1530.7	54383.0

$\Delta t = 10^{-3}$

Table 11: 3D manufactured results for `diagonalIion`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–30						Fit $N = 20$–30							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.985	1.985	1.985	1.985	1.985	1.985	1.993	1.993	1.993	1.993	1.993	1.993	122.72	4051.0
`NO--na--NO`	hexa	0.1	1.438	1.438	1.293	1.293	1.302	1.302	1.413	1.413	1.211	1.211	1.220	1.220	157.38	6313.0
`NO--na--NO`	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	201.73	5025.8
`NO--na--NO`	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	592.24	12057.2
`NO--CCp1--p1`	hexa	0.0	1.986	1.986	1.989	2.167	1.986	2.246	1.993	1.993	1.995	2.208	1.993	2.188	161.77	4760.6
`NO--CCp1--p1`	hexa	0.1	1.438	1.438	1.298	2.167	1.302	2.230	1.413	1.413	1.212	2.208	1.220	2.158	290.23	6574.4
`NO--CCp1--p1`	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	386.53	6193.2
`NO--CCp1--p1`	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	669.90	12252.5
`NO--CCp1--p2`	hexa	0.0	1.986	1.986	1.991	2.098	1.986	2.090	1.993	1.993	1.995	2.130	1.993	2.062	223.61	6202.6
`NO--CCp1--p2`	hexa	0.1	1.438	1.438	1.300	2.098	1.303	2.086	1.413	1.413	1.212	2.130	1.220	2.055	302.02	7373.6
`NO--CCp1--p2`	tetra unstr	0.0	0.663	0.663	0.377	1.924	0.375	0.459	0.829	0.829	0.617	1.896	0.617	0.670	403.89	6652.1
`NO--CCp1--p2`	tetra unstr	0.1	0.677	0.677	0.400	1.928	0.398	0.484	0.828	0.828	0.629	1.903	0.629	0.684	710.39	12503.9
`NO--CCp1--p3`	hexa	0.0	1.986	1.986	1.991	2.094	1.986	2.088	1.993	1.993	1.995	2.127	1.993	2.060	305.02	9028.5
`NO--CCp1--p3`	hexa	0.1	1.438	1.438	1.300	2.095	1.303	2.085	1.413	1.413	1.212	2.127	1.220	2.054	442.70	10162.7
`NO--CCp1--p3`	tetra unstr	0.0	0.663	0.663	0.377	1.922	0.375	0.462	0.829	0.829	0.617	1.906	0.617	0.671	474.23	7668.9
`NO--CCp1--p3`	tetra unstr	0.1	0.677	0.677	0.400	1.926	0.398	0.487	0.828	0.828	0.629	1.912	0.629	0.685	749.34	13003.3
`NO--CCp2--p1`	hexa	0.0	1.977	1.977	2.002	3.171	1.978	2.610	1.989	1.989	1.999	3.091	1.989	2.376	231.17	5533.9
`NO--CCp2--p1`	hexa	0.1	1.447	1.447	1.258	3.172	1.313	2.174	1.428	1.428	1.175	3.092	1.236	1.620	346.79	6924.0
`NO--CCp2--p1`	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	772.64	9414.5
`NO--CCp2--p1`	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	960.39	13641.3
`NO--CCp2--p2`	hexa	0.0	1.969	1.969	2.004	3.247	1.970	2.745	1.985	1.985	1.998	3.146	1.985	2.481	293.43	6950.3
`NO--CCp2--p2`	hexa	0.1	1.457	1.457	1.251	3.248	1.326	2.314	1.443	1.443	1.170	3.147	1.253	1.712	405.66	8181.0
`NO--CCp2--p2`	tetra unstr	0.0	0.663	0.663	0.377	2.079	0.375	0.467	0.829	0.829	0.617	1.272	0.617	0.673	833.91	9946.2
`NO--CCp2--p2`	tetra unstr	0.1	0.677	0.677	0.400	2.127	0.398	0.488	0.828	0.828	0.629	1.333	0.629	0.684	1015.5	14096.3
`NO--CCp2--p3`	hexa	0.0	1.969	1.969	2.004	3.252	1.970	2.751	1.985	1.985	1.998	3.149	1.985	2.488	427.37	9779.4
`NO--CCp2--p3`	hexa	0.1	1.457	1.457	1.251	3.253	1.326	2.323	1.443	1.443	1.170	3.150	1.253	1.720	576.86	10941.5
`NO--CCp2--p3`	tetra unstr	0.0	0.663	0.663	0.378	1.991	0.375	0.469	0.829	0.829	0.617	1.240	0.617	0.676	814.59	11030.6
`NO--CCp2--p3`	tetra unstr	0.1	0.677	0.677	0.400	2.038	0.398	0.489	0.828	0.828	0.629	1.295	0.629	0.686	1051.9	14990.0
`NO--CCp3--p1`	hexa	0.0	1.980	1.980	1.993	4.033	1.980	2.052	1.992	1.992	1.995	4.072	1.992	2.002	427.29	7432.8
`NO--CCp3--p1`	hexa	0.1	1.453	1.453	1.241	4.015	1.318	1.271	1.431	1.431	1.167	4.028	1.238	1.184	537.11	8767.6
`NO--CCp3--p1`	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	1766.1	18179.3
`NO--CCp3--p1`	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	1720.8	22209.3
`NO--CCp3--p2`	hexa	0.0	1.975	1.975	1.992	3.981	1.975	2.098	1.990	1.990	1.995	4.053	1.990	2.007	508.10	8849.4
`NO--CCp3--p2`	hexa	0.1	1.469	1.469	1.230	3.966	1.336	1.300	1.449	1.449	1.160	4.019	1.258	1.185	645.50	9995.3
`NO--CCp3--p2`	tetra unstr	0.0	0.663	0.663	0.378	0.891	0.375	0.420	0.829	0.829	0.617	0.675	0.617	0.658	1800.1	18709.2
`NO--CCp3--p2`	tetra unstr	0.1	0.677	0.677	0.400	0.930	0.398	0.443	0.828	0.828	0.629	0.689	0.629	0.671	1826.2	22660.9
`NO--CCp3--p3`	hexa	0.0	1.975	1.975	1.992	3.973	1.975	2.099	1.991	1.991	1.995	4.055	1.990	2.007	682.75	11674.3
`NO--CCp3--p3`	hexa	0.1	1.469	1.469	1.230	3.959	1.336	1.300	1.449	1.449	1.160	4.021	1.258	1.184	835.59	12809.8
`NO--CCp3--p3`	tetra unstr	0.0	0.663	0.663	0.378	0.888	0.375	0.434	0.829	0.829	0.617	0.684	0.617	0.666	1936.3	19795.1
`NO--CCp3--p3`	tetra unstr	0.1	0.677	0.677	0.400	0.927	0.398	0.456	0.828	0.828	0.629	0.698	0.629	0.678	1866.5	23532.0
`NO--GP--p1`	hexa	0.0	1.985	1.985	1.992	2.071	1.992	2.077	1.993	1.993	1.996	2.038	1.996	2.041	299.85	4511.0
`NO--GP--p1`	hexa	0.1	1.461	1.461	1.992	2.073	1.992	2.079	1.431	1.431	1.996	2.038	1.995	2.041	345.03	6562.4
`NO--GP--p1`	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	254.13	5262.5
`NO--GP--p1`	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	586.17	12250.4
`NO--GP--p2`	hexa	0.0	1.984	1.984	1.984	2.014	1.984	2.018	1.992	1.992	1.992	2.007	1.993	2.009	979.42	5953.0
`NO--GP--p2`	hexa	0.1	1.483	1.483	1.984	2.014	1.985	2.018	1.450	1.450	1.992	2.007	1.992	2.009	849.58	7217.6
`NO--GP--p2`	tetra unstr	0.0	0.663	0.663	1.832	0.862	0.518	0.860	0.829	0.829	1.792	0.867	0.653	0.866	441.41	5718.4
`NO--GP--p2`	tetra unstr	0.1	0.677	0.677	1.843	0.883	0.547	0.882	0.828	0.828	1.812	0.884	0.668	0.883	880.58	12497.6
`NO--GP--p3`	hexa	0.0	1.984	1.984	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.992	2.008	1784.3	8779.5
`NO--GP--p3`	hexa	0.1	1.483	1.483	1.984	2.011	1.984	2.015	1.450	1.450	1.992	2.005	1.992	2.008	1811.7	9791.1
`NO--GP--p3`	tetra unstr	0.0	0.663	0.663	1.828	0.864	0.514	0.862	0.829	0.829	1.790	0.869	0.652	0.868	946.10	6720.5
`NO--GP--p3`	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.543	0.884	0.828	0.828	1.810	0.886	0.667	0.885	1222.1	12996.5
`p1--na--NO`	hexa	0.0	1.967	2.095	1.924	1.924	1.927	1.927	1.920	2.125	1.833	1.833	1.840	1.840	206.83	7134.2
`p1--na--NO`	hexa	0.1	2.067	2.093	2.044	2.044	2.044	2.044	2.067	2.124	2.028	2.028	2.029	2.029	219.86	8694.8
`p1--na--NO`	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	655.41	10344.2
`p1--na--NO`	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	654.84	11878.0
`p1--CCp1--p1`	hexa	0.0	1.967	2.095	1.924	2.166	1.926	2.192	1.920	2.125	1.833	2.207	1.840	2.114	252.70	7284.9
`p1--CCp1--p1`	hexa	0.1	2.067	2.093	2.043	2.167	2.044	2.224	2.067	2.124	2.028	2.208	2.029	2.167	281.56	8859.1
`p1--CCp1--p1`	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	847.66	11240.4
`p1--CCp1--p1`	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	760.10	12504.4
`p1--CCp1--p2`	hexa	0.0	1.967	2.095	1.924	2.098	1.926	2.083	1.920	2.125	1.833	2.130	1.840	2.049	296.92	8201.4

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p1--CCp1--p2	hexa	0.1	2.067	2.093	2.043	2.098	2.044	2.090	2.067	2.124	2.028	2.130	2.029	2.061	325.52	9504.8
p1--CCp1--p2	tetra unstr	0.0	1.475	1.866	1.142	1.973	1.144	1.493	1.768	1.927	1.422	1.984	1.432	1.592	880.24	11704.0
p1--CCp1--p2	tetra unstr	0.1	1.612	1.909	1.265	1.973	1.267	1.622	1.884	1.962	1.558	1.983	1.561	1.723	772.60	12931.7
p1--CCp1--p3	hexa	0.0	1.967	2.095	1.924	2.094	1.926	2.081	1.920	2.125	1.833	2.127	1.840	2.048	399.00	11002.1
p1--CCp1--p3	hexa	0.1	2.067	2.093	2.043	2.094	2.044	2.088	2.067	2.124	2.028	2.127	2.029	2.059	398.19	11445.5
p1--CCp1--p3	tetra unstr	0.0	1.475	1.866	1.142	1.965	1.144	1.506	1.768	1.927	1.422	1.982	1.432	1.599	908.84	12636.4
p1--CCp1--p3	tetra unstr	0.1	1.612	1.909	1.265	1.965	1.267	1.633	1.884	1.962	1.558	1.982	1.561	1.729	805.41	13737.5
p1--CCp2--p1	hexa	0.0	1.966	2.095	1.924	3.169	1.926	2.040	1.920	2.125	1.827	3.089	1.840	1.840	312.61	7660.9
p1--CCp2--p1	hexa	0.1	2.065	2.093	2.048	3.170	2.043	2.196	2.066	2.124	2.028	3.090	2.028	2.085	386.88	8866.5
p1--CCp2--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	1202.6	14517.9
p1--CCp2--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	1051.4	15428.2
p1--CCp2--p2	hexa	0.0	1.966	2.095	1.924	3.244	1.925	2.092	1.920	2.125	1.824	3.143	1.840	1.841	373.41	8935.8
p1--CCp2--p2	hexa	0.1	2.064	2.093	2.050	3.245	2.042	2.261	2.065	2.124	2.028	3.145	2.028	2.109	392.88	9631.5
p1--CCp2--p2	tetra unstr	0.0	1.475	1.866	1.141	2.917	1.145	1.201	1.768	1.927	1.422	2.912	1.432	1.469	1212.9	14965.3
p1--CCp2--p2	tetra unstr	0.1	1.612	1.909	1.264	2.928	1.267	1.339	1.884	1.962	1.558	2.938	1.561	1.617	1068.2	15829.5
p1--CCp2--p3	hexa	0.0	1.966	2.095	1.924	3.250	1.925	2.095	1.920	2.125	1.824	3.146	1.840	1.843	520.09	11754.7
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.050	3.251	2.042	2.265	2.065	2.124	2.028	3.148	2.028	2.111	534.05	12101.7
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.141	2.922	1.145	1.213	1.768	1.927	1.422	2.906	1.432	1.479	1211.2	15837.4
p1--CCp2--p3	tetra unstr	0.1	1.612	1.909	1.264	2.938	1.267	1.350	1.884	1.962	1.558	2.938	1.561	1.627	1100.2	16782.6
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.919	4.067	1.926	1.874	1.920	2.125	1.826	4.139	1.840	1.725	511.31	9564.9
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.043	4.057	2.043	2.038	2.067	2.124	2.026	4.119	2.029	2.002	575.29	9935.4
p1--CCp3--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	2161.6	23052.8
p1--CCp3--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	1778.9	24017.3
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.917	4.010	1.926	1.850	1.921	2.125	1.823	4.112	1.841	1.677	723.19	10810.6
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.042	4.000	2.043	2.035	2.067	2.124	2.025	4.092	2.029	1.988	668.21	11136.3
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.141	2.935	1.145	1.205	1.768	1.927	1.422	2.111	1.432	1.475	2163.9	23477.6
p1--CCp3--p2	tetra unstr	0.1	1.612	1.909	1.264	3.163	1.267	1.335	1.884	1.962	1.558	2.438	1.561	1.616	1804.9	24406.4
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.917	4.003	1.926	1.849	1.921	2.125	1.823	4.114	1.841	1.676	932.98	13646.1
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.042	3.992	2.043	2.035	2.067	2.124	2.025	4.094	2.029	1.988	879.35	13988.5
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.141	2.836	1.145	1.243	1.768	1.927	1.422	2.067	1.432	1.500	2330.3	24391.1
p1--CCp3--p3	tetra unstr	0.1	1.612	1.909	1.264	3.084	1.267	1.378	1.884	1.962	1.558	2.395	1.561	1.647	1858.2	25370.1
p1--GP--p1	hexa	0.0	1.968	2.095	1.992	2.157	1.999	2.164	1.920	2.125	1.997	2.201	2.001	2.200	357.41	7338.5
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.001	2.162	2.067	2.124	1.997	2.199	2.004	2.199	375.47	8901.2
p1--GP--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	610.55	10400.1
p1--GP--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	644.92	11931.5
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.921	2.125	1.993	2.129	1.996	2.128	863.68	8084.6
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.092	1.991	2.096	2.067	2.124	1.993	2.128	1.998	2.126	852.43	9548.3
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.768	1.927	1.953	1.877	1.777	1.880	810.89	10858.2
p1--GP--p2	tetra unstr	0.1	1.612	1.909	1.920	1.897	1.790	1.900	1.884	1.962	1.955	1.921	1.849	1.923	831.86	12353.3
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.921	2.125	1.993	2.126	1.996	2.124	1939.9	10771.9
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.067	2.124	1.993	2.124	1.998	2.123	1752.9	11162.2
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.873	1.768	1.927	1.951	1.889	1.775	1.892	1182.8	11857.3
p1--GP--p3	tetra unstr	0.1	1.612	1.909	1.916	1.900	1.786	1.903	1.884	1.962	1.953	1.927	1.847	1.929	1142.1	13225.7
p2--na--NO	hexa	0.0	1.947	3.025	1.942	1.942	1.946	1.946	2.001	2.766	2.000	2.000	2.003	2.003	345.81	25850.3
p2--na--NO	hexa	0.1	1.946	3.053	1.941	1.941	1.946	1.946	2.000	2.807	1.999	1.999	2.002	2.002	333.19	23777.0
p2--na--NO	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1218.3	18239.1
p2--na--NO	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1233.7	21605.4
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.993	2.168	2.001	2.183	2.035	2.884	2.031	2.209	2.036	2.145	445.26	13056.2
p2--CCp1--p1	hexa	0.1	2.008	3.133	1.997	2.167	2.006	2.192	2.037	2.925	2.033	2.209	2.038	2.152	454.48	15812.8
p2--CCp1--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1728.9	19037.9
p2--CCp1--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1458.2	21897.2
p2--CCp1--p2	hexa	0.0	2.040	3.141	2.024	2.098	2.036	2.087	2.059	2.936	2.052	2.131	2.060	2.062	516.68	13704.3
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.033	2.098	2.046	2.088	2.064	2.974	2.056	2.131	2.065	2.062	505.95	16457.1
p2--CCp1--p2	tetra unstr	0.0	2.105	2.705	1.863	1.971	1.864	1.960	2.382	2.719	2.113	1.982	2.122	1.974	1831.4	19519.9
p2--CCp1--p2	tetra unstr	0.1	2.348	2.824	2.014	1.971	2.018	1.970	2.721	2.901	2.364	1.982	2.374	1.986	2025.1	22291.9
p2--CCp1--p3	hexa	0.0	2.040	3.141	2.024	2.095	2.036	2.085	2.059	2.936	2.052	2.127	2.060	2.061	646.23	15346.1
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.033	2.095	2.046	2.086	2.064	2.974	2.056	2.127	2.065	2.061	625.23	17752.0
p2--CCp1--p3	tetra unstr	0.0	2.105	2.705	1.863	1.963	1.864	1.958	2.382	2.719	2.113	1.981	2.122	1.973	1812.3	20472.8
p2--CCp1--p3	tetra unstr	0.1	2.347	2.824	2.014	1.963	2.018	1.968	2.721	2.901	2.364	1.981	2.374	1.984	1955.3	23168.1
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.998	3.170	2.002	2.075	2.035	2.884	2.035	3.091	2.037	2.071	534.33	13059.3
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.003	3.171	2.007	2.094	2.037	2.925	2.037	3.091	2.039	2.079	573.07	15812.8
p2--CCp2--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1963.6	22309.5
p2--CCp2--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	2075.7	24373.9
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.034	3.246	2.037	2.170	2.059	2.936	2.059	3.146	2.061	2.125	604.10	13794.7
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.044	3.246	2.047	2.206	2.064	2.975	2.064	3.146	2.066	2.143	549.34	16456.9
p2--CCp2--p2	tetra unstr	0.0	2.105	2.705	1.863	2.944	1.864	1.932	2.384	2.720	2.116	2.972	2.122	2.121	2247.5	22778.6
p2--CCp2--p2	tetra unstr	0.1	2.350	2.825	2.016	2.945	2.018	2.104	2.727	2.903	2.369	2.972	2.376	2.408	2176.2	24847.1
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.034	3.251	2.037	2.173	2.059	2.936	2.059	3.149	2.061	2.126	722.33	15901.9
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.044	3.251	2.047	2.209	2.064	2.975	2.064	3.149	2.066	2.144	682.04	17752.8
p2--CCp2--p3	tetra unstr	0.0	2.105	2.705	1.863	2.958	1.864	1.943	2.384	2.720	2.116	2.980	2.122	2.134	2286.3	23691.0
p2--CCp2--p3	tetra unstr	0.1	2.350	2.825	2.016	2.959	2.018	2.111	2.727	2.903	2.369	2.981	2.376	2.414	2260.1	25797.0
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.997	4.087	2.002	2.020	2.035	2.884	2.035	4.142	2.037	2.056	721.83	13838.5
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.002	4.082	2.007	2.026	2.037	2.925	2.037	4.137	2.039	2.058	666.51	15813.9
p2--CCp3--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	2875.2	30961.1
p2--CCp3--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	2895.9	32396.6
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.033	4.022	2.037	2.072	2.060	2.936	2.059	4.109	2.061	2.100	857.26	14932.9
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.043	4.017	2.047	2.084	2.064	2.975	2.064	4.103	2.066	2.107	785.14	16538.8
p2--CCp3--p2	tetra unstr	0.0	2.105	2.705	1.863	3.893	1.864	1.897	2.384	2.720	2.116	3.841	2.123	2.129	3178.2	31359.2
p2--CCp3--p2	tetra unstr	0.1	2.350	2.825	2.016	3.911	2.018	2.048	2.727	2.903	2.369	3.890	2.376	2.392	3167.7	32884.7
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.033	4.014	2.037	2.073	2.060	2.936	2.059	4.110	2.061	2.101	990.02	17499.1
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.043	4.010	2.047	2.085	2.064	2.975	2.064	4.104	2.066	2.107	1022.1	18181.9
p2--CCp3--p3	tetra unstr	0.0	2.105	2.705	1.863	3.888	1.864	1.929	2.384	2.720	2.116	3.831	2.123	2.148	3323.1	32183.5

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Ion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p2--CCp3--p3	tetra unstr	0.1	2.350	2.825	2.016	3.907	2.018	2.074	2.727	2.903	2.369	3.884	2.376	2.406	3223.8	33736.6
p2--GP--p1	hexa	0.0	2.004	3.107	2.011	3.133	2.844	3.122	2.035	2.884	2.004	3.004	2.447	2.998	590.22	26061.8
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.886	3.131	2.037	2.925	2.004	3.021	2.473	3.014	514.55	23988.2
p2--GP--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1154.9	18312.9
p2--GP--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1382.5	21701.0
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.897	3.215	2.060	2.936	2.001	3.092	2.431	3.082	1167.8	26712.3
p2--GP--p2	hexa	0.1	2.051	3.166	2.004	3.236	2.920	3.221	2.064	2.975	2.001	3.105	2.439	3.094	1039.5	24632.4
p2--GP--p2	tetra unstr	0.0	2.105	2.705	1.928	2.794	2.100	2.795	2.384	2.720	1.960	2.779	2.102	2.783	1472.4	18701.6
p2--GP--p2	tetra unstr	0.1	2.350	2.825	1.928	2.831	2.146	2.833	2.727	2.903	1.960	2.862	2.167	2.866	1684.2	21997.0
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.896	3.220	2.060	2.936	2.000	3.095	2.430	3.085	2138.9	28004.9
p2--GP--p3	hexa	0.1	2.051	3.166	2.003	3.241	2.919	3.226	2.064	2.975	2.000	3.108	2.438	3.097	1870.2	25931.7
p2--GP--p3	tetra unstr	0.0	2.105	2.705	1.924	2.801	2.082	2.802	2.384	2.720	1.958	2.768	2.089	2.773	1914.1	19696.0
p2--GP--p3	tetra unstr	0.1	2.350	2.825	1.924	2.846	2.128	2.847	2.727	2.903	1.958	2.864	2.153	2.868	2017.0	22649.7
p3--na--NO	hexa	0.0	2.313	3.204	2.303	2.303	2.307	2.307	2.193	2.830	2.189	2.189	2.192	2.192	886.77	77052.6
p3--na--NO	hexa	0.1	2.278	3.207	2.269	2.269	2.273	2.273	2.169	2.827	2.166	2.166	2.168	2.168	879.82	77137.1
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2511.9	45194.0
p3--na--NO	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2695.5	53563.8
p3--CCp1--p1	hexa	0.0	2.758	3.637	2.675	2.166	2.727	2.263	2.547	3.524	2.476	2.207	2.525	2.199	1245.6	44827.9
p3--CCp1--p1	hexa	0.1	2.713	3.646	2.635	2.166	2.687	2.261	2.497	3.532	2.433	2.207	2.481	2.197	1364.9	45948.3
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	3533.2	45819.6
p3--CCp1--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	3589.5	53125.0
p3--CCp1--p2	hexa	0.0	3.096	3.790	2.987	2.097	3.017	2.094	2.864	3.829	2.773	2.130	2.782	2.063	1235.1	45475.2
p3--CCp1--p2	hexa	0.1	3.064	3.800	2.965	2.097	2.994	2.093	2.807	3.842	2.734	2.130	2.739	2.063	1437.7	46594.9
p3--CCp1--p2	tetra unstr	0.0	2.979	3.699	2.857	1.970	2.857	1.968	3.035	3.633	2.949	1.982	2.960	1.961	3474.5	46128.5
p3--CCp1--p2	tetra unstr	0.1	3.071	3.771	2.960	1.971	2.963	1.968	3.012	3.707	3.038	1.982	3.053	1.960	3711.0	53360.2
p3--CCp1--p3	hexa	0.0	3.096	3.790	2.987	2.094	3.017	2.092	2.864	3.829	2.773	2.127	2.782	2.062	1365.3	46770.8
p3--CCp1--p3	hexa	0.1	3.064	3.800	2.965	2.094	2.994	2.091	2.807	3.842	2.734	2.127	2.739	2.062	1565.9	47889.1
p3--CCp1--p3	tetra unstr	0.0	2.978	3.699	2.856	1.963	2.857	1.966	3.034	3.633	2.949	1.981	2.960	1.960	3499.2	46754.4
p3--CCp1--p3	tetra unstr	0.1	3.071	3.772	2.959	1.963	2.963	1.966	3.011	3.707	3.037	1.981	3.053	1.960	3720.5	53945.5
p3--CCp2--p1	hexa	0.0	2.782	3.644	2.745	3.172	2.752	2.948	2.574	3.538	2.546	3.092	2.553	2.790	1235.5	44827.1
p3--CCp2--p1	hexa	0.1	2.739	3.653	2.708	3.172	2.715	2.943	2.524	3.546	2.503	3.092	2.509	2.778	1453.5	45947.7
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	3723.2	47059.4
p3--CCp2--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	3912.0	53592.4
p3--CCp2--p2	hexa	0.0	3.335	3.812	3.202	3.247	3.220	3.232	3.264	3.882	3.065	3.147	3.095	3.171	1292.4	45473.8
p3--CCp2--p2	hexa	0.1	3.352	3.823	3.223	3.247	3.241	3.244	3.280	3.896	3.083	3.147	3.114	3.186	1418.8	46596.0
p3--CCp2--p2	tetra unstr	0.0	3.198	3.777	2.984	2.942	2.981	2.996	3.463	3.810	3.182	2.970	3.187	3.073	3489.9	47450.9
p3--CCp2--p2	tetra unstr	0.1	3.452	3.863	3.162	2.942	3.161	3.045	3.775	3.921	3.433	2.970	3.439	3.121	3872.8	53952.5
p3--CCp2--p3	hexa	0.0	3.335	3.812	3.202	3.253	3.220	3.231	3.264	3.882	3.065	3.150	3.095	3.171	1464.5	46767.1
p3--CCp2--p3	hexa	0.1	3.352	3.823	3.223	3.253	3.241	3.243	3.279	3.896	3.083	3.150	3.114	3.185	1614.2	47891.1
p3--CCp2--p3	tetra unstr	0.0	3.198	3.777	2.984	2.957	2.981	3.007	3.463	3.810	3.182	2.980	3.187	3.076	3546.9	48213.3
p3--CCp2--p3	tetra unstr	0.1	3.451	3.863	3.162	2.957	3.160	3.055	3.776	3.921	3.433	2.980	3.439	3.123	3932.9	54665.7
p3--CCp3--p1	hexa	0.0	2.781	3.644	2.742	4.034	2.751	2.764	2.573	3.538	2.546	4.068	2.552	2.578	1433.7	44827.5
p3--CCp3--p1	hexa	0.1	2.737	3.653	2.705	4.034	2.713	2.735	2.523	3.546	2.502	4.068	2.508	2.535	1461.2	45807.1
p3--CCp3--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	4234.3	53364.4
p3--CCp3--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	4611.4	58671.2
p3--CCp3--p2	hexa	0.0	3.332	3.813	3.194	3.978	3.216	3.106	3.260	3.882	3.069	4.042	3.091	3.018	1557.5	45473.6
p3--CCp3--p2	hexa	0.1	3.349	3.823	3.214	3.978	3.236	3.135	3.275	3.896	3.087	4.042	3.109	3.036	1885.2	46590.2
p3--CCp3--p2	tetra unstr	0.0	3.198	3.777	2.985	3.943	2.981	3.010	3.463	3.810	3.182	3.960	3.187	3.210	4203.9	53808.9
p3--CCp3--p2	tetra unstr	0.1	3.452	3.863	3.163	3.943	3.161	3.182	3.775	3.921	3.433	3.960	3.438	3.453	4665.1	59130.7
p3--CCp3--p3	hexa	0.0	3.332	3.813	3.194	3.971	3.216	3.106	3.260	3.882	3.069	4.044	3.091	3.018	2220.1	46767.6
p3--CCp3--p3	hexa	0.1	3.349	3.823	3.214	3.971	3.236	3.136	3.275	3.896	3.087	4.044	3.109	3.037	2072.9	47885.4
p3--CCp3--p3	tetra unstr	0.0	3.197	3.777	2.984	3.944	2.980	3.056	3.462	3.810	3.182	3.962	3.186	3.240	3881.6	54514.6
p3--CCp3--p3	tetra unstr	0.1	3.451	3.863	3.162	3.944	3.160	3.224	3.775	3.921	3.433	3.962	3.438	3.483	4625.4	59967.2
p3--GP--p1	hexa	0.0	2.781	3.644	2.003	3.789	2.716	3.807	2.573	3.538	2.000	3.679	2.354	3.683	1524.3	77272.0
p3--GP--p1	hexa	0.1	2.738	3.653	2.003	3.796	2.706	3.814	2.523	3.546	2.000	3.687	2.345	3.692	1268.9	77349.1
p3--GP--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	3571.4	45305.4
p3--GP--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	4078.1	53641.4
p3--GP--p2	hexa	0.0	3.332	3.813	1.994	3.892	2.454	3.910	3.260	3.882	1.996	3.955	2.171	3.965	2174.1	77922.8
p3--GP--p2	hexa	0.1	3.350	3.823	1.994	3.896	2.447	3.914	3.275	3.896	1.996	3.961	2.167	3.971	1884.3	77844.3
p3--GP--p2	tetra unstr	0.0	3.198	3.777	1.927	3.839	2.032	3.841	3.463	3.810	1.961	3.829	1.977	3.835	3649.2	45565.9
p3--GP--p2	tetra unstr	0.1	3.452	3.863	1.927	3.878	2.028	3.879	3.775	3.921	1.961	3.893	1.974	3.898	4018.1	53890.5
p3--GP--p3	hexa	0.0	3.332	3.813	1.994	3.884	2.452	3.904	3.260	3.882	1.995	3.956	2.170	3.967	3319.0	79065.9
p3--GP--p3	hexa	0.1	3.349	3.823	1.993	3.889	2.446	3.908	3.275	3.896	1.995	3.962	2.167	3.973	3166.2	79290.2
p3--GP--p3	tetra unstr	0.0	3.197	3.777	1.923	3.838	2.021	3.840	3.462	3.810	1.958	3.826	1.974	3.833	3682.2	46152.2
p3--GP--p3	tetra unstr	0.1	3.451	3.863	1.923	3.879	2.017	3.880	3.775	3.921	1.958	3.893	1.971	3.900	4260.6	54387.8

11.3.3 JFNK

$\Delta t = 10^{-2}$

Table 12: 3D manufactured results for `JFNK`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–30						Fit $N = 20$–30							
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`NO--na--NO`	hexa	0.0	1.985	1.985	1.994	1.994	1.994	1.994	1.993	1.993	2.009	2.009	2.009	2.009	10.903	4081.0
`NO--na--NO`	hexa	0.1	1.436	1.436	1.297	1.297	1.306	1.306	1.411	1.411	1.214	1.214	1.223	1.223	26.673	6311.1
`NO--na--NO`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	46.900	5184.6
`NO--na--NO`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	125.86	12050.8
`NO--CCp1--p1`	hexa	0.0	1.985	1.985	1.998	2.166	1.995	2.246	1.992	1.992	2.010	2.208	2.009	2.189	29.636	4914.5
`NO--CCp1--p1`	hexa	0.1	1.436	1.436	1.287	2.167	1.306	2.230	1.411	1.411	1.205	2.208	1.224	2.159	34.670	6578.7
`NO--CCp1--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	150.50	6443.0
`NO--CCp1--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	221.53	12246.8
`NO--CCp1--p2`	hexa	0.0	1.985	1.985	1.999	2.098	1.994	2.090	1.992	1.992	2.011	2.130	2.009	2.062	36.531	6570.2
`NO--CCp1--p2`	hexa	0.1	1.436	1.436	1.283	2.098	1.307	2.087	1.411	1.411	1.201	2.130	1.224	2.056	43.424	8032.6
`NO--CCp1--p2`	tetra unstr	0.0	0.664	0.664	0.375	1.924	0.373	0.457	0.826	0.826	0.617	1.896	0.616	0.669	145.87	7077.8
`NO--CCp1--p2`	tetra unstr	0.1	0.677	0.677	0.398	1.928	0.396	0.483	0.823	0.823	0.629	1.903	0.628	0.683	234.85	12496.9
`NO--CCp1--p3`	hexa	0.0	1.985	1.985	1.999	2.094	1.994	2.088	1.992	1.992	2.011	2.127	2.009	2.061	51.184	9863.2
`NO--CCp1--p3`	hexa	0.1	1.436	1.436	1.283	2.095	1.307	2.085	1.411	1.411	1.201	2.127	1.224	2.054	54.930	11329.7
`NO--CCp1--p3`	tetra unstr	0.0	0.664	0.664	0.375	1.923	0.373	0.460	0.826	0.826	0.617	1.906	0.616	0.670	160.88	8341.1
`NO--CCp1--p3`	tetra unstr	0.1	0.677	0.677	0.398	1.926	0.396	0.485	0.823	0.823	0.629	1.912	0.628	0.684	232.20	12995.8
`NO--CCp2--p1`	hexa	0.0	1.977	1.977	2.009	3.171	1.987	2.618	1.989	1.989	2.014	3.091	2.005	2.390	49.078	5690.6
`NO--CCp2--p1`	hexa	0.1	1.447	1.447	1.257	3.172	1.318	2.180	1.426	1.426	1.176	3.092	1.239	1.626	58.805	7158.6
`NO--CCp2--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	258.79	9806.2
`NO--CCp2--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	286.75	14003.1
`NO--CCp2--p2`	hexa	0.0	1.969	1.969	2.010	3.247	1.979	2.753	1.985	1.985	2.013	3.146	2.001	2.494	58.558	7339.2
`NO--CCp2--p2`	hexa	0.1	1.457	1.457	1.248	3.248	1.333	2.320	1.441	1.441	1.170	3.147	1.257	1.718	65.331	8808.0
`NO--CCp2--p2`	tetra unstr	0.0	0.664	0.664	0.375	2.080	0.373	0.465	0.826	0.826	0.617	1.273	0.616	0.673	242.81	10440.8
`NO--CCp2--p2`	tetra unstr	0.1	0.677	0.677	0.398	2.128	0.396	0.486	0.823	0.823	0.629	1.333	0.628	0.683	294.45	14639.7
`NO--CCp2--p3`	hexa	0.0	1.969	1.969	2.010	3.252	1.979	2.758	1.985	1.985	2.013	3.149	2.001	2.501	76.531	10633.9
`NO--CCp2--p3`	hexa	0.1	1.457	1.457	1.248	3.253	1.333	2.328	1.441	1.441	1.170	3.150	1.257	1.726	93.257	12108.2
`NO--CCp2--p3`	tetra unstr	0.0	0.664	0.664	0.375	1.992	0.373	0.466	0.826	0.826	0.617	1.241	0.616	0.675	284.94	11706.3
`NO--CCp2--p3`	tetra unstr	0.1	0.677	0.677	0.398	2.039	0.396	0.487	0.823	0.823	0.629	1.295	0.628	0.686	307.72	15907.4
`NO--CCp3--p1`	hexa	0.0	1.980	1.980	2.001	4.033	1.989	2.062	1.991	1.991	2.011	4.072	2.007	2.017	96.802	7587.6
`NO--CCp3--p1`	hexa	0.1	1.452	1.452	1.244	4.015	1.323	1.276	1.429	1.429	1.170	4.028	1.241	1.186	112.52	9061.8
`NO--CCp3--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	550.88	18575.4
`NO--CCp3--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	510.66	22778.3
`NO--CCp3--p2`	hexa	0.0	1.976	1.976	2.000	3.980	1.985	2.107	1.990	1.990	2.011	4.053	2.005	2.022	102.30	9238.3
`NO--CCp3--p2`	hexa	0.1	1.469	1.469	1.233	3.966	1.343	1.305	1.447	1.447	1.163	4.019	1.261	1.187	129.37	10707.1
`NO--CCp3--p2`	tetra unstr	0.0	0.664	0.664	0.375	0.890	0.373	0.418	0.826	0.826	0.617	0.674	0.616	0.657	624.23	19203.7
`NO--CCp3--p2`	tetra unstr	0.1	0.677	0.677	0.398	0.930	0.396	0.441	0.823	0.823	0.629	0.689	0.628	0.670	524.97	23405.3
`NO--CCp3--p3`	hexa	0.0	1.976	1.976	2.000	3.973	1.985	2.108	1.990	1.990	2.011	4.054	2.005	2.022	122.31	12535.0
`NO--CCp3--p3`	hexa	0.1	1.469	1.469	1.233	3.959	1.343	1.305	1.447	1.447	1.163	4.021	1.261	1.186	143.65	14002.8
`NO--CCp3--p3`	tetra unstr	0.0	0.664	0.664	0.375	0.888	0.373	0.432	0.826	0.826	0.617	0.684	0.616	0.665	573.46	20474.6
`NO--CCp3--p3`	tetra unstr	0.1	0.677	0.677	0.398	0.926	0.396	0.454	0.823	0.823	0.629	0.697	0.628	0.677	559.54	24674.4
`NO--GP--p1`	hexa	0.0	1.984	1.984	1.992	2.072	1.992	2.077	1.992	1.992	1.996	2.038	1.997	2.041	50.383	4665.5
`NO--GP--p1`	hexa	0.1	1.459	1.459	1.992	2.073	1.993	2.079	1.429	1.429	1.996	2.038	1.996	2.041	45.835	6563.2
`NO--GP--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	48.848	5486.9
`NO--GP--p1`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	127.67	12249.4
`NO--GP--p2`	hexa	0.0	1.983	1.983	1.984	2.014	1.985	2.018	1.992	1.992	1.992	2.007	1.993	2.009	130.18	6314.9
`NO--GP--p2`	hexa	0.1	1.482	1.482	1.984	2.015	1.985	2.018	1.447	1.447	1.992	2.007	1.993	2.010	110.28	7695.8
`NO--GP--p2`	tetra unstr	0.0	0.664	0.664	1.832	0.861	0.516	0.859	0.826	0.826	1.792	0.867	0.652	0.866	134.81	6127.4
`NO--GP--p2`	tetra unstr	0.1	0.677	0.677	1.844	0.883	0.546	0.881	0.823	0.823	1.812	0.884	0.668	0.883	173.59	12494.7
`NO--GP--p3`	hexa	0.0	1.983	1.983	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.993	2.008	326.06	9615.9
`NO--GP--p3`	hexa	0.1	1.482	1.482	1.984	2.011	1.985	2.015	1.447	1.447	1.992	2.006	1.992	2.008	241.31	11005.2
`NO--GP--p3`	tetra unstr	0.0	0.664	0.664	1.828	0.863	0.513	0.862	0.826	0.826	1.790	0.868	0.651	0.868	256.29	7394.5
`NO--GP--p3`	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.542	0.884	0.823	0.823	1.810	0.885	0.667	0.885	252.73	12996.4
`p1--na--NO`	hexa	0.0	1.967	2.095	1.927	1.927	1.930	1.930	1.919	2.125	1.838	1.838	1.845	1.845	73.508	7146.7
`p1--na--NO`	hexa	0.1	2.067	2.093	2.047	2.047	2.048	2.048	2.066	2.124	2.033	2.033	2.035	2.035	66.261	8682.2
`p1--na--NO`	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	201.56	10384.5
`p1--na--NO`	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	201.05	11873.0
`p1--CCp1--p1`	hexa	0.0	1.967	2.095	1.926	2.166	1.930	2.193	1.919	2.125	1.836	2.207	1.845	2.115	78.985	7316.5
`p1--CCp1--p1`	hexa	0.1	2.067	2.093	2.046	2.167	2.048	2.225	2.066	2.124	2.033	2.208	2.035	2.168	95.760	8858.9
`p1--CCp1--p1`	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	320.24	11428.9
`p1--CCp1--p1`	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	274.69	12624.6
`p1--CCp1--p2`	hexa	0.0	1.967	2.095	1.925	2.098	1.930	2.083	1.919	2.125	1.835	2.130	1.845	2.049	104.65	8760.1
`p1--CCp1--p2`	hexa	0.1	2.067	2.093	2.046	2.098	2.047	2.090	2.066	2.124	2.033	2.130	2.035	2.061	121.22	9611.3
`p1--CCp1--p2`	tetra unstr	0.0	1.474	1.866	1.139	1.973	1.142	1.493	1.771	1.928	1.419	1.984	1.428	1.590	306.60	12068.1
`p1--CCp1--p2`	tetra unstr	0.1	1.613	1.909	1.262	1.973	1.264	1.621	1.889	1.963	1.556	1.983	1.558	1.721	294.54	13192.7
`p1--CCp1--p3`	hexa	0.0	1.967	2.095	1.925	2.094	1.930	2.081	1.919	2.125	1.835	2.127	1.845	2.048	118.20	12040.5
`p1--CCp1--p3`	hexa	0.1	2.067	2.093	2.046	2.094	2.047	2.088	2.066	2.124	2.033	2.127	2.035	2.060	107.73	12507.1
`p1--CCp1--p3`	tetra unstr	0.0	1.474	1.866	1.139	1.965	1.142	1.505	1.771	1.928	1.419	1.982	1.428	1.597	329.54	13330.6
`p1--CCp1--p3`	tetra unstr	0.1	1.613	1.909	1.262	1.965	1.264	1.632	1.889	1.963	1.556	1.982	1.558	1.727	289.57	14324.8
`p1--CCp2--p1`	hexa	0.0	1.966	2.095	1.926	3.169	1.929	2.043	1.919	2.125	1.831	3.089	1.845	1.844	113.82	7872.0
`p1--CCp2--p1`	hexa	0.1	2.065	2.093	2.050	3.170	2.046	2.200	2.066	2.124	2.033	3.090	2.034	2.090	107.02	8897.9
`p1--CCp2--p1`	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	396.23	14798.2
`p1--CCp2--p1`	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	343.20	15617.1
`p1--CCp2--p2`	hexa	0.0	1.965	2.095	1.925	3.244	1.929	2.095	1.919	2.125	1.828	3.143	1.845	1.846	107.93	9524.3
`p1--CCp2--p2`	hexa	0.1	2.064	2.093	2.051	3.245	2.045	2.265	2.065	2.124	2.033	3.145	2.034	2.115	116.68	9993.9
`p1--CCp2--p2`	tetra unstr	0.0	1.475	1.866	1.139	2.917	1.142	1.199	1.771	1.928	1.418	2.913	1.428	1.465	424.91	15428.2
`p1--CCp2--p2`	tetra unstr	0.1	1.613	1.909	1.262	2.928	1.264	1.336	1.889	1.963	1.555	2.938	1.558	1.614	385.65	16245.6

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p1--CCp2--p3	hexa	0.0	1.965	2.095	1.925	3.250	1.929	2.098	1.919	2.125	1.828	3.146	1.845	1.848	125.46	12814.8
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.051	3.251	2.045	2.269	2.065	2.124	2.033	3.148	2.034	2.117	116.80	13283.4
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.139	2.922	1.142	1.211	1.771	1.928	1.418	2.906	1.428	1.476	436.87	16692.6
p1--CCp2--p3	tetra unstr	0.1	1.613	1.909	1.262	2.938	1.264	1.348	1.889	1.963	1.555	2.938	1.558	1.624	358.67	17518.9
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.922	4.067	1.929	1.877	1.919	2.125	1.830	4.138	1.845	1.729	151.89	9769.3
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.046	4.057	2.047	2.042	2.066	2.124	2.032	4.119	2.034	2.007	144.90	10198.8
p1--CCp3--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	604.20	23563.3
p1--CCp3--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	550.67	24379.4
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.920	4.009	1.930	1.853	1.920	2.125	1.827	4.112	1.846	1.682	158.35	11421.3
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.045	4.000	2.047	2.039	2.066	2.124	2.031	4.092	2.034	1.993	155.38	11864.5
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.139	2.936	1.142	1.202	1.771	1.928	1.418	2.110	1.428	1.472	647.90	24195.5
p1--CCp3--p2	tetra unstr	0.1	1.613	1.909	1.262	3.163	1.264	1.332	1.889	1.963	1.555	2.437	1.558	1.614	545.13	25023.2
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.920	4.003	1.930	1.852	1.920	2.125	1.827	4.114	1.846	1.680	216.54	14712.4
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.045	3.992	2.047	2.039	2.066	2.124	2.031	4.094	2.034	1.993	188.45	15149.0
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.139	2.837	1.142	1.240	1.771	1.928	1.418	2.066	1.428	1.496	650.11	25455.3
p1--CCp3--p3	tetra unstr	0.1	1.613	1.909	1.262	3.084	1.264	1.375	1.889	1.963	1.555	2.395	1.558	1.644	534.40	26289.6
p1--GP--p1	hexa	0.0	1.967	2.095	1.992	2.157	1.999	2.164	1.919	2.125	1.997	2.201	2.001	2.201	94.746	7340.6
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.002	2.162	2.066	2.124	1.997	2.199	2.005	2.199	86.296	8892.3
p1--GP--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	207.54	10483.4
p1--GP--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	211.38	11964.7
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.920	2.125	1.993	2.129	1.997	2.128	170.47	8587.1
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.093	1.991	2.096	2.066	2.124	1.993	2.128	1.998	2.126	146.84	9578.2
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.771	1.928	1.953	1.877	1.777	1.880	282.21	11109.8
p1--GP--p2	tetra unstr	0.1	1.613	1.909	1.920	1.897	1.790	1.900	1.889	1.963	1.955	1.921	1.848	1.922	322.57	12472.4
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.920	2.125	1.993	2.126	1.996	2.125	316.14	12162.0
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.066	2.124	1.993	2.124	1.998	2.123	275.06	12472.4
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.874	1.771	1.928	1.951	1.889	1.775	1.892	354.44	12384.3
p1--GP--p3	tetra unstr	0.1	1.613	1.909	1.916	1.900	1.786	1.903	1.889	1.963	1.953	1.927	1.846	1.929	367.74	13559.1
p2--na--NO	hexa	0.0	1.947	3.025	1.944	1.944	1.948	1.948	2.001	2.766	2.003	2.003	2.005	2.005	150.14	25853.8
p2--na--NO	hexa	0.1	1.946	3.053	1.943	1.943	1.947	1.947	2.000	2.807	2.002	2.002	2.005	2.005	121.96	21776.5
p2--na--NO	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	311.22	18260.0
p2--na--NO	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	393.20	21620.4
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.995	2.168	2.003	2.183	2.035	2.884	2.035	2.209	2.040	2.147	195.99	13057.6
p2--CCp1--p1	hexa	0.1	2.008	3.133	2.000	2.167	2.008	2.193	2.037	2.925	2.037	2.209	2.042	2.153	171.78	15809.6
p2--CCp1--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	424.47	19182.3
p2--CCp1--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	410.00	21990.9
p2--CCp1--p2	hexa	0.0	2.040	3.142	2.028	2.098	2.039	2.087	2.059	2.936	2.057	2.131	2.064	2.063	170.77	13778.1
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.037	2.098	2.049	2.088	2.063	2.975	2.062	2.131	2.069	2.063	166.93	16457.9
p2--CCp1--p2	tetra unstr	0.0	2.103	2.703	1.863	1.971	1.864	1.960	2.383	2.718	2.113	1.982	2.121	1.974	419.96	19827.3
p2--CCp1--p2	tetra unstr	0.1	2.347	2.823	2.013	1.971	2.016	1.970	2.727	2.903	2.362	1.982	2.372	1.985	409.58	22505.6
p2--CCp1--p3	hexa	0.0	2.040	3.142	2.028	2.095	2.039	2.085	2.059	2.936	2.057	2.127	2.064	2.061	168.67	16260.0
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.037	2.095	2.049	2.086	2.063	2.975	2.062	2.127	2.069	2.061	192.63	17828.6
p2--CCp1--p3	tetra unstr	0.0	2.103	2.703	1.863	1.963	1.864	1.958	2.383	2.718	2.112	1.981	2.121	1.972	468.73	21087.9
p2--CCp1--p3	tetra unstr	0.1	2.347	2.823	2.013	1.963	2.016	1.968	2.727	2.903	2.362	1.981	2.372	1.983	412.36	23563.6
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.999	3.170	2.004	2.077	2.035	2.884	2.038	3.091	2.040	2.074	184.42	13081.2
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.005	3.171	2.009	2.096	2.037	2.925	2.040	3.091	2.043	2.083	193.86	15814.8
p2--CCp2--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	574.74	22548.2
p2--CCp2--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	474.93	24551.7
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.036	3.246	2.040	2.173	2.059	2.936	2.063	3.146	2.065	2.129	211.08	14044.3
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.046	3.246	2.050	2.209	2.064	2.975	2.068	3.146	2.070	2.147	173.89	16455.3
p2--CCp2--p2	tetra unstr	0.0	2.103	2.704	1.863	2.944	1.864	1.932	2.384	2.719	2.114	2.972	2.121	2.120	588.74	23176.3
p2--CCp2--p2	tetra unstr	0.1	2.349	2.824	2.014	2.945	2.016	2.103	2.732	2.904	2.367	2.972	2.373	2.405	471.63	25125.6
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.036	3.251	2.040	2.176	2.059	2.936	2.063	3.149	2.065	2.130	221.97	16914.1
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.046	3.251	2.050	2.212	2.064	2.975	2.068	3.149	2.070	2.149	206.85	17986.4
p2--CCp2--p3	tetra unstr	0.0	2.103	2.704	1.863	2.958	1.864	1.943	2.384	2.719	2.114	2.980	2.121	2.133	539.90	24442.6
p2--CCp2--p3	tetra unstr	0.1	2.349	2.824	2.014	2.959	2.016	2.110	2.732	2.904	2.367	2.981	2.373	2.411	504.22	26263.2
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.999	4.087	2.004	2.022	2.035	2.884	2.038	4.142	2.040	2.059	219.13	14040.2
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.004	4.082	2.009	2.028	2.037	2.925	2.040	4.137	2.043	2.062	240.33	15818.3
p2--CCp3--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	769.58	31316.2
p2--CCp3--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	642.00	32700.4
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.035	4.022	2.040	2.074	2.059	2.936	2.063	4.109	2.065	2.104	235.26	15527.0
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.045	4.017	2.050	2.087	2.064	2.975	2.068	4.103	2.070	2.111	268.32	16648.2
p2--CCp3--p2	tetra unstr	0.0	2.103	2.704	1.863	3.893	1.864	1.896	2.384	2.719	2.114	3.841	2.121	2.127	785.01	31949.9
p2--CCp3--p2	tetra unstr	0.1	2.349	2.824	2.014	3.911	2.016	2.046	2.732	2.904	2.367	3.890	2.373	2.389	665.64	33324.8
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.035	4.014	2.040	2.075	2.059	2.936	2.063	4.110	2.065	2.105	273.20	18513.7
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.045	4.010	2.050	2.087	2.064	2.975	2.068	4.104	2.070	2.112	261.59	18886.3
p2--CCp3--p3	tetra unstr	0.0	2.103	2.704	1.863	3.888	1.864	1.929	2.384	2.719	2.114	3.831	2.121	2.146	808.64	33209.9
p2--CCp3--p3	tetra unstr	0.1	2.349	2.824	2.014	3.907	2.016	2.072	2.732	2.904	2.367	3.883	2.373	2.403	644.26	34591.0
p2--GP--p1	hexa	0.0	2.003	3.107	2.011	3.133	2.845	3.122	2.035	2.884	2.004	3.005	2.449	2.999	175.28	26059.4
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.888	3.131	2.037	2.925	2.004	3.021	2.476	3.015	162.51	21986.0
p2--GP--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	358.27	18397.1
p2--GP--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	366.28	21760.9
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.898	3.215	2.059	2.936	2.001	3.092	2.432	3.083	259.13	26708.4
p2--GP--p2	hexa	0.1	2.050	3.166	2.004	3.236	2.920	3.222	2.064	2.975	2.001	3.105	2.440	3.095	232.31	22636.7
p2--GP--p2	tetra unstr	0.0	2.103	2.704	1.928	2.794	2.100	2.796	2.384	2.719	1.960	2.779	2.102	2.784	382.90	19041.3
p2--GP--p2	tetra unstr	0.1	2.349	2.824	1.928	2.831	2.146	2.833	2.732	2.904	1.960	2.861	2.166	2.866	390.35	22132.5
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.897	3.220	2.059	2.936	2.000	3.095	2.432	3.085	436.66	28002.5
p2--GP--p3	hexa	0.1	2.050	3.166	2.003	3.242	2.920	3.227	2.064	2.975	2.000	3.108	2.439	3.097	357.00	24079.3
p2--GP--p3	tetra unstr	0.0	2.103	2.704	1.924	2.802	2.082	2.802	2.384	2.719	1.958	2.769	2.088	2.773	406.97	20303.3
p2--GP--p3	tetra unstr	0.1	2.349	2.824	1.924	2.847	2.128	2.847	2.732	2.904	1.958	2.864	2.152	2.868	453.30	23026.1
p3--na--NO	hexa	0.0	2.313	3.204	2.308	3.208	2.312	2.312	2.193	2.830	2.197	2.197	2.200	2.200	872.23	77059.3
p3--na--NO	hexa	0.1	2.278	3.207	2.274	2.274	2.278	2.278	2.169	2.826	2.174	2.174	2.176	2.176	725.16	72597.5

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1102.0	45158.7
p3--na--NO	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1239.7	53472.5
p3--CCp1--p1	hexa	0.0	2.761	3.638	2.698	2.166	2.745	2.264	2.550	3.526	2.508	2.207	2.552	2.199	938.71	44823.5
p3--CCp1--p1	hexa	0.1	2.717	3.647	2.659	2.166	2.706	2.261	2.501	3.534	2.465	2.207	2.509	2.198	936.37	45946.8
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1491.7	45872.4
p3--CCp1--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1295.9	53128.9
p3--CCp1--p2	hexa	0.0	3.129	3.794	3.046	2.097	3.078	2.094	2.914	3.838	2.857	2.130	2.869	2.063	914.47	45474.8
p3--CCp1--p2	hexa	0.1	3.103	3.804	3.033	2.097	3.064	2.093	2.862	3.851	2.828	2.130	2.835	2.063	936.18	46596.0
p3--CCp1--p2	tetra unstr	0.0	3.009	3.711	2.897	1.970	2.893	1.968	3.092	3.661	3.022	1.982	3.021	1.961	1555.3	46211.2
p3--CCp1--p2	tetra unstr	0.1	3.120	3.786	3.019	1.970	3.015	1.968	3.099	3.741	3.149	1.982	3.147	1.960	1320.1	53378.0
p3--CCp1--p3	hexa	0.0	3.129	3.794	3.046	2.094	3.078	2.092	2.914	3.838	2.857	2.127	2.869	2.062	843.74	46768.3
p3--CCp1--p3	hexa	0.1	3.103	3.804	3.033	2.094	3.064	2.091	2.862	3.851	2.828	2.127	2.835	2.062	877.74	47889.0
p3--CCp1--p3	tetra unstr	0.0	3.009	3.711	2.897	1.963	2.892	1.966	3.091	3.661	3.021	1.981	3.021	1.960	1563.1	46935.0
p3--CCp1--p3	tetra unstr	0.1	3.120	3.786	3.018	1.963	3.015	1.966	3.099	3.741	3.149	1.981	3.147	1.960	1265.7	53955.0
p3--CCp2--p1	hexa	0.0	2.781	3.643	2.755	3.172	2.763	2.955	2.572	3.538	2.566	3.092	2.573	2.804	857.17	44820.2
p3--CCp2--p1	hexa	0.1	2.738	3.653	2.719	3.172	2.726	2.951	2.523	3.546	2.524	3.092	2.530	2.794	909.48	45950.9
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1585.8	46884.2
p3--CCp2--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1324.3	53666.8
p3--CCp2--p2	hexa	0.0	3.332	3.812	3.211	3.247	3.230	3.235	3.260	3.882	3.083	3.147	3.113	3.176	872.52	45473.9
p3--CCp2--p2	hexa	0.1	3.350	3.823	3.234	3.247	3.253	3.247	3.275	3.896	3.103	3.147	3.135	3.190	914.55	46594.9
p3--CCp2--p2	tetra unstr	0.0	3.196	3.775	2.986	2.942	2.983	2.997	3.461	3.809	3.184	2.970	3.189	3.072	1570.5	47660.3
p3--CCp2--p2	tetra unstr	0.1	3.450	3.862	3.163	2.942	3.162	3.045	3.777	3.920	3.434	2.970	3.440	3.120	1383.0	54049.4
p3--CCp2--p3	hexa	0.0	3.332	3.812	3.211	3.253	3.230	3.234	3.260	3.882	3.083	3.150	3.113	3.175	870.79	46768.6
p3--CCp2--p3	hexa	0.1	3.350	3.823	3.234	3.253	3.253	3.246	3.275	3.896	3.103	3.150	3.135	3.189	942.03	47892.3
p3--CCp2--p3	tetra unstr	0.0	3.195	3.776	2.985	2.957	2.983	3.007	3.461	3.809	3.184	2.980	3.188	3.076	1477.5	48574.3
p3--CCp2--p3	tetra unstr	0.1	3.450	3.862	3.163	2.957	3.161	3.054	3.777	3.921	3.434	2.980	3.440	3.122	1312.3	54827.0
p3--CCp3--p1	hexa	0.0	2.780	3.644	2.753	4.033	2.762	2.775	2.572	3.538	2.566	4.067	2.573	2.597	848.74	44827.5
p3--CCp3--p1	hexa	0.1	2.737	3.653	2.716	4.033	2.725	2.746	2.522	3.545	2.523	4.068	2.530	2.555	924.30	45949.8
p3--CCp3--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1771.8	53595.6
p3--CCp3--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1473.3	58804.8
p3--CCp3--p2	hexa	0.0	3.330	3.812	3.205	3.978	3.228	3.112	3.256	3.881	3.086	4.042	3.110	3.026	838.54	45472.8
p3--CCp3--p2	hexa	0.1	3.347	3.823	3.226	3.978	3.250	3.143	3.271	3.895	3.106	4.042	3.131	3.046	962.87	46597.6
p3--CCp3--p2	tetra unstr	0.0	3.196	3.775	2.987	3.943	2.983	3.011	3.461	3.808	3.184	3.959	3.188	3.209	1352.6	54152.0
p3--CCp3--p2	tetra unstr	0.1	3.450	3.862	3.164	3.942	3.162	3.181	3.777	3.920	3.434	3.959	3.440	3.451	1423.9	59374.8
p3--CCp3--p3	hexa	0.0	3.330	3.812	3.205	3.971	3.228	3.112	3.256	3.881	3.086	4.044	3.110	3.027	887.10	46768.3
p3--CCp3--p3	hexa	0.1	3.347	3.823	3.226	3.971	3.250	3.143	3.271	3.895	3.106	4.044	3.131	3.046	959.95	47890.5
p3--CCp3--p3	tetra unstr	0.0	3.195	3.776	2.986	3.944	2.982	3.056	3.460	3.809	3.184	3.961	3.188	3.240	1351.6	55284.1
p3--CCp3--p3	tetra unstr	0.1	3.449	3.862	3.163	3.944	3.161	3.224	3.777	3.920	3.434	3.961	3.439	3.481	1400.7	60458.9
p3--GP--p1	hexa	0.0	2.780	3.644	2.003	3.792	2.717	3.810	2.572	3.538	2.000	3.687	2.355	3.691	1126.1	77279.7
p3--GP--p1	hexa	0.1	2.737	3.653	2.003	3.799	2.707	3.817	2.522	3.546	2.000	3.695	2.347	3.700	777.00	72810.8
p3--GP--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1407.6	45295.2
p3--GP--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1255.5	53554.9
p3--GP--p2	hexa	0.0	3.331	3.812	1.994	3.893	2.454	3.912	3.257	3.881	1.996	3.958	2.171	3.969	1014.5	77925.3
p3--GP--p2	hexa	0.1	3.348	3.823	1.994	3.897	2.447	3.916	3.272	3.896	1.996	3.964	2.167	3.975	750.17	69196.2
p3--GP--p2	tetra unstr	0.0	3.196	3.775	1.927	3.844	2.032	3.845	3.461	3.808	1.961	3.840	1.977	3.846	1367.1	45574.8
p3--GP--p2	tetra unstr	0.1	3.450	3.862	1.927	3.883	2.028	3.884	3.777	3.920	1.961	3.904	1.974	3.909	1245.5	53807.3
p3--GP--p3	hexa	0.0	3.331	3.812	1.994	3.886	2.452	3.905	3.257	3.881	1.995	3.960	2.170	3.970	1172.7	79220.7
p3--GP--p3	hexa	0.1	3.348	3.823	1.993	3.890	2.446	3.909	3.272	3.896	1.995	3.966	2.167	3.976	870.62	70486.9
p3--GP--p3	tetra unstr	0.0	3.195	3.776	1.923	3.842	2.021	3.844	3.460	3.809	1.958	3.837	1.974	3.844	1258.6	46320.8
p3--GP--p3	tetra unstr	0.1	3.450	3.862	1.923	3.884	2.017	3.885	3.777	3.921	1.958	3.904	1.971	3.911	1397.1	54335.8

$\Delta t = 10^{-3}$

Table 13: 3D manufactured results for `JFNK`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run.

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
NO--na--NO	hexa	0.0	1.985	1.985	1.985	1.985	1.985	1.985	1.993	1.993	1.993	1.993	1.993	1.993	72.358	4083.5
NO--na--NO	hexa	0.1	1.438	1.438	1.293	1.293	1.302	1.302	1.413	1.413	1.211	1.211	1.220	1.220	103.30	6309.8
NO--na--NO	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	166.32	5142.7
NO--na--NO	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	351.54	12049.4
NO--CCp1--p1	hexa	0.0	1.986	1.986	1.989	2.167	1.986	2.246	1.993	1.993	1.995	2.208	1.993	2.188	146.88	4921.8
NO--CCp1--p1	hexa	0.1	1.438	1.438	1.298	2.167	1.302	2.230	1.413	1.413	1.212	2.208	1.220	2.158	141.64	6578.8
NO--CCp1--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	397.40	6398.6
NO--CCp1--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	587.73	12245.9
NO--CCp1--p2	hexa	0.0	1.986	1.986	1.991	2.098	1.986	2.090	1.993	1.993	1.995	2.130	1.993	2.062	207.40	6567.7
NO--CCp1--p2	hexa	0.1	1.438	1.438	1.300	2.098	1.303	2.086	1.413	1.413	1.212	2.130	1.220	2.055	200.35	8038.6
NO--CCp1--p2	tetra unstr	0.0	0.663	0.663	0.377	1.924	0.375	0.459	0.829	0.829	0.617	1.896	0.617	0.670	423.96	7036.9
NO--CCp1--p2	tetra unstr	0.1	0.677	0.677	0.400	1.928	0.398	0.484	0.828	0.828	0.629	1.903	0.629	0.684	613.00	12498.1
NO--CCp1--p3	hexa	0.0	1.986	1.986	1.991	2.094	1.986	2.088	1.993	1.993	1.995	2.127	1.993	2.060	288.40	9876.9
NO--CCp1--p3	hexa	0.1	1.438	1.438	1.300	2.095	1.303	2.085	1.413	1.413	1.212	2.127	1.220	2.054	282.07	11332.3
NO--CCp1--p3	tetra unstr	0.0	0.663	0.663	0.377	1.922	0.375	0.462	0.829	0.829	0.617	1.906	0.617	0.671	469.85	8302.9
NO--CCp1--p3	tetra unstr	0.1	0.677	0.677	0.400	1.926	0.398	0.487	0.828	0.828	0.629	1.912	0.629	0.685	615.47	13000.4
NO--CCp2--p1	hexa	0.0	1.977	1.977	2.002	3.171	1.978	2.610	1.989	1.989	1.999	3.091	1.989	2.376	216.25	5692.4
NO--CCp2--p1	hexa	0.1	1.447	1.447	1.258	3.172	1.313	2.174	1.428	1.428	1.175	3.092	1.236	1.620	226.43	7165.9

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
NO--CCp2--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	773.69	9761.4
NO--CCp2--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	936.72	14003.0
NO--CCp2--p2	hexa	0.0	1.969	1.969	2.004	3.247	1.970	2.745	1.985	1.985	1.998	3.146	1.985	2.481	275.43	7341.0
NO--CCp2--p2	hexa	0.1	1.457	1.457	1.251	3.248	1.326	2.314	1.443	1.443	1.170	3.147	1.253	1.712	331.22	8813.8
NO--CCp2--p2	tetra unstr	0.0	0.663	0.663	0.377	2.079	0.375	0.467	0.829	0.829	0.617	1.272	0.617	0.673	824.29	10398.8
NO--CCp2--p2	tetra unstr	0.1	0.677	0.677	0.400	2.127	0.398	0.488	0.828	0.828	0.629	1.333	0.629	0.684	919.93	14641.5
NO--CCp2--p3	hexa	0.0	1.969	1.969	2.004	3.252	1.970	2.751	1.985	1.985	1.998	3.149	1.985	2.488	407.86	10640.1
NO--CCp2--p3	hexa	0.1	1.457	1.457	1.251	3.253	1.326	2.323	1.443	1.443	1.170	3.150	1.253	1.720	489.56	12109.3
NO--CCp2--p3	tetra unstr	0.0	0.663	0.663	0.377	1.991	0.375	0.469	0.829	0.829	0.617	1.240	0.617	0.676	921.20	11665.9
NO--CCp2--p3	tetra unstr	0.1	0.677	0.677	0.400	2.038	0.398	0.489	0.828	0.828	0.629	1.295	0.629	0.686	992.95	15908.5
NO--CCp3--p1	hexa	0.0	1.980	1.980	1.993	4.033	1.980	2.052	1.992	1.992	1.995	4.072	1.992	2.002	420.83	7592.3
NO--CCp3--p1	hexa	0.1	1.453	1.453	1.241	4.015	1.318	1.271	1.431	1.431	1.167	4.028	1.238	1.184	548.92	9065.3
NO--CCp3--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	2082.3	18530.6
NO--CCp3--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	1750.0	22782.3
NO--CCp3--p2	hexa	0.0	1.975	1.975	1.992	3.981	1.975	2.098	1.990	1.990	1.995	4.053	1.990	2.007	505.78	9240.4
NO--CCp3--p2	hexa	0.1	1.469	1.469	1.230	3.966	1.336	1.300	1.449	1.449	1.160	4.019	1.258	1.185	655.78	10710.5
NO--CCp3--p2	tetra unstr	0.0	0.663	0.663	0.378	0.891	0.375	0.420	0.829	0.829	0.617	0.675	0.617	0.658	2065.6	19164.9
NO--CCp3--p2	tetra unstr	0.1	0.677	0.677	0.400	0.930	0.398	0.443	0.828	0.828	0.629	0.689	0.629	0.671	1777.1	23407.9
NO--CCp3--p3	hexa	0.0	1.975	1.975	1.992	3.973	1.975	2.099	1.991	1.991	1.995	4.055	1.990	2.007	668.97	12534.2
NO--CCp3--p3	hexa	0.1	1.469	1.469	1.230	3.959	1.336	1.300	1.449	1.449	1.160	4.021	1.258	1.184	846.23	14010.0
NO--CCp3--p3	tetra unstr	0.0	0.663	0.663	0.378	0.888	0.375	0.434	0.829	0.829	0.617	0.684	0.617	0.666	2134.2	20432.8
NO--CCp3--p3	tetra unstr	0.1	0.677	0.677	0.400	0.927	0.398	0.456	0.828	0.828	0.629	0.698	0.629	0.678	1875.4	24675.8
NO--GP--p1	hexa	0.0	1.985	1.985	1.992	2.071	1.992	2.077	1.993	1.993	1.996	2.038	1.996	2.041	274.71	4669.2
NO--GP--p1	hexa	0.1	1.461	1.461	1.992	2.073	1.992	2.079	1.431	1.431	1.996	2.038	1.995	2.041	286.77	6561.2
NO--GP--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	172.50	5446.4
NO--GP--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	358.09	12247.7
NO--GP--p2	hexa	0.0	1.984	1.984	1.984	2.014	1.984	2.018	1.992	1.992	1.992	2.007	1.993	2.009	916.82	6341.8
NO--GP--p2	hexa	0.1	1.483	1.483	1.984	2.014	1.985	2.018	1.450	1.450	1.992	2.007	1.992	2.009	749.00	7704.6
NO--GP--p2	tetra unstr	0.0	0.663	0.663	1.832	0.862	0.518	0.860	0.829	0.829	1.792	0.867	0.653	0.866	507.51	6090.2
NO--GP--p2	tetra unstr	0.1	0.677	0.677	1.843	0.883	0.547	0.882	0.828	0.828	1.812	0.884	0.668	0.883	549.05	12497.3
NO--GP--p3	hexa	0.0	1.984	1.984	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.992	2.008	2080.8	9677.3
NO--GP--p3	hexa	0.1	1.483	1.483	1.984	2.011	1.984	2.015	1.450	1.450	1.992	2.005	1.992	2.008	1710.3	11009.2
NO--GP--p3	tetra unstr	0.0	0.663	0.663	1.828	0.864	0.514	0.862	0.829	0.829	1.790	0.869	0.652	0.868	907.09	7355.9
NO--GP--p3	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.543	0.884	0.828	0.828	1.810	0.886	0.667	0.885	915.26	12995.6
p1--na--NO	hexa	0.0	1.967	2.095	1.924	1.924	1.927	1.927	1.920	2.125	1.833	1.833	1.840	1.840	156.75	7136.1
p1--na--NO	hexa	0.1	2.067	2.093	2.044	2.044	2.044	2.044	2.067	2.124	2.028	2.028	2.029	2.029	152.25	8682.5
p1--na--NO	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	382.68	10354.9
p1--na--NO	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	450.48	11878.2
p1--CCp1--p1	hexa	0.0	1.967	2.095	1.924	2.166	1.926	2.192	1.920	2.125	1.833	2.207	1.840	2.114	195.46	7316.9
p1--CCp1--p1	hexa	0.1	2.067	2.093	2.043	2.167	2.044	2.224	2.067	2.124	2.028	2.208	2.029	2.167	235.13	8860.8
p1--CCp1--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	654.04	11401.3
p1--CCp1--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	639.72	12625.4
p1--CCp1--p2	hexa	0.0	1.967	2.095	1.924	2.098	1.926	2.083	1.920	2.125	1.833	2.130	1.840	2.049	255.15	8764.0
p1--CCp1--p2	hexa	0.1	2.067	2.093	2.043	2.098	2.044	2.090	2.067	2.124	2.028	2.130	2.029	2.061	320.59	9608.3
p1--CCp1--p2	tetra unstr	0.0	1.475	1.866	1.142	1.973	1.144	1.493	1.768	1.927	1.422	1.984	1.432	1.592	708.35	12038.6
p1--CCp1--p2	tetra unstr	0.1	1.612	1.909	1.265	1.973	1.267	1.622	1.884	1.962	1.558	1.983	1.561	1.723	668.04	13190.7
p1--CCp1--p3	hexa	0.0	1.967	2.095	1.924	2.094	1.926	2.081	1.920	2.125	1.833	2.127	1.840	2.048	364.01	12046.0
p1--CCp1--p3	hexa	0.1	2.067	2.093	2.043	2.094	2.044	2.088	2.067	2.124	2.028	2.127	2.029	2.059	375.87	12503.0
p1--CCp1--p3	tetra unstr	0.0	1.475	1.866	1.142	1.965	1.144	1.506	1.768	1.927	1.422	1.982	1.432	1.599	782.50	13300.4
p1--CCp1--p3	tetra unstr	0.1	1.612	1.909	1.265	1.965	1.267	1.633	1.884	1.962	1.558	1.982	1.561	1.729	688.38	14322.8
p1--CCp2--p1	hexa	0.0	1.966	2.095	1.924	3.169	1.926	2.040	1.920	2.125	1.827	3.089	1.840	1.840	273.24	7875.3
p1--CCp2--p1	hexa	0.1	2.065	2.093	2.048	3.170	2.043	2.196	2.066	2.124	2.028	3.090	2.028	2.085	333.30	8900.6
p1--CCp2--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	994.37	14770.4
p1--CCp2--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	941.43	15615.3
p1--CCp2--p2	hexa	0.0	1.966	2.095	1.924	3.244	1.925	2.092	1.920	2.125	1.824	3.143	1.840	1.841	342.59	9527.4
p1--CCp2--p2	hexa	0.1	2.064	2.093	2.050	3.245	2.042	2.261	2.065	2.124	2.028	3.145	2.028	2.109	372.72	9995.8
p1--CCp2--p2	tetra unstr	0.0	1.475	1.866	1.141	2.917	1.145	1.201	1.768	1.927	1.422	2.912	1.432	1.469	1121.4	15397.2
p1--CCp2--p2	tetra unstr	0.1	1.612	1.909	1.264	2.928	1.267	1.339	1.884	1.962	1.558	2.938	1.561	1.617	1006.7	16247.0
p1--CCp2--p3	hexa	0.0	1.966	2.095	1.924	3.250	1.925	2.095	1.920	2.125	1.824	3.146	1.840	1.843	468.05	12820.1
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.050	3.251	2.042	2.265	2.065	2.124	2.028	3.148	2.028	2.111	477.61	13286.9
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.141	2.922	1.145	1.213	1.768	1.927	1.422	2.906	1.432	1.479	1152.0	16666.6
p1--CCp2--p3	tetra unstr	0.1	1.612	1.909	1.264	2.938	1.267	1.350	1.884	1.962	1.558	2.938	1.561	1.627	1051.6	17517.1
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.919	4.067	1.926	1.874	1.920	2.125	1.826	4.139	1.840	1.725	490.32	9775.7
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.043	4.057	2.043	2.038	2.067	2.124	2.026	4.119	2.029	2.002	549.92	10198.1
p1--CCp3--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	2065.7	23533.8
p1--CCp3--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	1927.3	24383.1
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.917	4.010	1.926	1.850	1.921	2.125	1.823	4.112	1.841	1.677	651.67	11428.5
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.042	4.000	2.043	2.035	2.067	2.124	2.025	4.092	2.029	1.988	614.83	11868.1
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.141	2.935	1.145	1.205	1.768	1.927	1.422	2.111	1.432	1.475	2041.6	24166.4
p1--CCp3--p2	tetra unstr	0.1	1.612	1.909	1.264	3.163	1.267	1.335	1.884	1.962	1.558	2.438	1.561	1.616	1820.0	25022.9
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.917	4.003	1.926	1.849	1.921	2.125	1.823	4.114	1.841	1.676	904.04	14721.7
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.042	3.992	2.043	2.035	2.067	2.124	2.025	4.094	2.029	1.988	852.94	15150.3
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.141	2.836	1.145	1.243	1.768	1.927	1.422	2.067	1.432	1.500	2162.4	25434.4
p1--CCp3--p3	tetra unstr	0.1	1.612	1.909	1.264	3.084	1.267	1.378	1.884	1.962	1.558	2.395	1.561	1.647	1927.8	26288.7
p1--GP--p1	hexa	0.0	1.968	2.095	1.992	2.157	1.999	2.164	1.920	2.125	1.997	2.201	2.001	2.200	321.20	7342.0
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.001	2.162	2.067	2.124	1.997	2.199	2.004	2.199	312.60	8892.4
p1--GP--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	523.84	10459.1
p1--GP--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	524.88	11970.4
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.921	2.125	1.993	2.129	1.996	2.128	829.14	8590.9
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.092	1.991	2.096	2.067	2.124	1.993	2.128	1.998	2.126	807.62	9577.6
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.768	1.927	1.953	1.877	1.777	1.880	711.22	11085.8

			Fit $N = 10$–30						Fit $N = 20$–30							
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
p1--GP--p2	tetra unstr	0.1	1.612	1.909	1.920	1.897	1.790	1.900	1.884	1.962	1.955	1.921	1.849	1.923	907.71	12474.6
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.921	2.125	1.993	2.126	1.996	2.124	1859.7	12031.6
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.067	2.124	1.993	2.124	1.998	2.123	1756.2	12472.6
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.873	1.768	1.927	1.951	1.889	1.775	1.892	1090.2	12361.6
p1--GP--p3	tetra unstr	0.1	1.612	1.909	1.916	1.900	1.786	1.903	1.884	1.962	1.953	1.927	1.847	1.929	1269.1	13565.9
p2--na--NO	hexa	0.0	1.947	3.025	1.942	1.942	1.946	1.946	2.001	2.766	2.000	2.000	2.003	2.003	262.11	25852.2
p2--na--NO	hexa	0.1	1.946	3.053	1.941	1.941	1.946	1.946	2.000	2.807	1.999	1.999	2.002	2.002	239.10	21777.7
p2--na--NO	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	749.44	18258.0
p2--na--NO	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	925.06	21620.3
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.993	2.168	2.001	2.183	2.035	2.884	2.031	2.209	2.036	2.145	347.00	13057.2
p2--CCp1--p1	hexa	0.1	2.008	3.133	1.997	2.167	2.006	2.192	2.037	2.925	2.033	2.209	2.038	2.152	340.08	15812.1
p2--CCp1--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1129.9	19181.2
p2--CCp1--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1164.1	21994.5
p2--CCp1--p2	hexa	0.0	2.040	3.141	2.024	2.098	2.036	2.087	2.059	2.936	2.052	2.131	2.060	2.062	444.36	13775.5
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.033	2.098	2.046	2.088	2.064	2.974	2.056	2.131	2.065	2.062	452.39	16460.4
p2--CCp1--p2	tetra unstr	0.0	2.105	2.705	1.863	1.971	1.864	1.960	2.382	2.719	2.113	1.982	2.122	1.974	1185.2	19827.8
p2--CCp1--p2	tetra unstr	0.1	2.348	2.824	2.014	1.971	2.018	1.970	2.721	2.901	2.364	1.982	2.374	1.986	1139.6	22504.4
p2--CCp1--p3	hexa	0.0	2.040	3.141	2.024	2.095	2.036	2.085	2.059	2.936	2.052	2.127	2.060	2.061	558.69	16265.4
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.033	2.095	2.046	2.086	2.064	2.974	2.056	2.127	2.065	2.061	527.65	17829.2
p2--CCp1--p3	tetra unstr	0.0	2.105	2.705	1.863	1.963	1.864	1.958	2.382	2.719	2.113	1.981	2.122	1.973	1232.7	21091.2
p2--CCp1--p3	tetra unstr	0.1	2.347	2.824	2.014	1.963	2.018	1.968	2.721	2.901	2.364	1.981	2.374	1.984	1258.2	23563.2
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.998	3.170	2.002	2.075	2.035	2.884	2.035	3.091	2.037	2.071	458.92	13080.6
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.003	3.171	2.007	2.094	2.037	2.925	2.037	3.091	2.039	2.079	456.09	15807.8
p2--CCp2--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1770.4	22550.0
p2--CCp2--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1534.1	24551.7
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.034	3.246	2.037	2.170	2.059	2.936	2.059	3.146	2.061	2.125	522.93	14047.8
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.044	3.246	2.047	2.206	2.064	2.975	2.064	3.146	2.066	2.143	550.47	16461.9
p2--CCp2--p2	tetra unstr	0.0	2.105	2.705	1.863	2.944	1.864	1.932	2.384	2.720	2.116	2.972	2.122	2.121	1788.3	23183.3
p2--CCp2--p2	tetra unstr	0.1	2.350	2.825	2.016	2.945	2.018	2.104	2.727	2.903	2.369	2.972	2.376	2.408	1509.6	25124.1
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.034	3.251	2.037	2.173	2.059	2.936	2.059	3.149	2.061	2.126	678.82	16916.0
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.044	3.251	2.047	2.209	2.064	2.975	2.064	3.149	2.066	2.144	636.70	17987.3
p2--CCp2--p3	tetra unstr	0.0	2.105	2.705	1.863	2.958	1.864	1.943	2.384	2.720	2.116	2.980	2.122	2.134	1767.1	24450.4
p2--CCp2--p3	tetra unstr	0.1	2.350	2.825	2.016	2.959	2.018	2.111	2.727	2.903	2.369	2.981	2.376	2.414	1630.4	26269.5
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.997	4.087	2.002	2.020	2.035	2.884	2.035	4.142	2.037	2.056	658.77	14040.1
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.002	4.082	2.007	2.026	2.037	2.925	2.037	4.137	2.039	2.058	633.51	15815.6
p2--CCp3--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	2738.8	31317.0
p2--CCp3--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	2433.5	32698.6
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.033	4.022	2.037	2.072	2.060	2.936	2.059	4.109	2.061	2.100	755.42	15529.7
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.043	4.017	2.047	2.084	2.064	2.975	2.064	4.103	2.066	2.107	757.51	16648.0
p2--CCp3--p2	tetra unstr	0.0	2.105	2.705	1.863	3.893	1.864	1.897	2.384	2.720	2.116	3.841	2.123	2.129	2923.3	31951.7
p2--CCp3--p2	tetra unstr	0.1	2.350	2.825	2.016	3.911	2.018	2.048	2.727	2.903	2.369	3.890	2.376	2.392	2520.1	33327.7
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.033	4.014	2.037	2.073	2.060	2.936	2.059	4.110	2.061	2.101	996.82	18508.9
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.043	4.010	2.047	2.085	2.064	2.975	2.064	4.104	2.066	2.107	983.63	18888.2
p2--CCp3--p3	tetra unstr	0.0	2.105	2.705	1.863	3.888	1.864	1.929	2.384	2.720	2.116	3.831	2.123	2.148	2915.4	33215.1
p2--CCp3--p3	tetra unstr	0.1	2.350	2.825	2.016	3.907	2.018	2.074	2.727	2.903	2.369	3.884	2.376	2.406	2557.2	34591.3
p2--GP--p1	hexa	0.0	2.004	3.107	2.011	3.133	2.844	3.122	2.035	2.884	2.004	3.004	2.447	2.998	475.17	26058.0
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.886	3.131	2.037	2.925	2.004	3.021	2.473	3.014	434.84	21987.3
p2--GP--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	923.33	18403.9
p2--GP--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1077.7	21759.6
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.897	3.215	2.060	2.936	2.001	3.092	2.431	3.082	1125.6	26708.6
p2--GP--p2	hexa	0.1	2.051	3.166	2.004	3.236	2.920	3.221	2.064	2.975	2.001	3.105	2.439	3.094	919.07	22635.8
p2--GP--p2	tetra unstr	0.0	2.105	2.705	1.928	2.794	2.100	2.795	2.384	2.720	1.960	2.779	2.102	2.783	1131.3	19048.7
p2--GP--p2	tetra unstr	0.1	2.350	2.825	1.928	2.831	2.146	2.833	2.727	2.903	1.960	2.862	2.167	2.866	1319.3	22134.4
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.896	3.220	2.060	2.936	2.000	3.095	2.430	3.085	2346.5	28002.5
p2--GP--p3	hexa	0.1	2.051	3.166	2.003	3.241	2.919	3.226	2.064	2.975	2.000	3.108	2.438	3.097	1865.3	24076.1
p2--GP--p3	tetra unstr	0.0	2.105	2.705	1.924	2.801	2.082	2.802	2.384	2.720	1.958	2.768	2.089	2.773	1507.3	20304.2
p2--GP--p3	tetra unstr	0.1	2.350	2.825	1.924	2.846	2.128	2.847	2.727	2.903	1.958	2.864	2.153	2.868	1828.5	23037.4
p3--na--NO	hexa	0.0	2.313	3.204	2.303	2.303	2.307	2.307	2.193	2.830	2.189	2.189	2.192	2.192	816.96	68118.1
p3--na--NO	hexa	0.1	2.278	3.207	2.269	2.269	2.273	2.273	2.169	2.827	2.166	2.166	2.168	2.168	892.18	72598.4
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	1833.8	45160.4
p3--na--NO	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2087.5	53471.6
p3--CCp1--p1	hexa	0.0	2.758	3.637	2.675	2.166	2.727	2.263	2.547	3.524	2.476	2.207	2.525	2.199	1199.2	44824.7
p3--CCp1--p1	hexa	0.1	2.713	3.646	2.635	2.166	2.687	2.261	2.497	3.532	2.433	2.207	2.481	2.197	1167.3	45948.4
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2843.6	45874.6
p3--CCp1--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2627.8	53130.2
p3--CCp1--p2	hexa	0.0	3.096	3.790	2.987	2.097	3.017	2.094	2.864	3.829	2.773	2.130	2.782	2.063	1176.3	45473.1
p3--CCp1--p2	hexa	0.1	3.064	3.800	2.965	2.097	2.994	2.093	2.807	3.842	2.734	2.130	2.739	2.063	1260.0	46595.7
p3--CCp1--p2	tetra unstr	0.0	2.979	3.699	2.857	1.970	2.857	1.968	3.035	3.633	2.949	1.982	2.960	1.961	3055.9	46211.8
p3--CCp1--p2	tetra unstr	0.1	3.071	3.771	2.960	1.971	2.963	1.968	3.012	3.707	3.038	1.982	3.053	1.960	2685.8	53376.4
p3--CCp1--p3	hexa	0.0	3.096	3.790	2.987	2.094	3.017	2.092	2.864	3.829	2.773	2.127	2.782	2.062	1264.3	46766.6
p3--CCp1--p3	hexa	0.1	3.064	3.800	2.965	2.094	2.994	2.091	2.807	3.842	2.734	2.127	2.739	2.062	1457.9	47888.7
p3--CCp1--p3	tetra unstr	0.0	2.978	3.699	2.856	1.963	2.857	1.966	3.034	3.633	2.949	1.981	2.960	1.960	2976.5	46930.7
p3--CCp1--p3	tetra unstr	0.1	3.071	3.772	2.959	1.963	2.963	1.966	3.011	3.707	3.037	1.981	3.053	1.960	2727.5	53953.5
p3--CCp2--p1	hexa	0.0	2.782	3.644	2.745	3.172	2.752	2.948	2.574	3.538	2.546	3.092	2.553	2.790	1126.2	44825.4
p3--CCp2--p1	hexa	0.1	2.739	3.653	2.708	3.172	2.715	2.943	2.524	3.546	2.503	3.092	2.509	2.778	1332.6	45949.2
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	3195.5	47169.4
p3--CCp2--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2955.2	53666.3
p3--CCp2--p2	hexa	0.0	3.335	3.812	3.202	3.247	3.220	3.232	3.264	3.882	3.065	3.147	3.095	3.171	1216.2	45473.7
p3--CCp2--p2	hexa	0.1	3.352	3.823	3.223	3.247	3.241	3.244	3.280	3.896	3.083	3.147	3.114	3.186	1389.5	46594.7
p3--CCp2--p2	tetra unstr	0.0	3.198	3.777	2.984	2.942	2.981	2.996	3.463	3.810	3.182	2.970	3.187	3.073	3139.1	47660.2
p3--CCp2--p2	tetra unstr	0.1	3.452	3.863	3.162	2.942	3.161	3.045	3.775	3.921	3.433	2.970	3.439	3.121	3022.9	54043.4
p3--CCp2--p3	hexa	0.0	3.335	3.812	3.202	3.253	3.220	3.231	3.264	3.882	3.065	3.150	3.095	3.171	1407.3	46767.6

			Fit $N = 10$–30						Fit $N = 20$–30							
`Vm-states-Ion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]
`p3--CCp2--p3`	hexa	0.1	3.352	3.823	3.223	3.253	3.241	3.243	3.279	3.896	3.083	3.150	3.114	3.185	1572.6	47888.4
`p3--CCp2--p3`	tetra unstr	0.0	3.198	3.777	2.984	2.957	2.981	3.007	3.463	3.810	3.182	2.980	3.187	3.076	3223.3	48661.1
`p3--CCp2--p3`	tetra unstr	0.1	3.451	3.863	3.162	2.957	3.160	3.055	3.776	3.921	3.433	2.980	3.439	3.123	3093.8	54824.9
`p3--CCp3--p1`	hexa	0.0	2.781	3.644	2.742	4.034	2.751	2.764	2.573	3.538	2.546	4.068	2.552	2.578	1388.8	44828.8
`p3--CCp3--p1`	hexa	0.1	2.737	3.653	2.705	4.034	2.713	2.735	2.523	3.546	2.502	4.068	2.508	2.535	1433.8	45950.0
`p3--CCp3--p1`	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	3971.2	53601.4
`p3--CCp3--p1`	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	3675.5	58844.0
`p3--CCp3--p2`	hexa	0.0	3.332	3.813	3.194	3.978	3.216	3.106	3.260	3.882	3.069	4.042	3.091	3.018	1500.8	45473.3
`p3--CCp3--p2`	hexa	0.1	3.349	3.823	3.214	3.978	3.236	3.135	3.275	3.896	3.087	4.042	3.109	3.036	1588.8	46598.1
`p3--CCp3--p2`	tetra unstr	0.0	3.198	3.777	2.985	3.943	2.981	3.010	3.463	3.810	3.182	3.960	3.187	3.210	3382.6	54155.5
`p3--CCp3--p2`	tetra unstr	0.1	3.452	3.863	3.163	3.943	3.161	3.182	3.775	3.921	3.433	3.960	3.438	3.453	3690.8	59374.8
`p3--CCp3--p3`	hexa	0.0	3.332	3.813	3.194	3.971	3.216	3.106	3.260	3.882	3.069	4.044	3.091	3.018	1721.3	46770.7
`p3--CCp3--p3`	hexa	0.1	3.349	3.823	3.214	3.971	3.236	3.136	3.275	3.896	3.087	4.044	3.109	3.037	1751.7	47890.0
`p3--CCp3--p3`	tetra unstr	0.0	3.197	3.777	2.984	3.944	2.980	3.056	3.462	3.810	3.182	3.962	3.186	3.240	3477.8	55286.0
`p3--CCp3--p3`	tetra unstr	0.1	3.451	3.863	3.162	3.944	3.160	3.224	3.775	3.921	3.433	3.962	3.438	3.483	3867.2	60459.8
`p3--GP--p1`	hexa	0.0	2.781	3.644	2.003	3.789	2.716	3.807	2.573	3.538	2.000	3.679	2.354	3.683	1453.9	77276.2
`p3--GP--p1`	hexa	0.1	2.738	3.653	2.003	3.796	2.706	3.814	2.523	3.546	2.000	3.687	2.345	3.692	1107.5	68550.1
`p3--GP--p1`	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2698.7	45294.4
`p3--GP--p1`	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2581.7	53555.5
`p3--GP--p2`	hexa	0.0	3.332	3.813	1.994	3.892	2.454	3.910	3.260	3.882	1.996	3.955	2.171	3.965	2161.7	77923.7
`p3--GP--p2`	hexa	0.1	3.350	3.823	1.994	3.896	2.447	3.914	3.275	3.896	1.996	3.961	2.167	3.971	1552.4	69194.7
`p3--GP--p2`	tetra unstr	0.0	3.198	3.777	1.927	3.839	2.032	3.841	3.463	3.810	1.961	3.829	1.977	3.835	2892.2	45571.9
`p3--GP--p2`	tetra unstr	0.1	3.452	3.863	1.927	3.878	2.028	3.879	3.775	3.921	1.961	3.893	1.974	3.898	2807.7	53808.9
`p3--GP--p3`	hexa	0.0	3.332	3.813	1.994	3.884	2.452	3.904	3.260	3.882	1.995	3.956	2.170	3.967	3441.1	79067.7
`p3--GP--p3`	hexa	0.1	3.349	3.823	1.993	3.889	2.446	3.908	3.275	3.896	1.995	3.962	2.167	3.973	2516.6	70489.4
`p3--GP--p3`	tetra unstr	0.0	3.197	3.777	1.923	3.838	2.021	3.840	3.462	3.810	1.958	3.826	1.974	3.833	2931.1	46322.2
`p3--GP--p3`	tetra unstr	0.1	3.451	3.863	1.923	3.879	2.017	3.880	3.775	3.921	1.958	3.893	1.971	3.900	3400.9	54328.9

12 Notes

The current result blocks are intentionally restricted to `crankNicolson` and the `consistent` mass matrix, so time-scheme or mass-matrix comparisons should be read from older auxiliary data rather than from these tables. The reconstructed quantities V_m^G, u_1^G, and u_2^G are quadrature-point/reconstructed diagnostics; the cell-centred fields are the values used in the main finite-volume update and in the L_2 convergence tables.